RECOVERY OF MATES ON MARTÍN NORMS III: FINITE BASE SUPPORT AND TRUNCATION

Abstract. This is the third part of a study of the density of norm-attaining operators from Martín's renormings of c_0 into ℓ_2^2. Part I reduced density, for a designed operator and N blocks, to an approximation statement (Lemma Z) about first rows, and Part II split Lemma Z into a statement about first rows with finite base support and an approximation statement (LSC-trunc) for infinite base support; at finite base support it left open the rows of the residual (C*). Here we settle the finite-support half for a new designed operator. We show that the lever hypothesis of Part II is a property of the first row and not of the design, and replace it by a choice of *generic* weights: if the weights are algebraically independent over the field generated by the other data of the design, then the threshold value is never a local-minimum value of the floor statuses on an exact zero set, which gives status coherence with design constants. We construct violation-tolerant d-consistent engineered approximants, whose per-piece bounds have radius proportional to the scale with window-independent constants (for a fixed lower bound of the near-threshold gaps), and average at the norm-attaining approximant itself. We then fix one operator T_{fin}, independent of N, in two versions: with option (a) it has norm $c_1 < 1$ and satisfies all other conclusions of Martín's Lemma B, so that it defines Martín-type norms; with option (b) it has norm one and satisfies the conclusion of Martín's Lemma B. We prove Master Theorem VI: with option (a), for the norms p_N built with T_{fin}, every first row with finite base support is recoverable, for every $N \geq 1$ (with option (b): for $N = 1$, and for $N \geq 2$ at the rows with $\varrho_1(f) \neq 1$). Consequently, for T_{fin} with option (a) and every N (with option (b): for $N = 1$), density of $\mathrm{NA}((c_0, p_N), \ell_2^2)$ is equivalent to $(\text{LSC-trunc})_N$. For infinite base support we prove a lowering transfer for contact truncations, an engineered bound and a windowed recovery theorem at arbitrary base support, and, for a variant $T_{\mathrm{fin}}^{\infty}$ of T_{fin} (option (a), over the doubly exponential base of Part II, with finitely many additional design quantities per stage), cushion domination at deep-raised rows and uniform solvability of the d-consistency system, and obtain a transport theorem for $T_{\mathrm{fin}}^{\infty}$: $(\text{LSC-trunc})_N$ holds at every triple (f, g, ρ) that satisfies a joint push condition and an A-regularity condition at infinitely many main stages. Triples not covered satisfy, at almost all main stages, one of two explicit conditions, (W*) or (A*/U*). The density of $\mathrm{NA}((c_0, p_N), \ell_2^2)$, for any N, and of $\mathrm{NA}((c_0, p), \ell_2^2)$ remains open.

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1. Introduction

1.1. **The problem.** Martín [8] constructed renormings p of c_0 whose norm-attaining functionals form an algebraically trivial set. The construction starts from the canonical base norm q of c_0 (a smoothing of the supremum norm in the

sense of Debs, Godefroy and Saint-Raymond [2]) and an injective operator T of norm one given by his Lemma B, which rests on [6, Proposition 2.8]; we call such an operator *admissible* [I, Definition 1.3] ($q^*(Te_{n,m}) \leq 1$ with equality for some (n,m), injectivity, $Y \cap c_{00} = \{0\}$, and density of the normalized vectors in every block). For a set I of blocks, p_N denotes the norm built with $I = \{1,\dots,N\}$ and p the norm built with $I = \mathbb{N}$. The question studied in Part I [I], Part II [II] and here is whether $\mathrm{NA}((c_0,p_N),\ell_2^2)$ and $\mathrm{NA}((c_0,p),\ell_2^2)$ are norm dense in the corresponding spaces of operators. It is a test case for Lindenstrauss' property B [7] of the two-dimensional Hilbert space (compare [4, 5]): a negative answer for one N and one admissible operator would give a two-dimensional range space without property B [I, Section 9].

An operator into ℓ_2^2 is a pair (f,g) of functionals. For a *first row* $f \in S_{p^*}$ the *mate fibre* is $\mathcal{C}(f) = \{g : (f,g) \text{ is contractive}\}$, and the *recoverable set* is

$$\mathcal{R}(p) := \{f \in S_{p^*} : \ (f,g) \in \mathrm{cl}\,\mathrm{NA}((c_0,p),\ell_2^2) \text{ for every } g \in \mathcal{C}(f)\}$$

[I, Definition 1.15]. Density holds if and only if $\mathcal{R}(p) = S_{p^*}$ [I, Proposition 1.13], and density for p_N for infinitely many N implies density for p [I, Lemma 1.5].

1.2. **What Parts I and II proved.** Part I proved the transport of finite certificates along sequences of first rows and an averaging theorem over scales [I, Theorems 3.8 and 4.1]; the resonance obstruction, an admissible operator and a first row at which the intrinsically recoverable mates lie on one line while $\mathcal{C}(f)$ is infinite dimensional [I, Theorem 6.6]; engineered recovery at first rows with finite base support [I, Theorem 7.18]; and, for signature-ladder operators, recovery under signature room [I, Theorem 8.18] and the reduction of density to an approximation statement [I, Theorem 8.20, Problem 8.21]:

> *Lemma Z.* For every $f \in S_{p_N^*}$, $g \in \mathcal{C}(f)$, $\rho < 1$ and $\varepsilon > 0$ there is a first row f' with finite base support and signature room such that $p_N^*(f'-f) < \varepsilon$ and $\mathrm{dist}(\rho g, \mathcal{C}(f')) < \varepsilon$.

Part II fixed one admissible operator T_{final}, independent of N, with a doubly exponentially decaying diagonal base, bounded-gap signature sets and windows split into pigeonhole sub-windows at which every rate object of a first row is either tiny or robust ([II, Definition 3.5, Theorem 3.12, Theorem 3.19]). For T_{final} it proved that a first row with finite base support which is not recoverable is a (C*) row, at which a near-exact coherent shift resonance occurs at every clean sub-window of every large level [II, Theorem 5.15, Corollary 5.17], and, for a modified design with absorber targets, recovery whenever at infinitely many main stages some activity class has no active shift or admits a $\varkappa$-neutral lever ($\mathrm{KN}_{w,a}$) [II, Theorem 5.68]. It split Lemma Z into its two halves [II, Proposition 6.48]: density of $\mathrm{NA}((c_0,p_N),\ell_2^2)$ is equivalent to recovery at first rows with finite base support together with

> (LSC-trunc)$_N$: for every $f \in S_{p_N^*}$, $g \in \mathcal{C}(f)$, $\rho \in (0,1)$ and $\varepsilon > 0$ there is $f' \in S_{p_N^*}$ with finite base support such that $p_N^*(f'-f) < \varepsilon$ and $\mathrm{dist}(\rho g, \mathcal{C}(f')) < \varepsilon$.

Lemma Z, and density for every N and for p, were left open in Part II; the open part consisted of the (C*) rows at finite base support and of (LSC-trunc) for the rows with infinite base support not covered by [II, Section 6].

1.3. **Main results: finite base support.** *Status coherence by generic weights (Section 2).* Inside one block the exact shifted coarse system and the relative positions of all switching carriers depend on the block only through the ratios $u_l(\hat{z})/\varkappa$ (Lemma 2.1), so peak pushes alone cannot achieve exactness and threshold protection independently; Part II supplied the second lever by the hypothesis $(\mathrm{KN}_{w,a})$. We show that (KN) is a property of the first row, not of the design (Lemma 2.6, Remark 2.7), and that a second lever is nevertheless always available. Writing the inward rows of the coarse cone at the *floor* of the relative positions removes the weights of the near-threshold switching carriers from the exact zero set of the tiny minors and from the floor functions (Lemma 2.17). If all weights of the design are algebraically independent over the countable field generated by its other data, then the threshold value 1 is never a local-minimum value of a floor status on an exact zero set (Lemmas 2.20 and 2.22); the proof uses the Tarski–Seidenberg principle [11, 10], the semialgebraic Sard theorem [1] and Puiseux expansions [12]. Compactness and the Łojasiewicz inequality turn this into status coherence with design constants (Lemma 2.24), and Proposition 2.32 is the resulting exactification with threshold protection, without (KN).

The violation-tolerant d-consistent engineered approximant (Section 3). To average over a window of scales at a row near the target row, one needs per-piece bounds with radius proportional to the scale and window-independent constants. The obstruction recorded in [II, Remark 2.31(c)], the junction mismatch between the θ-decomposition and the $\pm$-decompositions of a piece (Proposition 3.7), is removed exactly: at the approximant the normalized values of the finitely many switching carriers are re-tuned to their values at the base row by small levers (d-consistency, Lemma 3.10, proved with the Sherman–Morrison formula and the Poincaré–Miranda theorem [9]). This gives the uniform engineered bound (Theorem 3.15), windowed recovery at the norm-attaining approximant itself (Theorem 3.17), the threshold comparison (Proposition 3.21), and recovery of every activity class under (KN) (Theorem 3.26).

The operator T_{fin} and Master Theorem VI (Section 4). T_{fin} is the design of [II, Definition 5.48] on the ladder of [II, Definition 3.5] with the changes listed in Remark 4.4, chiefly: the base is $\mu_s = 2^{-s^2-1}$; there are no fresh-dominated stages; the weights are generic; the rate scheme and the design factor contain the objects of Sections 2 and 3; and the window factor $Q(w)$ does not contain Part II's $\bar{D}(w)^3$ and has exponent $\omega(l) + l^2 + 20$ (Definitions 4.1–4.3). The recursion is well founded and does not refer to N (Lemma 4.6), and T_{fin} satisfies (T-b)–(T-d) of [I, Definition 1.3]; with option (b) it is admissible and satisfies the conclusion of Martín's Lemma B; with option (a) its norm is $c_1 < 1$ (the only departure from Lemma B), so the norms it defines are Martín-type norms (Theorem 4.7). The results of Parts I and II used here are transferred to T_{fin} through an explicit list of interface properties (Lemmas 4.9 and 4.10, Proposition 4.11). The recovery follows a single route for every kind of activity class (route V, Subsection 4.3): the companion is the exactification of Proposition 2.32, the approximant is the d-consistent engineered approximant of Section 3, at which the block scalars are frozen up to the weight of the fine carriers (Lemma 4.17), and the averaging is Theorem 3.17. The main result of this part is the following.

> *Master Theorem VI* (Theorem 4.25). Let p_N be the norm on c_0 built with T_{fin} and the blocks $\{1, \dots, N\}$. With option (a), for every $N \geq 1$ every

$f \in S_{p_N^*}$ with finite base support lies in $\mathcal{R}(p_N)$. With option (b), the same holds for $N = 1$, and for $N \geq 2$ at every f with finite base support and $\varrho_1(f) \neq 1$, where $\varrho_1(f)$ is the relative position of the first stage.

The operator is fixed before N and before the row; this is legitimate because the coincidence sets of the generic-weight argument contain no datum of a row (Lemma 4.5). Combined with the splitting of [II, Proposition 6.48] this gives:

> *Density criterion* (Theorem 4.26, Corollary 4.27). Let T_{fin} be taken with option (a). For every $N \geq 1$, $\mathrm{NA}((c_0, p_N), \ell_2^2)$ is dense if and only if $(\text{LSC-trunc})_N$ holds. If $(\text{LSC-trunc})_N$ holds for infinitely many N, then $\mathrm{NA}((c_0, p), \ell_2^2)$ is dense for the Martín-type norm p built with T_{fin}.

The same equivalence holds with option (b) for $N = 1$. Thus, for this operator, the density problem is reduced exactly to the approximation of arbitrary first rows by first rows with finite base support while keeping the mate approximately in the fibre.

1.4. **Main results: infinite base support.** Section 5 treats (LSC-trunc). Truncating the base support to finitely many coordinates costs nothing at finitely many scales fixed in advance (the lowering transfer, Lemma 5.2 and Corollary 5.4); the factor 2 in the lowering transfer is sharp, and the cushion loss can be linear in the scale r rather than quadratic (Remark 5.3), so truncation is free only at finitely many scales fixed in advance and must be built into the approximant rather than performed afterwards. Inside the violation-tolerant engineered approximant the truncated support becomes a set of contacts whose two-sided use is a tolerated violation; this gives an engineered bound and a windowed recovery theorem with raise transfer at arbitrary base support (Theorems 5.10 and 5.12), so that (LSC-trunc) at (f, g, ρ) follows from window families with side cushion domination and exact d-consistency at base rows of arbitrary support. For the design $T_{\mathrm{fin}}^{\infty}$ of Definition 5.42 (T_{fin} with option (a), over the doubly exponential diagonal base of Part II, with finitely many additional quantities per stage), the route-V data at deep-raised rows are cushion dominated with an absolute constant $C_+ = 5$ on the whole support of the companion (Proposition 5.19); after a cheap preparation of the base row the d-consistency system is uniformly solvable at every row (Theorem 5.31); and Step A of the companion construction is solvable whenever a cone condition (A-regularity, Definition 5.36) holds, while the obstructions to it have a rigid sign structure (Lemma 5.37, Theorem 5.39, Corollary 5.40). The result is:

> *Transport theorem* (Theorem 5.51). Let the design be $T_{\mathrm{fin}}^{\infty}$, $N \geq 1$, $f \in S_{p_N^*}$ with finite or infinite base support, $g \in \mathcal{C}(f)$ and $\rho \in (0, 1)$. If for infinitely many main stages L there are a clean sub-window w of L and an activity class a with at least $n(w)/D_{\mathrm{cl}}(w)$ scales such that the push condition (H1*) holds and Step A is A-regular (H2) at (L, w, a), then $(f, \rho g)$ is a limit of norm-attaining pairs whose first rows have finite base support; in particular $(\text{LSC-trunc})_N$ holds at (f, g, ρ).

Explicit sufficient conditions are given in Corollary 5.52; for instance, every first row whose base support meets no signature set beyond $s_{\max}(L)$, for infinitely many main stages L, lies in $\mathcal{R}(p_N)$ for the norms p_N built with $T_{\mathrm{fin}}^{\infty}$.

1.5. **What remains open.** *The density of* $\mathrm{NA}((c_0, p_N), \ell_2^2)$*, for every* $N \geq 1$*, and of* $\mathrm{NA}((c_0, p), \ell_2^2)$ *remains open, for* T_{fin}*, for* $T_{\mathrm{fin}}^{\infty}$ *and for every admissible operator,*

in particular for the operators of Martín's Lemma B. For T_{fin} with option (a) the problem is equivalent to (LSC-trunc)$_N$ (Theorem 4.26), and for $T_{\mathrm{fin}}^{\infty}$ the triples (f, g, ρ) at which (LSC-trunc)$_N$ is not established are described by Proposition 5.55: if the hypothesis of the transport theorem fails, then at every clean sub-window of all but finitely many main stages and for every admissible class one of the following holds.

- (W*) The push condition (H1*) fails: the active weak peaks of the blocks of kind (T) cannot be pushed jointly below the threshold by admissible moves of the allowed size.
- (A*/U*) (H1*) holds, and for every pre-placement there is a near-certificate of Step A with a robustly negative common mode γ; it is of type (A*) if $\gamma \le -\gamma_0$, and of type (U*) if $-\gamma_0 < \gamma \le -c_{\mathrm{A}}\nu_1/(8H(L))$, in which case the coefficients of the carriers whose near coordinates are all thick carry ℓ_2-mass at least $c_{\mathrm{A}}/2$. Here c_{A} is the Step-A constant of Theorem 5.39 and $\gamma_0 = c_{\mathrm{A}}/(4\|h\|_2)$ is the constant of Definition 5.54; the type may depend on the pre-placement.

In the case (A*/U*) the thick part of the base support at design depth agrees, up to an error $\varepsilon\,\mathrm{Des}(L)/|\varkappa_\Psi|$ (ε the size of the certificate), with a fixed multiple $\Psi_y/\varkappa_\Psi$ of the carrier combination of the certificate, with a robustly negative factor $\varkappa_\Psi$ (Proposition 5.55(i)). These are necessary conditions for not being covered, not characterizations; it is not known whether rows satisfying them at all large main stages exist, nor whether such rows carry a mate that is not recovered (Remark 5.56). At finite base support one case remains open for option (b): $N \ge 2$ and $\varrho_1(f) = 1$ (Remark 4.28(b)). Master Theorem VI is proved for the designed operator only, not for an arbitrary admissible operator. No counterexample to density is known, and none of the verifications produced one.

1.6. **Verification status.** The results of Sections 2 and 3 on which Master Theorem VI rests were checked by one referee each and then re-derived in a second, independent verification organized in four clusters (the route-V companion; the generic-weight chain; the engineered approximant and the averaging; the transfer to T_{fin} and the window bookkeeping); the combination in Section 4 was checked by two referees and re-derived in the same verification. The corrections introduced in the second verification (the common pull window, the transfer interface of Lemma 4.10 and Proposition 4.11, the s-free coefficients, the gap criterion of Lemma 3.4(a), the bounds of Lemmas 3.10 and 3.14(g)) were each made and checked once. Proposition 3.7, Lemma 3.23, Corollary 3.27 and Lemma 4.21 were checked by one referee. The proof of Master Theorem VI given here uses Lemma 4.21; a version that does not use it is described in Remark 4.29. The present text was then read by three further referees, whose corrections are incorporated; these corrections have not been through a second independent verification. The results of Section 5 were each checked by one adversarial referee, and the corrections found there are incorporated; they have not yet been through a second independent verification. They are proved for the design $T_{\mathrm{fin}}^{\infty}$; they use the route-V companion and the window count of Section 4 for that design, that is, the transfer of Section 4 from the base μ to the base $\mu^\star$, which rests on the base independence of the window theorems of Part II and has not been re-verified line by line (Remark 5.57). Statements of earlier drafts that were found to be false or not proved as stated are not used: the master theorem without a lever in the design of an earlier draft (see [II, Remark 5.70]), the

two-sidedness of a bank placed with an offset at the row where the d-consistency system is solved (Lemma 5.25), the sufficiency of a push condition for each weak peak separately, characterizations of the non-covered rows by carrier types (Remark 5.56(e)), and levers for tuned zero-value absorbers (an earlier Lemma LV(d); see Remarks 4.13 and 3.28); the exact route with zero-value absorbers of [II, Theorem 5.68] does not transfer to T_{fin} and is not used (Remark 4.12). Finite numerical models were used as sanity checks of several identities; they are evidence only and are reported as such.

1.7. **Organization and conventions.** Section 2 proves status coherence by generic weights, Section 3 constructs the violation-tolerant d-consistent engineered approximant, Section 4 defines T_{fin} and proves Master Theorem VI and the density criterion, Section 5 treats infinite base support, and Section 6 summarizes the status and states the open problems.

The notation of Parts I and II is used without comment, in particular the forced data $(\xi, q_0, a, w, F, z, \hat{z}, e, \nu)$ of a first row [I, Proposition 1.9], two-piece data [I, Definition 7.7], the signature-ladder notation of [I, Section 8], and the windows, rate objects and clean sub-windows of [II, Section 3]. References of the form [I, Theorem 8.20] and [II, Proposition 6.48] are to Part I [I] and Part II [II]. Constants written C_f depend on the first row and on N (and on the mate where said) but not on the level, the sub-window or the scale; "design constants" are powers of the design factor $\mathrm{Des}(L)$. Every lemma, proposition, theorem and corollary is given with a proof; material that is only sketched or heuristic appears only in remarks that say so explicitly.

2. Generic weights and status coherence

This section removes the lever hypothesis ($\mathrm{KN}_{w,a}$) of [II, Definition 5.55] from the exactification with threshold protection of [II, Proposition 5.56]. The difficulty, isolated in [II, Proposition 5.46] and [II, Remark 5.47], is that inside one block the exact shifted coarse system and the relative positions of all switching carriers depend on the block only through the ratios $u_l(\hat{z})/\varkappa$; peak pushes act on all of them in the same way, so exactness and threshold protection cannot be achieved independently by peak pushes (Subsection 2.2). The hypothesis (KN) supplied a second, $\varkappa$-neutral lever. We show that (KN) is a property of the first row and not of the design (Lemma 2.6), and that a second lever is nevertheless always available: after the inward rows of the coarse cone are written at the *floor* of the relative positions (Definition 2.13), the exact zero set of the tiny minors and the floor functions contain no weight of the near-threshold switching carriers (Lemma 2.17). These weights then enter only as denominators of the floor statuses, and if all weights of the design are algebraically independent over the countable field of its other data, the threshold value 1 is never a local-minimum value of the floor statuses on an exact zero set (Lemmas 2.20 and 2.22). Compactness, the Łojasiewicz inequality and Puiseux expansions turn this into status coherence with design constants (Lemma 2.24), and Proposition 2.32 is the resulting exactification.

The section incorporates the corrections found in the two independent verifications of this material: the floor convention for active weak peaks (Lemma 2.4(b) and Definition 2.8), the weight-free compact region (Definition 2.11), the exact hold of every non-active coarse strict non-peak of a (T)-block (Proposition 2.32(b) and

Remark 2.33), the common pull window of the realization (Lemma 2.29), the Lipschitz constant of the floor statuses in the design thresholds (Definition 2.27), and the s-free aggregated coefficients of the rate objects (Lemma 2.17(c)).

2.1. **Setting and notation.**

Blocks. We use the block notation of [I, Lemma 1.7] and [II, Subsection 2.2]. For a first row f' and a block m let $\zeta = \hat{\zeta}_m = R^{**}_m \hat{z}'$, $|\zeta| = |\zeta|_m$, let $\bar{\vartheta} = \bar{\vartheta}(\zeta)$ be the zero of the threshold equation [II, Lemma 2.6], $M = \bar{\vartheta}/(\bar{\vartheta} + |\zeta|)$, $C = |\zeta|/(\bar{\vartheta} + |\zeta|)$ (so $M + C = 1$ and $\bar{\vartheta} = |\zeta| M/C$), $\mathsf{v}_k = |\zeta(k)|/\Phi(k)^2$ and $\varrho_k = \mathsf{v}_k/\bar{\vartheta}$ the relative position of the coordinate k; $P = \{k : \varrho_k \geq 1\}$ is the set of peaks and Q its complement, the strict non-peaks. We index carriers by their stages l, with block $m(l)$, coordinate $k(l)$, weight $\Phi_l := \Phi_{m(l)}(k(l)) = 2^{-m(l)-k(l)} c_l$, $\lambda_l := m(l)\Phi_l$ and normalized target vector u_l; then $|\zeta(k(l))| = \lambda_l |u_l(\hat{z}')|$ and $\mathsf{v}_{k(l)}\Phi_l = m(l)|u_l(\hat{z}')|$, and we write $\varrho_l := \varrho_{k(l)}$. For a set S of coordinates, $\Phi_S^2 := \sum_{k\in S} \Phi(k)^2$. Put

$$a := \frac{|\zeta|}{\bar{\vartheta}} = \frac{C}{M}.$$

The block scalar for an arbitrary finite switching set. For a finite set Ω of coordinates of block m (not necessarily contained in Q) put, as in [II, Subsection 2.2],

$$\varkappa(\zeta;\Omega) := \frac{|\zeta|\big(|\zeta| + \bar{\vartheta}\big) - \sum_{k\in\Omega} \mathsf{v}_k^2 \Phi(k)^2}{\bar{\vartheta}},$$

and, for the carriers l with $k(l) \in \Omega$, the *signed ratios* $r_l := u_l(\hat{z}')/\varkappa(\zeta;\Omega)$ (when $\varkappa \neq 0$), together with

$$R^2 := m^2 \sum_{k(l)\in\Omega} r_l^2, \qquad \mathsf{k} := \frac{\varkappa(\zeta;\Omega)}{\bar{\vartheta}}, \qquad s_Q := \frac{1}{\bar{\vartheta}^2} \sum_{k\in Q\setminus\Omega} \mathsf{v}_k^2 \Phi(k)^2.$$

The ratios of [II, Corollary 2.11] are $\mathsf{r}_l = m|r_l|$. For $s > 0$ and $x \in [0,1)$ let

$$\mathbf{K}(s,x) := \frac{s + \sqrt{s^2 x + s(1-x)}}{1-x}.$$

Windows. In Subsections 2.4–2.7 the following is fixed: $N \geq 1$, the design T_{fin} of Section 4.1 (besides the properties of T_{final} and $D^{U1'}$ used through Part II, namely (T-a), (W1), (W5), the signature sets with bounded gaps, the window properties of [II, Corollary 3.20] and the tools of [II, Subsections 4.2–4.3 and 5.6], the new requirements are listed in Definition 2.27), a row $f \in S_{p^*_N}$, a main stage $L \geq l_f$, a sub-window w of L that is clean for f ([II, Definition 3.18]), and an activity class a of w ([II, Lemma 5.53]). We write $b := b(w)$, $u := u(w)$, $T_{\mathrm{lo}} := T_{\mathrm{lo}}(w)$, $\mathrm{Des} := \mathrm{Des}(L)$, $D := D(L)$, and

$$\eta' := \eta'(w) := \frac{T_{\mathrm{lo}}(w)^4}{L\,\mathrm{Des}(L)}.$$

Carriers $l \leq L$ are *coarse*, carriers $l > L$ are *fine*. By cleanness [II, Corollary 3.20(b)] every coarse carrier of a block $m \leq N$ has $\varrho_l(f) \in [0,b] \cup [u, 1-u] \cup [1-b, 1+b] \cup [1+u, \infty)$. A coarse carrier is *nearly neutral* if $\varrho_l(f) \leq b$, *near-threshold* if $|\varrho_l(f) - 1| \leq b$, a *weak peak* if $\varrho_l(f) \in [1, 1+b]$ and a *robust peak* if $\varrho_l(f) \geq 1 + u$. For a block m let $P_{\mathrm{rob}}(m)$ be its robust peaks and $\sigma_{\mathrm{rob}}(m) := \Phi^2_{P_{\mathrm{rob}}(m)}$. Constants C, c are numerical, C_f, c_f depend only on f (and N), and K denotes quantities of the form $C_f(\mathrm{Des}/u)^C$ as in [II, Subsection 5.6].

2.2. **Why two levers are needed.**

Lemma 2.1 (Status rigidity in ratio space)**.** *Let ζ be the block vector of block m of a first row, let $\Omega \subseteq Q$ be finite, $\varkappa := \varkappa(\zeta;\Omega)$, and let r_l, R, a, k, s_Q be as in Subsection* 2.1*. Then:*

(a) *$a^2 = \Phi_P^2 + \mathsf{k}^2R^2 + s_Q$, $\mathsf{k} = a + \Phi_P^2 + s_Q$, and $\varrho_l = m|r_l|\mathsf{k}/\Phi_l$ for every carrier l with $k(l) \in \Omega$; moreover $R < 1$ and $a > \mathsf{k}R$.*

(b) *For fixed (Φ_P^2, s_Q) the pair (a, k) is a real-analytic function of R^2, and $d\mathsf{k}/d(R^2) = \mathsf{k}^2/(2(a - \mathsf{k}R^2)) > 0$.*

(c) *After division by $\varkappa$, the coupling row* (X3) *of [II, Definition 5.52] reads $\Delta'_m = \sum_{k(l)\in\Omega} r_l\gamma_l$; hence the exact shifted coarse system depends on block m only through the ratios (r_l), and by* (a), (b) *so do the relative positions of all carriers of Ω, for fixed (Φ_P^2, s_Q).*

(d) *A push of a peak, with the values on Ω fixed, multiplies all ratios of the block by one common factor; if the ratios are restored afterwards, every ϱ_l, $k(l) \in \Omega$, returns to its old value up to the change of (Φ_P^2, s_Q).*

Proof. (a) is [II, Corollary 2.11] with $\mathsf{r}_l = m|r_l|$. (b) With $x := R^2$ and $c_0 := \Phi_P^2 + s_Q > 0$ fixed, a is the root of $G(a) := a^2 - (a+c_0)^2x - c_0$ on the half-line $\{a > (a + c_0)\sqrt{x}\}$, where $G'(a) = 2(a - \mathsf{k}x) \geq 2(a - \mathsf{k}\sqrt{x}) > 0$; the implicit function theorem gives analyticity, and differentiating $a^2 = c_0 + \mathsf{k}^2x$ with $da = d\mathsf{k}$ gives $2a\,d\mathsf{k} = \mathsf{k}^2dx + 2\mathsf{k}x\,d\mathsf{k}$. (c) is [II, Proposition 5.46(i)], and (d) is [II, Proposition 5.46(ii)]. □

Remark 2.2 (Two levers)*.* Lemma 2.1 explains the gap of the one-lever exactification recorded in [II, Remark 5.47]: if the ratios of the active switching carriers are moved to an exact point of the tiny minors and $\varkappa$ is set by one peak push per block, all relative positions of the switching carriers are determined by the exact point and by (Φ_P^2, s_Q), and peak pushes cannot change them independently. Threshold protection therefore needs a second, independent lever. In [II, Proposition 5.56] this lever is an inactive switching carrier of robust relative position, whose inward push raises $\bar\vartheta$ at first order without moving $\varkappa$ ([II, Proposition 5.46(iii)]); its existence is the hypothesis ($\mathrm{KN}_{w,a}$) of [II, Definition 5.55]. Lemma 2.6 below shows that (KN) is a property of the row f, which no move made at a companion can create. In this section the second lever is the position of the exact point on the exact zero set, decoupled from the statuses by generic weights.

2.3. **The block identities.**

Lemma 2.3 (The α-mass identity)**.** (a) *For every $\zeta \in \ell_1 \setminus \{0\}$,*

$$\sum_{k\in P}(\varrho_k - 1)\Phi(k)^2 = \frac{|\zeta|}{\bar\vartheta} = a = \frac{C}{M}.$$

In particular every block has a peak k with $\varrho_k > 1$.

(b) *In the window setting of Subsection* 2.1 *let $m \leq N$ and assume*

$$\frac{b}{12} + \frac{2C_U\,c_{L+1}}{3\bar\vartheta_m(f)} < a_m(f), \qquad C_U := \sup_l |u_l(\hat z)|, \tag{1}$$

where $\hat z$, $\bar\vartheta_m(f)$, $a_m(f)$ refer to f. Then $P_{\mathrm{rob}}(m) \neq \emptyset$, and $\sigma_{\mathrm{rob}}(m) \geq \Phi_{\min}(L)^2$ with $\Phi_{\min}(L) := \min_{l\leq L}\Phi_l$. Condition (1) *holds for every block $m \leq N$ as soon as L exceeds an f-constant, which we include in l_f.*

Proof. (a) By [II, Lemma 2.6(c)], $\sum_{k\in P}\Phi(k)^2(\mathsf{v}_k-\bar\vartheta)=|\zeta|$; divide by $\bar\vartheta$ and use $\varrho_k=\mathsf{v}_k/\bar\vartheta$ and $|\zeta|/\bar\vartheta=C/M$ ([II, Lemma 2.6(b)]). Since $|\zeta|>0$, some term is positive. (Equivalently: pairing the representation $\zeta/|\zeta|=\alpha+D^2w/C$ of [I, Lemma 1.7] with w gives $1=M\|\alpha\|_1+C$, so $\|\alpha\|_1=1$, and $\varrho_k-1=|\alpha(k)|C/(M\Phi(k)^2)$ on P.)

(b) Split P into the robust coarse peaks, the weak coarse peaks and the fine peaks. A weak peak contributes $(\varrho_k-1)\Phi(k)^2\le b\Phi(k)^2$, and $\|\Phi_m\|_2^2\le\sum_k 4^{-m-k}\le\frac{1}{12}$, so the weak peaks contribute at most $b/12$. A fine peak l contributes at most $\varrho_l\Phi_l^2=|\zeta(k(l))|/\bar\vartheta_m=\lambda_l|u_l(\hat z)|/\bar\vartheta_m\le C_U\lambda_l/\bar\vartheta_m$; as $\lambda_l=m(l)2^{-m(l)-k(l)}c_l\le c_l/2$ and $c_{l+1}\le c_l/4$, the fine carriers contribute at most $\frac{2}{3}C_Uc_{L+1}/\bar\vartheta_m$. By (a) and (1) the robust peaks carry a positive mass, so $P_{\rm rob}(m)\ne\emptyset$ and $\sigma_{\rm rob}(m)\ge\Phi_{\min}(L)^2$. C_U is finite because the vectors u_l are bounded in ℓ_1 and $\hat z$ is bounded; $b=b(w)\le\eta'(w)\to0$ and $c_{L+1}\to0$ as $L\to\infty$, while $a_m(f)>0$ and $\bar\vartheta_m(f)>0$ do not depend on L. □

Lemma 2.4 (Closed form of $\varkappa/\bar\vartheta$)**.** *Let ζ be the block vector of block m of a first row, let Ω be a finite set of coordinates of block m, and put*

$$\sigma':=\Phi_{P\setminus\Omega}^2+\sum_{k\in\Omega\cap P}\bigl(1-\varrho_k^2\bigr)\Phi(k)^2+s_Q.$$

(a) $a^2=\sum_k\Phi(k)^2\min(\varrho_k,1)^2\le\|\Phi_m\|_2^2\le\frac{1}{12}$ *and* $\sigma'\le\|\Phi_m\|_2^2$*; if* $\Omega\subseteq Q$*, then* $\sigma'=\sigma:=\Phi_P^2+s_Q$.

(b) $\mathsf{k}=a+\sigma'$ *and* $a^2=\sigma'+\mathsf{k}^2R^2$*; if* $\varkappa(\zeta;\Omega)\ne0$*, then* $\varrho_l=m|r_l|\mathsf{k}/\Phi_l$ *for every carrier l with $k(l)\in\Omega$.*

(c) *If* $\sigma'>0$*, then* $R<1$*,* $a>\mathsf{k}R$*,* $\mathsf{k}=\mathbf{K}(\sigma',R^2)$ *and*

$$1-R^2=\frac{\sigma'(1+2a+\sigma')}{(a+\sigma')^2}\ge\sigma'.$$

(d) $\mathbf{K}$ *is real-analytic on a neighbourhood of* $(0,\infty)\times[0,1)$*; with* $\mathbf{A}:=\mathbf{K}-s$ *one has* $\mathbf{A}>0$*,* $\mathbf{A}^2=s+\mathbf{K}^2x$*,* $\partial_s\mathbf{K}>0$ *and* $\partial_x\mathbf{K}=\mathbf{K}^2/(2(\mathbf{A}-\mathbf{K}x))>0$ *there.*

(e) *For* $0<s_0\le1$ *and* (s,x) *in* $\mathcal{K}(s_0):=\{s_0\le s\le1,\ 0\le x\le1-s_0/4\}$*:*

$$2s\le\mathbf{K}\le\frac{10}{s_0},\qquad0<\partial_s\mathbf{K}\le\frac{16}{s_0^2},\qquad0<\partial_x\mathbf{K}\le\frac{1000}{s_0^4}.$$

Proof. (a) Divide the second identity of [II, Lemma 2.6(c)] by $\bar\vartheta^2$: $a^2=\Phi_P^2+\sum_{k\in Q}\varrho_k^2\Phi(k)^2$, which is the first formula. By (T-a), $\Phi_m(k)\le2^{-m-k}$, so $\|\Phi_m\|_2^2\le4^{-m}/3$. Each of the three terms of σ' is at most the corresponding part of $\sum_k\Phi(k)^2$ (note $1-\varrho_k^2\le1$ and $\varrho_k<1$ on Q).

(b) By [II, Lemma 2.6(c)], $|\zeta|^2=\bar\vartheta^2\Phi_P^2+\sum_{k\in Q}\mathsf{v}_k^2\Phi(k)^2$. Inserting this into the definition of $\varkappa$ and splitting $\Phi_P^2=\Phi_{P\setminus\Omega}^2+\Phi_{\Omega\cap P}^2$, $\sum_Q=\sum_{Q\setminus\Omega}+\sum_{\Omega\cap Q}$,

$$\bar\vartheta\varkappa=|\zeta|\bar\vartheta+\bar\vartheta^2\Phi_P^2+\sum_{k\in Q}\mathsf{v}_k^2\Phi(k)^2-\sum_{k\in\Omega}\mathsf{v}_k^2\Phi(k)^2=|\zeta|\bar\vartheta+\bar\vartheta^2\sigma';$$

divided by $\bar\vartheta^2$ this is $\mathsf{k}=a+\sigma'$. Next, $\sum_{k\in\Omega}\mathsf{v}_k^2\Phi(k)^2=\sum_{k(l)\in\Omega}m^2u_l(\hat z)^2=\varkappa^2R^2$, i.e. $\sum_{k\in\Omega}\varrho_k^2\Phi(k)^2=\mathsf{k}^2R^2$, and the same splitting in the formula of (a) gives $a^2=\sigma'+\sum_{k\in\Omega}\varrho_k^2\Phi(k)^2=\sigma'+\mathsf{k}^2R^2$. Finally $\varrho_l=\mathsf{v}_{k(l)}/\bar\vartheta=m|u_l(\hat z)|/(\Phi_l\bar\vartheta)=m|r_l|\mathsf{k}/\Phi_l$.

(c) $a>0$ because $\zeta\ne0$. If $\sigma'>0$, then $\mathsf{k}>a>0$ and $a^2>\mathsf{k}^2R^2$, so $\mathsf{k}R<a<\mathsf{k}$ and $R<1$. Substituting $a=\mathsf{k}-\sigma'$ into $a^2=\sigma'+\mathsf{k}^2R^2$ gives $(1-R^2)\mathsf{k}^2-2\sigma'\mathsf{k}+\sigma'^2-\sigma'=0$, with roots $\mathsf{k}_\pm:=[\sigma'\pm(\sigma'^2R^2+\sigma'(1-R^2))^{1/2}]/(1-R^2)$.

The inequality $\mathsf{k}_- \le \sigma'$ is equivalent to $\sigma' R^2 \le (\sigma'^2 R^2 + \sigma'(1-R^2))^{1/2}$, i.e. to $0 \le \sigma'(1-R^2)(1+\sigma' R^2)$, which holds; so k_- would give $a \le 0$, and $\mathsf{k} = \mathsf{k}_+ = \mathbf{K}(\sigma', R^2)$. From $R^2 = (a^2 - \sigma')/(a+\sigma')^2$ we get the formula for $1-R^2$; by (a), $a + \sigma' \le 12^{-1/2} + \frac{1}{12} < 1$, hence $(a+\sigma')^2 \le a + \sigma' \le 1 + 2a + \sigma'$, which gives $1 - R^2 \ge \sigma'$.

(d) For $s > 0$ and $x \in [0,1)$ the radicand $s^2 x + s(1-x)$ is positive, so $\mathbf{K}$ is real-analytic near such points, and $\mathbf{A} = [sx + (s^2 x + s(1-x))^{1/2}]/(1-x) > 0$. The identity $\mathbf{A}^2 = s + \mathbf{K}^2 x$ is the quadratic equation of (c). The s-derivative of the numerator of $\mathbf{K}$ is $1 + (2sx + 1 - x)/(2(s^2 x + s(1-x))^{1/2}) > 0$. Differentiating $\mathbf{A}^2 = s + \mathbf{K}^2 x$ in x (note $\partial_x \mathbf{A} = \partial_x \mathbf{K}$) gives $2\mathbf{A}\,\partial_x \mathbf{K} = \mathbf{K}^2 + 2\mathbf{K} x\,\partial_x \mathbf{K}$; and $\mathbf{A}^2 > \mathbf{K}^2 x$ gives $\mathbf{A} > \mathbf{K}\sqrt{x} \ge \mathbf{K} x$.

(e) On $\mathcal{K}(s_0)$ the radicand lies in $[s^2,\, s^2 + s] \subseteq [s^2, 2]$ (for $s \le 1$, $s(1-x) \ge s^2(1-x)$), so the numerator lies in $[2s, \frac{5}{2}]$ and $1 - x \in [s_0/4, 1]$; this gives the bounds on $\mathbf{K}$. Since the radicand is $\ge s(1-x) \ge s_0^2/4$ and $2sx + 1 - x \le 3$, $\partial_s \mathbf{K} \le (1 + 3/s_0) \cdot 4/s_0 \le 16/s_0^2$. Finally $\mathbf{A} - \mathbf{K}x \ge \mathbf{A} - \mathbf{K}\sqrt{x} = s/(\mathbf{A} + \mathbf{K}\sqrt{x}) \ge s/(2\mathbf{K})$ (as $\mathbf{A} < \mathbf{K}$), so $\partial_x \mathbf{K} \le \mathbf{K}^3/s \le 1000/s_0^4$. □

Remark 2.5 (Convention for active weak peaks). Lemma 2.4(b) holds for every finite Ω; this is used with the *pattern's* switching set (Definition 2.8), which contains the active weak peaks of f. For a weak peak, $1 - \varrho_k^2 \in [-3b, 0]$, so $\sigma' \ge \sigma_{\mathrm{rob}}(m) - 3b\|\Phi_m\|_2^2 \ge \sigma_{\mathrm{rob}}(m) - b/4$ whenever $P \setminus \Omega \supseteq P_{\mathrm{rob}}(m)$ and $\Omega \cap P$ consists of weak peaks. If instead the active weak peaks were treated as peaks ("peak convention"), all ratios of the block would change by a relative amount of order Φ_l^2/k, which is not small compared with the margins below, and their weights would enter the peak traces of the cone (Lemma 2.17); the formula-1 convention with the pattern's Ω avoids both.

Lemma 2.6 (Floor lemma)**.** *Let ζ be the block vector of block m of a first row, $\Omega \subseteq Q$ finite with $\varkappa(\zeta;\Omega) \ne 0$, $\Omega_a \subseteq \Omega$, and $P_0 \subseteq P$ with $\sigma_0 := \Phi_{P_0}^2 > 0$. Put $R_a^2 := m^2 \sum_{k(l)\in\Omega_a} r_l^2$. Then $R_a < 1$, and for every carrier l with $k(l) \in \Omega_a$*

$$\varrho_l \;\ge\; \varrho_l^0 := \frac{m|r_l|\,\mathbf{K}(\sigma_0, R_a^2)}{\Phi_l}.$$

If $r_l \ne 0$, equality holds if and only if $P = P_0$, $s_Q = 0$ and $r_{l'} = 0$ for every l' with $k(l') \in \Omega \setminus \Omega_a$.

Proof. By Lemma 2.4(a)–(c) with $\Omega \subseteq Q$: $\sigma = \Phi_P^2 + s_Q \ge \sigma_0 > 0$, $R < 1$ and $\varrho_l = m|r_l|\mathbf{K}(\sigma, R^2)/\Phi_l$. Moreover $R^2 = R_a^2 + m^2 \sum_{k(l')\in\Omega\setminus\Omega_a} r_{l'}^2 \ge R_a^2$. $\mathbf{K}$ is strictly increasing in both variables (Lemma 2.4(d)), and $\sigma = \sigma_0$ iff $P = P_0$ and $s_Q = 0$. □

Remark 2.7 ((KN) is a property of the row). (a) Let f' be any first row (for instance a companion of f) at which block m has $P(f') \supseteq P_{\mathrm{rob}}(m)$, $\Omega \subseteq Q(f')$, and at which the ratios of the active carriers Ω_a equal a given vector r. By Lemma 2.6 no active relative position at f' is below its floor $\varrho_l^0(r) = m|r_l|\mathbf{K}(\sigma_{\mathrm{rob}}(m), R_a(r)^2)/\Phi_l$. An inactive carrier, a fine carrier or an additional peak only adds to σ or to R^2; a lever that is created at a companion (a carrier with value 0 at f that is given a value and then pushed inward) can therefore lower the active statuses at fixed active ratios only back towards the floor. At fixed active ratios the largest possible decrease of k is $I_m(f') := \mathbf{K}(\sigma(f'), R(f')^2) - \mathbf{K}(\sigma_{\mathrm{rob}}(m), R_a^2)$, which at $f' = f$ is computed from

f. If every inactive carrier of Ω_m is nearly neutral and no weak peak of block m lies outside Ω, then $I_m(f) \le C\sigma_{\mathrm{rob}}(m)^{-5}(b + c_{L+1}^2)$ (proof of Lemma 2.9 below); a lever of robust size exists at f only if some inactive carrier of Ω_m has $\varrho(f) \ge u$, which is the condition of [II, Definition 5.55] for block m. In this sense (KN) is a property of f.

(b) *Heuristic.* We do not see how a design could supply such levers for every row: fine carriers have capacity at most of order $\bar{\vartheta}c_{L+1}^2$, far below the displacements of the Łojasiewicz step; levers tuned at a companion give back what they add, by (a); and families of near-parallel "sibling" carriers leave the coarsest member of every family without a sibling. This is not used anywhere; the rest of the section shows that no lever of this kind is needed.

2.4. **Patterns, kinds of blocks and the floor-form cone.**

Definition 2.8 (Switching set, kinds of blocks, pattern)**.** In the window setting of Subsection 2.1:

(a) For a block m let Ω_m be the set of coordinates $k(l)$ of the coarse carriers l of block m with $\varrho_l(f) \le 1 + b$, i.e. of all coarse carriers except the robust peaks, and $\Omega := \bigcup_m \Omega_m$. The ratios of the pattern are always computed with $\varkappa(\cdot\,;\Omega_m)$ (formula of Subsection 2.1), also when Ω_m contains weak peaks. The class a declares each component γ_l ($k(l) \in \Omega$) and each shift Δ'_m active or inactive ([II, Lemma 5.53]); $\Omega_a(m)$ denotes the active carriers of Ω_m, and $\mathcal{N}(m) \subseteq \Omega_a(m)$ the active near-threshold ones.

(b) A block m is of kind (U) if Δ'_m is inactive (this includes the blocks without active components); (KN) if Δ'_m is active and some inactive carrier of Ω_m has $\varrho(f) \ge u$; (T) if Δ'_m is active and every inactive carrier of Ω_m is nearly neutral. By cleanness every carrier of Ω has $\varrho(f) \in [0, b] \cup [u, 1 + b]$, so every block has exactly one kind. Since Ω_m contains the weak peaks of block m, an inactive weak peak makes a block with active shift of kind (KN); in particular every weak peak of a block of kind (T) is active. Put $\mathcal{N}_T := \bigcup\{\mathcal{N}(m) : m \text{ of kind (T)}\}$.

(c) The *pattern* κ of (f, w, a) consists of the finite combinatorial data: the sets Ω_m, $\Omega_a(m)$, $\mathcal{N}(m)$, $P_{\mathrm{rob}}(m)$, the signs vs_l of the coarse carriers, the classes R/G and the contact/free types of the coordinates of $E_c(w)$, the type of each aggregated signature row (sign row, equality row or absent), the kinds of the blocks, and the labels *tiny*/*robust* of the rate objects of Definition 2.13. Its *level* is L. There are finitely many patterns of each level, and they do not depend on N (blocks $m > N$ are absent and contribute no row).

The kind (U) needs no lever (Lemma 2.16), the kind (KN) is handled by the lever of [II, Proposition 5.56], and only the kind (T) is new.

Lemma 2.9 (Floor estimate in (T)-blocks)**.** *Let m be a block of kind* (T)*, let $r(f)$ be its ratio vector (computed with $\varkappa(\hat{\zeta}_m(f);\Omega_m)$), and assume $b \le \sigma_{\mathrm{rob}}(m)/8$. For every carrier l of $\Omega_a(m)$ put $\varrho_l^0(r) := m|r_l|\mathbf{K}(\sigma_{\mathrm{rob}}(m), R_a(m)(r)^2)/\Phi_l$, where $R_a(m)(r)^2 := m^2 \sum_{l\in\Omega_a(m)} r_l^2$. Then:*

(a) $|\varrho_l(f) - \varrho_l^0(r(f))| \le C\sigma_{\mathrm{rob}}(m)^{-6}\big(b + c_{L+1}^2\big)$;

(b) $1 - R_a(m)(r(f))^2 \ge \sigma_{\mathrm{rob}}(m) - b/4$.

Proof. Write $\sigma_0 := \sigma_{\mathrm{rob}}(m)$. At f, $P \setminus \Omega_m$ consists of $P_{\mathrm{rob}}(m)$ and the fine peaks, $\Omega_m \cap P$ consists of weak peaks, and $Q \setminus \Omega_m$ consists of fine carriers. Hence, in Lemma 2.4,

$$\sigma' - \sigma_0 = \Phi^2_{\text{fine peaks}} + s_Q + \sum_{k \in \Omega_m \cap P} (1 - \varrho_k^2)\Phi(k)^2 \in \Big[-\frac{b}{4},\ \sum_{l>L} \Phi_l^2 \Big],$$

and $\sum_{l>L} \Phi_l^2 \le \sum_{l>L} c_l^2/16 \le c_{L+1}^2/8$ (using $1 - \varrho^2 \ge -3b$ on weak peaks and $\|\Phi_m\|_2^2 \le \frac{1}{12}$). So $\sigma' \ge \frac{7}{8}\sigma_0 > 0$, and by Lemma 2.4(c) $1 - R^2 \ge \sigma' \ge \sigma_0 - b/4$; as $R_a \le R$ this gives (b). Also $\mathsf{k} = a + \sigma' \ge \sigma_0/2$. The inactive carriers of Ω_m are nearly neutral, so $R^2 - R_a^2 = \sum_{\text{inactive}} \varrho^2 \Phi^2/\mathsf{k}^2 \le b^2/(3\sigma_0^2)$. Both points (σ', R^2) and (σ_0, R_a^2) lie in the convex set $\mathcal{K}(\sigma_0/2)$ of Lemma 2.4(e) (since $1 - R_a^2 \ge 1 - R^2 \ge \sigma_0/2$ and $\sigma_0, \sigma' \le 1$), so

$$|\mathbf{K}(\sigma', R^2) - \mathbf{K}(\sigma_0, R_a^2)| \le \frac{64}{\sigma_0^2}\Big(\frac{b}{4} + c_{L+1}^2\Big) + \frac{16000}{\sigma_0^4} \cdot \frac{b^2}{3\sigma_0^2} \le C\sigma_0^{-5}(b + c_{L+1}^2),$$

using $b \le \sigma_0/8$. Finally $\varrho_l(f) = (m|r_l|/\Phi_l)\mathbf{K}(\sigma', R^2)$ by Lemma 2.4(b),(c), and $m|r_l|/\Phi_l = \varrho_l(f)/\mathbf{K}(\sigma', R^2) \le (1 + b)/(2\sigma') \le 2/\sigma_0$ (Lemma 2.4(e): $\mathbf{K} \ge 2s$). Multiplying gives (a). □

The hypothesis $b \le \sigma_{\mathrm{rob}}(m)/8$ holds for $L \ge l_f$: by Lemma 2.3(b), $\sigma_{\mathrm{rob}}(m) \ge \Phi_{\min}(L)^2 \ge D(L)^{-2}$, while $b \le \eta' \le T_{\mathrm{lo}}^4$ and $\mathrm{Des}(L)^2 T_{\mathrm{lo}} \le 1$ ([II, Corollary 3.20(c)]).

Lemma 2.10 (Effective columns)**.** *Let f' be a first row at which the coarse peaks of every block m are exactly $P_{\mathrm{rob}}(m)$ and every carrier of Ω is a strict non-peak, and let $\mathcal{L}$ be the coarse certificate of [II, Definition 5.52] at f',*

$$\mathcal{L}(\Delta', \gamma) = \Big(\sum_{k(l) \in \Omega} \gamma_l u_l - \sum_m \Delta'_m \tau_m \Big)\Big|_{E_c(w) \setminus F'}, \qquad \tau_m := \sum_{p \in P_{\mathrm{rob}}(m)} \mathrm{vs}_p \lambda_p u_p.$$

If (Δ', γ) satisfies (X3) *and* (X5) *of [II, Definition 5.52], then*

$$\mathcal{L}(\Delta', \gamma) = \sum_{l \in \Omega_a} \gamma_l \big(u_l - r_l \tau_{m(l)}\big)\Big|_{E_c(w) \setminus F'}, \qquad \Omega_a := \bigcup_m \Omega_a(m),$$

with the ratios r_l of f'. The column of γ_l depends only on u_l, on its own ratio r_l and on $\tau_{m(l)}$, which contains the weights of the robust peaks of block $m(l)$ only; the weights of the carriers of Ω do not occur.

Proof. By [II, Lemma 2.22(a)] and the definition of $\mathcal{L}$ in [II, Definition 5.52], the coefficient of a coarse peak p is $-\Delta'_m \mathrm{vs}_p \lambda_p$, which produces $-\Delta'_m \tau_m$. By (X5), $\gamma_l = 0$ for inactive carriers, and $\Delta'_m = 0$ for inactive shifts; by (X3) divided by $\varkappa_m$ (Lemma 2.1(c)), $\Delta'_m = \sum_{l \in \Omega_a(m)} r_l \gamma_l$ for active shifts. For an inactive shift the term $\Delta'_m \tau_m$ vanishes, and (X3) with $\Delta'_m = 0$ gives $\sum_{l \in \Omega_a(m)} r_l \gamma_l = 0$, so $\sum_l \gamma_l r_l \tau_m = 0$ as well. Substitute. □

Definition 2.11 (The weight-free region)**.** For a pattern κ let

$$\mathcal{K}_\kappa := \Big\{ r \in \mathbb{R}^{\Omega_a} :\ R_a(m)(r)^2 \le 1 - \tfrac{1}{4}\sigma_{\mathrm{rob}}(m) \text{ for every block } m \text{ with } \Omega_a(m) \ne \emptyset \Big\},$$

where $R_a(m)(r)^2 := m^2 \sum_{l \in \Omega_a(m)} r_l^2$. $\mathcal{K}_\kappa$ is compact (on it $|r_l| \le 1/m(l)$) and semialgebraic, and it is defined by the robust peak weights only. For a carrier $l \in \Omega_a$ let $\varrho_l^0(r) := m|r_l|\mathbf{K}(\sigma_{\mathrm{rob}}(m), R_a(m)(r)^2)/\Phi_l$ ($m = m(l)$) be its floor status,

and let Λ_κ be the largest Lipschitz constant on $\mathcal{K}_\kappa$ of the functions ϱ_l^0, $l \in \Omega_a$; by Lemma 2.4(e), $\Lambda_\kappa \le C \max_l m(l)\sigma_{\rm rob}(m(l))^{-4}\Phi_l^{-1} < \infty$.

Lemma 2.12 (Interior points)**.** *Assume $b \le \sigma_{\rm rob}(m)/4$ for every block m. Then $1 - R_a(m)(r(f))^2 \ge \sigma_{\rm rob}(m) - b/4$ for every block m, where $r(f)$ is the ratio vector of the pattern at f. Here and in Lemmas 2.24, 2.26 and Proposition 2.32, $|\cdot|$ denotes the Euclidean norm (on $\mathbb{R}^{\Omega_a}$, resp. $\mathbb{R}^n$). If $\rho_0 > 0$ and $3L\rho_0 \le \sigma_{\rm rob}(m)/4$ for every m, then every r with $|r - r(f)| \le \rho_0$ satisfies $R_a(m)(r)^2 \le 1 - \sigma_{\rm rob}(m)/2$ for every m; in particular $r \in \mathcal{K}_\kappa$.*

Proof. For every block, $P \setminus \Omega_m \supseteq P_{\rm rob}(m)$ and $\Omega_m \cap P$ consists of weak peaks, so $\sigma' \ge \sigma_{\rm rob}(m) - b/4 > 0$ (Remark 2.5); Lemma 2.4(c) gives $1 - R_a^2 \ge 1 - R^2 \ge \sigma'$. The function $r \mapsto R_a(m)(r) = m\|r^{(m)}\|_2$ is m-Lipschitz and $m \le L$, so for $|r - r(f)| \le \rho_0$ (and $L\rho_0 \le 1$), $R_a(m)(r)^2 \le R_a(m)(r(f))^2 + 3L\rho_0 \le 1 - \sigma_{\rm rob}(m) + b/4 + \sigma_{\rm rob}(m)/4$. □

Definition 2.13 (The floor-form cone and its rate objects)**.** Let κ be a pattern of level L with class a. For $r \in \mathcal{K}_\kappa$, the polyhedral cone $\Gamma^\flat(\kappa, a; r)$ in the variables $(\Delta', \gamma) \in \mathbb{R}^I \times \mathbb{R}^\Omega$ is defined by the rows (X1), (X3), (X5) of [II, Definition 5.52], with the coarse peaks $P_{\rm rob}(m)$, the switching set Ω of Definition 2.8, the ratio variables r in (X3), and the aggregated rows of (X1) on $S_l \cap (L, \infty)$ taken with the s-free coefficient $\alpha_l := \|v_l 1_{S_l \cap (L,\infty)}\|_1$; and with (X4) of [II, Definition 5.52] replaced, for every carrier $l \in \mathcal{N}(m)$, by

- (X4^0) $\mathrm{vs}_l\gamma_l + \Delta'_m\, m^2|r_l|\,\mathbf{K}\big(\sigma_{\rm rob}(m), R_a(m)(r)^2\big) \ge 0$ if m is of kind (T);
- (X4) $\mathrm{vs}_l\gamma_l/\lambda_l + \Delta'_m \ge 0$ if m is of kind (KN) (as in [II, Definition 5.52]);
- (X4^U) $\mathrm{vs}_l\gamma_l \ge 0$ if m is of kind (U).

The *matrix of* $\Gamma^\flat(\kappa, a; r)$ is formed in the active variables only: the columns of the components declared inactive by (X5) (the inactive γ_l and the inactive shifts Δ'_m) are deleted and the (X5) rows are omitted. The cone, as a set in the active coordinates, and its Hoffman constant are those of $\Gamma^\flat$. The *rate objects of* κ are the minors of this matrix and the minors of the matrices g_J formed by the Δ'-parts of the cofactor vectors of its $(n-1)$-row subsystems (the cofactor polynomials of [II, Definition 5.52]), as functions of $r \in \mathcal{K}_\kappa$. They are continuous semialgebraic functions on $\mathcal{K}_\kappa$ ($\mathbf{K}$ is semialgebraic: its graph is $\{(s, x, k) : (1-x)k^2 - 2sk + s^2 - s = 0,\ k > s\}$). The pattern labels each of them *tiny* or *robust* (its modulus at $r(f)$ is $\le b$, resp. $\ge u$; cleanness, [II, Definition 3.18, Theorem 3.19], concerns moduli). The *exact zero set* of κ is

$$Z_\kappa := \{r \in \mathcal{K}_\kappa :\ \text{every tiny rate object of } \kappa \text{ vanishes at } r\}.$$

In (X4^U) the shift Δ'_m is inactive, i.e. 0 by (X5), so (X4^U) is the row (X4) of [II] multiplied by $\lambda_l > 0$. The only modification of the cone of [II] is therefore (X4^0) in the blocks of kind (T).

Lemma 2.14 (The floor-form inward row)**.** (a) (Meaning.) *Let f' be a first row, m a block, l a carrier of block m with $k(l) \in Q(f')$ and $u_l(\hat{z}') \ne 0$, $\mathrm{vs}_l := \operatorname{sign} w'_m(k(l))$, and let $\Delta_m \in \mathbb{R}$, $\Delta\omega(l) \in \mathbb{R}$, $\gamma_l := \lambda_l(\Delta\omega(l) - \Delta_m w'_m(k(l)))$, $\Delta'_m := \Delta_m M'_m$. Then*

$$\lambda_l \mathrm{vs}_l \Delta\omega(l) = \mathrm{vs}_l\gamma_l + \Delta'_m\lambda_l\varrho_l(f'), \qquad \lambda_l\varrho_l(f') = m^2|r_l|\,\mathsf{k}(f').$$

So (X4^0) *says "$\mathrm{vs}_l\Delta\omega(l) \ge 0$, with k replaced by its floor $\mathbf{K}(\sigma_{\rm rob}, R_a^2)$".*

(b) (The data satisfy (X4^0).) *Let m be of kind (T), $l \in \mathcal{N}(m)$, and assume $b \le \sigma_{\rm rob}(m)/8$ and $\Lambda_\kappa \le \mathrm{Des}(L)$. For every scale $t \in \mathcal{W}(w)$ and every $r \in \mathcal{K}_\kappa$*

whose block-m coordinates satisfy $|r^{(m)} - r(f)^{(m)}| \le 2\eta'$, the coarse decomposition data $(\Delta'_{\mathrm{dec}}(t), \gamma_{\mathrm{dec}}(t))$ of [II, Subsection 5.6] satisfy

$$\mathrm{vs}_l \gamma_{\mathrm{dec},l} + \Delta'_{\mathrm{dec},m}\, m^2 |r_l|\, \mathbf{K}\big(\sigma_{\mathrm{rob}}(m), R_a(m)(r)^2\big) \;\ge\; -Kt.$$

(c) (Inward direction at a companion.) *Let $f^{\#}$ be a first row at which block m (of kind* (T)*) has the coarse peaks $P_{\mathrm{rob}}(m)$, every carrier of Ω_m is a strict non-peak, every carrier of $\Omega_m \setminus \Omega_a(m)$ has value 0, and the active ratios (computed with $\varkappa(\hat{\zeta}_m^{\#}; \Omega_m)$) form a vector $r^{\#} \in \mathcal{K}_\kappa$. Then for $l \in \mathcal{N}(m)$ with $r_l^{\#} \neq 0$,*

$$0 \le \varrho_l(f^{\#}) - \varrho_l^0(r^{\#}) \le \sigma_{\mathrm{rob}}(m)^{-3} c_{L+1}^2,$$

and every (Δ', γ) satisfying ($\mathrm{X4}^0$) *at $r^{\#}$ satisfies*

$$\lambda_l \mathrm{vs}_l \Delta\omega^{\#}(l) \ge -|\Delta'_m| \lambda_l \sigma_{\mathrm{rob}}(m)^{-3} c_{L+1}^2,$$

where $\Delta\omega^{\#}(l) := \gamma_l/\lambda_l + \Delta_m w_m^{\#}(k(l))$ and $\Delta_m := \Delta'_m / M_m^{\#}$.

Proof. (a) By [II, Lemma 2.6(b)], $w'_m(k) = \mathrm{sign}(\zeta(k)) M' \min(1, \varrho_k)$, so $w'_m(k(l)) = \mathrm{vs}_l M' \varrho_l$ off P. Hence $\lambda_l \Delta\omega(l) = \gamma_l + \Delta_m \lambda_l w'_m(k(l)) = \gamma_l + \Delta'_m \lambda_l \mathrm{vs}_l \varrho_l$; multiply by vs_l. By Lemma 2.4(b), $\lambda_l \varrho_l = m\Phi_l \cdot m|r_l|\mathsf{k}/\Phi_l$.

(b) The proof of [II, Proposition 5.54(a)] (via [I, Lemma 8.8] at strict non-peaks of f and [I, (16)] at peaks of f) gives $\mathrm{vs}_l \gamma_{\mathrm{dec},l}/\lambda_l + \Delta'_{\mathrm{dec},m} \ge -(4\,\mathrm{gap}(f)/t + C_f b)$ with $\mathrm{gap}(f) \le bM \le b$. Multiply by λ_l and write

$$\mathrm{vs}_l \gamma_{\mathrm{dec}} + \Delta'_{\mathrm{dec}} \lambda_l \varrho_l^0(r) = \big(\mathrm{vs}_l \gamma_{\mathrm{dec}} + \Delta'_{\mathrm{dec}} \lambda_l\big) + \Delta'_{\mathrm{dec}} \lambda_l \big(\varrho_l^0(r) - 1\big).$$

Here $|\Delta'_{\mathrm{dec}}| \le C_f$ ([II, Subsection 5.6]) and

$$|\varrho_l^0(r) - 1| \le \Lambda_\kappa |r - r(f)| + |\varrho_l^0(r(f)) - \varrho_l(f)| + |\varrho_l(f) - 1| \le 2\mathrm{Des}\,\eta' + C\sigma_{\mathrm{rob}}^{-6}(b + c_{L+1}^2) + b$$

by Lemma 2.9(a); here only the block-m coordinates of r enter, because ϱ_l^0 depends on the ratios of block m alone, and Λ_κ bounds its Lipschitz constant in these coordinates. Since $t \ge T_{\mathrm{lo}}$, $b \le \eta' \le T_{\mathrm{lo}}^4$, $\mathrm{Des}\,\eta' \le T_{\mathrm{lo}}^4$, $c_{L+1} \le b$ and $\sigma_{\mathrm{rob}}^{-1} \le D(L)^2$, all terms are $\le C_f \lambda_l D(L)^{12} t \le Kt$.

(c) By Lemma 2.6 with $P_0 := P_{\mathrm{rob}}(m)$, $\varrho_l(f^{\#}) \ge \varrho_l^0(r^{\#})$. As the carriers of $\Omega_m \setminus \Omega_a(m)$ have value 0, $R^2 = R_a(m)(r^{\#})^2$, and $\sigma - \sigma_{\mathrm{rob}}(m) = \Phi^2_{\text{fine peaks}} + s_Q \le \sum_{l' > L} \Phi_{l'}^2 \le c_{L+1}^2/8$. The points (σ, R^2) and $(\sigma_{\mathrm{rob}}, R^2)$ lie in $\mathcal{K}(\sigma_{\mathrm{rob}})$ because $r^{\#} \in \mathcal{K}_\kappa$, so by Lemma 2.4(e) the two values of $\mathbf{K}$ differ by at most $2c_{L+1}^2/\sigma_{\mathrm{rob}}^2$; and $m|r_l|/\Phi_l = \varrho_l^0/\mathbf{K}(\sigma_{\mathrm{rob}}, R^2) \le 1/(2\sigma_{\mathrm{rob}})$, since $\varrho_l^0 \le \varrho_l(f^{\#}) < 1$ and $\mathbf{K} \ge 2s$. This gives the bound. For the last claim apply (a) at $f^{\#}$: $\lambda_l \mathrm{vs}_l \Delta\omega^{\#}(l) = \big(\mathrm{vs}_l \gamma_l + \Delta'_m \lambda_l \varrho_l^0(r^{\#})\big) + \Delta'_m \lambda_l (\varrho_l(f^{\#}) - \varrho_l^0(r^{\#}))$, and the first bracket is ≥ 0 by ($\mathrm{X4}^0$), since $\lambda_l \varrho_l^0(r^{\#}) = m^2 |r_l^{\#}| \mathbf{K}(\sigma_{\mathrm{rob}}, R_a^2)$. □

Remark 2.15 (Use of Lemma 2.14). Part (b) replaces, in (T)-blocks, the data estimate for the row (X4) in the proof of [II, Proposition 5.54(a)]; for $\Delta'_{\mathrm{dec}} < 0$ only the inequality $\varrho^0 \le \varrho$ of Lemma 2.6 is needed, while for $\Delta'_{\mathrm{dec}} > 0$ the smallness of the inactive mass of a (T)-block (Lemma 2.9) is used. Part (c) shows that at the companion of Proposition 2.32 every point of the cone, and every increment, is inward at the protected carriers up to $C_f \lambda_l \sigma_{\mathrm{rob}}^{-3} c_{L+1}^2$; with the shift trick this makes these coordinates of kind [3] (the kinds are defined in [II, Subsection 2.4]; the shift trick itself is in the proof of [II, Proposition 4.24(iii)]). For the + side one needs $\mathrm{gap}(f) \le \frac{3}{2}\,\mathrm{gap}^{\#}$, which holds because $\mathrm{gap}(f) \le Mb \le M\mu/8$ and $\mathrm{gap}^{\#} \ge M\mu/4$ (Proposition 2.32 and Lemma 2.34); the inequality "$Mb \le \mathrm{gap}^{\#}/4$" used in an

earlier version is not needed and is not claimed. The assembly with the true shift is carried out in Section 4.3.

Lemma 2.16 (Rescaling in (U)-blocks)**.** *Let m be a block of kind* (U) *with $\Omega_a(m) \neq \emptyset$, and write $r = (r^{(m)}, r^{\mathrm{rest}})$ with $r^{(m)} := (r_l)_{l\in\Omega_a(m)}$.*

(a) *Every minor of the matrix of $\Gamma^\flat(\kappa, a; r)$ is homogeneous of degree 0 or 1 in $r^{(m)}$, and every cofactor polynomial is homogeneous in $r^{(m)}$ of a degree $e \leq n$, where n is the number of variables of the cone. Hence for $s \in (0, 1]$ the substitution $r^{(m)} \mapsto sr^{(m)}$ preserves the vanishing of every rate object and multiplies every rate object by $s^e \geq s^n$; it maps Z_κ into itself and does not change the functions ϱ^0_d, $d \in \mathcal{N}_T$.*

(b) *If at a first row the carriers of Ω_m are strict non-peaks and the active ratios of block m are $sr^{(m)}$, then every active status of block m is at most s times the status obtained, with the same σ and the same inactive ratios, at the ratios $r^{(m)}$.*

Proof. (a) The shift Δ'_m is inactive, so its column is not part of the matrix (Definition 2.13). Then $r^{(m)}$ occurs only in the row $(\mathrm{X3})_m$, $\sum_{l\in\Omega_a(m)} r_l\gamma_l = 0$, whose entries are linear in $r^{(m)}$: the rows $(\mathrm{X4}^U)$ contain no ratio, the rows (X3) and $(\mathrm{X4}^0)$ of the other blocks contain only their own ratios, and the (X1) rows contain no ratio at all (their entries are $u_l(j)$, $-\tau_m(j)$, α_l and $-\mathrm{vs}_l\lambda_l\alpha_l$); the ratios enter only through (X3) and $(\mathrm{X4}^0)$. A minor is a sum of products with one entry from each of its rows, hence homogeneous of degree 1 if it uses the row $(\mathrm{X3})_m$ and of degree 0 otherwise. The cofactor vectors of an $(n-1)$-row subsystem J have as components $(n-1)$-minors, so each column of g_J is homogeneous of a degree $e_J \in \{0, 1\}$, and each minor of g_J is homogeneous of degree $\sum e_J \leq n$. The functions ϱ^0_d ($d \in \mathcal{N}_T$) involve only ratios of blocks of kind (T). (b) By Lemma 2.4(b),(c), $\varrho_l = m|r_l|\mathbf{K}(\sigma, R^2)/\Phi_l$; replacing $r^{(m)}$ by $sr^{(m)}$ multiplies $|r_l|$ by s and does not increase R^2, and $\mathbf{K}$ is increasing in R^2. □

Lemma 2.17 (Weight-freeness)**.** *Let κ be a pattern of level L with class a, and for $d \in \mathcal{N}_T$ put*

$$\psi_d(r) := m(d)\,|r_d|\,\mathbf{K}\big(\sigma_{\mathrm{rob}}(m(d)), R_a(m(d))(r)^2\big) \qquad (r \in \mathcal{K}_\kappa),$$

so that $\varrho^0_d = \psi_d/\Phi_d$.

(a) *Every entry of the matrix of $\Gamma^\flat(\kappa, a; r)$, every rate object of κ, the region $\mathcal{K}_\kappa$, the set Z_κ and the functions ψ_d are defined, by explicit formulas, from: the values $u_l(j)$ of target vectors and the coefficients α_l; signs and the combinatorial data of κ; the weights of the robust peaks of the blocks with an active component; the weights of the carriers of the blocks of kind (KN); and the variables r. In particular no weight Φ_d with $d \in \mathcal{N}_T$ occurs in any of them.*

(b) *The only row of the cone of [II, Definition 5.52] that contains the weight of a near-threshold switching carrier is the row (X4), through $1/\lambda_l$; in $\Gamma^\flat$ it is kept only in blocks of kind (KN). An active weak peak d of f belongs to Ω, so its weight occurs neither in $\tau_{m(d)}$ nor in its aggregated signature row.*

(c) *For the clean sub-window w, let $\Gamma^\#$ be the cone with the aggregated coefficients*

$$\|v_l 1_{S_l\cap(L, s_{\mathrm{far}}(w))}\|_1$$

actually used at w ([II, Definition 5.52], [II, Lemma 5.50]). Then $\Gamma^{\#}$ arises from $\Gamma^{\flat}$ by multiplying each aggregated row by a factor $t_l \in [1-\varepsilon_w, 1]$, where

$$\varepsilon_w := \max_l 2^{1+s_1(l)-s_{\mathrm{far}}(w)}$$

and $s_1(l)$ is the least element of S_l beyond L. Every rate object of $\Gamma^{\#}$ equals the corresponding one of $\Gamma^{\flat}$ times a product of at most n^2 factors t_l; in particular the zero sets coincide, and robust values are multiplied by a factor $\geq (1-\varepsilon_w)^{n^2} \geq \frac{1}{2}$.

Proof. (a) Row by row. (X1): the column of γ_l has entries $u_l(j)$; the column of Δ'_m has entries $-\tau_m(j)$, with τ_m as in Lemma 2.10 (robust peak weights); an aggregated row has the single entry α_l (multiplying γ_l if $k(l) \in \Omega$) or $-\mathrm{vs}_l\lambda_l\alpha_l$ (multiplying Δ'_m if $l \in P_{\mathrm{rob}}(m)$). (X3) in ratio variables: entries 1 and $-r_l$ (the (X5) rows and the columns of the inactive variables are not part of the matrix, Definition 2.13). (X4^0): entries vs_l and $m^2|r_l|\mathbf{K}(\sigma_{\mathrm{rob}}(m), R_a(m)(r)^2)$, with $\sigma_{\mathrm{rob}}(m)$ a sum of squares of robust peak weights. (X4) in (KN)-blocks: entries vs_l/λ_l and 1, with l a carrier of a block of kind (KN). (X4^U): entries vs_l. Minors and cofactor polynomials are polynomials in these entries; $\mathcal{K}_\kappa$, Z_κ and ψ_d are defined from them and from σ_{rob}. A carrier $d \in \mathcal{N}_T$ is not a robust peak, and it is not a carrier of a block of kind (KN), since each carrier belongs to exactly one block. (b) is read off from the list in (a) and Definition 2.8(a). (c) For l with $S_l \cap (L,\infty) \neq \emptyset$, $\alpha_l \geq v_l(s_1(l)) = \delta_l 2^{-s_1(l)}/n_l$ and $\alpha_l - \|v_l 1_{S_l\cap(L,s_{\mathrm{far}}]}\|_1 \leq \sum_{s>s_{\mathrm{far}}} \delta_l 2^{-s}/n_l \leq 2\delta_l 2^{-s_{\mathrm{far}}}/n_l$, so the actual coefficient is $t_l\alpha_l$ with t_l as stated. Multiplying a row by $t_l > 0$ multiplies every minor containing it by t_l, every cofactor vector of a subsystem containing it by t_l, and hence every minor of g_J by a product of at most n such column factors, each a product of at most n factors t_l. Finally $s_{\mathrm{far}}(w)$ is the least s with $2^{-s} \leq c_{L+1}^2$ ([II, Lemma 5.50]); the weight bound (W5) of the design (Section 4.1) gives $c_{L+1} \leq 2^{-\min S_{L+1}} = 2^{-3\cdot 2^{L+1}}$, so $s_{\mathrm{far}}(w) \geq 12\cdot 2^L$, while $s_1(l) \leq L + 3\cdot 2^L$ for $l \leq L$. So $\varepsilon_w \leq 2^{1+L-9\cdot 2^L}$, and $n \leq 2L$ ([II, Lemma 5.53]) gives $(1-\varepsilon_w)^{n^2} \geq 1 - 4L^2\varepsilon_w \geq \frac{1}{2}$. □

Remark 2.18 (Why the s-free coefficients). The rate objects of level L must be computed before the sub-windows of level L, while $s_{\mathrm{far}}(w)$ depends on c_{L+1}, which is chosen after them. Lemma 2.17(c) removes this forward reference: the rate objects are those of $\Gamma^{\flat}$, and the cone actually used at w has the same zero sets, robust values within a factor 2, and Hoffman and Łojasiewicz constants within a factor 2. The same one-line change applies to the rate objects of [II, Definition 5.52].

2.5. **Generic weights.** Fix a pattern κ of level L with $\mathcal{N}_T \neq \emptyset$, and let ψ_d ($d \in \mathcal{N}_T$) be as in Lemma 2.17. For $\Phi = (\Phi_d)_{d\in\mathcal{N}_T} \in (0,\infty)^{\mathcal{N}_T}$ put

$$F_\Phi(r) := \max_{d\in\mathcal{N}_T} \frac{\psi_d(r)}{\Phi_d} \qquad (r \in Z_\kappa).$$

At the true weights, $F_\Phi(r) = \max_{d\in\mathcal{N}_T} \varrho_d^0(r)$ is the largest floor status of the protected carriers.

Definition 2.19 (Coincidences)**.** $\Phi \in (0,\infty)^{\mathcal{N}_T}$ is a *coincidence* for κ if some $r_0 \in Z_\kappa$ with $F_\Phi(r_0) = 1$ is a (not necessarily strict) local minimum of F_Φ on Z_κ. $B_\kappa \subseteq (0,\infty)^{\mathcal{N}_T}$ denotes the set of coincidences.

By Lemma 2.17(a), Z_κ and the ψ_d do not depend on Φ; this is what makes the following two lemmas possible.

Lemma 2.20 (The threshold is not a local-minimum value)**.** *B_κ is a semialgebraic set of dimension $< |\mathcal{N}_T|$.*

Proof. Z_κ is a compact semialgebraic set and the ψ_d are continuous semialgebraic functions on it, all independent of Φ. The set

$$B_\kappa = \big\{\Phi :\ \exists r_0 \in Z_\kappa\ \big[F_\Phi(r_0) = 1\ \wedge\ \exists \varepsilon > 0\ \forall r \in Z_\kappa\ (|r - r_0| < \varepsilon \Rightarrow F_\Phi(r) \geq 1)\big]\big\}$$

is first-order definable, hence semialgebraic (Tarski–Seidenberg [11, 10]).

Take a finite partition of Z_κ into connected Nash submanifolds S compatible with the sets $\{r_d > 0\}$, $\{r_d = 0\}$, $\{r_d < 0\}$, $d \in \mathcal{N}_T$ [1, Proposition 9.1.8]. On a stratum S each $|r_d|$ equals r_d, $-r_d$ or 0, and $R_a^2 < 1$ on $\mathcal{K}_\kappa$, so each $\psi_d|_S$ is a Nash function (Lemma 2.4(d)). For a nonempty $D \subseteq \mathcal{N}_T$ and a stratum $S \subseteq \{r_d \neq 0 : d \in D\}$ let $\Psi_{D,S} := (\psi_d)_{d\in D}|_S : S \to \mathbb{R}^D$ and let $CV_{D,S}$ be its set of critical values; by the semialgebraic Sard theorem [1, Theorem 9.6.2], $\dim CV_{D,S} < |D|$ (this includes the case $\dim S < |D|$, where every point is critical).

Let $\Phi \in B_\kappa$, with a witness r_0 lying in the stratum S, and $D := \{d : \psi_d(r_0) = \Phi_d\}$. D is nonempty because $F_\Phi(r_0) = 1$, and $r_{0,d} \neq 0$ for $d \in D$ because $\psi_d(r_0) = \Phi_d > 0$. A local minimum of F_Φ on Z_κ is a local minimum on S. Suppose r_0 were a regular point of $\Psi_{D,S}$. Then there is $v \in T_{r_0}S$ with $d\psi_d(r_0)v = -1$ for all $d \in D$, and a Nash arc γ in S with $\gamma(0) = r_0$, $\gamma'(0) = v$. For small $s > 0$, $\psi_d(\gamma(s)) = \Phi_d - s + o(s) < \Phi_d$ for $d \in D$, and $\psi_d(\gamma(s)) < \Phi_d$ for $d \notin D$ by continuity; so $F_\Phi(\gamma(s)) < 1$, contradicting the local minimality. Hence $(\Phi_d)_{d\in D} \in CV_{D,S}$, and

$$B_\kappa \subseteq \bigcup_{D,S} \big\{\Phi :\ (\Phi_d)_{d\in D} \in CV_{D,S}\big\},$$

a finite union of semialgebraic sets of dimension $< |D| + |\mathcal{N}_T \setminus D| = |\mathcal{N}_T|$. □

Remark 2.21. The proof covers isolated points of Z_κ (zero-dimensional strata, at which every point is a local minimum: for an isolated point r^* the corresponding Φ satisfy $\max_d \psi_d(r^*)/\Phi_d = 1$, a set of codimension one), kinks of the maximum (through the active set D), minima on the boundary of $\mathcal{K}_\kappa$ (boundary strata are strata of Z_κ), non-strict minima, and the non-smoothness of $|r_d|$ at $r_d = 0$ (witnesses have $r_d \neq 0$ for $d \in D$, and for $d \notin D$ only continuity is used). If the region were defined with bounds involving the weights Φ_d, $d \in \mathcal{N}_T$, the set Z_κ would depend on Φ and the argument would fail; this is why $\mathcal{K}_\kappa$ is defined by the robust peak weights only.

Lemma 2.22 (Algebraically independent weights avoid all coincidences)**.** *Let $k_0 \subseteq \mathbb{R}$ be a subfield containing all values $u_l(j)$ of the target vectors and all coefficients α_l, let κ be a pattern with $\mathcal{N}_T \neq \emptyset$, and let $k := k_0(\Phi_l :\ l \leq L,\ l \notin \mathcal{N}_T)$. If the weights $(\Phi_d)_{d\in\mathcal{N}_T}$ are algebraically independent over k, then $(\Phi_d)_{d\in\mathcal{N}_T} \notin B_\kappa$.*

Proof. Put $n := |\mathcal{N}_T|$. By Lemma 2.17(a), $\mathcal{K}_\kappa$, Z_κ and the ψ_d are defined by first-order formulas whose parameters are values $u_l(j)$, coefficients α_l, rational numbers (signs; the polynomial defining the graph of $\mathbf{K}$) and weights of carriers outside $\mathcal{N}_T$; so B_κ is defined by a first-order formula φ with parameters in k. Let $k^{\mathrm{rc}} \subseteq \mathbb{R}$ be the field of real numbers algebraic over k; it is real closed. Let $B_0 \subseteq (k^{\mathrm{rc}})^n$ be the set defined by φ in k^{rc}. By the Tarski–Seidenberg transfer, B_κ is the extension of B_0 to $\mathbb{R}$, and $\dim B_0 = \dim B_\kappa < n$ by Lemma 2.20 and the invariance of dimension under extension of real closed fields [1, Chapter 5]. The Zariski closure of B_0 has the same dimension [1, Section 2.8], so some nonzero $P \in k^{\mathrm{rc}}[X_1, \ldots, X_n]$ vanishes on B_0; by

transfer of the sentence $\forall x\,(\varphi(x) \Rightarrow P(x) = 0)$, P vanishes on B_κ. The coefficients of P lie in a finite extension $k' \subseteq k^{\mathrm{rc}}$ of k. Let $\sigma_1 = \mathrm{id}, \sigma_2, \dots, \sigma_q$ be the k-embeddings of k' into an algebraic closure $\bar{k} \supseteq k'$, and $\mathcal{N}P := \prod_i \sigma_i(P)$. Its coefficients are fixed by every automorphism of $\bar{k}$ over k, hence lie in k (characteristic 0); $\mathcal{N}P \neq 0$ as a product of nonzero polynomials; and $\mathcal{N}P(x) = P(x)\prod_{i\geq 2}\sigma_i(P)(x) = 0$ for every $x \in B_\kappa$. If $(\Phi_d)_{d\in\mathcal{N}_T}$ were in B_κ, then $\mathcal{N}P((\Phi_d)_d) = 0$ would be a nontrivial algebraic relation over k. □

Remark 2.23 (Order of choices). If a family $(c_l)_{l\in\Lambda}$ is algebraically independent over k_0, then for disjoint finite sets $A, B \subseteq \Lambda$ the family $(c_a)_{a\in A}$ is algebraically independent over $k_0(c_b : b \in B)$, and the same holds for $\Phi_l = 2^{-m(l)-k(l)}c_l$. So Lemma 2.22 applies to every pattern simultaneously, whatever the order in which the weights were chosen; what matters is that k_0 contains no weight. In particular, no datum of the row f enters B_κ: f only selects the pattern κ and the point $r(f)$. This is why weights fixed in the design, before f, avoid the coincidence sets of every row.

2.6. **Quantitative coherence.**

Lemma 2.24 (Quantitative coherence)**.** *Let κ be a pattern with $\mathcal{N}_T \neq \emptyset$ and $Z_\kappa \neq \emptyset$, and let $\Phi = (\Phi_d)_{d\in\mathcal{N}_T} \notin B_\kappa$. Write $F := F_\Phi$, $Y_1 := \{r \in Z_\kappa : F(r) \leq 1\}$ and, for $h > 0$ and $r \in Z_\kappa$ ($|\cdot|$ the Euclidean norm),*

$$G_h(r) := \inf\{F(r'') :\ r'' \in Z_\kappa,\ |r'' - r| < h\}.$$

(a) *$\mu_\kappa(h) := 1 - \max_{r\in Y_1} G_h(r) > 0$ for every $h > 0$ ($\mu_\kappa :\equiv 1$ if $Y_1 = \emptyset$). For every $r \in Y_1$ there is $r'' \in Z_\kappa$ with $|r'' - r| < h$ and $F(r'') \leq 1 - \frac{3}{4}\mu_\kappa(h)$.*

(b) *If $Y_1 = \emptyset$, then $F \geq 1 + m_0$ on Z_κ for some $m_0 > 0$. If $Y_1 \neq \emptyset$, there are $C_Y > 0$ and $\alpha_Y \in (0, 1]$ with $\mathrm{dist}(r, Y_1) \leq C_Y(F(r) - 1)_+^{\alpha_Y}$ for all $r \in Z_\kappa$.*

(c) *$h \mapsto \mu_\kappa(h)$ is semialgebraic, positive, nondecreasing and ≤ 1, and there are $c_\kappa, \beta_\kappa, h_\kappa > 0$ with $\mu_\kappa(h) \geq c_\kappa h^{\beta_\kappa}$ for $0 < h \leq h_\kappa$.*

Proof. (a) G_h is upper semicontinuous on Z_κ: if $r_n \to r$ and $r'' \in Z_\kappa$ has $|r'' - r| < h$, then $|r'' - r_n| < h$ for large n, so $\limsup_n G_h(r_n) \leq F(r'')$, and taking the infimum over r'' gives $\limsup_n G_h(r_n) \leq G_h(r)$. Y_1 is compact. For $r \in Y_1$: if $F(r) < 1$, then $G_h(r) \leq F(r) < 1$; if $F(r) = 1$, then r is not a local minimum of F on Z_κ (as $\Phi \notin B_\kappa$), so there are points $r'' \in Z_\kappa$ arbitrarily close to r with $F(r'') < 1$, and $G_h(r) < 1$. An upper semicontinuous function attains its maximum on a compact set, so $\max_{Y_1} G_h < 1$. The second statement follows from $G_h(r) \leq 1 - \mu_\kappa(h) < 1 - \frac{3}{4}\mu_\kappa(h)$ and the definition of the infimum (which need not be attained on the open ball).

(b) If $Y_1 = \emptyset$, $F - 1$ is positive and continuous on the compact set Z_κ. Otherwise $f_1 := \mathrm{dist}(\cdot, Y_1)$ and $f_2 := (F - 1)_+$ are continuous semialgebraic functions on the compact semialgebraic set Z_κ with $f_2^{-1}(0) = Y_1 = f_1^{-1}(0)$; by the Łojasiewicz inequality [1, Corollary 2.6.7] there are an integer $N_Y \geq 1$ and $C > 0$ with $f_1^{N_Y} \leq C f_2$ on Z_κ. Put $\alpha_Y := 1/N_Y$, $C_Y := C^{1/N_Y}$.

(c) μ_κ is defined by a first-order formula (with the parameters of κ and Φ), hence semialgebraic; it is positive by (a), nondecreasing because G_h is nonincreasing in h, and ≤ 1 because $F \geq 0$. A semialgebraic function of one variable that is positive on some $(0, h_1)$ has, near 0^+, a convergent Puiseux expansion $\mu_\kappa(h) = ch^{p/q}(1 + o(1))$ with $c > 0$ and $p/q \geq 0$ rational (the real closure of $\mathbb{R}(t)$ is the field of algebraic

Puiseux series; [12, Chapter 7]). Take $c_\kappa := c/2$, $\beta_\kappa := \max(p/q, 1)$ and $h_\kappa \in (0, 1]$ small. □

Remark 2.25 (Definite constants). The design needs definite values of the constants of Lemma 2.24; for instance N_Y is the least admissible integer, C_Y the least admissible constant for it, p/q the Puiseux exponent, c_κ half the leading coefficient and $h_\kappa := \min\{1, \sup\{h' : \mu_\kappa \geq c_\kappa h^{\beta_\kappa} \text{ on } (0, h']\}\}$; the same device fixes the constants C_L, N_L of Lemma 2.26. Only their existence is used. The normalization of (b) is dist $\leq C_Y(F-1)_+^{\alpha_Y}$; it enters the threshold (D2) of Definition 2.27 in exactly this form. The exponent β_κ is not explicit, and $\mu_\kappa(h)$ may be much smaller than h^p for any exponent p fixed in advance (for instance $p = 4$); this is the reason for the exact holds in Proposition 2.32 (Remark 2.33).

Lemma 2.26 (Łojasiewicz displacement on a compact semialgebraic set)**.** *Let $\mathcal{K} \subseteq \mathbb{R}^n$ be compact and semialgebraic and let $\pi_1, \ldots, \pi_s$ be continuous semialgebraic functions on $\mathcal{K}$. There are $C_L \geq 1$ and an integer $N_L \geq 1$ such that for every $S \subseteq \{1, \ldots, s\}$, every $\beta \in [0, 1]$ and every $v \in \mathcal{K}$ with $\max_{i\in S} |\pi_i(v)| \leq \beta$, at least one of the following holds:* (i) $\beta \geq 1/C_L$*;* (ii) *there is $v' \in \mathcal{K}$ with $\pi_i(v') = 0$ for all $i \in S$ and $|v' - v| \leq C_L \beta^{2/N_L}$ (Euclidean norm).*

Proof. The proof of [II, Lemma 5.3] applies verbatim with $[-2, 2]^n$ replaced by $\mathcal{K}$ and $Z_S := \Pi_S^{-1}(0) \cap \mathcal{K}$, $\Pi_S := \sum_{i\in S} \pi_i^2$: the Łojasiewicz inequality [1, Corollary 2.6.7] holds for continuous semialgebraic functions on a compact semialgebraic set, and a nearest point of Z_S lies in $\mathcal{K}$. □

2.7. **Exactification with threshold protection.**

Definition 2.27 (Requirements on the design at level L)**.** Let $\mathfrak{P}(L)$ be the set of patterns of level L; under option (b) ($c_1 = 1$) only the patterns with $1 \notin \mathcal{N}_T(\kappa)$ are included. For $\kappa \in \mathfrak{P}(L)$ let Lip_κ^* be the largest of: the Lipschitz constants on $\mathcal{K}_\kappa$ of the rate objects of κ; the constant Λ_κ of Definition 2.11 (which dominates the Lipschitz constant of F_Φ); and the Lipschitz constants of the functions $r \mapsto m|r_l|\mathbf{K}(\sigma, R_a(m)(r)^2 + \rho)/\Phi_l$, $l \in \Omega_a$, on $\{r : R_a(m)(r)^2 + \rho \leq 1 - \sigma_{\mathrm{rob}}(m)/8\}$, uniformly in $\sigma \in [\sigma_{\mathrm{rob}}(m)/2, 1]$ and $\rho \geq 0$ (these are the actual statuses at fixed σ and fixed inactive ratios). Let $C_L(\kappa)$, $N_L(\kappa)$ be the constants of Lemma 2.26 for the rate objects of κ on $\mathcal{K}_\kappa$, and, when $\mathcal{N}_T(\kappa) \neq \emptyset$ and $Z_\kappa \neq \emptyset$, let $m_0, C_Y, \alpha_Y, \mu_\kappa, c_\kappa, \beta_\kappa, h_\kappa$ be the constants of Lemma 2.24 at the actual weights (with the conventions $m_0 := \infty$ if $Y_1 \neq \emptyset$, $C_Y := 0$ if $Y_1 = \emptyset$, and the choices of Remark 2.25). The design satisfies the *QC requirements at level L* if:

- (QC1) (genericity) there is a countable subfield $k_0 \subseteq \mathbb{R}$, containing every value $u_l(j)$ and every coefficient α_l, over which the family $(c_l)_{l\geq 2}$ (option (b)), resp. $(c_l)_{l\geq 1}$ (option (a)), is algebraically independent;
- (QC2) (rate objects) the rate objects of every $\kappa \in \mathfrak{P}(L)$ belong to the rate scheme of level L, with values computed at the ratio vector of the row;
- (QC3) (constants) $\mathrm{Des}(L) \geq D(L) \geq \Phi_{\min}(L)^{-1}$, $\mathrm{Des}(L) \geq C_*(L)$, and for every $\kappa \in \mathfrak{P}(L)$: $\mathrm{Des}(L) \geq \max\{\mathrm{Lip}_\kappa^*, C_L(\kappa), N_L(\kappa), C_Y, 1/m_0, 1/c_\kappa, \beta_\kappa, 1/h_\kappa\}$; here $C_*(L) := |T(L)| + 2$ is the bound, in units of values, of [II, Definition 3.6] (see also (A1) of [II, Theorem 5.7] and Step 0 of the proof of [II, Proposition 5.56]): the closing moves (C1), (C2) of the assembly ([II, Subsection 4.3]) change every value $\mathrm{val}_{l''}$ by at most $C_*(L)b$. We put $C_{**}(L) := \mathrm{Des}(L)^3$; for $L \geq l_f$ the closing moves change the active ratio

vector (in the Euclidean norm) and every status of a coarse carrier by at most $C_{**}(L)b$ (Step 0 of the proof of Proposition 2.32);

(QC4) (thresholds) for every sub-window w of L, with $\mathrm{Lip}^* := \max_\kappa \mathrm{Lip}^*_\kappa$, $C_L := \max_\kappa C_L(\kappa)$, $N_L := \max_\kappa N_L(\kappa)$, $\beta(b) := (1 + \mathrm{Lip}^* C_{**}(L))b$ and

$$\delta_0(b) := \beta(b) + C_0 D(L)^{12} b + \mathrm{Lip}^* C_L \beta(b)^{2/N_L}$$

(C_0 twice the numerical constant of Lemma 2.9(a)), the number $b = b(w)$ satisfies $b \le \eta'(w)$, $\beta(b) < 1/C_L$, $\beta(b) \le \eta'(w)/3$ and

(D1) $C_L \beta(b)^{2/N_L} \le \eta'(w)/3$;

(D2) for every $\kappa \in \mathfrak{P}(L)$ with $\mathcal{N}_T \neq \emptyset$: $\delta_0(b) \le m_0/2$, $C_Y \delta_0(b)^{\alpha_Y} \le \eta'(w)/3$, and $b \le \mu_\kappa(\eta'(w)/3)/8$;

(QC5) (weights after L) $c_{L+1} \le b(w)^2$ for every sub-window w of L.

Remark 2.28 (Consistency of the requirements). (i) The constants in (QC3) are finite: Λ_κ and the Lipschitz constants of the status functions in Lip^*_κ are $\le CL\,D(L)^9$ by Lemma 2.4(e) and Lemma 2.3(b); the Lipschitz constants of the rate objects (minors of matrices of size at most $2L$ whose entries are Lipschitz on the compact set $\mathcal{K}_\kappa$) are finite; the Łojasiewicz constants exist by Lemma 2.26; the QC constants by Lemma 2.24, which applies because, by (QC1) and Lemma 2.22 (Remark 2.23), the actual weights are not a coincidence for any $\kappa \in \mathfrak{P}(L)$. A pattern of level L involves only carriers $\le L$, so only $c_1, \dots, c_L$ enter; these are chosen before the level-L data. (ii) In (QC4) the left sides are continuous, increasing in b and tend to 0 as $b \to 0$, while the right sides are positive numbers computed before $b(w)$; so the admissible b form an interval $(0, b_{\mathrm{QC}}(w)]$ with $b_{\mathrm{QC}}(w) > 0$. The clause $b \le \mu_\kappa(\eta'/3)/8$ is positive exactly because no pattern is a coincidence. The number $b(w)$ enters later quantities only through upper bounds, so the non-explicit exponents $1/\alpha_Y$, $N_L/2$ need not be dominated by Des. (iii) $\eta'/3 \le 1/\mathrm{Des}(L) \le h_\kappa$ by (QC3), so Lemma 2.24(c) gives $\mu_\kappa(\eta'/3) \ge c_\kappa(\eta'/3)^{\beta_\kappa} \ge \mathrm{Des}^{-1}(\eta'/3)^{\mathrm{Des}}$. (iv) That T_{fin} satisfies (QC1)–(QC5) at every main stage is part of its construction in Section 4.1 ((QC1) is Lemma $\mathrm{GEN}_{\mathrm{fin}}$ there, (QC5) is the weight bound (W2)).

Lemma 2.29 (Common pull window)**.** *In the situation of [II, Lemma 4.15], let η_* be a number with $\|x\|_\infty \le \eta_* \le c_T$, and choose the pull coordinate $j_{l''}$ as the largest element of $S_{l''}$ with $v_{l''}(j_{l''}) \ge \eta_*$. Then the conclusions of [II, Lemma 4.15] hold with ε_x replaced by η_*; in particular $v_{l''}(j_{l''}) \in [\eta_*, 2^{G(l)}\eta_*)$ for every $l'' \in L_0$, the banks $\mathfrak{n}_{l''}$ are positive, and the pull masses are $\mathfrak{p}_{l''} = 24\lambda_{l''} v_{l''}(j_{l''}) \ge 24\lambda_{l''}\eta_*$. For $\eta_* := \max(\|x\|_\infty, c_{L+1})$ the pulls also satisfy $\mathfrak{p}_{l''} \ge 24\lambda_{l''} c_{L+1}$.*

Proof. The proof of [II, Lemma 4.15] uses the level ε_x only through the inequality $|x_{l''}| \le \varepsilon_x$ and through the window bounds of the pull coordinates; both hold with η_* in place of ε_x. For the positivity of the banks we recall the mechanism. After the pulls, the bank equation of carrier l'' reads $\mathfrak{n}_{l''} = (\Delta_{l''}\mathcal{N} + \beta_{l''}(\mathcal{N} - \nu_p))/((\mu^\star_{j'})^2 v_{l''}(j'))$, where $\Delta_{l''} = x_{l''} + 2v_{l''}(j_{l''}) + O(C_T\eta_*^2)$ is the correction still to be made, $\beta_{l''} = \epsilon_{l''}\langle U^* u_{l''}, e^p\rangle$ has either sign, ν_p is the Hilbert norm after the pulls and $\mathcal{N} \ge \nu_p$ the norm after all banks ([II, Lemma 4.14]), with $\mathcal{N} - \nu_p \le C_T\eta_*^2$. Since $v_{l''}(j_{l''}) \ge \eta_* \ge |x_{l''}|$, $\Delta_{l''} \ge \eta_*/2$ and hence $\mathfrak{n}_{l''} \ge c_f\eta_* > 0$ for $\eta_* \le c_T$. □

Remark 2.30 (Why a common window). If the pull of a carrier is placed at a level adapted to its own first-order correction (for instance at the larger of that correction and c_{L+1}), a carrier whose correction is tiny (such as a nearly neutral carrier that is to be held at value 0) gets $\Delta_{l''}$ of the order of c_{L+1}, while $\beta_{l''}(\mathcal{N} - \nu_p)$ is of the

order of $\mathrm{Des}\,\eta'^2$, created by the other carriers' banks; if $\beta_{l''} < 0$, the bank equation then demands a negative mass, which is not admissible. The common window $[\eta_*, 2^{G(l)}\eta_*)$ of Lemma 2.29 removes this. The lower bound $\mathfrak{p} \geq 24\lambda c_{L+1}$ is the one needed for the side condition (U3) at the supports created by the companion (Section 4.3).

Lemma 2.31 (Statuses at exact ratios)**.** *Let m be a block of kind* (T) *and $f^{\#}$ a first row such that:* (i) *the coarse carriers of block m outside Ω_m are the robust peaks $P_{\mathrm{rob}}(m)$, and they are peaks of $f^{\#}$;* (ii) *every carrier of $\Omega_m \setminus \Omega_a(m)$ has value 0 at $f^{\#}$;* (iii) $\varkappa^{\#} := \varkappa(\hat{\zeta}_m^{\#}; \Omega_m) > 0$ *and the active ratios $r^{\#}$ of block m at $f^{\#}$ lie in $\mathcal{K}_\kappa$;* (iv) $\varrho_l^0(r^{\#}) + \sigma_{\mathrm{rob}}(m)^{-3} c_{L+1}^2 < 1$ *for every $l \in \Omega_a(m)$. Then every carrier of Ω_m is a strict non-peak of $f^{\#}$, and $\varrho_l^0(r^{\#}) \leq \varrho_l(f^{\#}) \leq \varrho_l^0(r^{\#}) + \sigma_{\mathrm{rob}}(m)^{-3} c_{L+1}^2$ for every $l \in \Omega_a(m)$.*

Proof. Apply Lemma 2.4 at $f^{\#}$ with $\Omega := \Omega_m$. By (i), $P \setminus \Omega_m$ consists of $P_{\mathrm{rob}}(m)$ and of fine peaks, and $Q \setminus \Omega_m$ of fine carriers, so $\sigma' \leq \sigma_0 + c_{L+1}^2/8$ with $\sigma_0 := \sigma_{\mathrm{rob}}(m)$ (the terms $1 - \varrho_k^2$, $k \in \Omega_m \cap P$, are ≤ 0). By (ii), $R^2 = R_a(m)(r^{\#})^2 \leq 1 - \sigma_0/4$. If $\sigma' \leq 0$, then $\mathsf{k} = a + \sigma' \leq a$ and $\mathsf{k} > 0$ by (iii), so $a^2 = \sigma' + \mathsf{k}^2 R^2 \leq a^2 R^2$, which forces $R \geq 1$; this is impossible. So $\sigma' > 0$, and $\mathsf{k} = \mathbf{K}(\sigma', R^2) \leq \mathbf{K}(\sigma_0 + c_{L+1}^2/8, R^2)$. By Lemma 2.4(e) on $\mathcal{K}(\sigma_0)$, this exceeds $\mathbf{K}(\sigma_0, R^2)$ by at most $2c_{L+1}^2/\sigma_0^2$, and $m|r_l|/\Phi_l = \varrho_l^0(r^{\#})/\mathbf{K}(\sigma_0, R^2) \leq 1/(2\sigma_0)$ by (iv). Hence $\varrho_l(f^{\#}) = m|r_l^{\#}|\mathsf{k}/\Phi_l \leq \varrho_l^0(r^{\#}) + \sigma_0^{-3} c_{L+1}^2 < 1$ for $l \in \Omega_a(m)$, and $\varrho_l(f^{\#}) = 0$ for the other carriers of Ω_m. So $\Omega_m \subseteq Q(f^{\#})$, and Lemma 2.6 with $P_0 := P_{\mathrm{rob}}(m)$ gives the lower bound. □

Proposition 2.32 (Exactification with threshold protection without (KN))**.** *Let the design satisfy the QC requirements of Definition 2.27 at the main stage L. Let $N \geq 1$, let $f \in S_{p_N^*}$ have finite base support, let $L \geq l_f$, let w be a sub-window of L that is clean for f, let a be an activity class of w, and let κ be the pattern of (f, w, a); under option* (b) *assume $1 \notin \mathcal{N}_T(\kappa)$. Put $\mu := \mu_\kappa(\eta'/3)$ if $\mathcal{N}_T \neq \emptyset$. Then there is a companion $f^{(1a)}$ of f with the following properties.*

(a) (Exactness.) *For every block m, $\varkappa(\hat{\zeta}_m^{(1a)}; \Omega_m) = \varkappa_m^* > 0$, where $\varkappa_m^*$ is the value after the closing moves; the active ratios r'' of $f^{(1a)}$ lie in Z_κ, every robust rate object of κ has modulus $\geq u/4$ at r'', $|r'' - r(f)| \leq 5L\mathrm{Des}(L)\eta'$, and in blocks of kind* (T) *and* (KN) *the block coordinates satisfy $|r''^{(m)} - r(f)^{(m)}| \leq 2\eta'$.*

(b) (Statuses.) *In every block of kind* (T)*: every carrier of $\mathcal{N}(m)$ is a strict non-peak with $\varrho \leq 1 - \mu/2$, every carrier of $\Omega_m \setminus \Omega_a(m)$ has value exactly 0, and the coarse peaks are exactly $P_{\mathrm{rob}}(m)$. In every block of kind* (KN)*: the conclusions of [II, Proposition 5.56(b)] hold with η' in place of $\eta(w)$. In every block of kind* (U)*: every active carrier of Ω_m has $\varrho \leq 1 - \eta'$, and every inactive near-threshold carrier is a strict non-peak with $\varrho \leq 1 - \eta'$, hence with gap $\geq \eta' M$. Every other coarse carrier keeps its robust status; relative positions of coarse carriers move by at most $C\mathrm{Des}(L)\eta'$ except where stated.*

(c) (Cost.) $p^*(f^{(1a)} - f) \leq C_f \mathrm{Des}(L)^3 \eta' \log(1/\eta') = o(T_{\mathrm{lo}}(w)^2)$.

Moreover $\mu \geq \mathrm{Des}(L)^{-1}(\eta'/3)^{\mathrm{Des}(L)}$ and $8b \leq \mu$.

Proof. Write $\sigma_0(m) := \sigma_{\mathrm{rob}}(m)$. For $L \geq l_f$ we have $b \leq \sigma_0(m)/8$ and $3L \cdot 2\eta' \leq \sigma_0(m)/4$ for every m (Lemma 2.3(b), (QC3), $\eta' \leq T_{\mathrm{lo}}^4$ and $\mathrm{Des}^2 T_{\mathrm{lo}} \leq 1$), so Lemmas 2.9 and 2.12 apply with $\rho_0 := 2\eta'$.

Step 0 (closing moves). Apply only the closing moves (C1), (C2) of the assembly ([II, Subsection 4.3]); there is no donor raise in active blocks, and the active weak peaks are *not* pushed at this stage. Let f_1 be the resulting row and r_1 its active ratio vector. The closing moves change every value by at most $C_*(L)b \le \mathrm{Des}(L)b$ (QC3). A ratio is $r_l = u_l(\hat z)/\varkappa_m$, and $\varkappa_m$, $\bar\vartheta_m$ are Lipschitz functions of the values with f-constants, bounded below by $1/C_f$ ([II, Lemma 2.7(a), Lemma 2.10]); a status carries in addition the factor $1/\Phi_l \le D(L)$. Hence

$$\Delta r_l \le \frac{\Delta u_l}{\varkappa_m} + \frac{|u_l|\,|\Delta\varkappa_m|}{\varkappa_m^2} \le C_f \mathrm{Des}\, b, \qquad \Delta\varrho_l \le C_f D(L)\mathrm{Des}\, b,$$

and since $|\Omega_a| \le 2L$, $C_f \le \mathrm{Des}(L)$ and $D(L) \le \mathrm{Des}(L)$ for $L \ge l_f$, we get $|r_1 - r(f)| \le C_{**}b$ (Euclidean norm) and statuses move by at most $C_{**}b$, $C_{**} = \mathrm{Des}^3$. By (QC4), $C_{**}b \le \beta(b) \le \eta'/3$. At $r(f)$ every tiny rate object has modulus $\le b$, so at r_1 its modulus is $\le \beta(b) = (1 + \mathrm{Lip}^* C_{**})b$. If $\mathcal{N}_T \ne \emptyset$, then for $d \in \mathcal{N}_T$, by Lemma 2.9(a) and $\varrho_d(f) \le 1 + b$,

$$\begin{aligned} F_\Phi(r_1) &\le \max_d \varrho_d^0(r(f)) + \Lambda_\kappa C_{**} b \\ &\le 1 + b + \tfrac12 C_0 D^{12}(b + c_{L+1}^2) + \mathrm{Lip}^* C_{**} b \le 1 + \beta(b) + C_0 D^{12} b, \end{aligned}$$

using $\sigma_{\mathrm{rob}}^{-1} \le D^2$, $c_{L+1}^2 \le b$ and $\Lambda_\kappa \le \mathrm{Lip}^*$.

Step 1 (Łojasiewicz point). $r_1 \in \mathcal{K}_\kappa$ by Lemma 2.12 with $\rho_0 = 2\eta'$, since $|r_1 - r(f)| \le C_{**}b \le \eta'/3$. Apply Lemma 2.26 on $\mathcal{K}_\kappa$ to the rate objects of κ, with S the tiny ones and $\beta := \beta(b)$. Alternative (i) is excluded by $\beta(b) < 1/C_L$ (QC4). So there is $r' \in Z_\kappa$ with $|r' - r_1| \le C_L \beta^{2/N_L} \le \eta'/3$ (D1), and $F_\Phi(r') \le F_\Phi(r_1) + \mathrm{Lip}^* C_L \beta^{2/N_L} \le 1 + \delta_0(b)$.

Step 2 (a point of Y_1). If $\mathcal{N}_T \ne \emptyset$: if Y_1 were empty, Lemma 2.24(b) would give $F_\Phi(r') \ge 1 + m_0 > 1 + \delta_0(b)$ by (D2). So $Y_1 \ne \emptyset$, and there is $r_1' \in Y_1$ with $|r_1' - r'| = \mathrm{dist}(r', Y_1) \le C_Y \delta_0(b)^{\alpha_Y} \le \eta'/3$ (D2). If $\mathcal{N}_T = \emptyset$, put $r_1' := r'$.

Step 3 (Puiseux point). If $\mathcal{N}_T \ne \emptyset$, the actual weights are not a coincidence for κ (Lemma 2.22 with (QC1); under option (b) because $1 \notin \mathcal{N}_T$), so Lemma 2.24(a) with $h := \eta'/3$ gives $r'' \in Z_\kappa$ with $|r'' - r_1'| < \eta'/3$ and $F_\Phi(r'') \le 1 - \frac34\mu$. If $\mathcal{N}_T = \emptyset$, put $r'' := r_1'$. Then $|r'' - r_1| < \eta'$ and, as $|r_1 - r(f)| \le C_{**}b \le \eta'/3$, $|r'' - r(f)| \le 2\eta'$; so $r'' \in \mathcal{K}_\kappa$ with $R_a(m)(r'')^2 \le 1 - \sigma_0(m)/2$ (Lemma 2.12). Every robust rate object has modulus $\ge u - 2\mathrm{Lip}^*\eta' \ge u/2$ at r'', every robust status moves by at most $2\mathrm{Des}\,\eta' \ll u$, and the signs of the ratios of the carriers of $\mathcal{N}_T$ are those of $r(f)$, because $|r_d(f)| = \varrho_d(f)\Phi_d/(m\mathsf{k}) \ge \Phi_{\min}(L)/(2L) \gg \eta'$ ($\mathsf{k} \le \frac12$ by Lemma 2.4(a)). By (D2), $8b \le \mu$; by Remark 2.28(iii), $\mu \ge \mathrm{Des}^{-1}(\eta'/3)^{\mathrm{Des}}$.

Step 4 (realization in one joint fixed point). The targets are:

- active carriers $l \in \Omega_a(m)$: value $u_l(\hat z^{(1a)}) = r_l'' \varkappa_m^*$ if m is of kind (T) or (KN), and $s_m r_l'' \varkappa_m^*$ with $s_m := 1 - 4\mathrm{Des}\,\eta'$ if m is of kind (U);
- *holds*: every carrier of $\Omega_m \setminus \Omega_a(m)$, m of kind (T), gets value exactly 0 (by (T) these are exactly the nearly neutral coarse strict non-peaks of the block other than the active ones);
- $\varkappa(\hat\zeta_m; \Omega_m) = \varkappa_m^*$ for every block m.

One-shot moves come first: the pulls of all tuned carriers, in the common window of Lemma 2.29 with $\eta_* := \max(\|x\|_\infty, c_{L+1})$, where x is the vector of corrections. The corrections include the (U)-block rescaling $(1 - s_m) r_l'' \varkappa_m^* = 4\mathrm{Des}\,\eta'\, r_l'' \varkappa_m^*$ and the displacements $|r'' - r_1|\varkappa_m^* \le \eta' \varkappa_m^*$ with $\varkappa_m^* \le C_f$, so $\eta_* \le C_f \mathrm{Des}\,\eta'$. In

blocks of kind (KN): if the lever d_0 (an inactive carrier of Ω_m with $\varrho_{d_0}(f) \geq u$, Definition 2.8(b)) is a weak peak of f ($\varrho_{d_0}(f) \in [1, 1+b]$), it is first pushed inward to $\varrho_{d_0} = 1-b$ by a private move; the value change is $\leq 2b\bar{\vartheta}_m \Phi_{d_0}/m$ and the cost is $O(b)$, and $\bar{\vartheta}_m$ and $\varkappa_m$ are continuous across the crossing of the threshold ([II, Lemma 2.7(a), Lemma 2.10(a)]). Thereafter d_0 is a strict non-peak with $\varrho_{d_0} \geq u$, as required in [II, Proposition 5.56]. Then the inward lever push of the proof of [II, Proposition 5.56] (Step 2), of size $s_0 := C' \mathrm{Des}\, \eta' \bar{\vartheta}_m \Phi^2_{P_m}/(\varrho_{d_0} M_m)$, is made, and the inactive near-threshold carriers of the block are pushed inward individually to relative position $\leq 1-\eta'$. In blocks of kind (U) the inactive near-threshold carriers are pushed inward individually to relative position $\leq 1-\eta'$. Since such a carrier has $\varrho(f_1) \leq 1 + (1+C_{**})b$ and $(1+C_{**})b \leq \eta'/3$ by (QC4), each push has size $\leq 2\eta'$ in relative position. The remaining unknowns are the banks of [II, Lemma 4.15] for the active carriers and for the held carriers of class G, the sizes of two-sided z-moves on free coordinates of $S_h^{\mathrm{nat}} \setminus (F \cup T(L))$ for the held carriers h of class R (their two signed rooms are $\geq u \| v_h 1_{S_h^{\mathrm{nat}}} \|_1 \gg b$, and the moves keep every moved coordinate free, so the E_c-types of the pattern do not change), and the size of one peak push per block. The equations are the targets above. The derivative of each tuned value with respect to its own bank is $\geq \mathrm{Des}^{-1/6}/2$, with respect to its own z-move it is $v_h(j)$, and with respect to the other unknowns it is $O(C\mathrm{Des}\, \eta_*)$ (second order, through the Hilbert normalization, [II, Lemma 4.14]); the derivative of $\varkappa_m$ with respect to its peak push lies in $[\frac{3}{4}, 1]$ ([II, Lemma 2.10(b)(i)], $X \leq CD^2 c_{L+1}^2$), and with respect to the values of Ω_m it is $O(X)$ ([II, Lemma 2.10(b)(iii)]); $\varkappa_m$ is continuous ([II, Lemma 2.10(a)]). So the Jacobian is diagonally dominant, and the explicit fixed point of [II, Lemma 4.15], extended by one scalar equation per block solved by the intermediate value theorem, converges, as in Step 2 of the proof of [II, Proposition 5.56]. Later re-solves change the corrections only by $O(\mathrm{Des}^2 \eta_*^2) \ll \eta_*$, which the banks absorb; the pulls are not moved again. At the resulting row $f^{(1a)}$ the active ratios equal r'' in blocks of kind (T) and (KN), and $(s_m r''^{(m)}, r''^{\mathrm{rest}})$ in blocks of kind (U); we call this vector r'' again. By Lemma 2.16(a) it still lies in Z_κ, robust rate objects are multiplied by at least $s_m^n \geq \frac{1}{2}$ (as $4n\mathrm{Des}\, \eta' \leq \frac{1}{2}$), hence have modulus $\geq u/4$, and F_Φ is unchanged. In blocks of kind (T) and (KN) the block coordinates of r'' are those of the Puiseux point, so $|r''^{(m)} - r(f)^{(m)}| \leq 2\eta'$ there. This proves (a).

Statuses. Blocks of kind (T): hypotheses (i)–(iii) of Lemma 2.31 hold by construction (the robust peaks move by $\ll u$; $\varkappa_m^* > 0$). For $d \in \mathcal{N}(m)$, $\varrho_d^0(r'') \leq F_\Phi(r'') \leq 1 - \frac{3}{4}\mu$; for the other active carriers l of the block, $\varrho_l(f) \leq 1-u$, so $\varrho_l^0(r'') \leq \varrho_l(f) + C_0 D^{12}(b + c_{L+1}^2) + 2\mathrm{Lip}^* \eta' < 1 - u/2$. Since $\sigma_0^{-3} c_{L+1}^2 \leq D^6 b^4 \leq b/8 \leq \mu/64$ (QC5), hypothesis (iv) holds, and Lemma 2.31 gives $\varrho_d(f^{(1a)}) \leq 1 - \frac{3}{4}\mu + \frac{\mu}{64} \leq 1 - \mu/2$ for $d \in \mathcal{N}(m)$, all carriers of Ω_m strict non-peaks, and coarse peaks exactly $P_{\mathrm{rob}}(m)$. Blocks of kind (KN): the proof of [II, Proposition 5.56](b) applies verbatim, the total displacement of the ratios being $\leq 2\eta'$ (including the Puiseux step) and the lever push dominating it by the choice of C'. Blocks of kind (U): by (QC3) the actual statuses are Des-Lipschitz in the ratios at fixed σ and fixed inactive ratios, so at the unscaled ratios the active statuses are $\leq 1 + C_{**} b + 2\mathrm{Des}\, \eta'$; by Lemma 2.16(b) the scaled ones are $\leq s_m(1 + 3\mathrm{Des}\, \eta') \leq 1 - \mathrm{Des}\, \eta' \leq 1 - \eta'$. The pushed inactive near-threshold carriers end at relative position $\leq 1 - \eta'$ up to second-order effects (see below), so they are strict non-peaks with gap $M(1-\varrho) \geq \eta' M$ ([II, Lemma 2.6(b)]);

the margin η' is kept because the push target is $1-\eta'-\eta'^{3/2}$, say, and the second-order effects are $O(\mathrm{Des}^2\eta'^2) \le \eta'^{3/2}$. Second-order cross effects are $O(\mathrm{Des}^2\eta'^2)$, small compared with the margins η' of (U)-blocks and $c_f\mathrm{Des}\,\eta' M$ of (KN)-blocks; in (T)-blocks they do not enter at all, because there the statuses depend only on the exact active ratios, the exact held values and the peak set (Lemma 2.31). This proves (b).

Cost. By [II, Lemma 4.15(c)] and Lemma 2.29, each pull has mass $\le 6\cdot 2^{G(L)}\eta_*$ and each bank mass $\le C_T\eta_*$, with $C_T \le C_f\mathrm{Des}$ and $2^{G(L)} \le \mathrm{Des}^{1/6}$; the z-moves of held carriers, the lever push and the inward pushes have size $\le C_f\mathrm{Des}\,\eta'$, and the weak-peak lever pushes have size $O(b) \le O(\eta')$. So every move has size $\le C_f\mathrm{Des}^2\eta'$ in units of masses and of values (as $\eta_* \le C_f\mathrm{Des}\,\eta'$). There are at most $|L_0| \le 2L$ tuned carriers (one pull and one bank each), at most $2L$ held or pushed carriers and one peak push per block, hence at most $6L$ moves, and $L \le \mathrm{Des}^{1/2}$. Therefore $\|a^{(1a)}-a\|_1 \le C_f\mathrm{Des}^{5/2}\eta'$ and $\Delta_m(\hat z^{(1a)}-\hat z) \le C_f\mathrm{Des}^{5/2}\eta'$ for every block m, and the counting bound of [II, Subsection 2.3] gives $c(\hat z^{(1a)}-\hat z) \le C_f\mathrm{Des}^{5/2}\eta'\log(e/\eta')$. By [II, Lemma 2.12(iv)], $p^*(f^{(1a)}-f) \le C_f\mathrm{Des}^3\eta'\log(1/\eta')$. Since $\eta' = T_{\mathrm{lo}}^4/(L\mathrm{Des})$, $\log(1/\eta') \le 5\log(1/T_{\mathrm{lo}})$ and $\mathrm{Des}^2T_{\mathrm{lo}} \le 1$, this is $\le C_fT_{\mathrm{lo}}^3\log(1/T_{\mathrm{lo}}) = o(T_{\mathrm{lo}}^2)$. □

Remark 2.33 (Why the holds must be exact). The margin $\mu = \mu_\kappa(\eta'/3)$ is only known to be $\ge c_\kappa(\eta'/3)^{\beta_\kappa}$ with a non-explicit exponent β_κ. Every quantity that separates the true status of a protected carrier from its floor $\psi_d(r'')/\Phi_d$ must therefore be dominated by μ and not merely by a power of η'. The active ratios are exact (fixed point), σ depends only on the peak set up to fine terms $\le c_{L+1}^2/8 \le b^4$, and the remaining term is the contribution of the non-active carriers of Ω_m to R^2. A non-active value of size v raises R^2 by an amount of order v^2, and the moves made for the other blocks change such values at second order, by $O(\mathrm{Des}^2\eta'^2)$ ([II, Lemma 4.14]); the resulting status error, of order $\mathrm{Des}^4\eta'^4$ times design factors, is not small compared with $c_\kappa(\eta'/3)^{\beta_\kappa}$ when $\beta_\kappa > 4$. Hence every non-active coarse strict non-peak of a (T)-block is held at value exactly 0 inside the joint fixed point, not merely made small. In (KN)- and (U)-blocks the margins are of order η' and second-order effects are harmless.

Lemma 2.34 (Persistence of the margin)**.** *In the situation of Proposition* 2.32*, let $f^\natural$ be obtained from $f^{(1a)}$ by moves that change the relative position of every carrier of $\mathcal{N}_T$ by at most $C_f\mathrm{Des}(L)c_{L+1}$. If $L \ge l_f$, then $\varrho_d(f^\natural) \le 1-\mu/4$ for every $d \in \mathcal{N}_T$.*

Proof. By (QC5), $c_{L+1} \le b^2$, and $b \le \eta' \le T_{\mathrm{lo}}^4/\mathrm{Des}$ gives $C_f\mathrm{Des}\,b \le C_fT_{\mathrm{lo}}^4 \le \frac12$ for $L \ge l_f$. Hence $C_f\mathrm{Des}\,c_{L+1} \le b/2 \le \mu/16 < \mu/4$, using $8b \le \mu$. □

Remark 2.35 (Scope). (i) Lemma 2.34 is the form in which the later steps of the companion (fine structure, release of far parts, and the passage from the reference shift to the true shift) use Proposition 2.32; there these moves change statuses, shifts and $\varkappa$ by at most $C_f\mathrm{Des}(L)c_{L+1}$ (Section 4.3). The lemma does not use any upper bound on μ. (ii) Nothing about f is assumed beyond the clean sub-window and finite base support; the class is fixed before the exactification, which is legitimate because classes are read off the decomposition data ([II, Lemma 5.53]). (iii) Under option (b), Proposition 2.32 excludes the patterns in which stage 1 (whose weight $c_1 = 1$ is rational) is an active near-threshold carrier of a block of kind (T). If this happens at clean sub-windows of infinitely many main stages, then $\varrho_1(f) = 1$. If

the weight-one carrier lies in a block $m(1) > N$, no pattern of p_N contains it; in the remaining case the exception is open. A slice version of Lemma 2.20 would apply if $2^{-m(1)-1}$ were a regular value of ψ_1 on every stratum; we do not know whether it can be a critical value identically in the transcendental parameters, and no such claim is made. (iv) The finite base support of f is used only through the realization of Step 4 (far coordinates outside F for pulls, banks and z-moves); the transfer of this section to infinite base support is not claimed here.

3. The violation-tolerant d-consistent engineered approximant

The engineered approximants of [I, Theorem 7.18] are built at a single row and at late stages of a construction sequence. To average over a window of scales at a row *near* the target row f one needs per-piece bounds whose radius is proportional to the scale of the piece, with constants that do not depend on the window, and one must tolerate small violations of the sign conditions of two-piece data. [II, Lemma 2.30] tolerates violations at a single stage; [II, Remark 2.31(c)] records the obstruction to a uniform use: the θ-decomposition and the $\pm$-decompositions of a piece are linked by a first-order term (the *junction mismatch*) that is not uniformly small along windows. In this section we remove it exactly: at the approximant f' the normalized values of the finitely many switching carriers are re-tuned to their values at the base row by small *levers* (*d-consistency*). With this, the bounds of [I, Theorem 7.18] and [II, Lemma 2.30] hold piece by piece with radius $c_{\mathrm{flat}}t$, and the averaging can be carried out at the norm-attaining approximant itself.

Standing conventions. Throughout this section I is finite and the base is *diagonal*: $Uk_j = \mu_j^\star e_j$, $U^*e_j^* = \mu_j^\star k_j$ with $\mu_j^\star > 0$ and (k_j) orthonormal, as for T_{final} [II, Definition 3.1]. For $A \in \ell_1 \setminus \{0\}$ and $z \in B_{\ell_\infty}$ with $z_j = \operatorname{sign} A_j$ on $\operatorname{supp} A$ we write $f[A,z]$ for the first row with prescribed data [II, Subsection 4.2]; its forced data are $e = U^*A/\|U^*A\|_H$, $\hat z = z + Ue$ and $w_m = J_m(R_m^{**}\hat z)$. For a row f_0 with forced data $(a_0, z_0, e_0, \nu_0, \hat z_0, \dots)$ and $F_0 := \operatorname{supp} a_0$ finite we put $\zeta_m := R_m^{**}\hat z_0$ and $A_m := |\zeta_m|_m = \sigma_m/q_0$. For a strict non-peak k of block m the *normalized value* is

$$\mathrm{nv}_k := \zeta_m(k)/A_m = \lambda_{k,m}u_{k,m}(\hat z_0)/A_m.$$

By [I, Lemma 1.7], $\mathrm{nv}_k = \Phi_m(k)^2 w_m(k)/C_m$, hence for every $\omega \in c_{00}$ vanishing on P_m

$$d_m(\omega) = \frac{\langle D_m w_m, D_m\omega\rangle}{C_m} = \sum_k \omega(k)\,\mathrm{nv}_k. \tag{2}$$

We also use the clamp variable $v_k := m|u_{k,m}(\hat z_0)|/A_m = |\mathrm{nv}_k|/\Phi_m(k)$ of the proof of [I, Lemma 7.14] ($\lambda_{k,m} = m\Phi_m(k)$).

Definition 3.1 (Pieces and violations)**.** Let $f_0 \in S_{p^*}$ have F_0 finite and contact set K_0. A *piece* at f_0 at scale $t \in (0,1]$ is a functional g with $g(\xi_0) = 0$ together with two pairs $(b^\pm, \omega^\pm)$ that represent g at f_0 in the sense of [I, Definition 7.7] ($\omega_m^\pm \in c_{00}$, $\operatorname{supp}\omega_m^\pm \subseteq Q_m$); they need not be side-admissible. Put $\Delta\omega := \omega^- - \omega^+$, $\Delta_m := d_m(\Delta\omega_m)$, $v := b^+ - b^- = \sum_m R_m^*(\Delta\omega_m - \Delta_m w_m)$, $b^\theta := \frac12(b^+ + b^-)$, $\omega^\theta := \frac12(\omega^+ + \omega^-)$. The *violation* at $j \notin F_0$ is $\mathrm{viol}(j) := (z_0(j)b_j^+)_- + (z_0(j)b_j^-)_+$ for $j \in K_0$ and $\mathrm{viol}(j) := |b_j^+| + |b_j^-|$ for $j \notin F_0 \cup K_0$, and the *violation mass* is $\epsilon := \sum_{j\notin F_0}\mathrm{viol}(j)$, as in [II, Lemma 2.30].

Let $f \in S_{p^*}$ with F finite be the *target row*, and fix $\rho \in (0,1)$, $\eta_0 \in (0,2]$ with $\rho^2(1+\eta_0) \le (1+\rho^2)/2$, $A_0, A_2 \ge 1$ and $\gamma_B \in (0,1]$. Put

$$\delta := \tfrac{1}{2}\big(1-\rho^2(1+\eta_0/2)\big) \;\ge\; (1-\rho^2)/8,$$

so $\rho^2\kappa \le 1-2\delta$ whenever $\kappa \le 1+\eta_0/2$. An *f-constant* is a number that depends only on $(f, I, T, \rho, \eta_0, A_0, A_2, \gamma_B)$ and on choices made at f (transfer data); a *window quantity* may in addition depend on a window and its data.

3.1. **Masses off the support, d-consistency, uniform neighbourhoods.** The next lemma complements [II, Lemma 4.14].

Lemma 3.2 (Masses and values)**.** *Let $a \in \ell_1 \setminus \{0\}$ with $F := \operatorname{supp} a$, $\nu := \|U^*a\|_H$, $e := U^*a/\nu$, $z \in B_{\ell_\infty}$ with $z = \operatorname{sign} a$ on F, and $\hat z := z + Ue$. Let J be finite, $c \in \mathbb{R}^J$, $A := a + \sum_{s\in J} c_s e_s^*$, $\nu_A := \|U^*A\|_H$, $e_A := U^*A/\nu_A$, $X := U^*(A-a) = \sum_{s\in J} c_s \mu_s^\star k_s$.*

(i) *If $J \cap F = \emptyset$, then $X \perp U^*a$, $\nu_A^2 = \nu^2 + \|X\|^2$, $0 \le \nu_A - \nu \le \|X\|^2/(2\nu)$, $\|e_A - e\| \le \|X\|/\nu$, and for every $u \in \ell_1$*

$$\langle U^*u, e_A - e\rangle = \frac{1}{\nu_A}\sum_{s\in J} c_s(\mu_s^\star)^2 u(s) + \Big(\frac{\nu}{\nu_A} - 1\Big)\langle U^*u, e\rangle.$$

(ii) *In general $\|e_A - e\| \le 2\|X\|/\nu$. If $s \in J \cap F$ and all other mass coordinates lie off F, the first-order change of $\langle U^*u, e_A\rangle$ caused by c_s is $c_s(\mu_s^\star)^2\big(u(s) - a_s\langle U^*u, e\rangle/\nu\big)/\nu$.*

(iii) *If $z^A \in B_{\ell_\infty}$ with $z^A = \operatorname{sign} A$ on $\operatorname{supp} A$ and $\hat x_A := z^A + Ue_A$, then for every $u \in \ell_1$*

$$|u(\hat x_A) - u(\hat z)| \le \|U^*u\|\,\|e_A - e\| + \sum_s |u(s)|\,|z_s^A - z_s|;$$

for a carrier $\|U^*u_{k,m}\| \le q^*(u_{k,m}) = 1$.

(iv) *For every $b \in \ell_1$, $\sum_{s\notin F}(\mu_s^\star)^2 b(s)^2 \le \nu h(b)$, where $h(b) = \|P^\perp U^*b\|^2/\nu$ ([I, Definition 3.1]). Consequently, if pieces $i = 1, \dots, n$ at $f[a,z]$ are given and the masses are $c_s := 4\rho s_1 z_s \max_i |b_i^\theta(s)|$ on a set J disjoint from F, then*

$$\|X\|^2 \le 16\rho^2 s_1^2\,\nu \sum_{i=1}^n h(b_i^\theta), \qquad \|X\| \le 4\rho s_1 \max_s \mu_s^\star \sum_{i=1}^n \|b_i^\theta\|_1.$$

Proof. (i) $U^*a = \sum_{s\in F} a_s\mu_s^\star k_s$ and $X \in \operatorname{span}\{k_s : s \in J\}$ with $J \cap F = \emptyset$, so $X \perp U^*a$; this gives $\nu_A^2 = \nu^2 + \|X\|^2$ and $\nu_A - \nu = \|X\|^2/(\nu_A + \nu)$. The identity is [II, (5)] rewritten with $e_A = (\nu e + X)/\nu_A$. As $\langle e_A, e\rangle = \nu/\nu_A$, $\|e_A - e\|^2 = 2(\nu_A - \nu)/\nu_A \le \|X\|^2/\nu^2$. (ii) $\|x/\|x\| - y/\|y\|\| \le 2\|x-y\|/\|y\|$; the first-order formula is the derivative of $(\nu e + c_s\mu_s^\star k_s)/\|\nu e + c_s\mu_s^\star k_s\|$ at $c_s = 0$, using $\langle k_s, e\rangle = a_s\mu_s^\star/\nu$. (iii) $\hat x_A - \hat z = (z^A - z) + U(e_A - e)$ and $u(Uh) = \langle U^*u, h\rangle$; $\|U^*u\| \le q^*(u)$. (iv) Write $U^*b = U^*(b1_{F^c}) + U^*(b1_F)$. The first summand lies in $\operatorname{span}\{k_s : s \notin F\}$, which is orthogonal to e and to $\operatorname{span}\{k_s : s \in F\}$; hence $P^\perp U^*b = U^*(b1_{F^c}) + P^\perp U^*(b1_F)$ is an orthogonal sum and $\|P^\perp U^*b\|^2 \ge \|U^*(b1_{F^c})\|^2 = \sum_{s\notin F}(\mu_s^\star)^2 b(s)^2$. For the masses, $\|X\|^2 = \sum_s c_s^2(\mu_s^\star)^2 \le 16\rho^2 s_1^2 \sum_i \sum_{s\notin F}(\mu_s^\star)^2 b_i^\theta(s)^2$, and $\|X\| \le \sum_s |c_s|\mu_s^\star$ with $\max_i|\cdot| \le \sum_i|\cdot|$. □

By (iv) and convexity of h, $h(b^\theta) \le (h(b^+) + h(b^-))/2 \le \Gamma_w/q_0$; so for data with $\Gamma_w \le 2$ the mass vector is $O(s_1(n\nu/q_0)^{1/2})$, uniformly in the scales of the pieces.

Definition 3.3 (d-consistency)**.** Let $f_0, f' \in S_{p^*}$ (data of f' primed) and let $\Omega = \bigcup_m \Omega_m$ be a finite set of strict non-peaks of f_0, $\Omega_m \subseteq Q_m$. f' is *d-consistent with f_0 on Ω* if every $k \in \Omega_m$ is a strict non-peak of f' and $\mathrm{nv}'_k = \mathrm{nv}_k$, i.e. $u_{k,m}(\hat{z}') = (A'_m/A_m)u_{k,m}(\hat{z}_0)$.

Lemma 3.4 (Identities of d-consistency)**.** *Let f' be d-consistent with f_0 on Ω.*

(a) *For every m and every ω supported in Ω_m: $d'_m(\omega) = d_m(\omega)$ and $H'_m(\omega) = (C_m/C'_m)H_m(\omega)$. For $k \in \Omega_m$: $w'_m(k) = (C'_m/C_m)w_m(k)$ and*

$$\mathrm{gap}'_m(k) - \mathrm{gap}_m(k) = \Big(1 - \frac{C'_m}{C_m}\Big)\big(1 - \mathrm{gap}_m(k)\big). \tag{3}$$

In particular $\mathrm{gap}'_m(k) \geq \mathrm{gap}_m(k)/2$ as soon as $|C'_m - C_m| \leq C_m \,\mathrm{gap}_m(k)/(2(1 - \mathrm{gap}_m(k)))$.

(b) *For every piece at f_0 with $\omega^{\pm}$ supported in Ω the junction terms $\kappa^{\pm}_m := \rho\,(d'_m - d_m)(\omega^{\pm}_m - \omega^{\theta}_m)$ of the proof of [I, Theorem 7.18] vanish.*

(c) (The block scalars are frozen up to the carriers outside Ω.) *Fix m, let $k_\natural \in P_m$ and $r > 0$ with $[C_m - r, C_m + r] \subseteq (0,1)$, $|C'_m - C_m| \leq r$, and $v_{k_\natural}, v'_{k_\natural} \geq \Phi_m(k_\natural)(1-c)/c$ for all $c \in [C_m - r, C_m + r]$. Put $c_* := 2\Phi_m(k_\natural)^2(1 - C_m - r)/(C_m + r)^3$ and $P^{\mathrm{rob}}_m := \{k \in P_m : v'_k \geq \Phi_m(k)(1 - C_m)/C_m\}$. Then*

$$|C'_m - C_m| \leq \frac{2(1 - C_m)}{C_m c_*} \sum_{k \notin \Omega_m \cup P^{\mathrm{rob}}_m} \Phi_m(k)\,|v'_k - v_k|.$$

Proof. (a) By (2) at both rows, $d'_m(\omega) = \sum \omega(k)\mathrm{nv}'_k = d_m(\omega)$; $H'_m(\omega) = (\|D_m\omega\|^2 - d'_m(\omega)^2)/C'_m = (C_m/C'_m)H_m(\omega)$. $\Phi^2 w'/C' = \mathrm{nv}' = \mathrm{nv} = \Phi^2 w/C$ gives $w'(k) = (C'/C)w(k)$. With $M = 1 - C$ and $|w(k)| = M - \mathrm{gap}(k)$: $\mathrm{gap}'(k) = M' - (C'/C)|w(k)| = (1 - C') - (C'/C)(1 - C - \mathrm{gap}(k))$, which is (3); then $\mathrm{gap}' \geq \mathrm{gap}/2$ iff $(C'/C - 1)(1 - \mathrm{gap}) \leq \mathrm{gap}/2$, which holds under the stated bound. (b) $\omega^{\pm} - \omega^{\theta} = \mp\frac{1}{2}\Delta\omega$ is supported in Ω; apply (a). (c) By the clamp formula of the proof of [I, Lemma 7.14], C_m and C'_m are roots of $F(c; v) = 1$ and $F(c; v') = 1$, $F(c; v) := \sum_k \min(\Phi_m(k)(1-c)/c, v_k)^2$, each nonincreasing in c. On the interval between C_m and C'_m the $k_\natural$-term of $F(\cdot; v')$ equals $\Phi_m(k_\natural)^2((1-c)/c)^2$, whose derivative is $\leq -c_*$; so $c_*|C'_m - C_m| \leq |F(C_m; v') - F(C'_m; v')| = |F(C_m; v') - F(C_m; v)|$. Put $x_k := \Phi_m(k)(1 - C_m)/C_m$. The k-th term of the last difference vanishes for $k \in \Omega_m$ ($v'_k = v_k$ by d-consistency) and for $k \in P^{\mathrm{rob}}_m$ (both $v_k \geq x_k$, since k is a peak of f_0, and $v'_k \geq x_k$); otherwise it is at most $2x_k|v'_k - v_k|$. □

Remark 3.5. If Ω_m contains every strict non-peak of f_0 in block m except a set Q^{fine}_m of carriers of small total weight, and the peaks of f_0 stay peaks with margin above their value changes, Lemma 3.4(c) bounds $|C'_m - C_m|$ by the moves of Q^{fine}_m alone; by (a) every carrier of Ω_m then keeps its gap and its relative position up to the same error. So d-consistency on all coarse strict non-peaks also protects the statuses of the switching carriers. (Numerically, with $\Omega_m = Q_m$ in a finite model, $|C'_m - C_m| \leq 1.1 \cdot 10^{-16}$; this is evidence only.)

Lemma 3.6 (Uniform neighbourhoods)**.** *Let $f \in S_{p^*}$ have F finite. In every block fix $\gamma_m \in (M_m/2, M_m)$ with $|w_m(k)| \neq \gamma_m$ for all k, put $\gamma_0 := \min_m(M_m - \gamma_m)$, fix a peak $k^{\natural}_m$ with $\alpha_m(k^{\natural}_m) \neq 0$ and margin $\mu^{\natural}_m > 0$, and, for $\eta_1 \in (0, \frac{1}{4}]$, transfer data $(\gamma_m, k^{\varsigma}_{*,m}, \Lambda^{\varsigma}_m)$ of f by [I, Lemma 7.4]. There is $r_f(\eta_1) > 0$ such that every $f'' \in S_{p^*}$ with $p^*(f'' - f) \leq r_f(\eta_1)$ (data double-primed) satisfies:*

(a) $|C_m''-C_m| \le C_m/8$, $|M_m''-M_m| \le \gamma_0/4$, $|\sigma_m''-\sigma_m| \le \sigma_m/8$, $|q_0''-q_0| \le q_0/8$, $|\nu''-\nu| \le \nu/8$, $F \subseteq \operatorname{supp} a''$ *with* $|a_j''| \ge a_{\min}/2$ *on* F ($a_{\min} := \min_F |a_j|$), *and* $k_m^\natural$ *is a peak of* f'' *with the same sign and margin* $\ge \mu_m^\natural/2$*;*
(b) *the transfer data persist:* $k_{*,m}^\varsigma$ *is a peak of* f'' *with* $w_m''(k_{*,m}^\varsigma) = M_m''$ *and positive margin,* $P_m'' \subseteq L_m''$*, and the quantities of [I, Proposition 7.3] built at* f'' *satisfy* $e_{y,m}''^\varsigma \ge \frac{1}{2}$, $\iota_m''^\varsigma \le 2\eta_1$, $\|D_m y_m''^\varsigma\|_2 \le K_y$ *and* $|Y_m''^\varsigma|/q_0'' \le K_Y$*, with* K_y, K_Y *independent of* η_1*.*

Proof. Otherwise there are $f_n \to f$ violating (a) or (b). By [I, Proposition 1.11], $a_n \to a$ in ℓ_1, the scalar data converge, $w_{n,m} \to w_m$ coordinatewise and $\xi_n \to \xi$ weak*; the strict threshold inequality of [I, Lemma 1.7] at $k_m^\natural$ and its margin are continuous, so $k_m^\natural$ is eventually a peak of f_n with margin $\ge \mu_m^\natural/2$; and [I, Lemma 7.5] gives the persistence and convergence of the transfer quantities. Since $e_y^\varsigma \ge 1-\eta_1 \ge \frac{3}{4}$ ([I, Lemma 7.4(d)]), $e'' \ge \frac{1}{2}$ eventually; $\|D_m y^\varsigma\|_2 \le \|\Phi_m\|_2 + \Lambda_0/m$ and $|Y^\varsigma|/q_0 \le \sigma_m e_y^\varsigma/q_0 + \Lambda_0$ do not depend on η_1. This contradicts the choice of f_n. □

All f-constants below are computed at f and are valid at every row of the ball $B_f := \{f'' : p^*(f''-f) \le r_f(\eta_1)\}$; rows of windows and their approximants lie in B_f eventually. The margin clause in (a) is what the slope argument of Lemma 3.4(c) needs on the whole interval between C_m and C_m'', and the bound on $|M_m''-M_m|$ is what [I, Lemma 7.2] needs at f'' (it is not implied by $|C_m''-C_m| \le C_m/8$ when $\gamma_0 < C_m/4$).

3.2. **The junction mismatch for valid data.** In the proof of [I, Theorem 7.18] the $\pm$-decompositions contain the terms $\kappa_m^\pm = \rho\,(d_m' - d_m)(\omega_m^\pm - \omega_m^\theta) = \mp\frac{\rho}{2}(d_m' - d_m)(\Delta\omega_m)$, which enter the first-order block terms $\mathfrak{l}_m'$ as $\tau\kappa_m^\pm$. The following bound corrects the order s_1/t^2 stated in [II, Remark 2.31(c)], which was computed for a datum scaled by $1/t$ whose coefficient Γ_w grows like t^{-2}.

Proposition 3.7 (Junction mismatch)**.** *Let* $f_0 \in S_{p^*}$ *have* F_0 *finite, and let* $(b^\pm, \omega^\pm)$ *be a piece at* f_0 *with* $\Gamma_w(b^\pm, \omega^\pm) \le \Gamma_{\max}$ (*computed at* f_0) *and* $\operatorname{supp}\omega_m^\pm \subseteq \Omega_m \subseteq Q_m$, Ω_m *finite. Let* $f' = f[A', z']$ *be a row such that every* $k \in \Omega_m$ *is a strict non-peak of* f'*, put* $\Pi_e := \|e' - e_0\|$ *and* $t_k := \|u_k 1_{\{z' \ne z_0\}}\|_1$ (*for the engineered approximant of [I, Definition 7.12],* $t_k = t_{k,m}(N'')$ *off the window masses*)*. Then for every block* m*:*

(a) $\|D_m\Delta\omega_m\|_2^2 = \Delta_m^2 + C_m H_m(\Delta\omega_m) \le \dfrac{4\Gamma_{\max}}{\sigma_m}\Big(C_m + \dfrac{C_m^3}{M_m^2\,\Phi_{P_m}^2}\Big)$, *where* $\Phi_{P_m}^2 := \sum_{k\in P_m}\Phi_m(k)^2$*;*
(b) *with* $\delta u_k := u_k(\hat z') - u_k(\hat z_0)$,

$$|\kappa_m^\pm| \le \frac{\rho m}{2}\,\|D_m\Delta\omega_m\|_2\,|\Omega_m|^{1/2}\max_{k\in\Omega_m}\frac{|\delta u_k|}{A_m'} + \frac{\rho}{2}|\Delta_m|\,\frac{|A_m' - A_m|}{A_m'};$$

(c) *consequently*

$$|\kappa_m^\pm| \le C\,|\Omega_m|^{1/2}\big(\Pi_e + \max_{k\in\Omega_m} t_k\big) + C\big(\Pi_e + \textstyle\sum_k \lambda_{k,m} t_k\big),$$

where C *depends only on* ρ*, on* $\Gamma_{\max}$ *and on the block constants* C_m, M_m, σ_m, Φ_{P_m} *and* A_m'*.*

Proof. (a) Decompose $D\Delta\omega = (\Delta/C)Dw + y$ with $y \perp Dw$; then $\|y\|^2 = \|D\Delta\omega\|^2 - \Delta^2 = CH(\Delta\omega)$. Since $\Delta\omega$ vanishes on P and $|w| = M$ there, $y = -(\Delta/C)Dw$

on P, so $\|y\| \geq |\Delta| M\Phi_P/C$ and $\Delta^2 \leq C^3 H(\Delta\omega)/(M^2\Phi_P^2)$. Finally $H(\Delta\omega) \leq 2H(\omega^+) + 2H(\omega^-) \leq 4\Gamma_{\max}/\sigma$. (b) By (2) at f_0 and at f' (the carriers of Ω_m are strict non-peaks of both rows), $(d'-d)(\Delta\omega) = \sum_{k\in\Omega} \Delta\omega(k)(\mathrm{nv}'_k - \mathrm{nv}_k)$ and $\mathrm{nv}'_k - \mathrm{nv}_k = \lambda_k \delta u_k/A' + \lambda_k u_k(\hat{z}_0)(1/A' - 1/A)$. The second parts sum to $d(\Delta\omega)(A-A')/A' = \Delta(A-A')/A'$. For the first, $\lambda_k = m\Phi_k$ and Cauchy–Schwarz give $|\sum \Delta\omega(k)\lambda_k \delta u_k| \leq m\|D\Delta\omega\|_2 (\sum_\Omega \delta u_k^2)^{1/2}$. (c) Lemma 3.2(iii): $|\delta u_k| \leq \Pi_e + t_k$; and $|A'-A| \leq \sum_k \lambda_k |\delta u_k|$ because $|\cdot|_m \leq \|\cdot\|_1$ [I, Remark 1.1(c)] and $\sum_k \lambda_k \leq 1$. Insert (a). □

Remark 3.8 (Why an exact cancellation is used; heuristic). By Lemma 3.2(iv), $\Pi_e = O(s_1(n\nu/q_0)^{1/2})$ for valid data, uniformly in the scales; so the mismatch is uniform in the scale t of a piece. It is *not* uniform along windows: $|\Omega_m|$ and the mass vector (which carries the masses of all n pieces of a window) grow with the window, while the rebalancing of [I, Proposition 7.3] absorbs a first-order level shift only with a fixed inefficiency $\iota \sim \eta_1$, which cannot be made window-dependent because [I, Lemma 7.2] needs $|\varepsilon|\Lambda \leq \gamma/2$ with $\Lambda = \Lambda(\eta_1)$ unbounded. The ratio of the resulting cost to the budget $\delta\tau^2$ at $|\tau| = s_1$ is of order $\eta_1 |\Omega|^{1/2} \|X\|/(\delta s_1)$. That this order is attained in general is a heuristic statement, supported by a finite-model computation in which the untuned mismatch is proportional to $\|X\|$ (and Proposition 3.7(b) is attained when $|\Omega_m| = 1$); it is not used below. This is why the mismatch is removed exactly, by d-consistency (Lemma 3.4(b)).

3.3. **Levers and exact d-consistency.** Let f_0 be a row with F_0 finite and $\Omega = \bigcup_m \Omega_m$ a finite set of strict non-peaks of f_0. A *pre-row* is a row $f_1 = f[A_1, z_1]$ obtained from f_0 by adding finitely many masses (of sign $z_0(j)$ at contacts $j \notin F_0$, or of the sign of $a_0(j)$ at $j \in F_0$) and by changing z_0 on a set Z_1; put $\nu_1 := \|U^* A_1\|_H$,

$$\delta_1 := \max_{k\in\Omega} |u_k(\hat{x}_1) - u_k(\hat{z}_0)|, \qquad \Delta_1 := \max_m \sum_k \lambda_{k,m} |u_{k,m}(\hat{x}_1) - u_{k,m}(\hat{z}_0)|.$$

Definition 3.9 (Levers)**.** Let $l \in \Omega$ be a carrier of block $m(l)$ and $\epsilon_l \in \{\pm 1\}$. Lever coordinates j of l satisfy $v_l(j) := u_l(j) > 0$. A *lever* of l is one of the following one-parameter families $P \mapsto (A, z)$, $P \in \mathbb{R}$, acting only at the lever coordinates Lev_l:

- (Z) *zero-data lever* at $j \notin \operatorname{supp} A_1$: $z(j) := \mathrm{clamp}_{[-1,1]}(z_1(j) + \epsilon_l P/v_l(j))$, and the overflow $|z_1(j) + \epsilon_l P/v_l(j)| - 1$, if positive, is placed as a bank at j of mass $\beta_j\, v_l(j) \cdot$ (overflow) with the sign of the contact reached, $\beta_j := \nu_1/((\mu_j^\star)^2 v_l(j))$;
- (TU) *tuning lever*: a pull coordinate $j_p \notin \operatorname{supp} A_1$ with $z_1(j_p) = \epsilon_l$ and a bank coordinate $j' \neq j_p$ with $z_1(j') = \epsilon_l$ and either $j' \notin \operatorname{supp} A_1$ or $\operatorname{sign} A_1(j') = \epsilon_l$; the pull is *fixed*: $z(j_p) := -\epsilon_l$ and a mass $-\epsilon_l \mathfrak{p}_l$ at j_p, $\mathfrak{p}_l := 24\lambda_l v_l(j_p)$; for $P \geq -2v_l(j_p)$ a bank of mass $\beta_{j'}(P + 2v_l(j_p))$ with sign ϵ_l at j'. The pull coordinate is taken from a set Pull_l of admissible pull coordinates of l (coordinates $j \notin \operatorname{supp} A_1$ with $z_1(j) = \epsilon_l$ and $v_l(j) > 0$), prescribed together with the lever.

A free coordinate with room and a contact with $z_1(j) = \epsilon_l$ (inward moves for $P \leq 0$, a bank for $P > 0$) are special cases of (Z). For a family of levers $(\mathrm{Lev}_l)_{l\in\Omega}$ write $f(P) := f[A(P), z(P)]$ for $P \in \mathbb{R}^\Omega$, with data $\hat{x}(P)$, $A_m(P) := |R_m \hat{x}(P)|_m$. The *separation condition* (SEP) is: (i) the sets Lev_l are pairwise disjoint and disjoint from $\operatorname{supp} A_1$, except TU bank coordinates with $\operatorname{sign} A_1(j') = \epsilon_l$; (ii)

$\mathrm{Lev}_l \cap \operatorname{supp} u_k = \emptyset$ for all $k \in \Omega \setminus \{l\}$. Zero-value absorber pairs tuned by a mass at a support coordinate are *not* levers in this sense (Remark 3.28).

Lever constants: $A_{\min} := \min_m A_m(f_0)$; G with $v_l(j_p) \in [r, 2^G r]$ for the pulls in use; $\beta_{\max} := \max \beta_j$ over the bank coordinates;

$$c_{\mathrm{tiny}} := \max_{l\in\Omega} \sum_{k\notin\Omega} \lambda_k \sum_{s\in \mathrm{Lev}_l^0} \frac{|u_k(s)|}{v_l(s)}, \qquad W_{\mathrm{pull}} := \sum_{k:\ \operatorname{supp} u_k \ni j_p \text{ for a pull } j_p,\ k\neq l(j_p)} \lambda_k,$$

where $\mathrm{Lev}_l^0 \subseteq \mathrm{Lev}_l$ consists of the (Z) coordinates and the TU bank coordinates of l (the TU pull coordinates are excluded: the pull is a fixed move, accounted for by W_{pull}). For $r_{\min} > 0$ let $\overline{W}_{\mathrm{pull}}(r_{\min})$ be the total weight $\sum \lambda_k$ of the carriers k for which there are $l \in \Omega \setminus \{k\}$ and $j \in \mathrm{Pull}_l$ with $v_l(j) \geq r_{\min}$ and $u_k(j) \neq 0$. It does not depend on the choice of the pulls, it is nonincreasing in $r_{\min}$, and $W_{\mathrm{pull}} \leq \overline{W}_{\mathrm{pull}}(r_{\min})$ whenever every pull in use has $v_l(j_p) \geq r_{\min}$; C_{lev} is the largest change of a carrier value per unit of $|P|_\infty$ caused by the lever moves at its own coordinates; $Y = (Y_{k,l})$ is the first-order effect on $\epsilon_k u_k(\hat{x})$ of the banks at support coordinates $j' \in \operatorname{supp} A_1$ through the normalization (Lemma 3.2(ii)): $|Y_{k,l}| \leq |A_1(j')|/(\nu_1 v_l(j'))$. Put $K_0 := 1 + 2/A_{\min}$ and $K_J := 2K_0$.

Lemma 3.10 (Exact d-consistency by levers)**.** *Let f_0, Ω, a pre-row f_1 and levers with* (SEP) *be given, and put, for $l \in \Omega$ of block m,*

$$F_l(P) := \epsilon_l \Big[u_l(\hat{x}(P)) - \frac{A_m(P)}{A_m(f_0)} u_l(\hat{z}_0) \Big].$$

There are $r_0 > 0$ and $C_2 < \infty$, depending only on the block data of f_0, Ω, the lever data (v_l and $\mu^\star$ on Lev_l, $\beta_{\max}$, G) and the status margins of the carriers of Ω and of the peaks of f_0, such that the following holds.

(i) $|F(0)|_\infty \leq \delta_1 + \dfrac{\Delta_1 + 2W_{\mathrm{pull}}}{A_{\min}} + C_2(2^{G+1}r)^2 + \dfrac{2^{G+2} r |\Omega| c_{\mathrm{tiny}}}{A_{\min}}$.

(ii) *Suppose*

$$K_J|F(0)|_\infty \leq \frac{r}{2}, \quad 3K_J C_2 r \leq \frac{1}{8}, \quad \frac{K_J|\Omega|c_{\mathrm{tiny}}}{A_{\min}} \leq \frac{1}{8}, \quad K_0\|Y\|_\infty \leq \frac{1}{2},$$
$$\delta_1 \leq r, \quad \Delta_1 \leq r, \quad r \leq r_0,$$

and $v_l(j_p) \in [r, 2^G r]$ for the TU levers. Then F has a zero P^* with $|P^*|_\infty \leq r$. At $f(P^*)$ every $l \in \Omega$ is a strict non-peak with $\mathrm{nv}_l(f(P^*)) = \mathrm{nv}_l(f_0)$, i.e. $f(P^*)$ is d-consistent with f_0 on Ω; every peak of f_0 whose margin, in value units, exceeds $C_{\mathrm{lev}}(r + \delta_1)$ plus the change of the threshold, is a peak of $f(P^*)$ with the same sign; the mass vector of the lever masses satisfies $\|X_{\mathrm{lev}}\| \leq C_2 r$; a carrier k meeting no lever coordinate moves by at most $|u_k(\hat{x}_1) - u_k(\hat{z}_0)| + \|e(P^*) - e_1\|$, and a carrier meeting lever coordinates by at most that plus $C_{\mathrm{lev}} r + 2|u_k(j_p)|$ (the last term for the pulls it meets).

(iii) *For $l \in \Omega_m$, $\mathrm{gap}_m(l)(f(P^*)) \geq \mathrm{gap}_m(l)(f_0)/2$ as soon as $|C_m(f(P^*)) - C_m(f_0)| \leq C_m \,\mathrm{gap}_m(l)/(2(1 - \mathrm{gap}_m(l)))$ (Lemma 3.4(a)).*

Proof. (1) Admissibility and continuity. For $|P|_\infty \leq r_0$ the pair $(A(P), z(P))$ is admissible: banks have the sign of z, clamped z-moves keep $|z| \leq 1$, a TU bank needs $P \geq -2v_l(j_p)$, which holds since $|P| \leq r \leq v_l(j_p)$. Finitely many coordinates move continuously, so $e(P) = U^*A(P)/\|U^*A(P)\|_H$, $\hat{x}(P) = z(P) + Ue(P)$ and $A_m(P)$ are continuous in P.

(2) First-order structure. By construction and Lemma 3.2(i), (ii) with base A_1: $\epsilon_l(u_l(\hat{x}(P)) - u_l(\hat{x}(0))) = P_l + E_l(P)$. A z-move at $j \in \mathrm{Lev}_l$ changes $\epsilon_l u_l$ by exactly P; a bank of mass $\beta_j x$ changes it by $x\,\nu_1/\nu_A$ up to $O(\|X_{\mathrm{lev}}\|^2/\nu_1^2)$; by (SEP)(ii) the moves of the other levers change u_l only through the normalization, at second order, except for the banks at support coordinates, whose first-order effect is $(YP)_l$. The bank slope is $\nu_1/\nu_A = 1 - O(r^2)$, so $|E_l(P) - (YP)_l| \le C_2'(r + |P|)|P|$ after the fixed moves; this is the form used in (3). Next, A_m is a norm with norming functional $w_m(P)$ at $R_m\hat{x}(P)$; with $\mathrm{d}\hat{x} := \hat{x}(P) - \hat{x}(0)$,

$$0 \le A_m(P) - A_m(0) - \langle w_m(0), R_m \mathrm{d}\hat{x}\rangle \le \langle w_m(P) - w_m(0), R_m \mathrm{d}\hat{x}\rangle.$$

For a carrier k whose status does not change, the clamp formula gives $|w_m(P)(k) - w_m(0)(k)| \le K_{\mathrm{cl}}(|u_k(\mathrm{d}\hat{x})| + |C_m(P) - C_m(0)|)$, so its contribution to the right side is of second order; a carrier that may change status contributes at most $2\lambda_k|u_k(\mathrm{d}\hat{x})|$, which is of second order unless k meets a lever coordinate. Further $\langle w_m(0), R_m\mathrm{d}\hat{x}\rangle = \sum_{k\in\Omega_m} \lambda_k w_m(0)(k)\epsilon_k(P_k + E_k) + \varrho_m$, where $|\varrho_m| \le |\Omega| c_{\mathrm{tiny}}|P|_\infty + O(|P|^2)$ collects the carriers outside Ω that meet lever coordinates. Finally $w_m(0)(k) = w_m(f_0)(k) + O(\delta_1 + 2^{G+1}r)$ and $u_l(\hat{x}(0)) = u_l(\hat{z}_0) + O(\delta_1 + (2^{G+1}r)^2)$.

(3) Linearization. With $p_l := \epsilon_l u_l(\hat{z}_0)/A_m(f_0)$ and $q_k := \epsilon_k\lambda_k w_m(f_0)(k)$ $(k, l \in \Omega_m)$ we obtain $F(P) = F(0) + (J + Y)P + R(P)$ with $J = I - pq^T$ blockwise and, for $|P|_\infty \le r_0$,

$$|R(P)|_\infty \le C_2(|P|_\infty + \delta_1 + r)|P|_\infty + \frac{|\Omega| c_{\mathrm{tiny}}}{A_{\min}}|P|_\infty.$$

By (2), $q^T p = \sum_{k\in\Omega_m} w_m(k)\mathrm{nv}_k = \sum_{\Omega_m} \Phi_m(k)^2 w_m(k)^2/C_m \le \|D_m w_m\|_2^2/C_m = C_m \le \|\Phi_m\|_2 < \frac{1}{2}$. By the Sherman–Morrison formula $(I - pq^T)^{-1} = I + pq^T/(1 - q^T p)$, and since $|u_l(\hat{z}_0)| \le q^*(u_l)q^{**}(\hat{z}_0) = 1$ and $\|q\|_1 \le \sum_k \lambda_k \le 1$, $\|J^{-1}\|_\infty \le 1 + 2\|p\|_\infty\|q\|_1 \le K_0$. As $K_0\|Y\|_\infty \le \frac{1}{2}$, $\|(J + Y)^{-1}\|_\infty \le 2K_0 = K_J$.

(4) Poincaré–Miranda. $G(P) := (J + Y)^{-1}F(P) = (J + Y)^{-1}F(0) + P + (J + Y)^{-1}R(P)$ is continuous on $[-r, r]^\Omega$. On the face $P_l = r$, using $\delta_1 \le r$,

$$G_l(P) \ge r - K_J|F(0)|_\infty - K_J\big(3C_2r^2 + |\Omega| c_{\mathrm{tiny}} r/A_{\min}\big) \ge r - \tfrac{r}{2} - \tfrac{r}{8} - \tfrac{r}{8} > 0,$$

and symmetrically $G_l < 0$ on $P_l = -r$. By the Poincaré–Miranda theorem [9] G has a zero P^* in the cube, and $F(P^*) = 0$.

(5) Statuses, sizes, and (i). Along the cube every value moves by at most $\delta_1 + C_{\mathrm{lev}}r$ (plus the fixed pull term for carriers meeting a pull), and C_m, $\bar{\vartheta}_m$ move by $O((r + \delta_1 + \Delta_1)/A_{\min}) = O(3r/A_{\min})$ through the clamp equation (here the hypotheses $\delta_1 \le r$ and $\Delta_1 \le r$ of (ii) are used; a small $|F(0)|$ alone does not bound Δ_1, because of cancellations in $A_m(0) - A_m(f_0)$); r_0 is chosen below the status margins of the carriers of Ω (in value units: $l \in \Omega_m$ stays a strict non-peak while $|u_l| < \bar{\vartheta}_m\Phi_m(l)/m$, a room of order $\Phi_m(l)\,\mathrm{gap}_m(l)A_m/(mC_m)$) and of the peaks considered, divided by C_{lev} and the Lipschitz constant of the threshold. So no status changes along the cube, which justifies (2). At P^*, $\epsilon_l u_l(\hat{x}) = \epsilon_l(A_m/A_m(f_0))u_l(\hat{z}_0)$, i.e. $\mathrm{nv}_l(P^*) = \mathrm{nv}_l(f_0)$. The lever masses are at most $\beta_{\max}(r + 2v_l(j_p)) \le \beta_{\max}2^{G+2}r$ and the pull masses at most $24\lambda_l 2^G r$, so $\|X_{\mathrm{lev}}\| \le C_2 r$ after enlarging C_2; the value bounds follow from Lemma 3.2(iii). For (i): at $P = 0$ the TU levers perform the fixed pull and a bank of mass $2\beta v_l(j_p)$; for l the flip $(-2v_l(j_p))$ and the bank $(+2v_l(j_p)\nu_1/\nu_A)$ cancel up to $C_2(2^{G+1}r)^2$, other carriers of Ω move only at second

order (SEP), and $|A_m(0) - A_m(f_0)| \le \Delta_1 + 2W_{\mathrm{pull}} + 2^{G+2}r|\Omega|c_{\mathrm{tiny}} + C_2(2^{G+1}r)^2$; divide by $A_{\min}$. (iii) is Lemma 3.4(a). □

Remark 3.11 (On the constants). (a) The TU pull is a *fixed* move: a carrier $k \neq l$ meeting j_p moves by $2|u_k(j_p)|$, not by $O(r)$; this is why W_{pull} appears in (i) and must be small in applications (there such carriers live at stages beyond j_p). The fixed pull and bank also contribute the second-order term $C_2(2^{G+1}r)^2$ to $F(0)$; in a finite model this term dominates $|F(0)|$ (evidence only). (b) The gap criterion of (iii) is sharp in the sense that the weaker criterion $|C_m(f(P^*)) - C_m(f_0)| \le \mathrm{gap}_m(l)/4$ does not imply $\mathrm{gap}(f(P^*)) \ge \mathrm{gap}(f_0)/2$ when $C_m < 1/2$; a finite-model example with $C_m = 0.243$, gap $= 0.01$ turns the carrier into a peak. (c) The rank-one structure is exact: a lever acting on one value changes the block norm, hence all normalized values of the block, and $1 - q^T p \ge 1 - C_m > \frac{1}{2}$; no second lever per block is needed for d-consistency.

3.4. **Window families and the d-consistent engineered approximant.**

Definition 3.12 (Window family)**.** Let f be the target row. A *base row* is $f_0 \in S_{p^*}$ with $p^*(f_0 - f) \le r_f(\eta_1)/2$ (Lemma 3.6; η_1 is fixed in Theorem 3.15), F_0 finite, $F \subseteq F_0$; put $E := F_0 \setminus F$. A *window family* at f_0 consists of pieces $(g_i; b_i^{\pm}, \omega_i^{\pm})$, $i = 1, \dots, n$, at scales $t_i \in [T, t_1]$, all supported in one finite set $\Omega = \bigcup_m \Omega_m$ of strict non-peaks of f_0, such that:

- (U1) $t_i\|b_i^{\pm}\|_1 \le A_0$ and $\Gamma_w^{(0)}(b_i^{\pm}, \omega_i^{\pm}) \le 1 + \eta_0/2$ (computed at f_0);
- (U2) $\operatorname{supp}\omega_{i,m}^{\pm} \subseteq \Omega_m$, and every $k \in \operatorname{supp}\omega_{i,m}^{\pm}$ is of kind [1′]: $\mathrm{gap}^0(k) > 0$ and $|\omega_i^{\pm}(k)| \le 2\,\mathrm{gap}^0(k)/t_i$; or [2]: $\mathrm{gap}^0(k) \ge \gamma_B$ and $|\omega_i^{\pm}(k)| \le A_2/t_i$; or [3]: $|\omega_i^{\pm}(k)| \le A_2/t_i$, $\mathrm{vs}_k\omega_i^{+}(k) \le 1.5\,\mathrm{gap}^0(k)/t_i$ and $\mathrm{vs}_k\omega_i^{-}(k) \ge -1.5\,\mathrm{gap}^0(k)/t_i$, $\mathrm{vs}_k := \operatorname{sign} w_m^0(k)$;
- (U3) for $j \in E$: $b_i^{\pm}(j) = \tilde{b}_i^{\pm}(j) + r_i^{\pm}(j)$ with $r_i^{+}(j) = -r_i^{-}(j)$, where $\tilde{b}_i^{\pm}(j)$ is contact-like ($z_0(j)\tilde{b}_i^{+}(j) \ge 0 \ge z_0(j)\tilde{b}_i^{-}(j)$) or satisfies $t_i|\tilde{b}_i^{\pm}(j)| \le K_3|a_0(j)|$, K_3 an f-constant;
- (U4) $b_i^{\theta}(j) = 0$ at every free coordinate $j \notin F_0 \cup K_0$ with $\mathrm{viol}_i(j) > 0$, and the *total violation* $\epsilon_i := \sum_{j \notin F_0} \mathrm{viol}_i(j) + 2\sum_{j \in E} |r_i^{+}(j)|$ is at most ϵ;
- (U5) for $I_- := \{m : \Delta_{i,m} < 0 \text{ for some } i\}$: $\mathrm{Scr}_m^{(0)}(x) \le C_S x^2$ for $0 < x \le x_0$ and $m \in I_-$ ([I, Definition 7.16] at f_0);
- (U6) every $l \in \Omega$ has a lever (Definition 3.9) at f_0, the levers satisfy (SEP), there is a window quantity $r_{\mathrm{pull}} \ge 0$ such that for every $r \in [r_{\mathrm{pull}}, r_0]$ every TU lever has a pull coordinate $j_p \in \mathrm{Pull}_l$ with $v_l(j_p) \in [r, 2^G r]$, and at the lever coordinates the data satisfy: at a (Z) coordinate j, $b_i^{\theta}(j) = 0$; at a TU bank coordinate j', the data are contact-like of sign ϵ_l up to an antisymmetric residue; at a TU pull coordinate j_p, $t_i|b_i^{\pm}(j_p)| \le (2A_2 + C_\Delta)\lambda_l v_l(j_p)$ up to an antisymmetric residue; all these residues and all $|b_i^{\pm}(j)|$ at (Z) coordinates are included in ϵ_i;
- (U7) $|\Delta_{i,m}| \le C_\Delta$, C_Δ an f-constant.

Coarse carriers are those of Ω and the peaks of f_0 with margin $\ge x_0$; all other carriers are *fine*. Window quantities: n, T, $\mathrm{gap}_{\min} := \min_{k \in \Omega} \mathrm{gap}^0(k)$, x_0, C_S, $|\Omega|$, the lever constants $\Lambda_{\mathrm{lev}} := \max(K_J, C_2, C_{\mathrm{lev}}, \beta_{\max}, 2^G, 1/r_0, 1/A_{\min}, 1/C_{\min})$, and

the fine weights

$$c_{\mathrm{fine}} := \sum_{k \text{ fine}} \lambda_k, \quad W_{\mathrm{lev}} := \sum_{\substack{k \notin \Omega \\ k \text{ meets a lever coordinate}}} \lambda_k,$$

and c_{tiny}, W_{pull} are as in Subsection 3.3.

Kind [1′] contains kind [1] of [I, Lemma 8.42] (constant 2, without the condition gap $\geq t^2$); it does not contain kind [1] of [II, Subsection 2.4], whose constant is 3, and the constant 2 is used in Step 3 of the proof of Theorem 3.15. Kind [1′] covers carriers pushed inward with $\Delta\omega = 0$.

Definition 3.13 (The d-consistent engineered approximant)**.** Given a window family, choose in this order:

(i) $N_w \geq \max F_0$ with $\mathrm{tail}_W := \max_i \big(\|b_i^{\pm} 1_{(N_w,\infty)}\|_1 + \|v_i 1_{(N_w,\infty)}\|_1\big)$ at most $\min(T^3, c_{\mathrm{late}}\delta T/A_0)$;

(ii) $s_1 \in (0,1]$; the *mass set* W consists of the contacts $j \in [1, N_w] \setminus F_0$ of f_0 and of the $j \in E$ with $\max_i |b_i^\theta(j)| > 0$; the masses are $\mathsf{m}_j := 4\rho s_1 \max_i |b_i^\theta(j)|$ with sign $z_0(j)$;

(iii) the pre-row $f_1 := f[A_1, z_1]$, $A_1 := a_0 + \sum_{j\in W} \mathsf{m}_j z_0(j) e_j^*$, $z_1 := z_0 1_{[1,N'']}$ (N'' from (iv)); $\Pi_X := \|e_1 - e_0\|$ (it does not depend on N''); with $r_{\min} := 32 K_J T^4 s_1$ and $\overline{W}_{\mathrm{pull}} := \overline{W}_{\mathrm{pull}}(r_{\min})$,

$$r := 16K_J\Big(1 + \frac{1}{A_{\min}}\Big)\big(\Pi_X + 2T^4 s_1\big) + \frac{32 K_J \overline{W}_{\mathrm{pull}}}{A_{\min}},$$

and then pull coordinates $j_p \in \mathrm{Pull}_l$ with $v_l(j_p) \in [r, 2^G r]$ (they exist by (U6) when $r \geq r_{\mathrm{pull}}$). Since $r \geq r_{\min}$, these pulls have $v_l(j_p) \geq r_{\min}$, so the actual W_{pull} satisfies $W_{\mathrm{pull}} \leq \overline{W}_{\mathrm{pull}}$; the definition of r is not circular;

(iv) $N'' > N_w$, larger than every lever coordinate, with

$$\mathfrak{t}(N'') \leq T^4 s_1^2, \qquad \max_{k\in\Omega} t_k(N'') \leq T^4 s_1, \qquad \sum_m \sum_k \Phi_m(k) 1[t_{k,m} > s_1 T^4] \leq s_1^2 T^4,$$

$$\rho V_{>N''} \leq \delta s_1/64, \qquad \|g_i 1_{(N'',\infty)}\|_1 \leq T^4 s_1 \ \text{ for every piece;}$$

(v) P^* by Lemma 3.10 at f_1, and $f' := f[A(P^*), z(P^*)]$, $a' := A(P^*)/q^*(A(P^*))$, $\hat{x}'$, $c' = 1/p(\hat{x}')$, w'_m, d'_m, σ'_m, ν' its data;

(vi) $g_i'' := b_i^\theta 1_{[1,N_w]} + \sum_m R_m^*(\omega_{i,m}^\theta - d'_m(\omega_{i,m}^\theta) w'_m)$, $c_i := g_i''(\hat{x}')$, $g_i' := \rho(g_i'' - c_i a')$.

The *perturbation size* is $\Pi := \Pi_X + r + \mathfrak{t}(N'')^{1/2}$.

f' is norm attaining ($A(P^*)$ and $z(P^*)$ are finitely supported, [I, Proposition 1.10(c)]), $g_i'(\hat{x}') = 0$, and, once Lemma 3.10 applies, f' is d-consistent with f_0 on Ω. A TU pull coordinate inside the window that carries a mass gets the total mass $-\epsilon_l(\mathfrak{p}_l + \mathsf{m}_j)$; this only enlarges $|a'_j|$.

Lemma 3.14 (Estimates at the approximant)**.** *There is a window constant*

$$K_w \leq C_f\Big(\frac{nA_0|\Omega|\Lambda_{\mathrm{lev}} C_S}{T\,\mathrm{gap}_{\min}\, x_0}\Big)^{C_{\mathrm{abs}}},$$

with C_f an f-constant and C_{abs} an absolute constant, such that the following holds. If $K_J 2^{G+3}|\Omega| c_{\mathrm{tiny}}/A_{\min} \leq \frac{1}{8}$, $K_0\|Y\|_\infty \leq \frac{1}{2}$ (Y computed at f_1), $r_{\mathrm{pull}} \leq 32 K_J T^4 s_1$ and $\Pi \leq 1/K_w$, then Lemma 3.10(ii) *applies at f_1 with the radius r of Definition 3.13(iii), and:*

(a) (sizes) $\Pi_X \le 8\rho s_1 \max_j \mu_j^\star\, nA_0/(T\nu_0)$; $q^*(A(P^*)-a_0) \le K_w(s_1+\overline{W}_{\rm pull})$; *and* $p^*(f'-f_0) \le K_w\big(\Pi\log(e/\Pi)+W_{\rm pull}+W_{\rm lev}\big)$;

(b) (values) *a carrier k meeting no lever coordinate satisfies* $|u_k(\hat{x}')-u_k(\hat{z}_0)| \le 2(\Pi_X+C_2r)+\mathfrak{t}_k(N'')$; *carriers of* Ω *satisfy* $\mathrm{nv}'_k=\mathrm{nv}^0_k$; *a carrier meeting lever coordinates moves by at most* $2(\Pi_X+C_2r)+\mathfrak{t}_k(N'')+C_{\rm lev}r+2\sum_{j_p}|u_k(j_p)|$;

(c) (block data) $|C'_m-C^0_m|+|M'_m-M^0_m|+|A'_m-A^0_m|+|c'-q^0_0|+|\nu'-\nu_0|+\|e'-e_0\| \le K_w\Pi$; *every peak of* f_0 *with margin* $> K_w\Pi$ *is a peak of* f' *with the same sign; every* $k\in\Omega$ *has* $\mathrm{gap}'(k)\ge \mathrm{gap}^0(k)/2$; *and* $d'_m(\omega)=d_m(\omega)$, $H'_m(\omega)=(C^0_m/C'_m)H_m(\omega)$ *for* ω *supported in* Ω_m;

(d) (Bregman terms) $0\le \mathrm{Bx}_m := \langle w'_m-w^0_m, R_m\hat{x}'\rangle \le K_w\Pi^2+4c_{\rm fine}(\Pi_X+C_2r)+2\mathfrak{t}(N'')+2C_{\rm lev}W_{\rm lev}r+4W_{\rm pull}$;

(e) (scrambling, $m\in I_-$) *with* A_m *the scrambled set of [I, Lemma 7.15] for* (w^0_m,w'_m),

$$\mathfrak{S}_m := \|D_m(w'_m-w^0_m)\|_2^2+\sum_{k\in A_m}\Big(\Phi_m(k)^2+\lambda_k\big(|w'_m(k)-w^0_m(k)|+(C'_m-C^0_m)_+\big)\Big)$$

satisfies $\mathfrak{S}_m \le K_w\Pi^2\log(e/\Pi)+4m\big(C_S(K_w\Pi)^2+s_1^2T^4\big)+4(W_{\rm lev}+W_{\rm pull})$;

(f) (base form) *for every piece and* $\diamond\in\{+,-,\theta\}$, *with* $\beta^\theta := b^\theta 1_{[1,N_w]}$, $\beta^\pm := b^\pm 1_{[1,N_w]}\pm\frac12 v1_{(N_w,\infty)}$ *and* $h'(\beta):=\|P'^\perp U^*\beta\|^2/\nu'$: $|c'h'(\beta^\diamond)-q^0_0h^0(b^\diamond)| \le C_f\big(A_0^2K_w\Pi/T^2+A_0\,\mathrm{tail}_W/T\big)$ *with* C_f *an* f*-constant;*

(g) (targets) $(g_i''-g_i)(\hat{x}') = -(b_i^\theta 1_{(N_w,\infty)})(\hat{x}')-\sum_m d^\theta_{i,m}\mathrm{Bx}_m$, *hence*

$$|c_i| \le (1+\|U\|)\mathrm{tail}_W+\|g_i1_{(N'',\infty)}\|_1+K_w(\Pi+W_{\rm pull})/t_i,$$

and $p^*(g'_i-\rho g_i)\le K_w(\Pi+c_{\rm fine}+W_{\rm lev}+W_{\rm pull}+\mathrm{tail}_W)/t_i$.

Proof. Lemma 3.10 applies. By Lemma 3.2(iii), $\delta_1\le\Pi_X+\max_\Omega t_k(N'')\le\Pi_X+T^4s_1$ and $\Delta_1\le\Pi_X+\mathfrak{t}(N'')\le\Pi_X+T^4s_1$. By Lemma 3.10(i), the choice of r and $W_{\rm pull}\le\overline{W}_{\rm pull}$, the four terms of $K_J|F(0)|_\infty$ are bounded by $K_J(\delta_1+\Delta_1/A_{\min})\le K_J(1+1/A_{\min})(\Pi_X+T^4s_1)\le r/16$, $2K_JW_{\rm pull}/A_{\min}\le r/16$, $K_JC_2 2^{2G+2}r\cdot r\le r/8$ and $K_J2^{G+2}r|\Omega|c_{\rm tiny}/A_{\min}\le r/16$, so $K_J|F(0)|_\infty\le 5r/16\le r/2$, as soon as $K_JC_22^{2G+2}r\le\frac18$; this, $3K_JC_2r\le\frac18$ and $r\le r_0$ follow from $r\le\Pi\le 1/K_w$. Also $\delta_1\le r$ and $\Delta_1\le r$, since $r\ge 16K_J(\Pi_X+T^4s_1)$. The pulls exist because $r\ge r_{\min}\ge r_{\rm pull}$ by (PULL) of Theorem 3.15.

(a) By Lemma 3.2(ii), (iv) (masses on $E\subseteq F_0$ cost the factor 2), with $t_i\ge T$ and $\|b_i^\theta\|_1\le A_0/t_i$. By Lemma 3.10, $|P^*|\le r$, bank masses are $\le\beta_{\max}2^{G+2}r$, pull masses $\le 24\lambda_l2^Gr$, z-moves $\le r/v_l$; and $r\le K_w(s_1+\overline{W}_{\rm pull})$ by (iii). For the cost use [II, Lemma 2.12(iv)]: $p^*(f'-f_0)\le(1+\|U\|)(\|a'-a_0\|_1+C_fc(\delta))$ with $\delta:=\hat{x}'-\hat{z}_0$; by (b), $\sum_k\min(\lambda_k,|u_k(\delta)|)\le\sum_k\min(\lambda_k,4K_w\Pi)+2\mathfrak{t}(N'')+W_{\rm lev}+W_{\rm pull}$, and [II, (1)] bounds the first sum by $C\Pi\log(e/\Pi)$; $\Delta_m(\delta)$ is bounded likewise. (b) is Lemma 3.2(iii) with $\|e'-e_0\|\le\|e'-e_1\|+\Pi_X$ and $\|e'-e_1\|\le 2\|X_{\rm lev}\|/\nu_1\le 2C_2r/\nu_1$, together with Lemma 3.10(ii). (c) $|A'_m-A^0_m|\le\sum_k\lambda_k|u_k(\hat{x}')-u_k(\hat{z}_0)|$; $|C'-C|$ by the clamp equation through $k^\natural_m$ (Lemma 3.4(c) with $\Omega=\emptyset$, i.e. [I, Lemma 7.14]'s argument, on the ball of Lemma 3.6); $q_0=1/(1+\sum_m A_m)$; ν',e' by Lemma 3.2. Peak margins are Lipschitz in the value and in $\bar{\vartheta}$ [I, Lemma 1.7]. The gap clause is Lemma 3.10(iii), because $K_w\Pi\le C_{\min}\mathrm{gap}_{\min}/4$ when $K_w\ge 4/(C_{\min}\mathrm{gap}_{\min})$ times the constant of the C-bound; the rest is Lemma 3.4(a). (d) By [I, Lemma 7.13(E5)], $0\le\mathrm{Bx}_m\le\langle w'_m-w^0_m,R_m(\hat{x}'-\hat{z}_0)\rangle=\sum_k\lambda_k(w'_m(k)-$

$w^0_m(k))u_k(\hat{x}' - \hat{z}_0)$. Coarse carriers have $|w' - w^0| \le K_w\Pi$ (peaks: $|M' - M|$; Ω: $|C'/C - 1|$) and values $\le 2\Pi$; fine carriers have $|w' - w^0| \le 2$ and contribute as displayed by (b). (e) As in the proof of [I, Lemma 7.17] with the perturbation size Π in place of s_1. Let K_* be the f-constant collecting the Lipschitz constants of values, thresholds and the clamp formula. By (b), (c), a peak of f_0 with margin $> K_*\Pi$ lies in S_1; a carrier of Ω lies in S_2 as soon as $\mathrm{gap}^0 \ge 2|C' - C|(1 + (M - \mathrm{gap}^0)/C)$, because $w' = (C'/C)w^0$ on Ω; a strict non-peak with Φ gap and gap $> K_*\Pi$ lies in S_2. So $A^{(1)} := \{k \in A_m : t_k \le \Pi\}$ consists of carriers meeting lever coordinates and of carriers with $\mu \le 2K_*\Pi$ or $\Phi\,\mathrm{gap} \le 2K_*\Pi$ or gap $\le 2K_*\Pi$, whose Φ-mass is $\le \mathrm{Scr}^{(0)}_m(2K_*\Pi) \le C_S(2K_*\Pi)^2$ by (U5) ($2K_*\Pi \le x_0$). As $\Pi \ge r \ge 16K_JT^4s_1$, $A^{(2)} \subseteq \{t_k > s_1T^4\}$, of Φ-mass $\le s_1^2T^4$ by (iv). With $|w' - w| \le 2$, $(C' - C)_+ \le 1$, $\Phi \le 1$ and $\lambda_k = m\Phi_m(k)$ the sum over A_m is at most $4m$ times the Φ-mass; carriers meeting lever coordinates contribute $\le 4\sum \lambda_k$. Finally $\|D(w' - w^0)\|^2 \le K_w\Pi^2\log(e/\Pi)$ by the last display of the proof of [I, Lemma 7.14] with K_Fs_1 replaced by $K_w\Pi$. (f) Three changes: $\beta^\diamond - b^\diamond$ has ℓ_1-norm $\le \mathrm{tail}_W$ and $|h(y) - h(y')| \le \|U\|^2(\|y\|_1 + \|y'\|_1)\|y - y'\|_1/\nu$; $|\|P'^\perp y\|^2 - \|P^\perp y\|^2| = |\langle y, e\rangle^2 - \langle y, e'\rangle^2| \le 2\|y\|^2\|e' - e\|$ with $\|y\| \le 2\|U\|A_0/T$ and $\|e' - e_0\| \le K_w\Pi$; and $|c'/\nu' - q^0_0/\nu_0| \le K_w\Pi$; the constants in front of $A_0^2K_w\Pi/T^2$ and $A_0\mathrm{tail}_W/T$ are f-constants ($\|U\|, 1/\nu_0 \le 2/\nu, 1/q^0_0$). (g) $g_i'' - g_i = -b^\theta_i 1_{(N_w,\infty)} + \sum_m d^\theta_{i,m}R^*_m(w^0_m - w'_m)$ by d-consistency ($d'_m(\omega^\theta) = d_m(\omega^\theta) = d^\theta_{i,m}$), and $R^*_m(w^0_m - w'_m)(\hat{x}') = \langle w^0_m - w'_m, R_m\hat{x}'\rangle = -\mathrm{Bx}_m$. So $c_i = (g_i'' - g_i)(\hat{x}') + g_i(\hat{x}' - \hat{z}_0)$ (as $g_i(\hat{z}_0) = 0$), with $\|\hat{x}'\|_\infty \le 1 + \|U\|$, $|d^\theta_{i,m}| \le \|D_m\omega^\theta\|_2 \le \max(2, A_2)\|\Phi_m\|_2/t_i$, and $g_i(\hat{x}' - \hat{z}_0) = \langle U^*g_i, e' - e_0\rangle + \sum_j g_i(j)(z'_j - z_0(j))$: beyond N'' this is $\le \|g_i 1_{(N'',\infty)}\|_1$; at a pull, $|g_i(j_p)| \le K(\lambda_l v_l(j_p) + W_{\mathrm{pull}})/t_i \le K(2^Gr + W_{\mathrm{pull}})/t_i$; at a (Z) coordinate the move is $\le r/v_l(j)$ and $|g_i(j)| \le |b_i(j)| + K\lambda_l v_l(j)/t_i + KW_{\mathrm{lev}}/t_i$. The bound for $p^*(g_i' - \rho g_i)$ follows from $p^*(\cdot) \le (1 + \|U\|)\|\cdot\|_1$, $\|R^*_m(w^0 - w')\|_1 \le \sum_k \lambda_k|w^0(k) - w'(k)| \le K_w\Pi + 2(c_{\mathrm{fine}} + W_{\mathrm{lev}} + W_{\mathrm{pull}})$ and $q^*(a') = 1$.

The form of K_w. Every estimate above, in Lemma 3.10 and in the proof of Theorem 3.15 is a fixed finite product of the quantities n, A_0/T, $1/\mathrm{gap}_{\min}$, $1/x_0$, C_S, $|\Omega|$ (entering linearly through sums over levers), the lever constants and f-constants; no estimate is iterated a window-dependent number of times (the Sherman–Morrison inverse is rank one per block, the Poincaré–Miranda argument involves no iteration, the clamp equation is used through one natural peak, and sums over Ω enter as factors $|\Omega|$). Hence K_w has the stated form. □

3.5. **The uniform violation-tolerant per-piece bound.**

Theorem 3.15 (Uniform engineered bound)**.** *There are f-constants $t_1 \in (0, 1]$, $c^f_{\mathrm{flat}} \in (0, \frac{1}{16}]$, $K_\sharp \ge 1$, $\eta_1 \in (0, \frac{1}{4}]$ and $c_{\mathrm{late}} > 0$, none of which depends on γ_B, with the following property. Put*

$$c_{\mathrm{flat}} := \min\big(c^f_{\mathrm{flat}},\ \gamma_B/(8\rho A_2)\big).$$

Let f_0 be a base row and let a window family (U1)–(U7) *with scales in $[T, t_1]$ be given at f_0, with window constant K_w of Lemma* 3.14*. Assume the* window conditions

(W-a) $c_{\mathrm{fine}} \le c_{\mathrm{late}}\delta T/\big(nA_0(1 + C_2K_J)(1 + 1/A_{\min})\big)$;
(W-b) $K_J2^{G+3}|\Omega|c_{\mathrm{tiny}}/A_{\min} \le \frac{1}{8}$ *and* $K_0\|Y\|_\infty \le \frac{1}{2}$ *at the pre-row;*
(W-c) N_w *is chosen as in Definition* 3.13(i)*;*

and let s_1 satisfy

$$\text{(VT$'$)}\ \ 64\rho\epsilon \le \delta s_1, \qquad \text{(FINE)}\ \ W_{\mathrm{lev}} + \overline{W}_{\mathrm{pull}} \le \frac{c_{\mathrm{late}}\delta A_{\min} T^4 s_1}{1 + C_{\mathrm{lev}}},$$

$$\text{(PULL)}\ \ r_{\mathrm{pull}} \le 32 K_J T^4 s_1, \qquad \text{(LATE)}\ \ s_1 \le s_{\mathrm{late}} := c_{\mathrm{late}}\Big(\frac{T\,\mathrm{gap}_{\min}\,x_0}{nA_0K_w}\Big)^7.$$

Then the approximant f' of Definition 3.13 *(for every N'' as in (iv)) is norm attaining and d-consistent with f_0 on Ω, $\Pi \le K_w s_1$, $p^*(f'-f_0) \le K_w\big(\Pi\log(e/\Pi) + s_1\big)$, and for every piece*

$$p^*(f' + \tau g_i') \le 1 + \frac{\tau^2}{2}(1-\delta) \qquad (0 < |\tau| \le c_{\mathrm{flat}} t_i), \tag{4}$$

$$p^*(g_i' - \rho g_i) \le K_w\big(K_w s_1 + c_{\mathrm{fine}} + \mathrm{tail}_W\big)/t_i.$$

Proof. Step 0 (constants). The order of choices is: $f \to K_\sharp \to \eta_1 \to$ transfer data and $r_f \to c^f_{\mathrm{flat}}, t_1, c_{\mathrm{late}} \to$ window $\to N_w \to s_1 \to$ pre-row $\to r \to$ pulls $\to N'' \to P^*$. For rows in B_f (Lemma 3.6), (U1) gives $h^0(b^\diamond) \le 4/q_0$ and $H_m(\omega^\diamond) \le 4/\sigma_m$. Put $K_\sharp := 4\max(1/q_0, \max_m 1/\sigma_m)(1+\eta_0) + 2$ and $\eta_1 := \min(\frac14, \delta/(192|I|K_\sharp))$; choose transfer data of f with this η_1 and $r_f := r_f(\eta_1)$. Let $a_{\min} := \min_F |a_j|$, $\gamma_0 := \min_m(M_m - \gamma_m)$, $\gamma_{\min} := \min_m \gamma_m$, $C_{\min} := \min_m C_m$, $M_{\min} := \min_m M_m$, $\Lambda_{\max} := \max \Lambda^\varsigma_m$. Let $K_W := C_W(A_0 + A_2 + C_\Delta)$ with an f-constant C_W such that $\|W^\natural_m - w'_m\|_\infty + \|D_m(W^\natural_m - w'_m)\|_2 \le (K_W/t_i)|\tau|$ in Step 2, $K_A := 2(1 + \|U\|)(2A_0+1) + 2$, and K' an f-constant times $(A_0 + A_2 + C_\Delta + 1)$ bounding the relative errors of [I, Lemma 3.3(b)] and [I, Lemma 3.4(d)] per unit c_{flat}. Then $c^f_{\mathrm{flat}} \le \frac{1}{16}$ is the largest number such that c_{flat} satisfies:

- (c1) $\rho c_{\mathrm{flat}} A_0 \le a_{\min}/32$; (c2) $\rho c_{\mathrm{flat}} K_3 \le \frac{1}{16}$; (c3) $\rho c_{\mathrm{flat}}(2A_2 + C_\Delta) \le 4$;
- (c4) $\rho c_{\mathrm{flat}} \le \frac{1}{16}$, $\rho c_{\mathrm{flat}}(A_2+2) \le M_{\min}/8$, $\rho c_{\mathrm{flat}}(A_2+2)\|\Phi_m\|_2 \le C_{\min}/8$, and $8\rho c_{\mathrm{flat}} A_2 \le \gamma_B$;
- (c5) $K_W c_{\mathrm{flat}} \le \min(\gamma_0/8, C_{\min}/4)$;
- (c6) $6|I|K_\sharp K_Y K_A c_{\mathrm{flat}} + 48 K_\sharp K_W K_y c_{\mathrm{flat}}/(C_{\min}/2) \le \delta/32$;
- (c7) $K' c_{\mathrm{flat}}(1+\rho^2) \le \delta/8$.

Only the last inequality of (c4) involves γ_B, and every condition is an upper bound on c_{flat}; so $c_{\mathrm{flat}} = \min(c^f_{\mathrm{flat}}, \gamma_B/(8\rho A_2))$. Then $t_1 \in (0,1]$ is chosen with

$$6K_\sharp c^2_{\mathrm{flat}} t_1^2(2+\Lambda_{\max}) \le \gamma_{\min}/2, \qquad 6K_\sharp c^2_{\mathrm{flat}} t_1^2 \le \gamma_0/8,$$
$$6|I|K_\sharp K_Y c^2_{\mathrm{flat}} t_1^2 \le 1, \qquad 36 K_\sharp^2 K_y^2 c^2_{\mathrm{flat}} t_1^2/(C_{\min}/2) \le \delta/32$$

(these hold for $\gamma_B \le 1$ when they hold with c^f_{flat}). Finally c_{late} is a small f-constant fixed in Steps 3–5.

Consequences of (LATE), (W-a)–(W-c), (FINE). Lemma 3.14(a) and the definition of r give $\Pi \le K_w s_1$ (enlarging K_w). As $K_w \ge 1$ and $T, \mathrm{gap}_{\min}, x_0 \le 1$, (LATE) with exponent 7 implies: $\Pi \le 1/K_w$; $2K_w\Pi \le x_0$; $K_w^3 s_1 \log(e/s_1) \le c_{\mathrm{late}}\delta$; $K_w^2 s_1 \le T^5$; $s_1\rho\max(2,A_2)/T \le \mathrm{gap}_{\min}/4$; $q^*(A(P^*)) \le 2$; $p^*(f'-f) \le r_f$; and $\rho c_{\mathrm{flat}} t_i |c_i| \le \frac12$ (by Lemma 3.14(g), (W-c) and (iv)). So Lemma 3.14 applies, f' lies in B_f, and Lemma 3.6 holds at f'. By Lemma 3.14(d), (e), $\Pi \le K_w s_1$, (W-a), (FINE), (LATE) and (iv):

$$\mathrm{Bx}_m + \mathfrak{S}_m \le c'_{\mathrm{late}}\delta s_1 \qquad (m \in I), \tag{5}$$

with c'_{late} as small as we wish when c_{late} is small (for $m \notin I_-$ only Bx_m is used). Indeed $K_w\Pi^2\log(e/\Pi) \le K_w^3 s_1^2 \log(e/s_1)$, $4c_{\mathrm{fine}}(\Pi_X + C_2 r) \le C c_{\mathrm{late}}\delta s_1$ by (W-a)

and the bounds of Definition 3.13 (iii), and $2C_{\mathrm{lev}}W_{\mathrm{lev}}r+4W_{\mathrm{pull}}+4(W_{\mathrm{lev}}+W_{\mathrm{pull}}) \le Cc_{\mathrm{late}}\delta s_1$ by (FINE).

Steps 1–2 (targets and decompositions). For $\diamond \in \{+,-,\theta\}$ put $B^\diamond := \rho(\beta^\diamond - c_i a')$, $\Omega_m^\theta := \rho(\omega_m^\theta - d'_m(\omega_m^\theta)w'_m)$ and $\Omega_m^\pm := \rho(\omega_m^\pm - d'_m(\omega_m^\theta)w'_m - (d_m^\pm - d_m^\theta)w_m^0)$, $d_m^\diamond := d_m(\omega_m^\diamond)$. As in Step 2 of the proof of [I, Theorem 7.18], $f' + \tau g'_i = (a' + \tau B^\diamond) + \sum_m R_m^*(w'_m + \tau\Omega_m^\diamond)$ for every τ; the identity uses only the two representations at f_0, so it holds for violated pairs ([II, Lemma 2.30], proof). By d-consistency (Lemma 3.14(c)), $\kappa_m^\pm = 0$ and

$$\Omega_m^\pm = \rho\big(\omega_m^\pm - d'_m(\omega_m^\pm)w'_m\big) + \rho(d_m^\pm - d_m^\theta)(w'_m - w_m^0).$$

We use $\diamond = \theta$ for $0 < |\tau| \le s_1$ and $\diamond = \operatorname{sign}\tau$ for $s_1 < |\tau| \le c_{\mathrm{flat}}t_i$ (if $c_{\mathrm{flat}}t_i \le s_1$ only the first regime occurs). The anchors V_m^{an} ($= w_m^0$ if $\Delta_m \ge 0$, $= y_{A,m}$ of [I, Lemma 7.15] for (w_m^0, w'_m) if $\Delta_m < 0$), the vectors $W_m^\natural$ and the remainder $\mathcal{Z} := \sum_{m\in I_-} r_m R_m^* Z_{A,m}$ are as there, with $r_m := \frac12|\tau|\rho|\Delta_m| \le |\tau|C_\Delta$. Since $|\omega^\diamond(k)| \le \max(2,A_2)/t_i$, $\|D\omega^\diamond\|_2 \le \max(2,A_2)\|\Phi_m\|_2/t_i$, $|d^\diamond| \le \|D\omega^\diamond\|_2$, $|\Delta_m| \le C_\Delta$ and $\|V_m^{\mathrm{an}} - w'_m\|_\infty \le 3$, the bound defining K_W holds.

Step 3: blocks. As $\omega_m^\diamond$ vanishes on P'_m (statuses of Ω and peaks preserved), [I, Lemma 3.2] at f' gives $\mathrm{l}'_m(W_m^\natural) = r_m\langle V_m^{\mathrm{an}} - w'_m, R_m x'\rangle/\sigma'_m$ (no κ term). Its modulus is at most $c'(\mathrm{Bx}_m + \mathfrak{S}_m)$ (for $y_{A,m}$ as in Step 3 of [I, Theorem 7.18], with $|(R_m x')(k)| \le \lambda_k c'$), so by (5) $|\mathrm{l}'_m| \le |\tau|C_\Delta(2/\sigma_m)c'_{\mathrm{late}}\delta s_1 \le \delta|\tau|s_1/(64|I|)$. For the excess write $W_{0,m} - r_m w'_m = (1-r_m)(w'_m + t'(\omega_m^\diamond - d'_m(\omega_m^\diamond)w'_m))$, $t' := \tau\rho/(1-r_m)$, $|t'| \le 2\rho c_{\mathrm{flat}}t_i$. Put $\mathrm{W} := w' + t'(\omega - d'(\omega)w')$. For k of kind [1′]: $|t'\omega(k)| \le 4\rho c_{\mathrm{flat}}\operatorname{gap}^0(k) \le \operatorname{gap}'(k)/2$, since $\operatorname{gap}' \ge \operatorname{gap}^0/2$ (Lemma 3.14(c)) and $\rho c_{\mathrm{flat}} \le \frac{1}{16}$. Kind [2]: $|t'\omega(k)| \le 2\rho c_{\mathrm{flat}}A_2 \le \gamma_B/4 \le \operatorname{gap}'(k)/2$. Kind [3]: with $\mathrm{vs} = \operatorname{sign} w'(k) = \operatorname{sign} w^0(k)$, the outward part satisfies $\mathrm{vs}\,t'\omega(k) \le 3\rho c_{\mathrm{flat}}\operatorname{gap}^0(k) \le \operatorname{gap}'(k)/2$, so $\mathrm{vs}\,\mathrm{W}(k) \le (1-d't')(M' - \operatorname{gap}') + \operatorname{gap}'/2 \le (1-d't')M'$ as $|d't'| \le \frac12$ by (c4); and $-\mathrm{vs}\,\mathrm{W}(k) \le |t'\omega(k)| \le 2\rho c_{\mathrm{flat}}A_2 \le (1-d't')M'$ by (c4) and $M' \ge \frac34 M$. Hence $\|\mathrm{W}\|_\infty = (1-d't')M'$, the conclusion of [I, Lemma 3.4(c)], whose proof uses only these coordinatewise bounds; then the computation of [I, Lemma 3.4(d)] (which needs only that conclusion and $|d't'|M'/C' \le \frac12$, given by (c4)) yields $N_m(\mathrm{W}) \le 1 + \frac{t'^2}{2}H'_m(\omega)(1 + 2|d't'|M'/C')$. For $\diamond = \theta$ and $|\tau| \le s_1$ the radius condition $|s_1\rho\omega^\theta(k)| \le \operatorname{gap}'(k)/2$ follows from (LATE). Therefore, as in [I, Theorem 7.18],

$$G'_m(W_m^\natural) \le \hat G_m := \frac{\rho^2\tau^2}{2}H'_m(\omega_m^\diamond)(1 + K'c_{\mathrm{flat}}) + r_m\mathfrak{g}_m, \qquad \mathfrak{g}_m \le K(\mathrm{Bx}_m + \mathfrak{S}_m),$$

with $H'_m(\omega) = (C_m^0/C'_m)H_m(\omega)$ and $r_m\mathfrak{g}_m = 0$ for θ.

Step 3: base. By [I, Lemma 7.1(b)] at f', $G'_b(A_\tau) = \mathrm{Exc}'(\tau B^\diamond + \mathcal{Z}) + \nu'\Psi'(U^*(\tau B^\diamond + \mathcal{Z})/\nu')$ and $\mathrm{Exc}'(\tau B^\diamond + \mathcal{Z}) \le \mathrm{Exc}'(\tau B^\diamond) + 2\|\mathcal{Z}\|_1$. We claim

(6)
$$\mathrm{Exc}'(\tau B^\pm) \le \tfrac12\rho|\tau|V_{>N''} + 2\rho|\tau|\epsilon_i \ \ (s_1 < |\tau|), \qquad \mathrm{Exc}'(\tau B^\theta) = 0 \ \ (0 < |\tau| \le s_1).$$

On $\operatorname{supp} a'$ the summand is $2(-\operatorname{sign}(a'_j)\tau B_j - |a'_j|)_+$ with $a'_j + \tau B_j = (1-\tau\rho c_i)a'_j + \tau\rho\beta_j$ and $1 - \tau\rho c_i \ge \frac12$; it vanishes if $|\tau\rho\beta_j| \le \frac12|a'_j|$ or if $\tau\rho\beta_j$ has the sign of a'_j, and the map $\beta_j \mapsto$ summand is $2\rho|\tau|$-Lipschitz. On F: $|a'_j| \ge a_{\min}/4$ and $|\tau\rho\beta_j| \le \rho c_{\mathrm{flat}}A_0 \le a_{\min}/32$ by (c1). On E: the residue costs at most $2\rho|\tau||r_i^+(j)|$ in the $\pm$ regimes and nothing in the θ regime ($r^\theta = 0$); for the rest, contact-like data have the sign of a'_j on the designated side, carry a mass m_j when $b^\theta(j) \ne 0$ (so $|\tau\rho\beta_j^\theta| \le \mathsf{m}_j/4 \le |a'_j|/2$ for $|\tau| \le s_1$), and the alternative gives $|\tau\rho\tilde b_j| \le$

$\rho c_{\mathrm{flat}} K_3 |a_0(j)| \le |a'_j|/8$ by (c2), since $|a'_j| \ge |a_0(j)|/2$. Window contacts with mass: for θ, $|\tau\rho\beta_j^\theta| \le \mathsf{m}_j/4 \le |a'_j|/2$; for $\pm$ the summand is $\le 2\rho|\tau|\mathrm{viol}(j)$ as in [II, Lemma 2.30]. TU pulls: $|\tau\rho\beta_j^\pm| \le \rho c_{\mathrm{flat}}(2A_2 + C_\Delta)\lambda_l v_l(j_p) + \rho|\tau||r(j_p)| \le 4\lambda_l v_l(j_p) + \rho|\tau||r(j_p)|$ by (c3), while $|a'_{j_p}| \ge \mathfrak{p}_l/q^*(A(P^*)) \ge 12\lambda_l v_l(j_p)$: no flip from the main part, and the residue is paid as a violation (only for $\pm$). TU banks: contact-like data of sign $\epsilon_l = \operatorname{sign} a'_{j'}$, protected by a mass in the θ regime when $b^\theta(j') \ne 0$ and $j' \le N_w$ (and $\beta_{j'}^\theta = 0$ for $j' > N_w$). (Z) coordinates: $\beta_j^\theta = 0$, so the θ summand vanishes whatever the state of j at f' (free, contact or bank), and the $\pm$ summand is $\le 2\rho|\tau||b_i^\pm(j)|$, included in ϵ_i. Off $\operatorname{supp} a'$, at window contacts without mass, free coordinates $j \le N_w$ (using (U4)), contacts in $(N_w, N'']$, free $j > N_w$ and contacts $j > N''$: exactly as in the proof of [II, Lemma 2.30]. Summing gives (6). Further $\|\mathcal{Z}\|_1 \le |\tau| C_\Delta \sum_{m\in I_-} \mathfrak{S}_m$ and $P'^\perp U^* a' = 0$, so by [I, Lemma 3.3]

$$G'_b(A_\tau) \le \hat{G}_b := \frac{\rho^2\tau^2}{2} h'(\beta^\diamond)(1 + K' c_{\mathrm{flat}}) + \tfrac{1}{2}\rho|\tau| V_{>N''} + 2\rho|\tau|\epsilon_i + 2|\tau| C_\Delta \sum_m \mathfrak{S}_m,$$

the last three terms being absent for θ. For $|\tau| > s_1$: $\frac{1}{2}\rho V_{>N''} \le \delta s_1/128$ (iv), $2\rho\epsilon_i \le \delta s_1/32$ (VT′) and $2C_\Delta \sum_m \mathfrak{S}_m \le \delta s_1/128$ by (5); so these terms are $\le \frac{\delta}{16}|\tau| s_1 \le \frac{\delta}{16}\tau^2$. By Lemma 3.14(c), (f), $H' \le 2H$ and $h' \le 2h^0$, hence $\hat{G}_b, \hat{G}_m \le K_\sharp \tau^2$ for $|\tau| \le c_{\mathrm{flat}} t_i$.

Step 4: rebalancing. Apply [I, Proposition 7.3] at f' with $h := \tau g'_i$ ($h(x') = 0$), the decomposition of Step 2, the bounds $\hat{G}_b, \hat{G}_m$, and the transfer vectors of f' (Lemma 3.6(b)). Then $|\varepsilon_m| \le 2(|\mathfrak{l}'_m| + \hat{G}_m + \hat{\Gamma}) \le 6K_\sharp\tau^2$ ($|\mathfrak{l}'_m| \le \tau^2$ for $|\tau| > s_1$, and $\mathfrak{l}'_m = 0$ for θ). Condition (R1) follows from [I, Lemma 7.2] with $\eta := (K_W/t_i)|\tau| \le K_W c_{\mathrm{flat}}$, by (c5) and the conditions on t_1 (here $|M'_m - M_m| \le \gamma_0/4$ is used), and $|\lambda| \le 6|I|K_\sharp K_Y \tau^2 \le 1$. The error satisfies

$$E \le 12|I|K_\sharp\eta_1\tau^2 + 6|I|K_\sharp K_Y|\tau|^3\frac{K_A}{t_i} + \frac{48K_\sharp K_W K_y|\tau|^3/t_i + 36K_\sharp^2 K_y^2\tau^4}{C_{\min}/2} \le \frac{\delta}{8}\tau^2,$$

where K_A/t_i bounds $(|q^*(A_\tau) - 1| + q^*(A_\tau - a'))/|\tau|$ (as $q^*(B^\diamond) \le (1 + \|U\|)\rho(2A_0/t_i + |c_i|)$, $|c_i| \le 1/t_i$ and $\|\mathcal{Z}\|_1/|\tau| \le 1$), by the choice of η_1, (c6), $|\tau| \le c_{\mathrm{flat}} t_i$ and $\tau^4 \le c_{\mathrm{flat}}^2 t_1^2 \tau^2$.

Step 5: levels. $\hat{\Gamma} = c'\hat{G}_b + \sum_m \sigma'_m \hat{G}_m \le \frac{\rho^2\tau^2}{2}\big(c'h'(\beta^\diamond) + \sum_m \sigma'_m H'_m(\omega_m^\diamond)\big)(1 + K' c_{\mathrm{flat}})$ + (linear terms). By Lemma 3.14(c), (f), the bracket is at most $\Gamma_\diamond^{(0)} + C_f(A_0^2 K_w \Pi/T^2 + A_0 \mathrm{tail}_W/T + K_w\Pi) \le \Gamma_\diamond^{(0)} + \delta/8$ by (LATE), (W-c) and the choice of c_{late}; and $\Gamma_\theta^{(0)} \le \max(\Gamma_+^{(0)}, \Gamma_-^{(0)}) \le 1 + \eta_0/2$ by convexity of Γ_w (which does not need admissibility). The linear terms are at most $\frac{\delta}{16}\tau^2$ (base) and $\sum_m \sigma'_m r_m \mathfrak{g}_m \le |\tau| C_\Delta K \sum_m (\mathrm{Bx}_m + \mathfrak{S}_m) \le \frac{\delta}{32}\tau^2$ (blocks) for $|\tau| > s_1$, and absent for θ. With (c7) and $\rho^2(1 + \eta_0/2) = 1 - 2\delta$, for $0 < |\tau| \le c_{\mathrm{flat}} t_i$:

$$p^*(f' + \tau g'_i) \le 1 + \hat{\Gamma} + E \le 1 + \frac{\tau^2}{2}\Big(1 - 2\delta + \frac{\delta}{8} + \frac{\delta}{8} + \frac{\delta}{8} + \frac{\delta}{16} + \frac{\delta}{4}\Big) \le 1 + \frac{\tau^2}{2}(1 - \delta).$$

The remaining assertions are Lemma 3.14(a), (c), (g) with $\Pi \le K_w s_1$ and (FINE). □

Remark 3.16 (What changed relative to [I, Theorem 7.18]). (a) The first-order junction mismatch is zero (exact d-consistency) instead of $K(\omega)s_1$, so $K_\sharp$ is an f-constant. (b) Every "late stage" statement is replaced by an explicit bound in

the perturbation size $\Pi \le K_w s_1$. (c) The scrambling hypothesis is used in the explicit form (U5). (d) Violations are tolerated as in [II, Lemma 2.30], uniformly in the piece: their cost $2\rho|\tau|\epsilon$ does not depend on t_i. (e) Lever coordinates are protected: zero data, contact-like data, or pull masses dominating $c_{\mathrm{flat}} t_i$ times the data. (f) The only interplay between s_1 and the window is the interval $[64\rho\epsilon/\delta, s_{\mathrm{late}}]$; (VT′) bounds s_1 from below by the violation mass, (LATE) from above by window quantities. (g) The radius $c_{\mathrm{flat}} t_i$ is proportional to the scale; c_{flat} depends on the window only through γ_B.

3.6. **Averaging at the norm-attaining approximant.**

Theorem 3.17 (Windowed recovery through d-consistent approximants)**.** *Let I be finite, $f \in S_{p^*}$ with F finite, $g \in \mathcal{C}(f)$, $\rho \in (0,1)$, $\eta_0 \in (0,2]$ with $\rho^2(1+\eta_0) \le (1+\rho^2)/2$, and δ as above. Let $f_j \to f$ be base rows, $\varepsilon_j := p^*(f_j - f)$. Suppose that for every j there are $T_j \in (0, t_1]$, $n_j \ge 1$, a nonempty set $\mathcal{S}_j \subseteq \{T_j 2^{1-i} : 1 \le i \le n_j\}$ with $n'_j := |\mathcal{S}_j|$, a number $T'_j \in (0, \min \mathcal{S}_j]$, a number $\gamma_{B,j} \in (0,1]$, $K_j \ge 1$, and a window family at f_j with scales $\mathcal{S}_j \subseteq [T'_j, t_1]$ (window parameter $T := T'_j$) satisfying* (U1)–(U7) *with* fixed A_0, A_2 *and with $\gamma_{B,j}$; let $c_{\mathrm{flat}}(j) := \min(c^f_{\mathrm{flat}}, \gamma_{B,j}/(8\rho A_2))$ and $K_{w,j}$ be the constants of Theorem* 3.15 *and Lemma* 3.14 *for this family. Assume:*

- (E-c) $p^*(g - g_{j,t}) \le K_j t$ *for* $t \in \mathcal{S}_j$*;*
- (E-d) $K_j T_j \to 0$ *and* $n'_j c_{\mathrm{flat}}(j)/K_j \to \infty$*;*
- (E-f) *the window conditions* (W-a)–(W-c) *of Theorem* 3.15 *hold at f_j, and there is $s_{1,j}$ satisfying* (VT′), (FINE) *and* (LATE) *at f_j, such that, with $\Pi_j \le K_{w,j} s_{1,j}$ the resulting perturbation size,*
$$K_{w,j}\big(K_{w,j} s_{1,j} + c_{\mathrm{fine},j} + \mathrm{tail}_{W,j}\big) \le K_j (T'_j)^2;$$
- (E-e) $\varepsilon_j + K_{w,j}\big(\Pi_j \log(e/\Pi_j) + s_{1,j}\big) \le \theta_j (T'_j)^2$ *with* $\theta_j / c_{\mathrm{flat}}(j)^2 \to 0$.

Then $(f, \rho g) \in \mathrm{cl\,NA}((c_0, p), \ell^2_2)$. If the hypotheses hold for every $\rho < 1$, then $(f, g) \in \mathrm{cl\,NA}((c_0, p), \ell^2_2)$.

Proof. Fix j large; write $n := n'_j$, $K := K_j$, $c := c_{\mathrm{flat}}(j)$, $\mathcal{S} := \mathcal{S}_j$, $T' := T'_j$. For large j, $p^*(f_j - f) \le r_f/2$. Theorem 3.15 at $f_0 := f_j$ with $s_1 := s_{1,j}$ and tail_W as in (E-f) gives a norm-attaining f'_j and functionals $g'_{j,t}$ ($t \in \mathcal{S}$) with

- (A) $p^*(f'_j + \tau g'_{j,t}) \le 1 + \frac{\tau^2}{2}(1-\delta)$ for $|\tau| \le ct$;

$\varepsilon'_j := p^*(f'_j - f) \le \theta_j T'^2$ by (E-e); and $p^*(g'_{j,t} - \rho g_{j,t}) \le K_{w,j}(K_{w,j} s_1 + c_{\mathrm{fine}} + \mathrm{tail}_W)/t \le KT'^2/t \le Kt$ by (E-f), so $p^*(g'_{j,t} - \rho g) \le (1+\rho)Kt \le 2Kt$. Since $g \in \mathcal{C}(f)$, [I, Proposition 1.13(c)] gives

- (B) $p^*(f'_j + r g'_{j,t}) \le p^*(f + r\rho g) + \varepsilon'_j + |r| p^*(g'_{j,t} - \rho g) \le s(\rho r) + \varepsilon'_j + 2K|r|t$ for all real r.

Put $\bar g_j := \frac{1}{n}\sum_{t\in\mathcal{S}} g'_{j,t}$; by convexity $p^*(f'_j + r\bar g_j) \le \frac{1}{n}\sum_t p^*(f'_j + r g'_{j,t})$. Fix $r_0 \in (0,1]$ with $r_0^2 \le \min(\delta, 1-\rho^2)$.

Case $|r| \le r_0$. Let $\mathcal{S}_r := \{t \in \mathcal{S} : ct < |r|\}$; as the scales are distinct dyadic numbers, $\sum_{t\in\mathcal{S}_r} t < 2|r|/c$. If $\mathcal{S}_r = \emptyset$, all terms obey (A), and $1 + \frac{r^2}{2}(1-\delta) \le 1 + \frac{r^2}{2} - \frac{r^4}{8} \le s(r)$ because $r^2 \le \delta$ and $\sqrt{1+x} \ge 1 + \frac{x}{2} - \frac{x^2}{8}$. Otherwise $|r| > c \min \mathcal{S} \ge cT'$, so $\varepsilon'_j \le \theta_j r^2/c^2$ and

$$p^*(f'_j + r\bar g_j) \le \max\Big\{1 + \frac{r^2}{2}(1-\delta),\, s(\rho r)\Big\} + \frac{\theta_j}{c^2} r^2 + \frac{2K|r|}{n} \cdot \frac{2|r|}{c}.$$

By (E-d), (E-e), for j large $\theta_j/c^2 + 4K/(nc) \le \min(\delta/4, (1-\rho^2)/6)$, and then the right side is $\le s(r)$, because $1 + \frac{r^2}{2}(1-\delta) + \frac{\delta r^2}{4} \le 1 + \frac{r^2}{2} - \frac{r^4}{8}$ (as $r^2 \le \delta$) and $s(\rho r) + (1-\rho^2)r^2/6 \le s(r)$ by [I, Lemma 3.6(a)] ($|r| \le 1$).

Case $|r| \ge r_0$. By (B) for all t, $p^*(f'_j + r\bar{g}_j) \le s(\rho r) + \varepsilon'_j + 2K|r|\frac{1}{n}\sum_{t\in\mathcal{S}} t \le s(\rho r) + \varepsilon'_j + 4K|r|T_j/n$. For j large, $\varepsilon'_j \le (1-\rho^2)r_0^2/6$ and $4KT_j/n \le 4KT_j \le (1-\rho^2)r_0/6$; since $\min(r^2, |r|) \ge r_0|r|$, [I, Lemma 3.6(a)] gives $p^*(f'_j + r\bar{g}_j) \le s(r)$.

Hence $\bar{g}_j \in \mathcal{C}(f'_j)$ and, f'_j being norm attaining, $(f'_j, \bar{g}_j) \in \mathrm{NA}$ by [I, Proposition 1.13(d)]. Finally $\|(f'_j, \bar{g}_j) - (f, \rho g)\| \le \varepsilon'_j + p^*(\bar{g}_j - \rho g) \le \varepsilon'_j + \frac{1}{n}\sum_t 2Kt \le \varepsilon'_j + 4KT_j/n \to 0$. The last sentence follows since $(f, g) = \lim_{\rho\to 1}(f, \rho g)$. □

Remark 3.18. (a) Unlike [II, Theorem 2.20] no exactness of the averaged data and no final application of [I, Corollary 7.21] are needed: the averaging is carried out at the norm-attaining approximant itself, with the per-piece bounds (A). This is what makes violated data usable along a sequence of windows. (b) The set of scales enters only through $\sum_{\mathcal{S}_r} t < 2|r|/c$, $\min \mathcal{S} \ge T'$ and $\frac{1}{n}\sum_{\mathcal{S}} t \le 2T_j/n$; so any subset of a dyadic window may be used, as in [II, Lemma 5.14]. (c) $c_{\mathrm{flat}}(j)$ enters only through the range of (A), the dyadic count and the threshold $|r| > c_{\mathrm{flat}}(j)T'_j$, all tracked by (E-d), (E-e); a window-dependent $\gamma_{B,j}$ is therefore allowed. (Numerically, in a model of inputs (A), (B) averaged over random half-subsets of a dyadic window, the averaged bound depends on c_{flat} only through $n'c_{\mathrm{flat}}/K$ and $\theta/c_{\mathrm{flat}}^2$; evidence only.)

3.7. **The threshold comparison.** The interval $[64\rho\epsilon/\delta, s_{\mathrm{late}}]$ of Theorem 3.15 is nonempty only if the violation mass is small compared with the lateness threshold. The violation masses produced by fine carriers are of the size of the first weight after the stage, and s_{late} is a power of window quantities; the comparison is arranged by a design condition.

Definition 3.19 (Design conditions (D-lev) and ($\mathrm{W}_{\mathrm{exp}}$))**.** For the design $D^{U1'}$ of [II, Definition 5.48] on T_{final}:

(D-lev) for every stage L, with $\sigma_2(L) := \max_{l\le L}$ (second element of $S_l \cap (s_{\max}(L), \infty)$),

$$\mathrm{Des}(L) \ge \max_{l\le L} \frac{n_l\, 2^{\sigma_2(L)}}{(\mu^\star_{\sigma_2(L)})^2 \delta_l};$$

($\mathrm{W}_{\mathrm{exp}}$) for every *main* stage L, the lower weight bound of stage $L+1$ is chosen with $c^{\mathrm{low}}_{L+1} \le \exp(-1/T_{\mathrm{lo}}(L, M(L)))$, so that $c_{L+1} \le \exp(-1/T_{\mathrm{lo}}(L, M(L)))$.

Lemma 3.20 (The additions are harmless)**.** *Both conditions are computable at the stage where they are imposed and do not depend on N or on f. Imposing them keeps the recursion well founded, keeps* (T-a)–(T-d), (P1), (P2) *and N-independence, and every result of Part II valid for $D^{U1'}$ remains valid. Moreover, under (D-lev) $(\mu^\star_j)^2 v_l(j) \ge 1/\mathrm{Des}(L)$ for every $l \le L$ and every $j \in S_l \cap (s_{\max}(L), \sigma_2(L)]$.*

Proof. (D-lev) is a lower bound on $\mathrm{Des}(L)$ by quantities fixed at stage L (signature sets, base entries, δ_l, n_l); ($\mathrm{W}_{\mathrm{exp}}$) is an upper bound on one weight per main stage, computed after all sub-windows of level L are fixed. Imposed at main stages only, it does not conflict with the cluster weights $c_{p+i} = c^{\mathrm{low}}_{p+1} 4^{1-i}$ of $D^{U1'}$ (imposed at every stage it would, since $\exp(-1/T_{\mathrm{lo}}) \ll c^{\mathrm{low}}_{p+1}/4$). The lower-bound uses of weights in Part II (the absorber length R, the least absorber entry and the cluster

capacity, [II, Definition 5.48, Lemma 5.51]) are computed from c^{low} after ($\mathrm{W_{exp}}$) is imposed, so they hold for the modified weights; the absorber targets are produced by the same rule from the new c^{low}. Enlarging Des enlarges $\bar{D}(w)$, and the condition $Q(w) \geq \bar{D}(w)^3$ of [II, Definition 5.48], on which the class and cube pigeonhole rests, is imposed after $\mathrm{Des}(L)$ is fixed. Every other window condition is monotone under enlarging Des and shrinking weights ([II, Lemma 3.22]); the other uses of weights in Part II are upper bounds. For the last claim: $v_l(j) = \delta_l 2^{-j}/n_l$ on S_l, so $(\mu^\star_j)^2 v_l(j)$ decreases in j and is $\geq (\mu^\star_{\sigma_2(L)})^2 \delta_l 2^{-\sigma_2(L)}/n_l \geq 1/\mathrm{Des}(L)$ for $j \leq \sigma_2(L)$. □

Proposition 3.21 (Threshold comparison)**.** *Let $C_0 \geq 1$. There is $T_* > 0$, depending only on C_0 and f-constants, such that the following holds for every window family at a base row with scales in $[T, t_1]$ and $T \leq T_*$. Suppose*

(i) *K_w, n, A_0/T, $1/\operatorname{gap}_{\min}$, $1/x_0$ and Λ_{lev} are at most $\exp(C_0 \log^2(1/T))$, and* (W-b) *holds;*
(ii) *ϵ, c_{fine}, $W_{\mathrm{lev}}+\overline{W}_{\mathrm{pull}}$ and r_{pull} are at most $\exp(C_0 \log^2(1/T))\, c_*$ for a number $c_* \leq \exp(-1/T)$;*
(iii) *N_w is chosen with $\mathrm{tail}_W \leq \exp(-1/T)$.*

Then $s_1 := \max(64\rho\epsilon/\delta, \exp(-1/(2T))) = \exp(-1/(2T))$ satisfies (VT′), (FINE), (PULL) *and* (LATE)*;* (W-a) *and* (W-c) *hold; and*

$$K_w\big(K_w s_1 + c_{\mathrm{fine}} + \mathrm{tail}_W\big) + K_w\big(\Pi \log(e/\Pi) + s_1\big) \leq \exp\big(-1/(3T)\big).$$

Proof. By (i), $s_{\mathrm{late}} = c_{\mathrm{late}}(T \operatorname{gap}_{\min} x_0/(nA_0K_w))^7 \geq c_{\mathrm{late}} \exp(-35 C_0 \log^2(1/T))$, which exceeds $\exp(-1/(2T))$ for T small; and

$$64\rho\epsilon/\delta \leq 64\rho e^{C_0 \log^2(1/T)} e^{-1/T}/\delta \leq e^{-1/(2T)}$$

for T small. So $s_1 = e^{-1/(2T)} \in [64\rho\epsilon/\delta, s_{\mathrm{late}}]$. (FINE) holds because

$$e^{C_0 \log^2(1/T)} e^{-1/T} \leq c_{\mathrm{late}} \delta A_{\min} T^4 e^{-1/(2T)}/(1 + C_{\mathrm{lev}})$$

for T small, as $1/A_{\min}, C_{\mathrm{lev}} \leq e^{C_0 \log^2(1/T)}$; (PULL) holds because $r_{\mathrm{pull}} \leq e^{C_0 \log^2(1/T)} e^{-1/T} \leq 32 K_J T^4 e^{-1/(2T)}$ for T small (note $K_J \geq 2$). (W-a) holds likewise, since for T small

$$\frac{c_{\mathrm{late}} \delta T}{nA_0(1 + C_2 K_J)(1 + 1/A_{\min})} \geq e^{-5C_0 \log^2(1/T)} c_{\mathrm{late}} \delta T;$$

and (W-c) holds as $e^{-1/T} \leq \min(T^3, c_{\mathrm{late}} \delta T/A_0)$. Finally $\Pi \leq K_w s_1$, and every term on the left of the display of the proposition is at most

$$3e^{3C_0 \log^2(1/T)} e^{-1/(2T)} \log(e/s_1) \leq e^{-1/(3T)} \qquad (T \text{ small}).$$

□

Remark 3.22 (Where the hypotheses come from). (a) By Lemma 3.14, $\log K_w \leq C_{\mathrm{abs}} \log(C_f n A_0 |\Omega| \Lambda_{\mathrm{lev}} C_S/(T \operatorname{gap}_{\min} x_0))$; when n, Des and $|\Omega|$ are at most $\log_2(1/T)$ and $1/\operatorname{gap}_{\min}, 1/x_0 \leq T^{-C\,\mathrm{Des}}$, this is $\leq C \log^2(1/T)$: the design quantity Des enters the base of a power, never the exponent of K_w. (b) At companions of a clean sub-window w of a main stage L, the violation mass, c_{fine}, c_{tiny}, W_{lev} and $\overline{W}_{\mathrm{pull}}$ produced by carriers at stages $> L$ are at most $C_f \mathrm{Des}(L) c_{L+1}$, and ($\mathrm{W_{exp}}$) with $T_{\mathrm{lo}}(L, M(L)) \leq T_{\mathrm{lo}}(w)$ gives $c_{L+1} \leq e^{-1/T}$; this is hypothesis (ii) with $c_* := c_{L+1}$. (c) The weaker condition $c_{L+1} \leq b(L, M(L))^2 \leq T_{\mathrm{lo}}(w)^8$ of T_{final}

does not suffice for the lateness threshold obtained here, which contains $\mathrm{gap}^7_{\min}$; whether a sharper threshold would make it suffice is not investigated.

3.8. **Fine carriers enter only through their weight.** Hypothesis (U5) of Theorem 3.15 controls the scrambling of *all* carriers of the negative blocks. It is used only in Lemma 3.14(e). When the fine carriers have total weight small compared with s_1, it can be replaced by a condition on coarse carriers only.

Lemma 3.23 (Coarse scrambling suffices)**.** *Replace* (U5) *in Definition* 3.12 *by*

(U5′) *every peak of f_0 that is not fine has margin $\ge x_0$, and every $k \in \Omega_m$ has $\Phi_m(k)\,\mathrm{gap}^0(k) \ge x_0$ and $\mathrm{gap}^0(k) \ge x_0$,*

(*no condition on fine carriers and on the set I_-*)*. Then Lemma* 3.14 *holds with* (e) *replaced by*

$$\mathfrak{S}_m \le K_w \Pi^2 \log(e/\Pi) + 4m s_1^2 T^4 + 4c_{\mathrm{fine}} + 4(W_{\mathrm{lev}} + W_{\mathrm{pull}}) \qquad (m \in I),$$

and Theorem 3.15 *holds with* (U5) *replaced by* (U5′) *if in addition $4c_{\mathrm{fine}} \le c_{\mathrm{late}} \delta s_1$.*

Proof. In the proof of Lemma 3.14(e), a coarse carrier with $t_k \le \Pi$ lies in S_1 or S_2 of [I, Lemma 7.15] once $2K_*\Pi \le x_0$ (margins $\ge x_0$; $\Phi\,\mathrm{gap}$, $\mathrm{gap} \ge x_0$; for Ω also by d-consistency). So $A^{(1)}$ contains only fine carriers and carriers meeting lever coordinates, and their contribution to $\mathfrak{S}_m$ is at most $\sum(\Phi_m(k)^2 + \lambda_k(2+1)) \le 4\sum \lambda_k$, as $\Phi_m(k) \le \lambda_k$. $A^{(2)}$ and $\|D_m(w'_m - w^0_m)\|_2^2$ are bounded as before (the clamp-formula bound of [I, Lemma 7.14] holds for every carrier, with value moves $\le 2\Pi + t_k$ off the lever coordinates and $|w' - w^0| \le 2$ on them). In the proof of Theorem 3.15, (U5) enters only through (5), which holds again when $4c_{\mathrm{fine}} \le c_{\mathrm{late}}\delta s_1$. □

Under the hypotheses of Proposition 3.21, $4c_{\mathrm{fine}} \le 4e^{C_0 \log^2(1/T)} c_* \le c_{\mathrm{late}}\delta e^{-1/(2T)}$ for T small; so in that regime Theorem 3.15 needs no information on the statuses of fine carriers. This is not a statement that fine carriers are harmless in general: only their total weight is used, against $s_1 \ge e^{-1/(2T)}$, and this is where ($\mathrm{W_{exp}}$) is essential.

3.9. **Activity classes in the violation-tolerant route.** We now apply the results above at the companions of [II, Subsection 5.6], for every activity class, one-signed or mixed. Throughout this subsection the design is $D^{U1'}$ of [II, Definition 5.48] on T_{final} with (D-lev) and ($\mathrm{W_{exp}}$) (Definition 3.19), $N \ge 1$ is fixed, $f \in S_{p_N^*}$ has finite base support, w is a clean sub-window of a main stage $L \ge l_f$, and a is an activity class ([II, Lemma 5.53]) satisfying ($\mathrm{KN}_{w,a}$) ([II, Definition 5.55]; for a class without active shift this is vacuous). We use the notation of [II, Subsection 5.6]: A the active blocks, $A_\mp := \{m \in A : \sigma_m = \mp 1\}$, Ω_{act}, $E_{\mathrm{c}}(w)$, J_{fine}, $\mathcal{V}(\Delta\omega)$, $\mathcal{L}(\Delta', \gamma)$, the true shift Δ'', and $\mathcal{S}$ a set of scales of class a in one cube of [II, Proposition 5.54(c)]. The new point compared with the proof of [II, Theorem 5.68] is that *no absorber carries a switching coefficient*: residues on the coarse coordinates and trace corrections are not cancelled but tolerated as violations.

Lemma 3.24 (Completion without absorber switching)**.** *Let $f^\#$ be obtained from f by the moves of [II, Definition 5.57] with the following changes:* (1a) *as there (*[II, Proposition 5.56]*, or* [II, Theorem 5.7] *for a class without active shift); steps* (1b), (1c) *are omitted (no absorber is tuned, and after* (1a) *no value of Ω_{act} and no $\varkappa_m$ has to be restored); in step* (2) *the recursion of [II, Lemma 5.60] is run*

with owner candidates of types (a), (b) *and* (d) *only, where* (d) *comprises every carrier at a stage* $> L$ *of an active block, absorbers of* $D^{U1'}$ *included, with its forced coefficient* $-\Delta''_m\lambda_k w^{\#}_m(k)$*; a type-*(d) *carrier of a block of* A_- *is self-aligned and one of a block of* A_+ *is anti-aligned* ($z_j := \operatorname{sign}(-\sigma_m\varsigma_k u_k(j))$ *on* $O(k)$, $\varsigma_k := \operatorname{sign} Y_k$)*, and unowned coordinates get* $z_j := 0$*. For* $t \in \mathcal{S}$ *let* $\Delta\omega(t)$ *have the components of the data in the proof of [II, Theorem 5.68] on* Ω_{act}*,* 0 *on the inactive* Ω*-carriers, and* no *absorber components; let* ω^+ *be the clamped decomposition used there,* $\omega^- := \omega^+ + \Delta\omega(t)$*,* $\mathcal{V} := \mathcal{V}(\Delta\omega(t))$*,* $\mathcal{V}_{\mathrm{proj}} := \mathcal{L}(\Delta''^{*}, \gamma(t))$ *on* $E_c(w) \setminus F^{\#}$ *and* 0 *elsewhere,* $\mathcal{V}_{\mathrm{fine}} := \mathcal{V} - \mathcal{V}_{\mathrm{proj}}$ *off* $F^{\#}$ *and* 0 *on* $F^{\#}$*, and*

$$b^+ := \beta + \chi\mathcal{V}_{\mathrm{proj}} + \tfrac{1}{2}\mathcal{V}_{\mathrm{fine}}, \qquad b^- := b^+ - \mathcal{V},$$

with χ *and* β *(supported in* $F^{\#}$*,* $b^+(\xi^{\#}) = 0$*) as in the proof of [II, Theorem 5.68]. Then for every* $t \in \mathcal{S}$*:*

(a) *every carrier at a stage* $> L$ *of a block of* A_- *is a peak of* $f^{\#}$ *with* $|u_k(\hat{z}^{\#})| = |Y_k| + \mathrm{own}_k \geq \delta^{\circ}_k$ *and margin* $\geq q_0^{\#}\delta^{\circ}_k/2$*; hence* $\mathrm{Scr}^{\#}_m(x) \leq C(x/q_0^{\#})^2$ *for* $m \in A_-$ *and* $0 < x \leq x_0 := c_f \operatorname{gap}_{\min}\Phi_{\min}(L)$*;*

(b) $(b^{\pm}, \omega^{\pm})$ *represent a functional* g_t *at* $f^{\#}$ *with* $p^*(g - g_t) \leq Kt$*,* $K = C_f(\mathrm{Des}(L)/u(w))^C$*,* $\Gamma^{\#}_w(b^{\pm}, \omega^{\pm}) \leq 1 + \eta_0/2$*, and the block parts are of kinds* [1′]–[3]*;*

(c) *the total violation satisfies* $\epsilon(t) \leq C_f\mathrm{Des}(L)c_{L+1} + C_f c^2_{L+1}/T_{\mathrm{lo}}(w)$*, and* $b^{\theta} = 0$ *at every violated free coordinate;*

(d) $p^*(f^{\#} - f) \leq \theta_w T_{\mathrm{lo}}(w)^2$ *with* $\theta_w \to 0$*, and every coarse carrier has the status and the gap of [II, Proposition 5.56(b)].*

Proof. (a) At a type-(d) carrier k of a block of A_- the rule gives $z_j = \varsigma_k \operatorname{sign} u_k(j)$ on $O(k) \supseteq S_k$, and $\hat{z}^{\#}_j = z_j$ on J_{fine} (diagonal base); so $u_k(\hat{z}^{\#}) = Y_k + \varsigma_k\mathrm{own}_k$ and $|u_k(\hat{z}^{\#})| = |Y_k| + \mathrm{own}_k \geq \delta^{\circ}_k$. The peak criterion and the margin follow from (W3) and [I, Lemma 1.7] as in [II, Lemma 5.60]; this computation does not involve the other blocks. Absorbers enter as carriers at stages $> L$; at absorber stages the weight conditions of $D^{U1'}$ give $c_p \leq (\delta_p^{\max}H_p/2)^2$, so they are covered. The scrambling bound is the proof of [II, Lemma 5.62]: coarse carriers do not contribute below x_0 (statuses and gaps of [II, Proposition 5.56(b)]; no absorber is a coarse carrier now), every other carrier of a block of A_- is a peak with margin $\geq q_0^{\#}\delta^{\circ}_k/2$, $\Phi_k \leq (\delta_k H_k)^2$ by (W3), and the super-exponential decay of $\delta_k H_k$ gives $\sum_{\delta_k H_k \leq X}(\delta_k H_k)^2 \leq 2X^2$.

(b) $b^- - b^+ = -\mathcal{V} = -\sum_m R^*_m(\Delta\omega_m - \Delta''_m w^{\#}_m)$ and $\omega^- - \omega^+ = \Delta\omega$, so both pairs represent the same functional ([II, Lemma 2.22]). Compared with the data of the proof of [II, Theorem 5.68], the absorber coefficients are removed and $\mathcal{V}_{\mathrm{fine}}$ is re-split; this changes g_t by at most $\|\mathcal{V}_{\mathrm{fine}}\|_1 + \sum_a |\gamma_a| \leq C_f\mathrm{Des}\, c_{L+1} + C_f c^2_{L+1}/t \leq Kt$ (the bound of $\|\mathcal{V}_{\mathrm{fine}}\|_1$ is the one of (c) below). The data change by a vector y with $q_0 h(y)$, $\sigma_m H_m(y) = O(c^2_{L+1})$, because $q_0 h(y) \leq \|U\|^2\|y\|_1^2/\nu$ and $\sigma_m H_m(x) \leq (\sum_k \lambda_k|x(k)|)^2/C_m$; since Γ_w is quadratic, the cross terms change Γ_w by $O(c_{L+1})$, and $\sqrt{\Gamma_w}$ (a seminorm) changes by $O(c_{L+1})$. This is absorbed in the slack of the proof of [II, Theorem 5.68], which bounds $\sqrt{\Gamma^{\#}_w}$ by $\sqrt{\Gamma_w(B_-, \Theta_-)} + Kt + C_f p\varepsilon$, since $c_{L+1} \leq Kt$. The remaining claims are those of the proof of [II, Theorem 5.68] (kinds: increments satisfy the inward rows (X4), so near-threshold carriers are of kind [3]).

(c) At a contact $j \in E_c(w) \setminus F^{\#}$: $z_j^{\#}\mathcal{V}_{\mathrm{proj}}(j) \geq 0$ (admissibility of $\mathcal{L}(\Delta''^{*},\gamma)$, row (X1), [II, Lemma 5.58(b)]) and $\chi \in [0,1]$ give $(z_j b_j^{+})_{-} \leq |\mathcal{V}_{\mathrm{fine}}(j)|/2$ and $(z_j b_j^{-})_{+} \leq |\mathcal{V}_{\mathrm{fine}}(j)|/2$. At a free coordinate of $E_c(w)$, $\mathcal{V}_{\mathrm{proj}}(j) = 0$ (row (X1)), so $b_j^{\pm} = \pm\frac{1}{2}\mathcal{V}_{\mathrm{fine}}(j)$, $b_j^{\theta} = 0$ and $\mathrm{viol}(j) = |\mathcal{V}_{\mathrm{fine}}(j)|$. On J_{fine}, $\mathcal{V}_{\mathrm{proj}} = 0$ and $b^{\pm} = \pm\frac{1}{2}\mathcal{V}$: on coordinates owned by owners of types (a), (b) or by negative type-(d) carriers the data are admissible by the dominance of [II, Lemma 5.61] (whose estimates do not use signs); unowned coordinates carry $\mathcal{V}(j) = 0$; on coordinates owned by positive type-(d) carriers $\mathrm{viol}(j) \leq |\mathcal{V}(j)|$. On $E_c(w)$, $\mathcal{V}_{\mathrm{fine}} = [\mathcal{L}(\Delta'',\gamma) - \mathcal{L}(\Delta''^{*},\gamma)] + (\text{carriers at stages} > L)$; the first part is the trace correction of [II, Lemma 5.58(b)], of ℓ_1-norm $\leq C_f \mathrm{Des}\, c_{L+1} + C_f c_{L+1}^2/t$, the second has ℓ_1-norm $\leq C_f \sum_{l>L}\lambda_l$ (coefficients $\leq C_f\lambda$, $\|u\|_1 \leq 1$); the same bound covers the coordinates owned by positive fine carriers. The residues at support coordinates are antisymmetric and are counted in ϵ in the same way.

(d) [II, Lemma 5.64]: moves (1b), (1c) are absent, and step (2) is as there. □

Lemma 3.25 (Levers at the companion)**.** *In the setting of Lemma* 3.24 *let* Ω *be the set of coarse strict non-peaks of* $f^{\#}$*, and for* $l \in \Omega$ *let* $j(l)$ *be the least element of* $S_l \cap (s_{\max}(L),\infty) \setminus F^{\#}$*. Then:*

(a) *every* $l \in \Omega_{\mathrm{act}}$ *has a TU lever with* ϵ_l *the sign of its switching coefficient (constant on* $\mathcal{S}$*), bank coordinate* $j' := j(l)$ *and set of pull coordinates* $\mathrm{Pull}_l := S_l \cap (j(l), s_{\mathrm{far}}(w)] \setminus F^{\#}$*, all contacts of* $f^{\#}$ *with* $z^{\#} = \epsilon_l$*; at* j' *the data are contact-like of sign* ϵ_l *up to the antisymmetric residue* $\pm\frac{1}{2}\mathcal{V}_{\mathrm{fine}}(j')$*, at a pull coordinate* $t|b^{\pm}(j_p)| \leq (2A_2 + C_\Delta)\lambda_l v_l(j_p)$ *up to such a residue, and, with* $G := 6G(L)$*,* $G(L) := 2^{L+1}$*, and* $r_{\mathrm{pull}} := 2^{2G}\max_l v_l(s_{\mathrm{far}}(w)) \leq \mathrm{Des}(L)^2 c_{L+1}^2$*, for every* $r \in [r_{\mathrm{pull}}, r_0]$ *there is* $j_p \in \mathrm{Pull}_l$ *with* $v_l(j_p) \in [r, 2^G r]$*;*

(b) *every* $l \in \Omega$ *with* $\gamma_l(t) = 0$ *for* $t \in \mathcal{S}$ *(the inactive carriers, including those of class R and those held at value* 0*) has a zero-data lever* (Z) *at* $j(l)$*, where* $b^{\theta}(j(l)) = 0$ *and* $b^{\pm}(j(l)) = \pm\frac{1}{2}\mathcal{V}_{\mathrm{fine}}(j(l))$*;*

(c) *the levers satisfy* (SEP)*; the lever constants satisfy*

$$\Lambda_{\mathrm{lev}} \leq C_f\big(\mathrm{Des}(L)/(u(w)\,\mathrm{gap}_{\min})\big)^{C};$$

and c_{tiny}*,* W_{lev} *and* $\overline{W}_{\mathrm{pull}}$ *are at most* $C_f \mathrm{Des}(L)^C c_{L+1}$*.*

Proof. Beyond $s_{\max}(L)$ the only carrier $\leq L$ whose vector is nonzero on S_l is l, with $u_l = v_l = \delta_l 2^{-s}/n_l$ there (allowedness, [II, Theorem 3.12(d)]); this gives (SEP)(ii), and (SEP)(i) holds because the sets S_l are pairwise disjoint and $j(l) \notin F^{\#}$. Inside $(s_{\max}(L), \sigma_2(L)]$ the support of $f^{\#}$ meets S_l at most in its first element (a bank of the assembly or of (1a)), so $j(l) \leq \sigma_2(L)$ and, by (D-lev) and Lemma 3.20, $\beta_{j(l)} = \nu/((\mu^{\star}_{j(l)})^2 v_l(j(l))) \leq \nu\,\mathrm{Des}(L)$. On $E_c(w) \cap S_l \cap (s_{\max}(L),\infty)$ the projected vector is $\mathcal{V}_{\mathrm{proj}} = \gamma_l v_l$, so the data are $(\chi, \chi-1)\gamma_l v_l \pm \frac{1}{2}\mathcal{V}_{\mathrm{fine}}$. (a) For $l \in \Omega_{\mathrm{act}}$, the sign of γ_l is fixed on the class ([II, Proposition 5.54(e)]) and the coordinates of $S_l \cap (s_{\max}(L), s_{\mathrm{far}}(w)] \setminus F^{\#}$ are contacts with $z^{\#} = \epsilon_l$ (closed by (C1) of [II, Definition 4.16], unchanged by (1a) and by the release); so the data at j' are contact-like of sign ϵ_l. As $|\Delta\omega(l)| \leq 2A_2/t$ and $|\Delta''| \leq C_\Delta$, $|\gamma_l| \leq \lambda_l(2A_2/t + C_\Delta)$, which gives the bound at the pulls. Consecutive elements of S_l differ by $2^{l+1} \leq 2^{L+1}$, so v_l drops by at most $2^{G(L)}$ between them. Hence for $r \in [r_{\mathrm{pull}}, r_0]$ (with r_0 decreased, if necessary, so that $2^G r_0 \leq \min_l v_l(j(l))$; this keeps the bound on Λ_{lev},

as $1/v_l(j(l)) \le \mathrm{Des}(L)$) the interval $[r, 2^G r]$ lies in $[v_l(s_{\mathrm{far}}(w)), v_l(j(l))]$ and contains at least six elements of $S_l \cap (j(l), s_{\mathrm{far}}(w)]$, while the moves of (1a) (the tuning pull and bank of [II, Lemma 4.15], a hold or lever pull, and the assembly bank) make at most five elements of each S_l support coordinates of $f^{\#}$; so Pull_l meets $[r, 2^G r]$. Finally $2^{-s_{\mathrm{far}}(w)} \le c_{L+1}^2$ and $2^{2G} = 2^{12G(L)} \le \mathrm{Des}(L)^2$ ([II, Lemma 4.1(d)]) give $r_{\mathrm{pull}} \le \mathrm{Des}(L)^2 c_{L+1}^2$. (b) Here $\mathcal{V}_{\mathrm{proj}}(j(l)) = \gamma_l v_l(j(l)) = 0$, so $b^\theta(j(l)) = (\chi - \frac{1}{2})\mathcal{V}_{\mathrm{proj}}(j(l)) = 0$. (c) Lever coordinates of l are met only by l among carriers $\le L$; carriers at later stages meeting them have total weight $\le C \sum_{k>L} \lambda_k \le C c_{L+1}$ and meet them with $|u_k(s)| \le 1$. The sum defining c_{tiny} runs over the (Z) and TU bank coordinates only, which are the coordinates $j(l)$, and $1/v_l(j(l)) \le \mathrm{Des}(L)$; the pull coordinates enter only through $\overline{W}_{\mathrm{pull}}$, which is the weight of carriers at stages $> s_{\max}(L) > L$ meeting Pull_l. The constants of Lemma 3.10 are bounded by $C_f \mathrm{Des}(L)^C$ times powers of $1/u(w)$ and of $1/\operatorname{gap}_{\min}$, which enter through r_0: the status radius of Lemma 3.10 is measured in value units, $r_0 \le c \min_{l \in \Omega} \Phi_m(l) \operatorname{gap}_m(l) A_{\min}/(m C_{\mathrm{lev}})$, and $r_0 \le c$ times the coarse peak margins $\ge (u/2)\bar{\vartheta}\Phi$ divided by C_{lev}. □

Theorem 3.26 (Activity classes are recovered under (KN))**.** *Let the design be $D^{U1'}$ with* (D-lev) *and* ($\mathrm{W}_{\mathrm{exp}}$)*, $N \ge 1$, and $f \in S_{p_N^*}$ with finite base support. Suppose that for infinitely many main stages L some clean sub-window w of L has at least $n(w)/\bar{D}(w)$ dyadic scales whose activity class a (without active shift, one-signed, or mixed) satisfies* ($\mathrm{KN}_{w,a}$)*. Then $f \in \mathcal{R}$: $(f, g) \in \mathrm{cl}\,\mathrm{NA}((c_0, p_N), \ell_2^2)$ for every $g \in \mathcal{C}(f)$.*

Proof. Fix $g \in \mathcal{C}(f)$ and $\rho < 1$. For each such (L, w) with $L \ge l_f$, the pigeonhole of the proof of [II, Theorem 5.68] gives a set $\mathcal{S}$ of at least $n(w)/\bar{D}(w)^2 \ge L 2^{L^3} (\mathrm{Des}/u)^C$ scales of one class and one cube (this count rests on the condition $Q(w) \ge \bar{D}(w)^3$ of [II, Definition 5.48], not on [II, Lemma 3.22]). Let $f^{\#}$ be the companion of Lemma 3.24 and Ω the coarse strict non-peaks of $f^{\#}$. The data of Lemma 3.24 form a window family at $f_0 := f^{\#}$: (U1), (U2) by (b); (U3) as in the proof of [II, Theorem 5.68] (at the banked, pulled and support coordinates of (1a) the data are contact-like or $t|b| \le K_3 |a^{\#}|$, up to antisymmetric residues counted in ϵ); (U4) by (c); (U5) by (a); (U6) by Lemma 3.25; (U7) by [II, Lemma 2.23]. Take $\gamma_B(w) := \min_m M_m u(w)/4$, so $c_{\mathrm{flat}}(w) \ge c_f u(w)$.

Threshold comparison. Use the window parameter $T := T_{\mathrm{lo}}(w) \le \min \mathcal{S}$ (scales lie in $[T, t_1]$). Since $T_{\mathrm{lo}}(w) = 2^{-n(w)} T_{\mathrm{hi}}(w)$, the number of pieces is at most $n(w) \le \log_2(1/T)$; moreover $L \le \mathrm{Des}(L) \le n(w)$, $|\Omega| \le 2L$, $1/u(w) \le n(w)$ ([II, Proposition 3.14(a)]), $\operatorname{gap}_{\min} \ge c_f \eta(w) M \ge c_f T^4/(L \,\mathrm{Des})$ ([II, Proposition 5.56(b)] with $\eta(w) = T_{\mathrm{lo}}(w)^4/(L\,\mathrm{Des})$; robust and kind-[1′] carriers have larger gaps), $x_0 \ge c_f \operatorname{gap}_{\min} \Phi_{\min}(L)$ with $1/\Phi_{\min}(L) \le \mathrm{Des}(L)$, $C_S \le C/q_0^2$, and $\Lambda_{\mathrm{lev}} \le C_f(\mathrm{Des}/(u \operatorname{gap}_{\min}))^C$ (Lemma 3.25). By Lemma 3.14, $K_w \le T^{-C'}$ for an absolute C' and T small; in particular hypothesis (i) of Proposition 3.21 holds with an absolute C_0. By Lemma 3.24(c) and Lemma 3.25(c), hypothesis (ii) holds with $c_* := c_{L+1}$, since $C_f \mathrm{Des}(L)^C \le e^{C_0 \log^2(1/T)}$, $C_f c_{L+1}^2/T \le c_{L+1}$ and $r_{\mathrm{pull}} \le \mathrm{Des}(L)^2 c_{L+1}^2 \le c_{L+1}$ (Lemma 3.25(a)); in particular the radius satisfies $r \ge 32 K_J T^4 s_1 \ge r_{\mathrm{pull}}$; and $c_{L+1} \le \exp(-1/T_{\mathrm{lo}}(L, M(L))) \le e^{-1/T}$ by ($\mathrm{W}_{\mathrm{exp}}$), because $T_{\mathrm{lo}}(L, M(L)) \le T_{\mathrm{lo}}(w)$. (W-b) follows from Lemma 3.25(c) and, for the s_1-dependent part of Y, from (LATE). Choosing N_w with $\mathrm{tail}_W \le e^{-1/T}$, for L large Proposition 3.21 applies, with $s_1 = e^{-1/(2T)}$.

Recovery. Apply Theorem 3.17 along the infinitely many pairs (L, w), with $f_j := f^{\#}$, $\mathcal{S}_j := \mathcal{S}$, $K_j := K$ of Lemma 3.24(b) and $s_{1,j} := e^{-1/(2T)}$. (E-c) is Lemma 3.24(b). (E-d): $KT_{\mathrm{hi}}(w) \to 0$ by [II, Lemma 3.22], and $n' c_{\mathrm{flat}}/K \geq L 2^{L^3} (\mathrm{Des}/u)^C c_f u/(C_f(\mathrm{Des}/u)^{C'}) \to \infty$ for the constants of [II, Theorem 5.68]. (E-f) and (E-e): Proposition 3.21 bounds the relevant quantities by $e^{-1/(3T)} \leq KT^2$, and $\varepsilon_j \leq \theta_w T_{\mathrm{lo}}(w)^2 \leq \theta_w T^2$ by Lemma 3.24(d); so (E-e) holds with $T'_j := T$ and $\theta_j := \theta_w + e^{-1/(3T)} T^{-2}$, and $\theta_j / c_{\mathrm{flat}}^2 \leq C(\theta_w + e^{-1/(3T)} T^{-2}) n(w)^2 \to 0$ since $\theta_w \leq C_f \mathrm{Des}\, T_{\mathrm{lo}}^2 \log(1/T_{\mathrm{lo}}) + C_f c_{L+1} \log(1/c_{L+1})/T_{\mathrm{lo}}^2$ and $n(w) \leq \log_2(1/T)$. Hence $(f, \rho g) \in \mathrm{cl}\,\mathrm{NA}$; let $\rho \to 1$. □

Corollary 3.27 (Step (2) is dispensable)**.** *In Lemma* 3.24 *the recursion of step* (2) *may be omitted: let* $f^{\#} := f^{(1a)}$ (*no release, z unchanged on* J_{fine}) *and put* $b^{\pm} := \pm\frac{1}{2}\mathcal{V}$ *on* J_{fine}*, the data being as in Lemma* 3.24 *elsewhere. Then* (b)–(d) *of Lemma* 3.24 *hold,* $\|\mathcal{V} 1_{J_{\mathrm{fine}}}\|_1 \leq C_f \mathrm{Des}(L) c_{L+1}$*, the violation on* J_{fine} *is at most* $\|\mathcal{V} 1_{J_{\mathrm{fine}}}\|_1$ *with* $b^{\theta} = 0$ *there, and* (U5′) *of Lemma* 3.23 *holds with* $x_0 := c_f \,\mathrm{gap}_{\min} \Phi_{\min}(L)$*. Consequently Theorem* 3.26 *holds with this companion, and no information on fine carriers* (*ownership, self-alignment, release*) *is used.*

Proof. A coordinate $j \in J_{\mathrm{fine}}$ lies in no coarse target and in no near part $S_l \cap [1, s_{\mathrm{far}}(w)]$, so coarse carriers contribute to $\mathcal{V}(j)$ only through $S_l \cap (s_{\mathrm{far}}(w), \infty)$, in total at most $C_f \mathrm{Des} \sum_l \lambda_l (\delta_l/n_l) 2^{1 - s_{\mathrm{far}}(w)} \leq C_f \mathrm{Des}\, c_{L+1}^2$ ($2^{-s_{\mathrm{far}}(w)} \leq c_{L+1}^2$); carriers at stages $> L$ have forced coefficients $\leq C_\Delta \lambda_k$ and $\|u_k\|_1 \leq 1$, in total $\leq C_\Delta \sum_{k>L} \lambda_k \leq C c_{L+1}$. At a free j, $b_j^{\theta} = 0$ and $\mathrm{viol}(j) = |\mathcal{V}(j)|$; at a contact, $\mathrm{viol}(j) = (z_j \mathcal{V}(j))_- \leq |\mathcal{V}(j)|$. The changes of g_t and Γ_w against Lemma 3.24(b) are bounded as there. (U5′): by [II, Proposition 5.56(b)] every coarse peak has relative margin $\geq u/2$, so margin $\geq (u/2)\bar{\vartheta}\Phi \geq x_0$, and every coarse strict non-peak has $\mathrm{gap} \geq \mathrm{gap}_{\min} \geq x_0$ and $\Phi\,\mathrm{gap} \geq \Phi_{\min}\,\mathrm{gap}_{\min} \geq x_0$; without step (2) there is no later drift. In the proof of Theorem 3.26 use Lemma 3.23: $4c_{\mathrm{fine}} \leq C c_{L+1} \leq c_{\mathrm{late}} \delta e^{-1/(2T)}$ for L large. □

Remark 3.28 (Absorbers in the violation-tolerant route). The zero-value absorber pairs of $D^{U1'}$ make coarse data *exact* ([II, Proposition 5.59]). They cannot be combined with the levers of Lemma 3.10: re-tuning an absorber by a mass at its tuning coordinate has a range of order $m_0(\mu^{\star}_{p_0})^2$, which is not a design quantity of level L; the coarse levers act at coordinates of $E_c(w)$ whose first absorber pair has $|u_a(s)| \geq 1/(8n_a)$, so a lever move, a pull flip or a window mass moves the absorber's value far beyond its room, which is of the size of its weight; untuned, the absorber becomes a peak with nonzero switching coefficient, which costs a first-order term in the θ regime. Therefore the statement that such absorbers can be re-tuned by levers is false in this application and is not used, and the mixed-class theorem with tuned absorbers is not claimed. In the violation-tolerant route absorbers carry no switching coefficient: they are ordinary fine carriers, and residues on coarse coordinates become tolerated violations of mass $O(\mathrm{Des}(L) c_{L+1})$ (Lemma 3.24(c)). A finite-model computation illustrates the obstruction (an absorber pushed 90–170 times beyond its room by a coarse lever move of size 10^{-6}); it is evidence only.

Remark 3.29 (What is not proved here). Theorem 3.26 needs $(\mathrm{KN}_{w,a})$. Without it the residual for finite base support is the status-coherence problem of [II, Remark 5.70]: rows at which, at all large main stages and all clean sub-windows, almost all scales lie in classes violating (KN), i.e. an active block with an active

near-threshold switching carrier and no inactive Ω-carrier of robust relative position. This section does not treat that problem, nor rows with infinite base support.

4. The operator T_{fin} and Master Theorem VI

This section assembles the results of Sections 2 and 3 into a statement about one operator. We define a single operator T_{fin}, independent of the number N of blocks, and prove that for the norms p_N built with it every first row with finite base support is recoverable, for every N, when T_{fin} is taken with option (a) of Definition 4.1 (with option (b): for $N = 1$, and for $N \geq 2$ at the rows with $\varrho_1(f) \neq 1$) (Theorem 4.25). By the reduction of [I, Section 1] and [II, Proposition 6.48] the density of $\mathrm{NA}((c_0, p_N), \ell_2^2)$ is then *equivalent* (option (a), every N; option (b), $N = 1$) to the approximation statement (LSC-trunc)$_N$ at infinite base support (Theorem 4.26); (LSC-trunc)$_N$ itself is not proved here.

The design is that of [II, Definition 5.48] (rounds of absorber pairs around main stages) on the ladder of [II, Definition 3.5], with the changes listed in Remark 4.4, chiefly: the base is $\mu_s = 2^{-s^2-1}$ instead of $\mu_s^\star$; there are no fresh-dominated stages; the weights are chosen *generically* (algebraically independent over a countable field fixed in advance); the rate scheme and the design factor contain the objects of Sections 2 and 3; and the window factor $Q(w)$ does not contain Part II's $\bar{D}(w)^3$ and has exponent $\omega(l) + l^2 + 20$. Because the results of Part II are proved for Part II's operator, we list the properties of the operator that their proofs use and verify them for T_{fin} (Subsection 4.2). The recovery itself follows the violation-tolerant route of Section 3 (Subsection 4.3): the companion is the exactification of Proposition 2.32; the approximant is the d-consistent engineered approximant of Definition 3.13, at which the block scalars are frozen up to the weight of the fine carriers (Lemma 4.17); and the averaging is Theorem 3.17.

The verification status of the argument is recorded in Remark 4.29. The route through exact data with zero-value absorbers ([II, Theorem 5.68]) is *not* used: its absorber tuning does not transfer to T_{fin} (Remark 4.12).

4.1. The design.

Throughout, $\|\cdot\|$ on ℓ_1-type objects and the notation of [I, Section 1] and [II, Section 3] are used: a stage l carries a target y_l, a signature $\delta_l h_l$, the normalized target vector u_l, a weight c_l, a block $m(l)$ and a position $k(l)$ in that block; $\Phi_l = 2^{-m(l)-k(l)} c_l$ and $\lambda_l = m(l)\Phi_l$.

Definition 4.1 (Fixed data of T_{fin})**.**

(i) *Base.* $H = \ell_2$ with orthonormal basis $(k_s)_{s\geq 1}$, $\mu_s := 2^{-s^2-1}$, and $U : H \to c_0$ the diagonal operator $Uk_s := \mu_s e_s$. Then U is compact with dense range, $\|U\| = \mu_1 = \frac{1}{4}$, and $q^*(a) = \|a\|_1 + (\sum_s \mu_s^2 a_s^2)^{1/2}$.

(ii) *Signatures.* $S_l := \{2^l(2i+1) : i \geq 1\}$, $Z_0 := \mathbb{N} \setminus \bigcup_l S_l$, h_l, H_l, $\delta_l^{\max}$ and $\iota_l = 2^{-l}$ as in [II, Definition 3.4] (with $\|U\| = \frac{1}{4}$). The sets S_l are pairwise disjoint, consist of even numbers and have bounded gaps $G_l = 2^{l+1}$.

(iii) *Target pool.* $Y_{\mathrm{pool}} := \{x/q^*(x) : 0 \neq x \in c_{00} \text{ with rational entries}\}$, enumerated as $(y^{(i)})_{i\geq 1}$ with $y^{(1)} := e_1^*/q^*(e_1^*)$; it is norm dense in S_{q^*}. The *absorber pool* A_{pool} is the set of all $x/q^*(x)$ with $x = \beta e_s^* + e_{p_0}^* + 4\sum_{r=1}^{R} 2^{-r} e_{p_r}^*$, $s \geq 1$, $\beta = \pm 1$, $R \geq 1$ and odd integers $p_0 < \cdots < p_R$. Both pools are countable.

(iv) *Schedules.* A block sequence $(m_r)_{r\ge1}$ with $m_1 := 2$ in which every $m \ge 1$ occurs infinitely often, and a sequence $(i_\nu)_{\nu\ge1}$ in which every positive integer occurs infinitely often.

(v) *Normalization option.* Under *option (a)* the weight c_1 is chosen generically in $[\frac12, 1)$ like all other weights; under *option (b)* $c_1 := 1$.

(vi) *The field k_0.* For $y \in Y_{\mathrm{pool}} \cup A_{\mathrm{pool}}$ and $l \ge 1$ let $\delta(y,l)$ be the value of δ_l produced by rule (GM) of Definition 4.3(S2) for the target y at a stage l ($\delta(y,1) := \delta_1^{\max}$), and $n(y,l) := q^*(y + \delta(y,l)h_l)$. Then k_0 is the subfield of $\mathbb{R}$ generated by all $y(j)$, $\delta(y,l)$, $n(y,l)$ ($y \in Y_{\mathrm{pool}} \cup A_{\mathrm{pool}}$, $l, j \ge 1$); it is countable.

The values $\delta(y,l)$ are well defined independently of the weights, because (GM) involves only y, H_l and $\delta_l^{\max}$; each of them is $\delta_l^{\max}$ or an endpoint of an excluded interval, hence lies in $\mathbb{Q}(y(j) : j \ge 1)$. The numbers $n(y,l)$ may be transcendental and are included by definition.

Definition 4.2 (Rounds)**.** The stages are grouped into rounds $r = 1, 2, \dots$; l_r denotes the number of stages before round r ($l_1 := 0$).

(r.1) *Candidate.* Let $m := m_r$, $\nu := \#\{r' \le r : m_{r'} = m\}$, $y := y^{(i_\nu)}$, $\Sigma(y) := \operatorname{supp} y \cap \bigcup_{l'} S_{l'}$ and $p := l_r + 2|\Sigma(y)| + 1$. The candidate is *allowed with margin* if (a) $\operatorname{supp} y \cap S_{l'} = \emptyset$ for all $l' \ge p$ and (c$^+$) $\operatorname{supp} y \cap S_{l'} \subseteq [1, p - 2|\operatorname{supp} y| - 1]$ for all $l' < p$. If it is allowed, $Y_r := y$; otherwise $Y_r := y^{(1)}$ and $p := l_r + 1$. The stage $p_r := p$ is the *main stage* of round r.

(r.2) *PRE stages $l_r + 1, \dots, p_r - 1$:* for each $s \in \Sigma(Y_r)$, in increasing order, two consecutive stages of block 1 carrying absorber targets for $(s, +1)$ and $(s, -1)$ (a *biased pair* for s).

(r.3) *MAIN stage p_r*: block m_r, target Y_r.

(r.4) *CLUSTER stages $p_r + 1, \dots, p_r + R_{\mathrm{cl}}(p_r)$*, where $E^{\mathrm{ch}}(p) := T(p) \cup \bigcup_{l'' \le p}(S_{l''} \cap [1,p])$, $T(p) := \bigcup_{l'' \le p} \operatorname{supp} y_{l''}$ and $R_{\mathrm{cl}}(p) := 2|E^{\mathrm{ch}}(p)|$: a biased pair of block 1 for every $s \in E^{\mathrm{ch}}(p)$. Then $l_{r+1} := p_r + R_{\mathrm{cl}}(p_r)$.

The absorber target of a PRE or CLUSTER stage l for (s, β) is $x/q^*(x) \in A_{\mathrm{pool}}$ with $p_0 < \cdots < p_R$ the least odd integers $> \max(9, \max T(l-1))$ and $R = R(l)$ the least integer ≥ 1 with $2^{3-R} \le \mu_{p_0}^2 \mathrm{lb}_l^2/4$, where lb_l is the a-priori lower bound of Definition 4.3(S4). For every stage l, $k(l) := 1 + \#\{l' < l : m(l') = m(l)\}$, and $\mathfrak{j}(k,m)$ denotes the k-th stage of block m.

In round 1 no candidate meeting a signature set is allowed (the bound in (c$^+$) is ≤ 0), so stage 1 is the main stage of round 1 and lies in block $m_1 = 2$. No fresh-dominated stages are used.

Definition 4.3 (Stage l of T_{fin})**.** Given all quantities of the stages $< l$, the following are formed in this order.

(S1) *Target y_l* by Definition 4.2; at an absorber stage first lb_l (see (S4)), then $R(l)$, then y_l.

(S2) *Signature size δ_l* by rule (GM) of [II, Definition 3.5(2)].

(S3) n_l, u_l, v_l, δ_l°, $\underline{y}_l$, $\Lambda^\circ(l)$ as in [II, Definition 3.5(3)].

(S4) *Weight.* At a PRE or MAIN stage, $c_l^{\max}$ is the minimum of (W1)–(W3), (W5), (W7) of [II, Definition 3.5(4)], of (W4″) of [II, Definition 5.48] (with $T_{\mathrm{lo}}(l-1, M(l-1))^3$), and, at MAIN stages $l \ge 2$, of (W6). At the i-th CLUSTER stage of the main stage p, $c_{p+i}^{\max} := \min\{c^{\mathrm{cl}}(p)4^{1-i},\ c_{p+i-1}/4\}$,

where $c^{\mathrm{cl}}(p)$ is the minimum of $c_p/4$, $b(p,M(p))^2$, $(\delta^{\max}_{p+1}H_{p+1}/2)^2$, (W4″) for every $s \in E^{\mathrm{ch}}(p)$ (with $T_{\mathrm{lo}}(p,M(p))^3$), (W5) and (W7) at stage $p+1$ with δ_{p+1} replaced by $\delta^{\max}_{p+1}/2$, and

$$(\mathrm{W}_{\exp}) \qquad \exp\big(-1/T_{\mathrm{lo}}(p,M(p))\big).$$

The a-priori lower bound is $\mathrm{lb}_l := (1-2^{-l})c_l^{\mathrm{low}}$ at a PRE stage, with c_l^{low} the minimum of $c_{l-1}/4$, $b(l-1,M(l-1))^2$, $(\delta^{\max}_l H_l/2)^2$, (W4″) at s, (W5) with $\delta^{\max}_l/2$ and (W7); and $\mathrm{lb}_{p+i} := (1-2^{-(p+i)})c^{\max}_{p+i}$ at a CLUSTER stage. *Generic choice:* c_l is a number in $[(1-2^{-l})c_l^{\max}, c_l^{\max}]$ that is transcendental over $k_0(c_1,\dots,c_{l-1})$ (for $l \geq 2$, and for $l=1$ under option (a) with $c_1^{\max} := 1$); under option (b) $c_1 := 1$. Then $\Phi_l := 2^{-m(l)-k(l)}c_l$ and $\lambda_l := m(l)\Phi_l$.

(S5) *Level data.* $T(l)$, $s_{\max}(l) := \max(T(l)\cup\{1\})$, $\sigma(l) := \max_{l''\leq l}\min(S_{l''}\cap(s_{\max}(l),\infty))$, $\sigma_2(l) :=$ $\max_{l''\leq l}$ (second element of $S_{l''}\cap(s_{\max}(l),\infty)$), $B_\mu(l) := 2^{\sigma(l)}/\mu^2_{\sigma(l)}$, $\delta_{\min}(l)$, $G(l) = 2^{l+1}$, $D_{\mathrm{room}}(l)$, and the design constants and rate objects of level l of [II, Definitions 3.6 and 3.7] with $B^\star(l)$ replaced by $B_\mu(l)$. In addition the rate scheme of level l contains, for every pattern κ and every class a of level l (patterns range over the carriers $l''\leq l$ and their blocks), the rate objects of the floor-form cone $\Gamma^\flat(\kappa,a)$ of Definition 2.13, whose aggregated signature rows carry the s-free coefficients $\alpha_{l''} = \|v_{l''}1_{S_{l''}\cap(l,\infty)}\|_1$; and the constants Lip^*_κ, $C_L(\kappa)$, $N_L(\kappa)$, m_0, C_Y, α_Y, c_κ, β_κ, h_κ of Definition 2.27, computed with the actual weights $c_1,\dots,c_l$; under option (b) only patterns with $1\notin\mathcal{N}_T(\kappa)$ are used. $\omega(l)$ is the number of rate objects. The *design factor* $\mathrm{Des}(l)$ is the design factor of [II, Definition 3.5(5)] (with B_μ), multiplied at main stages by $4^{R_{\mathrm{cl}}(l)}2^{k(l+R_{\mathrm{cl}}(l))}\underline{y}_l^{-1}$, by $\max_\kappa\max\{\mathrm{Lip}^*_\kappa, C_L(\kappa), N_L(\kappa), C_Y, 1/m_0, 1/c_\kappa, \beta_\kappa, 1/h_\kappa, 1\}$ (with $1/\infty := 0$), and by $\max_{l''\leq l} n_{l''}2^{\sigma_2(l)}/(\mu^2_{\sigma_2(l)}\delta_{l''})$ (condition (D-lev) of Definition 3.19 for the base μ).

(S6) *Sub-windows.* $M(l) := \omega(l)+1$; for $w=(l,i)$, $1\leq i\leq M(l)$, in lexicographic order: $u(w) := \frac{1}{4}$ for $w=(1,1)$ and $u(w) := b(w^-)$ otherwise;

$$Q(w) := \Big(\frac{4\,\mathrm{Des}(l)}{u(w)}\Big)^{\omega(l)+l^2+20};$$

$n(w)$, $T_{\mathrm{hi}}(w)$, $T_{\mathrm{lo}}(w)$ as in [II, Definition 3.5(6)]; $\eta'(w) := T_{\mathrm{lo}}(w)^4/(l\,\mathrm{Des}(l))$; and $b(w) := \min\{b_{\mathrm{II}}(w), b_{\mathrm{QC}}(w)\}$, where $b_{\mathrm{II}}(w)$ is the number $b(w)$ of [II, Definition 3.5(6)] with $\mathrm{Lip}(l)$ replaced by $\mathrm{Lip}^*(l)$, the maximum of $\mathrm{Lip}(l)$ and of the numbers Lip^*_κ, and $b_{\mathrm{QC}}(w)$ is the largest b satisfying (QC4) of Definition 2.27.

(S7) $T_{\mathrm{fin}}e_{k,m} := c_{\mathrm{j}(k,m)}u_{\mathrm{j}(k,m)}$, extended linearly and continuously to $\ell_1(\mathbb{N}\times\mathbb{N})$.

Remark 4.4 (Comparison with Part II). (a) Relative to [II, Definitions 3.5 and 5.48]: the base μ replaces $\mu^\star$, and $B_\mu(l)$, formed at the first element beyond $s_{\max}(l)$, replaces $B^\star(l)$, formed at the third; the fresh-dominated stages are omitted; the absorber length R is computed from lb_l, and the tuning coordinate is the least admissible odd integer (no condition of the form $(\mu^\star_{p_0})^{2^{2^{p_0}}-1} < 2^{-R}$); the cluster weights are generic below $c^{\mathrm{cl}}(p)4^{1-i}$ instead of being equal to it; the factor $\bar{D}(w)^3$ of [II, Definition 5.48] does not enter $Q(w)$, and the exponent of $Q(w)$ contains l^2 (this is used in Lemma 4.22); the rate scheme, the design factor and $b(w)$ contain the objects of Definitions 2.27 and 3.19. (b) The aggregated signature rows of the level-l

rate objects carry the coefficients $\alpha_{l''}$, which do not depend on $s_{\mathrm{far}}(w)$. The cones used at a sub-window w of a main stage L have the coefficients $\|v_{l''}1_{S_{l''}\cap(L,s_{\mathrm{far}}(w)]}\|_1$, and $s_{\mathrm{far}}(w)$ depends on c_{L+1}, which is chosen after the level-L data. The two matrices differ by positive row scalings in $[1-2^{1+L-9\cdot 2^L},1]$ (Lemma 2.17(c)). The scalings are ≤ 1, so the zero sets coincide, a rate object that is tiny for the rate scheme is tiny for the cone, and one that is robust ($\ge u$) for the rate scheme has modulus $\ge u/2$ for the cone, because the minors involve at most $R_\Gamma(L)$ scaled rows and $(1-2^{1+L-9\cdot 2^L})^{R_\Gamma(L)}\ge\frac12$; Hoffman [3] and Łojasiewicz constants agree up to a fixed factor. Writing the rate objects with $\alpha_{l''}$ removes a forward reference to c_{L+1} from the level-L data. (c) The Lipschitz constant Lip^* in b_{II} and in (QC4) dominates the Lipschitz constant of the floor-status function F_Φ, which carries the factor $1/\Phi_d$, and those of the floor statuses of all active coarse carriers; the Lipschitz constant of the determinantal objects alone does not dominate it.

Lemma 4.5 (Genericity of the weights)**.** (a) *Every datum of T_{fin} that is not a weight lies in k_0 or is an integer: $y_l\in Y_{\mathrm{pool}}\cup A_{\mathrm{pool}}$, $\delta_l=\delta(y_l,l)$, $n_l=n(y_l,l)$, every value $u_l(j)$, every μ_s and H_l, every coefficient α_l and every aggregated coefficient $\|v_l1_{S_l\cap(a,b]}\|_1$ ($a<b\le\infty$ integers), and the integers $m(l)$, $k(l)$, $R(l)$, p_i and the stage types.*

(b) *Under option* (a) *the family $(c_l)_{l\ge1}$, under option* (b) *the family $(c_l)_{l\ge2}$, is algebraically independent over k_0. Consequently, for every finite set D of stages (not containing* 1 *under option* (b)*), $(\Phi_d)_{d\in D}$ is algebraically independent over $k_0(\Phi_l:l\notin D)$, and for every pattern κ of every level (with $1\notin\mathcal{N}_T(\kappa)$ under option* (b)*), $(\Phi_d)_{d\in\mathcal{N}_T(\kappa)}\notin B_\kappa$.*

Proof. (a) Main targets lie in Y_{pool} and absorber targets in A_{pool}, whatever the integers s,β,R,p_i are; then $\delta_l=\delta(y_l,l)$ and $n_l=n(y_l,l)$ by definition, $u_l(j)=(y_l(j)+\delta_l2^{-j}1_{S_l}(j))/n_l$, and $\|v_l1_{S_l\cap(a,b]}\|_1=(\delta_l/n_l)\sum_{s\in S_l\cap(a,b]}2^{-s}$ with a rational sum; μ_s and H_l are rational. (b) If $c_1,\dots,c_n$ are algebraically independent over k_0 and c_{n+1} is transcendental over $k_0(c_1,\dots,c_n)$, then $c_1,\dots,c_{n+1}$ are algebraically independent over k_0; induction gives the first claim, and Φ_l is a nonzero rational multiple of c_l. A subfamily of an algebraically independent family is algebraically independent over the field generated by the complementary subfamily. For the last claim apply Lemma 2.22 with this k_0 (which contains every $u_l(j)$ and every α_l by (a)) and $k:=k_0(\Phi_l:l\le L,\ l\notin\mathcal{N}_T)$. By Lemma 2.17(a), B_κ contains no datum of a row f: a row only selects the pattern and the point $r(f)$. □

Lemma 4.6 (Well-foundedness, positivity, N-independence)**.** (a) *Every quantity of stage l is a function of the fixed data, of the quantities of the stages $<l$ and of the earlier steps of stage l.* (b) *Every set from which a choice is made is nonempty, and every quantity used as a divisor is finite and positive; in particular $c_l>0$, $R(l)$ exists, the constants of* (S5) *are finite and positive, and $b(w)>0$.* (c) *No rule refers to N.* (d) (Weights.) *$c_{l+1}\le c_l/4$ for every l. For every main stage L and every sub-window w of L:*

$$\sum_{l>L}\lambda_l\le\frac{c_{L+1}}{3},\qquad c_{L+1}\le b(w)^2\le T_{\mathrm{lo}}(w)^8,\qquad c_{L+1}\le\exp\big(-1/T_{\mathrm{lo}}(w)\big).$$

(e) (Lower bounds.) *$c_l\ge c_l^{\max}/2$ for every $l\ge2$; at a PRE stage $c_l/8\le\mathrm{lb}_l\le c_l$; at the i-th cluster stage of a main stage L, $c_{L+i}\ge c^{\mathrm{cl}}(L)4^{1-i}/2$.* (f) *$T_{\mathrm{fin}}$ satisfies the*

QC requirements (QC1)–(QC5) *of Definition* 2.27 *at every main stage, and* (D-lev) *of Definition* 3.19 *with* μ *in place of* $\mu^\star$.

Proof. (a) At stage l the order is: target (at absorber stages lb_l, which uses c_{l-1}, $b(l-1,M(l-1))$, $T_{\mathrm{lo}}(l-1,M(l-1))$, $\delta_l^{\max}$, H_l, s and the (W7)-data of stage $l-1$, or, at a cluster stage, the data of the main stage p and c_{p+i-1}; then $R(l)$; then y_l), δ_l, n_l, u_l, $c_l^{\max}$, the generic c_l (which uses only the field $k_0(c_1,\dots,c_{l-1})$), the level data of (S5) (computed with $c_1,\dots,c_l$), $\mathrm{Des}(l)$, and the sub-windows of (S6) in lexicographic order. The number $k(l+R_{\mathrm{cl}}(l))$ at a main stage l is known at stage l, because the stages $l+1,\dots,l+R_{\mathrm{cl}}(l)$ are block-1 cluster stages and $R_{\mathrm{cl}}(l)$ depends only on $T(l)$. Requirements about later stages, namely (W2), (W7) and ($\mathrm{W}_{\mathrm{exp}}$), are imposed at those stages. The aggregated rows of (S5) carry $\alpha_{l''}$, which does not depend on later weights (Remark 4.4(b)).

(b) The set of admissible δ in (GM) is $[\frac12\delta_l^{\max},\delta_l^{\max}]$ minus at most $2\cdot 3^{|\operatorname{supp} y_l|}$ open intervals of total length $\le \delta_l^{\max}/8$, a nonempty finite union of closed intervals; $c_l^{\max}$ is a minimum of finitely many positive numbers; the interval $[(1-2^{-l})c_l^{\max},c_l^{\max}]$ has positive length and $k_0(c_1,\dots,c_{l-1})$ is countable, so a transcendental c_l exists; $2^{3-R}\to 0$, so $R(l)$ exists. The objects of [II, Definition 3.5] are finite and positive as in [II, Lemma 3.10]; the constants of Definition 2.27 are finite and positive by Remark 2.28(i), which applies because no pattern of level l is a coincidence (Lemma 4.5(b); a pattern of level l involves only $c_1,\dots,c_l$, chosen before the level-l data). $b_{\mathrm{QC}}(w)>0$ by Remark 2.28(ii), and $b_{\mathrm{II}}(w)>0$ trivially.

(c) The rounds, the schedules, the targets and the weights make no reference to N, and the patterns range over all carriers $\le l$ and their blocks. For p_N the carriers of blocks $m>N$ are absent; a pattern of p_N is the design pattern in which these blocks are inactive (they contribute no row), which the design enumerates.

(d) $c_l\le c_{l-1}/4$ by (W1) at PRE and MAIN stages and by the cluster rule. Hence $\sum_{l>L}c_l\le\frac43 c_{L+1}$, and $\lambda_l\le c_l/4$ ($m2^{-m}\le\frac12$, $2^{-k}\le\frac12$). Stage $L+1$ is the first cluster stage of L ($R_{\mathrm{cl}}(L)\ge 2$, as $T(L)\neq\emptyset$), so $c_{L+1}\le c^{\mathrm{cl}}(L)\le b(L,M(L))^2$, and $b(L,M(L))\le b(w)$ because b decreases along the sub-windows of a level ($b(w)<u(w)=b(w^-)$); $b(w)\le\eta'(w)\le T_{\mathrm{lo}}(w)^4$. Finally $c_{L+1}\le\exp(-1/T_{\mathrm{lo}}(L,M(L)))\le\exp(-1/T_{\mathrm{lo}}(w))$, since T_{lo} decreases along the sub-windows of L.

(e) $c_l\ge(1-2^{-l})c_l^{\max}\ge c_l^{\max}/2$. At a PRE stage each entry of c_l^{low} is at most the corresponding entry of $c_l^{\max}$ (because $\delta_l\ge\delta_l^{\max}/2$) and at least a quarter of it, so $c_l/8\le(1-2^{-l})c_l^{\max}/4\le\mathrm{lb}_l\le c_l$. At cluster stages, by induction from $c_{L+i}^{\max}=\min(c^{\mathrm{cl}}(L)4^{1-i},c_{L+i-1}/4)$, $c_{L+i}\ge\prod_{j\le i}(1-2^{-(L+j)})c^{\mathrm{cl}}(L)4^{1-i}\ge(1-\sum_{n>L}2^{-n})c^{\mathrm{cl}}(L)4^{1-i}\ge c^{\mathrm{cl}}(L)4^{1-i}/2$.

(f) (QC1) is Lemma 4.5(b); (QC2) and (QC3) hold by (S5) ($\mathrm{Des}(L)$ contains $C_*(L)$ and $D_{\mathrm{room}}(L)\ge 1/\Phi_{\min}(L)$ through the factor of [II, Definition 3.5(5)], and the QC constants by the additional factor); (QC4) holds because $b(w)\le b_{\mathrm{QC}}(w)$ and $b(w)\le b_{\mathrm{II}}(w)\le\eta(w)=\eta'(w)$; (QC5) is (d). (D-lev) holds by the last factor of $\mathrm{Des}(L)$. □

Theorem 4.7 (Admissibility; Martín's Lemma B for T_{fin})**.** (a) $q^*(T_{\mathrm{fin}}e_{k,m})=c_{\mathrm{j}(k,m)}\le c_1$*; under option* (a) $c_1\in[\frac12,1)$*, under option* (b) $c_1=1$*, attained at* $(k,m)=(1,2)$.

(b) T_{fin} *satisfies* (T-b), (T-c) *and* (T-d) *of [I, Definition 1.3]. Under option* (b) *it is admissible. Under option* (a) *it satisfies* (T-a) *without its equality clause,* $\|T_{\mathrm{fin}}\|=c_1<1$.

(c) *The range* $Y := T_{\mathrm{fin}}(\ell_1(\mathbb{N}\times\mathbb{N}))$ *is a separable operator range with* $Y \cap \mathrm{NA}((c_0,q),\mathbb{R}) = \{0\}$*, and for every* m *the normalized vectors* $\{u_{k,m} : k \ge 1\}$ *are norm dense in* S_{q^*}*. Under option* (a)*, the norms* p_N *and* p *built with* T_{fin} *are Martín-type norms: they are given by Martín's construction* [8] *with an injective operator of norm* $c_1 \le 1$ *having these properties.*

Proof. (a) $q^*(u_l) = 1$ and $c_l \le c_{l-1}/4$ (Lemma 4.6(d)).

(b) *Allowedness.* Every target satisfies allowedness (a): main targets by (r.1). An absorber target for (s,β) at stage l has support $\{s, p_0, \dots, p_R\}$ with p_i odd, hence in Z_0, and s either lies in Z_0 or lies in some $S_{l''}$ with $l'' < l$: for a PRE pair of round r, $s \le p - 2|\operatorname{supp} Y_r| - 1 \le l_r < l$ by (c^+), and $l'' < \min S_{l''} \le s$; for a cluster pair of the main stage $p < l$, either $s \in S_{l''} \cap [1,p]$, so $l'' < s \le p$, or $s \in T(p) \cap S_{l''}$, and then l'' is smaller than the stage of the target containing s by allowedness (a) of that target. Consequently, for $s \in S_{l'}$: $u_l(s) = 0$ for $l < l'$, $u_{l'}(s) = \delta_{l'} 2^{-s}/n_{l'}$, and every target y_l with $l > l'$ and $s \in \operatorname{supp} y_l$ satisfies

$$c_l \le \iota_{l'} 2^{-2s} c_{l'} \delta_{l'}/2 :$$

at PRE and MAIN stages by (W4″); at a cluster stage $p+i$ because $c_{p+i} \le c^{\mathrm{cl}}(p)$, which contains (W4″) for every $s \in E^{\mathrm{ch}}(p)$, and the coordinate s of a cluster target lies in $E^{\mathrm{ch}}(p)$. Also $n_l \in [\frac{3}{4}, \frac{5}{4}]$, since $\delta_l q^*(h_l) \le \delta_l^{\max}(1 + \|U\|)H_l \le \frac{1}{4}$.

(T-b), (T-c). Let $x \in \ell_1(\mathbb{N}\times\mathbb{N})$, indexed by stages, with $Z := T_{\mathrm{fin}} x = \sum_l x_l c_l u_l \in c_{00}$, and suppose $x_{l'} \ne 0$. For $s \in S_{l'}$,

$$Z(s) = x_{l'} c_{l'} \delta_{l'} 2^{-s}/n_{l'} + \sum_{l>l',\ s\in\operatorname{supp} y_l} x_l c_l y_l(s)/n_l.$$

If the sum is nonempty with least index $L(s)$, then $\sum_{l\ge L(s)} c_l \le \frac{4}{3} c_{L(s)} \le \frac{2}{3}\iota_{l'} 2^{-2s} c_{l'}\delta_{l'}$, and $|y_l(s)| \le 1$, so

$$|Z(s)| \ge c_{l'}\delta_{l'} 2^{-s}\big(|x_{l'}|/n_{l'} - \tfrac{8}{9}\|x\|_\infty 2^{-s}\big) > 0$$

for all large $s \in S_{l'}$. This contradicts $Z \in c_{00}$. So $x = 0$, which is (T-b), and $Y \cap c_{00} = \{0\}$, which is (T-c).

(T-d). Fix m and i. Block m is the block of infinitely many rounds, and $i_\nu = i$ for infinitely many ν; at such a round the candidate is $y^{(i)}$, and it is allowed with margin as soon as the main stage p exceeds $1 + \max\operatorname{supp} y^{(i)} + 2|\operatorname{supp} y^{(i)}|$ (for (a): $S_{l'} \cap \operatorname{supp} y^{(i)} \ne \emptyset$ forces $l' < \min S_{l'} \le \max\operatorname{supp} y^{(i)} < p$; for (c^+): $\operatorname{supp} y^{(i)} \subseteq [1, p - 2|\operatorname{supp} y^{(i)}| - 1]$). Hence $y_l = y^{(i)}$ for infinitely many stages l of block m, and $q^*(u_l - y^{(i)}) \le |1/n_l - 1| + \delta_l q^*(h_l)/n_l \le 4(1 + \|U\|)2^{-l} H_l \to 0$ along them. Density of Y_{pool} in S_{q^*} gives (T-d). Under option (b) (T-a) holds with equality at $(1,2)$; under option (a) $\|T_{\mathrm{fin}}\| = \sup_l c_l = c_1 < 1$.

(c) Y is the range of a bounded operator on a separable space; $\mathrm{NA}((c_0,q),\mathbb{R}) = c_{00}$ [I, Proposition 1.10(c)] (the proof uses only that U is compact with dense range), so (T-c) gives $Y \cap \mathrm{NA}((c_0,q),\mathbb{R}) = \{0\}$. The density statement is (T-d). Martín's construction [I, Remarks 1.1 and 1.2] uses of the operator only injectivity, the condition on the range, the density of the normalized vectors in every block, and $\|T\| \le 1$ (through $\Phi_m(k) \le 2^{-m-k}$ and $\|\Phi_m\|_1 < 1$ in Martín's Lemma A). □

Remark 4.8 (The two options). Option (b) keeps T_{fin} inside the class of [I, Definition 1.3]; it costs the genericity of one weight, $c_1 = 1$, and leads to the exception in Theorem 4.25(b). Option (a) leaves that class only through the equality clause of

(T-a). The results of Part I used below remain valid, because their proofs use the operator only through (T-b)–(T-d) and the upper bound $q^*(Te_{k,m}) \le 1$ (Lemma 4.9). The normalized operator T_{fin}/c_1 of option (a) does not reduce option (b) to option (a): the block norms are not homogeneous in the weights (the unit ball of $|\cdot|_m$ is $B_{\ell_1} + D_m(B_{\ell_2})$), so the coincidence sets of the two options are unrelated.

4.2. **Transfer of the tools to T_{fin}.** The results of [I] are stated for admissible operators, and those of [II], Sections 3–5, for T_{final} and $D^{U1'}$. We now show that every result used below holds for T_{fin}. For [I] the point is the equality clause of (T-a); for [II] we list the properties of the operator through which the proofs of the results used below depend on it, and verify each for T_{fin}.

Lemma 4.9 (Part I under option (a))**.** *The following results of [I] hold for every operator satisfying* (T-b), (T-c), (T-d) *and $q^*(Te_{k,m}) \le 1$ for all (k,m), in particular for T_{fin} under both options: Propositions 1.9–1.13, Lemmas 1.5–1.7 and 1.16, Definition 3.1 and Lemmas 3.2–3.4 and 3.6, the results of Section 7 used in Section 3 (7.13–7.17, Theorem 7.18 and the lemmas used in its proof), Lemmas 8.8 and 8.10, and Lemma 8.27 (equation (16)). The same holds for the results of [I] used only in Section 5: Lemmas 1.6, 7.1, 8.7, 8.26 and 8.28 and equation (15).*

Proof. In [I] the operator enters through $\Phi_m(k) = 2^{-m-k} q^*(Te_{k,m})$, through $L^*w = T(\sum_m \sum_k m 2^{-m-k} w_m(k) e_{k,m})$ (an identity valid for every T), and through (T-b)–(T-d). The proofs of the listed results use, besides (T-b)–(T-d), only the upper bounds $\Phi_m(k) \le 2^{-m-k}$, $\lambda_{k,m} \le m2^{-m-k}$, $\|\Phi_m\|_1 \le 2^{-m} < 1$, $\|\Phi_m\|_2 < \frac{1}{2}$ and $\|R_m\| \le m2^{-m}$, all implied by $q^*(Te_{k,m}) \le 1$. The equality clause of (T-a) and lower bounds on $q^*(Te_{k,m})$ occur in [I] only in Definition 1.3, in the verifications that particular operators are admissible, and in the resonance example of [I, Section 6] and [I, Corollary 7.23]; none of these is used here. Lemma 8.27 (equation (16)) is derived from Lemma 8.8 without further use of the operator; Lemmas 8.7, 8.26 and 8.28 and equation (15) are statements about the forced data and the clamp formula whose proofs use T only through (T-b)–(T-d) and $q^*(Te_{k,m}) \le 1$ (via $\Phi_m(k) \le 2^{-m-k}$ and $\lambda_{k,m} \le m2^{-m-k}$). □

We write *MS* for "at the sub-windows of main stages"; windows are used only there.

Lemma 4.10 (The interface)**.** *T_{fin} has the following properties.*

- (B1) *The base is diagonal. Hence the identities of [II, Lemma 3.2(b)] and those of [II, Lemma 4.14] hold exactly.*
- (B2) *$\|U\| = \frac{1}{4}$, $\mu_s \in (0,1]$ is decreasing, $\nu \le \|U\|$.*
- (B3) (Bank efficiency.) *For every l, every $l'' \le l$ and $j' := \min(S_{l''} \cap (s_{\max}(l), \infty))$: $\mu_{j'}^2 v_{l''}(j') \ge \mathrm{Des}(l)^{-1/6}$, $(\mu_{j'} v_{l''}(j'))^{-2} \le \mathrm{Des}(l)^{1/3}$ and $v_{l''}(j') \ge \mathrm{Des}(l)^{-1/6}$.*
- (B4) (Lever efficiency.) *$\mu_j^2 v_l(j) \ge 1/\mathrm{Des}(L)$ for $l \le L$ and $j \in S_l \cap (s_{\max}(L), \sigma_2(L)]$.*
- (S1) *Allowedness (a) holds for every target, (c) with margin for main targets; $u_{l'}(s) = 0$ for $l' < l$ and $u_l = v_l$ on S_l beyond $\max \operatorname{supp} y_l$.*
- (S2) *$s_{\max}(l) \to \infty$.*
- (W1) *MS: $\sum_{l'>L} \lambda_{l'} \le c_{L+1}/3 \le b(L, M(L))^2/3$.*
- (W2) *$\Phi_k \le c_k \le (\delta_k H_k)^2$ for every stage $k \ge 2$.*

(W3) (W4″) *holds at every target with* $T_{\mathrm{lo}}(\cdot)^3$ *read at the preceding main stage at cluster stages.*
(W4) $c_{l+1} \le c_l/4$*; for consecutive stages* $l < l'$ *of block* 1, $\lambda_l \ge 8\lambda_{l'}$.
(W5) *MS:* $c_{L+1} \le \exp(-1/T_{\mathrm{lo}}(L, M(L)))$.
(W6) (Lower bounds.) *The only lower bounds on weights used by the results of Proposition* 4.11 *are: the least entry* η_k *of an absorber target satisfies* $\eta_k \ge \mu_{p_0}^2 c_k^2/2^{13}$*; and bounds computed from the actual weights after they are chosen* ($\lambda_l \ge 1/D(l)$, $\Phi_{\min}(L) \ge 1/\mathrm{Des}(L)$).
(D1) $\mathrm{Des}(l)$ *dominates every design constant used by the results of Proposition* 4.11 *as in [II, Lemma 4.1(c)] and [II, Proposition 3.16]:* l, $2^{G(l)}$, $1 + |T(l)|$, $D_{\mathrm{room}}(l)$, $G^{**}(l)$, $H_{\mathrm{tune}}(l)$, $H_{\mathrm{conf}}(l) \le \mathrm{Des}(l)^{1/6}$, *and* $\mathrm{Des}(l) \ge \mathrm{Lip}^*(l)$, $C_*(l)$, $C_L(l)$, $N_L(l)$, 2^{6l^3}, $\Lambda^\circ(l)$.
(D2) *MS:* $Q(w) \ge (4\mathrm{Des}(l)/u(w))^{\omega(l)+20}$*;* $n(w)$, $T_{\mathrm{hi}}(w)$, $T_{\mathrm{lo}}(w)$ *are defined from* $Q(w)$ *as in [II, Definition 3.5(6)];* $u(w^+) = b(w) < u(w)$*;* $b(w)$ *satisfies every* upper *bound imposed on* $b(w)$ *in [II].*
(D3) (GW1)–(GW4) *of [II, Lemma 3.22] hold MS.*
(A1) (First touchers.) *For a sub-window* w *of a main stage* L *and* $s \in E_c(w)$*, the first two carriers after* L *whose targets contain* s *form a biased pair, or there is no such carrier.*

Proof. (B1), (B2): Definition 4.1(i); $\nu = \|U^*A\|$ with $\|A\|_1 \le 1$. (B3) $j' \le \sigma(l)$ and $n_{l''} \le \frac{5}{4}$, so $v_{l''}(j') = \delta_{l''} 2^{-j'}/n_{l''} \ge \frac{4}{5}\delta_{\min}(l) 2^{-\sigma(l)}$, and, μ being decreasing, $\mu_{j'}^2 v_{l''}(j') \ge \frac{4}{5}\delta_{\min}(l)/B_\mu(l)$. The bracket of $\mathrm{Des}(l)$ ([II, Definition 3.5(5)] with B_μ) contains $l 2^{l^3} B_\mu(l)\delta_{\min}(l)^{-1}$ to the sixth power, and $l 2^{l^3} \ge 2$, so $\mathrm{Des}(l)^{1/6} \ge 2B_\mu(l)/\delta_{\min}(l)$, which gives the three inequalities ($B_\mu \ge 2^\sigma$ for the third). (B4) is the last factor of $\mathrm{Des}(L)$, as in Lemma 3.20. (S1) is the allowedness argument in the proof of Theorem 4.7. (S2) Every main stage p is followed by $R_{\mathrm{cl}}(p) \ge 2$ absorber stages, and an absorber target at stage l contains odd integers $p_R > \max T(l-1)$, so $s_{\max}(l) > s_{\max}(l-1)$ at every absorber stage. [This replaces the use of fresh-dominated stages in [II, Lemma 4.1(e)], the only use of those stages in the results below.] (W1), (W5): Lemma 4.6(d). (W2) At PRE and MAIN stages $k \ge 2$ by (W3) of [II] (at stage 1, which carries $c_1 \ge \frac{1}{2}$, the inequality fails and is not claimed; (W2) is used only for fine carriers $k > L \ge l_f$); at stage $p+1$ because $c^{\mathrm{cl}}(p) \le (\delta_{p+1}^{\max} H_{p+1}/2)^2 \le (\delta_{p+1} H_{p+1})^2$; at $p+i$, $i \ge 2$: $c_{p+i} \le b(p, M(p))^2 \le 2^{-8n(p,M(p))}$, while $(\delta_{p+i} H_{p+i})^2 \ge 2^{-2(p+i)-4-6\cdot 2^{p+i}}$, $p + i \le 3p + 2s_{\max}(p)$ and $n(p, M(p)) \ge p 2^{p^3} \mathrm{Des}(p) \ge p 2^{p^3} 2^{12 s_{\max}(p)^2}$ (as $B_\mu(p)^6 \ge 2^{12\sigma(p)^2}$). (W3) by (S4). (W4) by (W1) of [II] and the cluster rule; for consecutive block-1 stages k increases by 1 and c decreases by a factor ≥ 4. (W6) Reading the proofs of the results listed in Proposition 4.11 shows that lower bounds on weights enter only through the least absorber entry in [II, Lemma 5.61(ii)] (owners inside absorber blocks) and through the design-computed bounds; the absorber length of the zero-value tuning [II, Lemma 5.51] and the cluster capacity in [II, Proposition 5.59(c)] belong to the route not used here. For the least entry: R is minimal, so $2^{4-R} > \mu_{p_0}^2 \mathrm{lb}_k^2/4 \ge \mu_{p_0}^2 c_k^2/256$ by Lemma 4.6(e); and $q^*(x) \le 8$, so the least entry $4 \cdot 2^{-R}/q^*(x)$ is $\ge 2^{-R}/2 \ge \mu_{p_0}^2 c_k^2/2^{13}$. The chain of [II, Lemma 5.61(ii)], $n(p'-1) \ge \mathrm{Des}(p'-1) \ge (B_\mu(p'-1)D(p'-1))^6 \ge (\eta_k \lambda_k)^{-6}$ for the first stage p' after the absorber sequence, survives the factor 2^{13}: first, $\sigma(p'-1) > s_{\max}(p'-1) \ge p_R(k) = p_0 + 2R(k) > R(k)$; second, the definition of $R(k)$ gives $2^{R(k)} \ge 32/(\mu_{p_0}^2 \mathrm{lb}_k^2) \ge 2^{11}/(\mu_{p_0}^2 c_k^2)$, since

$\mathrm{lb}_k \geq c_k/8$; hence, with $1/\mu_\sigma^2 \geq 16$,

$$B_\mu(p'-1) = 2^{\sigma(p'-1)}/\mu^2_{\sigma(p'-1)} \geq 2^{15}/(\mu^2_{p_0} c_k^2),$$

and, as $D(p'-1) \geq 1/\Phi_k$, $\eta_k \geq \mu^2_{p_0} c_k^2/2^{13}$ and $\lambda_k \geq \Phi_k$,

$$B_\mu(p'-1)D(p'-1) \geq \frac{2^{15}}{\mu^2_{p_0} c_k^2 \Phi_k} \geq 4(\eta_k\lambda_k)^{-1}.$$

(Here $D(l) := D_{\mathrm{room}}(l)$ of [II, Subsection 5.6]; [II, Lemma 5.61(ii)] needs only $n(p'-1) \geq \log_2(4C_f/(T_{\mathrm{lo}}\eta_k\lambda_k))$, so the margin is large.) (D1) The bracket of $\mathrm{Des}(l)$ contains these factors (with B_μ for $B^\star$), and the remaining factors are products in $\mathrm{Des}(l)$. The results below use $G^{**}(l)$ ([II, Proposition 4.24]) and not $H_{\mathrm{conf}}(l)$ itself, which enters only Des and the bound $G^{**} \leq H_{\mathrm{conf}}$. (D2) The exponent $\omega(l) + l^2 + 20 \geq \omega(l) + 20$ and $4\mathrm{Des}/u \geq 1$. The bands are disjoint: $Q(w) \geq (4/u(w))^{20}$ gives $T_{\mathrm{hi}}(w) \leq u(w)^{20}4^{-20}$ and $b(w) \leq \eta'(w) \leq T_{\mathrm{hi}}(w)^4 < u(w)$. $b(w) \leq b_{\mathrm{II}}(w)$, and $b_{\mathrm{II}}(w)$ is Part II's $b(w)$ with a larger Lipschitz constant, hence smaller; no result of [II] uses a lower bound on $b(w)$ (it enters later quantities only through $c_{L+1} \leq b^2$, $u(w^+) = b(w)$ and $1/u$ in $Q(w^+)$). (D3) (GW1) follows from (D2) and $\mathrm{Des} \geq \Lambda^\circ$; (GW2) from (D2) and $\mathrm{Des}(l) \geq (l2^{l^3}\Lambda^\circ(l))^6$; (GW3) from (W1) and the box bound [I, Lemma 8.10]; (GW4) from the definition of T_{hi}; [II, Lemma 3.22(b)] uses (T-a) only through $q^*(Te_{k,m}) \leq 1$. (A1) Let $s \in E_c(w)$. If $s \in T(L)$ or $s \in S_{l''} \cap [1, L]$, then $s \in E^{\mathrm{ch}}(L)$, whose biased pair is in the cluster immediately after L, and the absorber targets of that cluster are the first targets after L that contain s. Otherwise $s \in S_{l''} \cap (L, s_{\mathrm{far}}(w)]$; later carriers whose targets contain s are PRE pairs of later main targets containing s or cluster pairs of later main stages $p \geq s$, because absorber targets contain, apart from their own s, only fresh odd integers, signature sets are disjoint, and $y^{(1)}$ is supported in Z_0; the first such carriers form a biased pair. □

Proposition 4.11 (Transfer list)**.** *The following results of [II] hold for T_{fin}, with* Des *the design factor of T_{fin}, μ in place of $\mu^\star$ and "level" read as "main stage": Lemmas 2.9, 2.10, 2.12, 2.22, 2.23, 2.30, Corollary 2.11 and Proposition 5.46* (*no property of the operator is used*)*; Lemmas 5.1, 5.3, 5.14 and Theorem 5.7* (*algebra, Łojasiewicz, averaging, and* (B3), (D1), (D2))*; Theorem 3.12, Proposition 3.14 and Lemma 3.22* (S1, W1, D2, D3)*; Theorem 3.19 and Corollary 3.20* (D2)*; Lemmas 4.7, 4.14, 4.15, 4.17–4.19, Definition 4.16 and Proposition 4.24* (B1–B3, D1, D2, S1, S2, W1)*; Lemma 5.50* (A1)*; Definition 5.52, Lemma 5.53 and Proposition 5.54(a)–(c),(e)* (D1, D2, B1)*; Proposition 5.56* (B3, D1, D2)*; Definition 5.57(2), Lemmas 5.58, 5.60, 5.61, 5.62 and 5.64* (W1–W4, W6, D1, D2)*. The counting bounds "$\leq C_f\bar{D}(w)$" of Lemma 5.53 and Proposition 5.54(d) are replaced by Lemma* 4.22*. Lemma 5.51 and Proposition 5.59(b)–(d)* (*zero-value absorbers*) *are not claimed for T_{fin} and are not used.*

Proof. For each listed result, the proof in [II] uses the operator only through the properties named in parentheses (and through the results of [I] covered by Lemma 4.9), and these hold for T_{fin} by Lemma 4.10. We record the points where the two operators differ quantitatively. (i) The base enters Lemma 4.15 (tuning) and the results built on it only through (B1)–(B3): with (B3), the contraction bound $|M'| \leq C_f\mathrm{Des}^{2/3}\varepsilon_x$ and the bound $\kappa_T \leq C_f\mathrm{Des}\,\varepsilon_x^2$ hold, the banks are positive and the fixed point exists for $\varepsilon_x \leq c_T = c_f\mathrm{Des}^{-1}$; this threshold contains the

factor $l^{-1}4^{-G(l)}(s'v')^2$ needed for positivity of the banks of shallow carriers, because $\mathrm{Des}^{1/2} \geq l4^{G(l)}$ and $\mathrm{Des}^{1/3} \geq (s'v')^{-2}$. The depth $s_{\mathrm{t}}(l)$ of Part II (third element beyond $s_{\max}$) is used there only for infinite base support; for finite base support the depths used are the first element beyond $s_{\max}$ (tuning banks, peak pushes, closing moves) and the second (σ_2, levers). (ii) $\|U\| = \frac{1}{4}$ instead of $\frac{1}{16}$ changes the donor bound of Lemma 4.17(i) to $\mathrm{Des}^{1/3}\Lambda/2$; Lemma 4.18 needs $N\mathrm{Des}^{1/3}\Lambda \leq 1$, true for main stages $L \geq l_f$. (iii) In Lemma 4.1, part (d) is (B3) and part (e) is (S2). (iv) Part II's $D^{U1'}$ is not transferred as a definition; its properties used below are Lemma 4.6, Theorem 4.7 and (A1). (v) The class and ray counts are the only window statements that are not of the form treated by Lemma 3.22: their window factor has an exponent growing with L, and Part II handles them by $Q(w) \geq \bar{D}(w)^3$; for T_{fin} they are handled by Lemma 4.22. □

Remark 4.12 (The exact route does not transfer). The zero-value tuning of absorbers [II, Lemma 5.51] needs $4e_1 \leq 2^{-R}$ for the perturbations $e_r := \mu_{p_r}^2\theta_a/\nu$ of the binary coordinates; Part II obtains it from the doubly exponential base and its rule for p_0. For T_{fin}, with the first absorber of the cluster of a main stage L, 2^{-R} is of the order $\mu_{p_0}^2 c_{L+1}^2$ while e_1 is of the order $\mu_{p_0+2}^2\theta_a/\nu$ with θ_a of the size of the residue to be absorbed, which contains the true-shift correction and is not small compared with c_{L+1} in general. So the intermediate-value sweep of [II, Lemma 5.51] can fail at the first used pair, and the route through exact data with zero-value absorbers ([II, Theorem 5.68]) is not available for T_{fin} as defined. Whether a different choice of the tuning coordinate p_0 repairs this is not examined here. The proof of Theorem 4.25 tunes no absorber and is not affected.

4.3. **Route V: the companion.**

Standing hypotheses. In this subsection and the next, T_{fin} is taken with option (a), or with option (b) and only patterns κ with $1 \notin \mathcal{N}_T(\kappa)$; $N \geq 1$; $f \in S_{p_N^*}$ has finite base support F; $g \in \mathcal{C}(f)$ and $\rho \in (0,1)$. The number $l_f = l_f(f, N, g, \rho)$ is a stage beyond which every f-constant is absorbed; precisely, l_f is so large that for every main stage $L \geq l_f$ and every sub-window w of L: the conclusions of Lemma 2.3(b) and the hypotheses $L \geq l_f$ of Section 2 hold; every block $m \leq N$ has a carrier $\leq L$ (so the pattern of f at w is a pattern of level L of the design, and $J := |\Omega| + N \leq 2L$); rows within $C_f T_{\mathrm{lo}}(w)^2$ of f lie in the ball B_f of Lemma 3.6; $T_{\mathrm{hi}}(w) \leq t_1$ (Theorem 3.15); $T_{\mathrm{lo}}(w) \leq T_*$ of Proposition 3.21 for the constant C_0 of Lemma 4.19; and $C_f N \leq 4\mathrm{Des}(L)/u(w)$ for the finitely many f-constants C_f named below. Each condition holds for large L because $T_{\mathrm{lo}}(w) \leq 2^{-L^3}$, $\mathrm{Des}(L) \to \infty$, and every block is served in infinitely many rounds.

Let $L \geq l_f$ be a main stage, w a sub-window of L that is clean for f ([II, Definition 3.18]), and a an activity class of w for the data of (g, ρ) ([II, Lemma 5.53]: the class of a scale is read off the decomposition data of g at f and is fixed before the companion is built), with pattern κ (Definition 2.8). We write $T := T_{\mathrm{lo}}(w)$, $b := b(w)$, $u := u(w)$, $\mathrm{Des} := \mathrm{Des}(L)$, $\eta' := \eta'(w)$, and

$$\mu_{\min}(w) := \min\{\mu_{\kappa'}(\eta'/3) : \kappa' \in \mathfrak{P}(L),\ \mathcal{N}_T(\kappa') \neq \emptyset\} \quad (:= 1 \text{ if there is none}),$$
$$g_{\min}(w) := \tfrac{1}{8}\min\{c_\flat\eta', \mu_{\min}(w)\},$$

where $c_\flat \in (0,1]$ is a constant depending only on f such that the gaps produced in blocks of kind (KN) and (U) by Proposition 2.32(b) are $\geq c_\flat\eta' M_m$: the minimum of

the numerical constant 1 for the inward pushes in (U)-blocks and the f-constant c_f of [II, Proposition 5.56(b)] for the inactive near-threshold carriers of (KN)-blocks. By Lemma 4.6(f), Remark 2.28(iii) and (QC4), $8b \le \mu_{\min}(w)$. A *class set* is a set $\mathcal{S}$ of dyadic scales of w that all have the class a.

Remark 4.13 (Route decision). For each (L, w, a) one companion is built, for every kind of class: without active shift, one-signed, or mixed. The violation-tolerant route needs from the coarse system only three things: the coarse statuses, the kind-[3] property of the near-threshold switching carriers with the true shift, and the exactness of the cone at the realized ratios. Proposition 2.32 supplies all three, and it treats each block separately, whatever the signs of the shifts of the other blocks; so mixed classes need no separate treatment. Absorbers are not tuned: they are ordinary fine carriers. (Tuned zero-value absorbers cannot follow the levers of Lemma 3.10, whose moves are much larger than the absorbers' value rooms; Remark 3.28.)

Lemma 4.14 (The route-V companion)**.** *Under the standing hypotheses there is a first row $f^{\#}$ with finite base support $F^{\#} \supseteq F$, built in the order: class a, Step A, Step B below, with the following properties. Let $\Omega^{\#}$ be the set of coarse strict non-peaks of $f^{\#}$.*

- (a) (Exactness.) *The realized active ratios r'' (the active ratios of $f^{(1a)}$, i.e. $u_l(\hat{z}^{\#})/\varkappa'_m$ with the block scalars $\varkappa'_m$ of $f^{(1a)}$) lie in Z_κ; every robust rate object has modulus $\ge u/4$ at r''; the ℓ_1-Hoffman constant of the cone $\Gamma^{\#}(\kappa, a)$ at r'' is $\le C(\mathrm{Des}/u)^C$, and the bound $A_{\max}$ of [II, Proposition 5.54] is $\le C_f(\mathrm{Des}/u)^C$.*
- (b) (Statuses.) *Every coarse carrier of $f^{\#}$ is either a peak with relative margin $\ge u/4$, or a strict non-peak with $\mathrm{gap}^{\#} \ge g_{\min}(w)$. In every block of kind (T), every carrier $d \in \mathcal{N}(m)$ has $\varrho_d(f^{\#}) \le 1 - \mu/4$ with $\mu := \mu_\kappa(\eta'/3)$, and every carrier of $\Omega_m \setminus \Omega_a(m)$ has $|u_l(\hat{z}^{\#})| \le 4c_{L+1}^2$.*
- (c) (Data.) *For every scale t of a class set $\mathcal{S}$ there are pieces $(g_t; b_t^{\pm}, \omega_t^{\pm})$ at $f^{\#}$, with $\omega_t^{\pm}$ supported in $\Omega^{\#}$, such that $p^*(g - g_t) \le K_{\mathrm{data}} t$ with K_{data} as in Lemma 4.22, $\Gamma_w^{\#}(b_t^{\pm}, \omega_t^{\pm}) \le 1 + \eta_0/2$, and every carrier of $\Omega^{\#}$ is of kind [1′], [2] (with $\gamma_B := \gamma(w) := \min_m M_m u/4$) or [3] of Definition 3.12, relative to $\mathrm{gap}^{\#}$.*
- (d) (Violations.) *The total violation of these pieces is $\epsilon(t) \le C_f \mathrm{Des}\, c_{L+1} + C_f c_{L+1}^2/T$, and $b_t^{\theta}(j) = 0$ at every free coordinate j with a violation.*
- (e) (Scrambling.) *Every fine carrier of a negative active block is a peak with margin $\ge q_0^{\#}\delta_k^{\circ}/2$, and for every negative active block m, $\mathrm{Scr}_m^{\#}(x) \le C(x/q_0^{\#})^2$ for $0 < x \le x_0(w) := c_f g_{\min}(w)\Phi_{\min}(L)$.*
- (f) (Cost.) $p^*(f^{\#} - f) \le C_f T^3 \log(1/T)$.

Proof. Step A is Proposition 2.32 applied to (f, w, a); it yields $f^{(1a)}$. Its joint fixed point holds every non-active carrier of Ω_m of a block of kind (T) at value exactly 0, i.e. *every* nearly neutral coarse strict non-peak of such a block other than the active ones (Remark 2.33 shows that holding fewer carriers is not enough), and all its pulls lie in the common window of Lemma 2.29 with $\eta_* = \max(\|x\|_\infty, c_{L+1})$, so every bank is positive and every pull has mass $\ge 24\lambda_l c_{L+1}$. No absorber is tuned.

Step B is step (2) of Lemma 3.24 (the fine-structure step of [II, Definition 5.57(2)] and [II, Lemma 5.60] without absorber tuning), applied to $f^{(1a)}$; steps (1b), (1c) of [II, Definition 5.57] do not occur, because the joint fixed point of Step A already

restores every $\varkappa_m$ and every active value. First the G1 release: the parts $S_{l''} \cap (s_{\mathrm{far}}(w), \infty)$ of the signature sets of the coarse carriers that do not own them are released into $J_{\mathrm{fine}} := \mathbb{N} \setminus (F^{(1a)} \cup E_c(w) \cup \{\text{coordinates fixed in Step A}\})$. Then the static-ownership recursion on J_{fine}: the owner candidates are (a) the active carriers of Ω, (b) the coarse peaks of the active blocks, and (d) every carrier at a stage $> L$ of an active block, absorbers included, with its forced coefficient $-\Delta''_m \lambda_k w(k)$; a coordinate gets the z-value of [II, Lemma 5.60] for its first owner (negative fine owners self-aligned, positive fine owners anti-aligned, coarse owners by the sign of their coefficient), and unowned coordinates get $z := 0$. The result is $f^{\#}$. The class enters Step B only through the signs of the shifts and of the active components, which are constant on the class ([II, Lemma 5.53]); no common shift value is used (Lemma 4.21).

(a) is Proposition 2.32(a) (with the robust bound $u/4$), and Step B does not change the active values; the change of the block scalars in Step B is carried by the true shift $\Delta''_m = \Delta'_m \varkappa'_m / \varkappa^{\#}_m$ in (c), so exactness is used only at r''. The Hoffman bound is [II, Lemma 5.1] with the minors $\geq u/4$, and the bound on $A_{\max}$ is [II, Proposition 5.54].

(b) At $f^{(1a)}$: Proposition 2.32(b) gives $\varrho_d \leq 1 - \mu/2$ for $d \in \mathcal{N}(m)$ in blocks of kind (T), held values 0, gaps $\geq c_\flat \eta' M$ in blocks of kind (KN) and (U), and robust statuses moved by $\leq C \mathrm{Des}\, \eta' \ll u$. Step B changes the block vector ζ_m only through the fine carriers (ℓ_1-mass $\leq \sum_{l>L} \lambda_l \leq c_{L+1}/3$) and through the released far parts of non-owners, which change a coarse value by $\leq 2\delta_l \sum_{s>s_{\mathrm{far}}} 2^{-s}/n_l \leq 4c_{L+1}^2$ (as $2^{-s_{\mathrm{far}}} \leq c_{L+1}^2$); it changes neither the base part a (hence not e) nor any active value. By the clamp equation, as in Lemma 3.4(c) with $\Omega = \emptyset$ and the natural peak of Lemma 3.6, the block scalar C_m moves by $\leq C_f c_{L+1}$; since $\varrho_k = \mathsf{v}_k C_m / (\Phi_k^2 M_m)$ for strict non-peaks, every relative position moves by $\leq C_f \mathrm{Des}\, c_{L+1}$, and $\varkappa$ moves by $\leq C_f c_{L+1}$. By (QC5) and Lemma 2.34, $\varrho_d(f^{\#}) \leq 1 - \mu/4$ for $d \in \mathcal{N}_T$; the held values end at $\leq 4c_{L+1}^2$, which changes the protected statuses only by $O(D^2 c_{L+1}^4) \ll \mu$. As $M_m \geq \frac{1}{2}$, $\mathrm{gap}^{\#}(d) = M_m(1 - \varrho_d) \geq \mu/8 \geq g_{\min}(w)$; robust margins and the gaps $\geq c_\flat \eta' M$ change by $\leq C_f \mathrm{Des}\, c_{L+1} \leq C_f \mathrm{Des}\, b^2 \ll g_{\min}(w)$.

(c), (d), (e) are Lemma 3.24(a)–(c) for the companion built with Step A in place of step (1a) there. The proof of Lemma 3.24, and of the results [II, Lemmas 5.58, 5.60–5.62, 5.64] it invokes, uses step (1a) only through: (i) the coarse statuses (for the scrambling threshold x_0, for the coefficients $|w| = M$ of coarse peak owners, and for the fact that in Step B only carriers at stages $> L$ change status); (ii) the kind-[3] property of the near-threshold carriers of $\Omega^{\#}$; (iii) the exactness of $\Gamma^{\#}(\kappa, a)$ at the realized ratios (projection with error $K_{\mathrm{data}} t$ and the Hoffman bound). Item (i) is (b). Item (iii) is (a). For item (ii): in blocks of kind (T) the decomposition data satisfy (X4^0) up to $K_{\mathrm{data}} t$ (Lemma 2.14(b), applied at the realized ratios r'', whose block coordinates satisfy $|r''^{(m)} - r(f)^{(m)}| \leq 2\eta'$ in blocks of kind (T) by Proposition 2.32(a)), and every point of the cone is inward at $f^{\#}$ with the *true* shift $\Delta''_m = \Delta'_m \varkappa'_m / \varkappa^{\#}_m$ ([II, Lemma 5.58(a)]; its middle term vanishes because Step B does not move active values): for a near-threshold carrier l, using (X4^0) at the cone point in the form $\mathrm{vs}_l \gamma_l + \Delta'_m \lambda_l \varrho_l^0(r'') \geq 0$ (Lemma 2.14(a)),

$$\begin{aligned}\lambda_l \mathrm{vs}_l \Delta\omega^{\#}(l) &= \mathrm{vs}_l \gamma_l + \Delta''_m \lambda_l \varrho_l^{\#} \\ &\geq -|\Delta''_m - \Delta'_m| \lambda_l \varrho_l^{\#} - |\Delta'_m| \lambda_l |\varrho_l^{\#} - \varrho_l^0(r'')| \geq -C_f \lambda_l (c_{L+1} + D^C c_{L+1}^2),\end{aligned}$$

which is $\geq -3\lambda_l \operatorname{gap}^{\#}(l)/t$ since $\operatorname{gap}^{\#} \geq M\mu/4$ and $c_{L+1} \leq b^2 \leq \mu^2/64$; on the + side, $\operatorname{vs}_l\omega^+(l) \leq \operatorname{gap}(f)/t$ with $\operatorname{gap}(f) \leq M_m b \leq M_m\mu/8 \leq 1.5\operatorname{gap}^{\#}(l)$. In blocks of kind (KN) the row (X4) of [II, Definition 5.52] with the true shift gives $\operatorname{vs}_l\Delta\omega^{\#}(l) \geq -|\Delta'^{*}_m|\operatorname{gap}^{\#}/M - C_f c_{L+1} \geq -3\operatorname{gap}^{\#}/t$; in blocks of kind (U), $\Delta''_m = 0$ and $\operatorname{vs}_l\Delta\omega = \operatorname{vs}_l\gamma_l/\lambda_l \geq 0$. Kinds of the other carriers of $\Omega^{\#}$: active robust carriers are of kind [2]; held carriers and inactive nearly neutral carriers have $\Delta\omega = \Delta''_m w^{\#} \approx 0$ and gap close to M, kind [2]; the lever carrier of a (KN)-block is of kind [2] in negative blocks (an active negative block has no inactive near-threshold carrier, by [I, Lemma 8.8] and [I, (16)]) and of kind [3] in positive blocks; the carriers of (U)-blocks pushed inward by η' have $\Delta\omega = 0$ and $|\omega| \leq 2\operatorname{gap}/t$, kind [1′] (their gap may be smaller than t^2; Theorem 3.15 uses only $|\omega| \leq 2\operatorname{gap}^0/t$ for them). For the representation and the violation count of (c), (d): on a contact of $E_c(w)$ the row (X1) at the data point gives $zV_{\mathrm{proj}} \geq 0$, so the violation is $\leq |V_{\mathrm{fine}}|/2$; on a free coordinate of $E_c(w)$, $V_{\mathrm{proj}} = 0$, $b^{\pm} = \pm V_{\mathrm{fine}}/2$ and $b^{\theta} = 0$; the trace correction $-\sum_m(\Delta''_m - \Delta'_m)P_m$ on $E_c(w)$ has ℓ_1-mass $\leq C_f c_{L+1}$ and is a tolerated violation (it is not absorbed, since no absorber is tuned); on J_{fine}, coarse owners and negative fine owners dominate the later coefficients ([II, Lemma 5.61], including owners in absorber blocks through the least-entry bound of Lemma 4.10(W6)), unowned coordinates have $V = 0$, and positive anti-aligned owners contribute violations of total mass $\leq C_f\sum_{l>L}\lambda_l$. With $\sqrt{\Gamma^{\#}_w}$ a seminorm and $\|V_{\mathrm{fine}}\|_1 \leq C_f(\operatorname{Des} c_{L+1} + c^2_{L+1}/t)$, the representation error and $\Gamma^{\#}_w \leq 1 + \eta_0/2$ follow as in [II, Proposition 5.54(e)]. For (e): a negative fine carrier k (any class, absorbers included) is self-aligned, so $|u_k(\hat{z}^{\#})| \geq \delta^{\circ}_k \gg \bar{\vartheta}\Phi_k$ by (W2), and k is a peak with margin $\geq q^{\#}_0\delta^{\circ}_k/2$; coarse peaks have margin $\geq q^{\#}_0\bar{\vartheta}\Phi_{\min}u/2 > x_0$ and coarse strict non-peaks have gap $\geq g_{\min}$ and $\Phi\operatorname{gap} \geq \Phi_{\min}g_{\min} > x_0$ by (b); so only fine carriers with $\delta^{\circ}_k \leq 2x/q^{\#}_0$ enter $\operatorname{Scr}_m(x)$, with $\Phi_k \leq (\delta_k H_k)^2$, and the super-exponential decay gives the bound ([II, Lemma 5.62]).

(f) Step A costs $\leq C_f\operatorname{Des}^3\eta'\log(1/\eta') \leq C_f T^3\log(1/T)$ (Proposition 2.32(c)). Step B changes the row only on J_{fine}, where the coarse carriers move by $\leq 4c^2_{L+1}$ and the fine carriers have total weight $\leq c_{L+1}/3$; by [II, Lemma 2.12] its cost is $\leq C_f c_{L+1}\log(1/c_{L+1}) \leq C_f T^4$ (Lemma 4.6(d)). □

Remark 4.15 (On Step B). The proof of Corollary 3.27 uses of the companion only the coarse statuses and the bound on $\mathcal{V}$ on J_{fine}; with (b) above it applies to Lemma 4.14. So Step B may be omitted if (U5) is replaced by (U5′) of Lemma 3.23: under ($\mathrm{W}_{\mathrm{exp}}$) the fine carriers then enter only through their total weight, and fine-origin misfits are tolerated violations of mass $O(\operatorname{Des} c_{L+1})$. We keep Step B in the main line because that is the version whose verification was carried out twice.

4.4. **Route V: base rows, frozen block scalars, threshold.** We keep the standing hypotheses of Subsection 4.3, let $\mathcal{S}$ be a class set of (L, w, a), and let $f^{\#}$ and the pieces $(g_t; b^{\pm}_t, \omega^{\pm}_t)$, $t \in \mathcal{S}$, be those of Lemma 4.14.

Proposition 4.16 (The companion is a base row)**.** *The pieces $(g_t; b^{\pm}_t, \omega^{\pm}_t)_{t\in\mathcal{S}}$ form a window family at $f_0 := f^{\#}$ in the sense of Definition* 3.12*, with window parameter $T = T_{\mathrm{lo}}(w)$, $\Omega := \Omega^{\#}$, an f-constant A_0, $A_2 = 22$, $\gamma_B := \gamma(w) = \min_m M_m u(w)/4$, an f-constant C_Δ, the levers of Lemma* 3.25 *with the pull window $[r, 2^G r]$, $G :=$*

$6G(L)$, and window quantities

$$n \le \log_2(1/T), \quad \mathrm{gap}_{\min} \ge g_{\min}(w), \quad x_0 = x_0(w), \quad C_S \le C/q_0^2, \quad c_{\mathrm{fine}} \le c_{L+1}/3,$$
$$c_{\mathrm{tiny}} + W_{\mathrm{lev}} + \overline{W}_{\mathrm{pull}} \le C_f \mathrm{Des}(L)^C c_{L+1}, \qquad \Lambda_{\mathrm{lev}} \le C_f \big(\mathrm{Des}(L)/(u(w) g_{\min}(w))\big)^C.$$

Moreover $c_{\mathrm{flat}}(w) \ge c_f u(w)$ for the number c_{flat} of Theorem 3.15.

Proof. The scales of $\mathcal{S}$ lie in $\mathcal{W}(w) \subseteq [T, t_1]$ ($L \ge l_f$), and $f^\#$ lies in B_f with $F \subseteq F^\#$ (Lemma 4.14(f) and Lemma 3.6(a)).

(U1), (U2): Lemma 4.14(c); the bound $t\|b_t^\pm\|_1 \le A_0$ and the constant $A_2 = 22$ are those of [II, Proposition 4.24(iii)], which apply by Proposition 4.11.

(U3): The coordinates of $E := F^\# \setminus F$ are created in Step A (tuning banks and pulls of active carriers and of held carriers, peak-push banks, lever moves in blocks of kind (KN), and the inward pushes of the inactive near-threshold carriers in blocks of kind (KN) and (U), realized by pulls in the common window); Step B creates none, and no absorber coordinate is a support coordinate. The data there are $b^\pm = \beta + (\chi, \chi - 1)\mathcal{V}_{\mathrm{proj}} \pm \frac{1}{2}\mathcal{V}_{\mathrm{fine}}$. The last term is an antisymmetric residue $r^\pm$, counted in the total violation. The remaining part is contact-like at banks (by (X2) the switching component of an active carrier has the sign ϵ_l of its bank, and the coefficient of a coarse peak has the sign of its trace), and at a pull j_p it satisfies $t|\tilde{b}^\pm(j_p)| \le (2A_2 + C_\Delta)\lambda_l v_l(j_p)$ ([II, Proposition 4.24(iv)], whose moves have the same types as those of Step A), while the pull mass is $|a^\#(j_p)| \ge c\lambda_l v_l(j_p)$; so $t|\tilde{b}^\pm| \le K_3|a^\#(j_p)|$ with an f-constant K_3. At a pull of a held carrier $\gamma_l = 0$, so $\tilde{b}^\pm(j_p) = 0$; the residue there is $\le C_f c_{L+1}$ in absolute value, while the pull mass is $\ge 24\lambda_l c_{L+1}$ by the common window (Lemma 2.29), so even without splitting off the residue, $t|b^\pm(j_p)| \le K_3|a^\#(j_p)|$ for $t \le T_{\mathrm{hi}}(w)$. The same argument applies at the pulls of the inward pushes: the pushed carriers are inactive in the class ($\gamma_l = 0$, Lemma 3.25(b)), so the data there are the residue $\le C_f c_{L+1}$, while the pull mass is $\ge 24\lambda_l c_{L+1}$.

(U4): Lemma 4.14(d). (U5): Lemma 4.14(e).

(U6): Lemma 3.25 at the companion of Lemma 4.14. Its proof uses: allowedness and the disjointness of the signature sets; that the support of $f^\#$ meets $S_l \cap (s_{\max}(L), \sigma_2(L)]$ at most in its first element; that the coordinates of $S_l \cap (s_{\max}(L), s_{\mathrm{far}}(w)] \setminus F^\#$ are contacts with $z^\# = \epsilon_l$ (closing move (C1) of [II, Definition 4.16], unchanged by Step A outside its own coordinates and by the release); the constancy of the signs on the class; (D-lev); and (W1). The first point holds because Step A places every bank at the first element of S_l beyond $s_{\max}(L)$, and its pulls have $v_l(j_p) \in [\eta_*, 2^{G(L)}\eta_*)$ with $\eta_* \le C_f \mathrm{Des}(L)\eta'$, far below v_l at the second element, which is $\ge 1/\mathrm{Des}(L)$ by (D-lev) while $\eta' \le T^4$. So the bank coordinate $j(l)$ of Lemma 3.25 is at most the second element, and (B4) gives $\beta_{j(l)} \le \nu\,\mathrm{Des}(L)$. For a held carrier l of class G, $\gamma_l = 0$, so it has the zero-data lever of Lemma 3.25(b); for a held carrier of class R the coordinate $j(l)$ is free and carries no data. Step A uses a bounded number of coordinates of each S_l (at most a Step-0 push, one tuning pull and one bank, a lever or inward push, and one peak push). The pull coordinates of Lemma 3.10 have $v_l(j_p)$ of the order of $r \le \Pi$; we choose them among the unused elements of S_l in a v-window of ratio $2^{6G(L)}$, which contains at least six elements of S_l because consecutive elements differ by $\le G(L)$; the factor $2^{6G(L)} \le \mathrm{Des}(L)$ enters only C_2. As $r \ge s_1 T^4 \ge e^{-1/(2T)}T^4 \gg c_{L+1} \ge 2^{-s_{\mathrm{far}}(w)/2}$, these coordinates lie below

$s_{\mathrm{far}}(w)$. Carriers outside Ω that meet lever coordinates are at stages $> L$: beyond $s_{\max}(L) = \max T(L)$ no coarse target meets S_l.

(U7): $|\Delta_{t,m}| \le C_\Delta$ by [II, Lemma 2.23], as the shifts of the class are bounded by an f-constant.

Window quantities. $n \le n(w) \le \log_2(1/T)$ since $T = 2^{-n(w)}T_{\mathrm{hi}}(w)$. $\mathrm{gap}_{\min} \ge g_{\min}(w)$ and the value of x_0 by Lemma 4.14(b), (e); C_S by Lemma 4.14(e). A carrier that is neither in $\Omega^\#$ nor a peak with margin $\ge x_0$ is fine in the sense of Definition 3.12; coarse peaks have margin $\ge q_0^\# \bar\vartheta \Phi_{\min}(L)u/4 \ge x_0$, so fine carriers are at stages $> L$ and $c_{\mathrm{fine}} \le \sum_{l>L} \lambda_l \le c_{L+1}/3$. The bounds on c_{tiny}, W_{lev}, $\overline{W}_{\mathrm{pull}}$ and Λ_{lev} are Lemma 3.25(c) with $\mathrm{gap}_{\min} \ge g_{\min}(w)$ and the extra factor $2^{6G(L)}$. Finally $c_{\mathrm{flat}} = \min(c^f_{\mathrm{flat}}, \gamma(w)/(8\rho A_2)) \ge c_f u(w)$, since $M_m \ge \frac12$ on B_f. □

Let now the window conditions of Theorem 3.15 hold for this family (they are verified in Lemma 4.19), and let f' be the d-consistent engineered approximant of Definition 3.13 at $f_0 = f^\#$, with lever parameter P^*; primes denote its data.

Lemma 4.17 (The block scalars are frozen)**.** *In this situation, for $L \ge l_f$:*

(i) $\mathrm{nv}'_k = \mathrm{nv}^\#_k$ *for every $k \in \Omega^\#$, in particular for every carrier of $\mathcal{N}_T$ and every held carrier;*

(ii) $|C'_m - C^\#_m| \le C_f c_{L+1}$ *for every block m;*

(iii) *for every $k \in \Omega^\#$ of block m,*

$$\frac{\varrho'_k}{\varrho^\#_k} = \frac{C'_m(1-C^\#_m)}{C^\#_m(1-C'_m)},$$

hence $|\varrho'_k - \varrho^\#_k| \le C_f c_{L+1}$, $\mathrm{gap}'(k) \ge \mathrm{gap}^\#(k)/2$, and $\varrho_d(f') \le 1 - \mu/8$ for every $d \in \mathcal{N}_T$;

(iv) *for every lever parameter P with $|P|_\infty \le r$ (not only at P^*) no carrier of $\Omega^\#$ and no coarse peak of $f^\#$ changes status; along the cube the carriers of $\Omega^\#$ move by at most $\delta_1 + C_{\mathrm{lev}} r$ and reach $\mathrm{nv}^\#$ at P^*; coarse peaks of $f^\#$ meet no lever coordinate and move only at second order, by less than their margins; among the carriers outside $\Omega^\#$, only fine carriers (stages $> L$, total weight $\le c_{L+1}/3$) meet lever coordinates.*

Proof. (i) is the d-consistency of f' with $f^\#$ on $\Omega^\#$ (Theorem 3.15).

(ii) Fix m and put $C := C^\#_m$. By the clamp formula ([I, Lemma 7.14]; Lemma 3.4(c)), $C^\#_m$ and C'_m are the roots of $F(c;v) = 1$, $F(c;v') = 1$ with $F(c;v) = \sum_k \min(\Phi_k(1-c)/c, v_k)^2$ and $v_k = |\mathrm{nv}_k|/\Phi_k$. Apply Lemma 3.4(c) with the natural peak $k^\natural_m$ of Lemma 3.6, which is a peak with margin $\ge \mu^\natural_m/2$ at every row of B_f, so that its term is $\Phi^2((1-c)/c)^2$ on an interval $[C-r', C+r']$ with r' an f-constant. In the sum of Lemma 3.4(c) the terms of $\Omega^\#$ vanish by (i), and so do those of the coarse peaks of $f^\#$: their relative margins are $\ge u/4$, while their values move by $\le K_w\Pi \ll u$ (Lemma 3.14(b)) and $|C'_m - C| \le K_w\Pi$ (Lemma 3.14(c)), so they belong to P^{rob}_m. Every remaining term belongs to a carrier at a stage $> L$ and is at most $2\Phi_k(1-C)C^{-1}|v'_k - v_k| \le 2(1-C)C^{-1}(|\mathrm{nv}'_k| + |\mathrm{nv}_k|)$, and $|\mathrm{nv}_k| = \lambda_k|u_k(\hat z)|/A_m \le (1+\|U\|)\lambda_k/A_{\min}$ at both rows. Summing, $|C'_m - C| \le C_f \sum_{l>L}\lambda_l \le C_f c_{L+1}$ (Lemma 4.6(d)); C and $1-C$ are bounded below by f-constants on B_f.

(iii) For a strict non-peak k of block m, by [I, Lemma 1.7],

$$\varrho_k = \mathsf{v}_k/\bar\vartheta = |\mathrm{nv}_k| C_m/(\Phi_k^2 M_m),$$

and $\mathrm{nv}'_k = \mathrm{nv}^{\#}_k$, which gives the identity and $|\varrho'_k - \varrho^{\#}_k| \le C_f c_{L+1}$. By Lemma 3.4(a), $\mathrm{gap}'(k) \ge \mathrm{gap}^{\#}(k)/2$ as soon as $|C'_m - C| \le C\,\mathrm{gap}^{\#}(k)/(2(1-\mathrm{gap}^{\#}(k)))$; this holds because $\mathrm{gap}^{\#}(k) \ge g_{\min}(w)$, $b \le \min(\eta', \mu_{\min}/8) \le 8g_{\min}(w)/c_\flat$, and so $C_f c_{L+1} \le C_f b^2 \le 64 C_f c_\flat^{-2} g_{\min}(w)^2 \le C^{\#}_m g_{\min}(w)/4$ for $L \ge l_f$ (as $g_{\min}(w) \to 0$ and $C^{\#}_m$ is bounded below on B_f). For $d \in \mathcal{N}_T$, Lemma 4.14(b) gives $\varrho^{\#}_d \le 1 - \mu/4$, and $C_f c_{L+1} \le C_f b^2 \le \mu/8$.

(iv) By Lemma 3.10(ii) and step (5) of its proof, along $|P|_\infty \le r$ the carriers of $\Omega^{\#}$ move by at most $\delta_1 + C_{\mathrm{lev}} r$, and they reach $\mathrm{nv}^{\#}$ at P^* by (i); a coarse peak meets no lever coordinate, so it moves only through the normalization, at second order; and the radius r_0 of Lemma 3.10 is, in value units, below $c \min_{l\in\Omega} \Phi_l \,\mathrm{gap}_l\, A_{\min}/(mC_{\mathrm{lev}})$ and below the coarse peak margins divided by C_{lev}; Theorem 3.15 uses $r \le r_0$ (proof of Lemma 3.14). The lever coordinates are private: by (SEP) they meet no other carrier of $\Omega^{\#}$, and by (U6) the other carriers meeting them are at stages $> L$. □

Remark 4.18 (Why the freezing matters). The margin $\mu = \mu_\kappa(\eta'/3)$ protecting the carriers of $\mathcal{N}_T$ is only known to be $\ge \mathrm{Des}^{-1}(\eta'/3)^{\mathrm{Des}}$; it cannot be compared with the size r of the lever moves or with $K_w\Pi$ in general. It does not have to be: on $\Omega^{\#}$ the levers act exactly (i), and the only quantity through which the other carriers act on a relative position of $\Omega^{\#}$ is the block scalar C_m, which (ii) freezes up to the weight $c_{L+1} \le \mu^2/64$ of the fine carriers. This requires the levers on *all* coarse strict non-peaks: a single coarse strict non-peak left untuned changes C_m at first order in the lever size. (In finite block models, with all of Ω tuned, $|C' - C|$ was at the working precision, and with one robust strict non-peak left untuned the protected carrier moved by 10^{-7} to 10^{-5} and in several instances became a peak; this is evidence only.)

Lemma 4.19 (Threshold comparison at the companion)**.** *There is an absolute constant C_0 such that, under the standing hypotheses, for the window family of Proposition* 4.16 *with $T = T_{\mathrm{lo}}(w)$:*

(i) $g_{\min}(w) \ge T^{5\mathrm{Des}(L)+2} \ge 2^{-6\log_2^2(1/T)}$ *and* $x_0(w) \ge T^{5\mathrm{Des}(L)+3}$*;*

(ii) *hypotheses* (i)–(iii) *of Proposition* 3.21 *hold with this C_0 and $c_* := c_{L+1}$, once N_w is chosen with* $\mathrm{tail}_W \le e^{-1/T}$*; in particular* (W-b) *holds;*

(iii) *consequently $s_1 := e^{-1/(2T)}$ satisfies* (VT′)*,* (FINE) *and* (LATE)*,* (W-a) *and* (W-c) *hold, and*

$$K_w\big(K_w s_1 + c_{\mathrm{fine}} + \mathrm{tail}_W\big) + K_w\big(\Pi\log(e/\Pi) + s_1\big) \le e^{-1/(3T)}.$$

Proof. (i) We have $L \le n(w)$ and $\mathrm{Des}(L) \le Q(w) \le n(w) \le \log_2(1/T)$, because $T = 2^{-n(w)} T_{\mathrm{hi}}(w)$. Hence $1/T \ge 2^{n(w)} \ge 3n(w)^2 \ge 3L\,\mathrm{Des}(L)$ and $\eta'/3 = T^4/(3L\,\mathrm{Des}) \ge T^5$. By Remark 2.28(iii) every $\mu_{\kappa'}(\eta'/3)$ is $\ge \mathrm{Des}^{-1}(\eta'/3)^{\mathrm{Des}} \ge T^{5\mathrm{Des}+1}$ (as $1/\mathrm{Des} \ge T$), and $c_\flat \eta' \ge c_\flat T^5 \ge T^6$, so $g_{\min}(w) \ge \frac{1}{8} T^{5\mathrm{Des}+1} \ge T^{5\mathrm{Des}+2}$ for $T \le \frac{1}{8}$; with $\mathrm{Des} \le \log_2(1/T)$ this is $\ge 2^{-6\log_2^2(1/T)}$ when $\log_2(1/T) \ge 2$. Finally $x_0 = c_f g_{\min} \Phi_{\min}(L)$ and $\Phi_{\min}(L) \ge 1/\mathrm{Des}(L) \ge T$.

(ii) By Proposition 4.16 and (i), with natural logarithms: $n \le \log_2(1/T)$; $A_0/T \le C_f/T$; $\log(1/\mathrm{gap}_{\min}) \le 9\log^2(1/T)$ and $\log(1/x_0) \le 11\log^2(1/T)$ for T small; $1/u(w) \le n(w)$ ([II, Proposition 3.14(a)]); $|\Omega| \le 2L \le 2n$; $C_S \le C/q_0^2$; and $\Lambda_{\mathrm{lev}} \le C_f(\mathrm{Des}/(u g_{\min}))^C \le e^{C\log^2(1/T)}$. Since K_w is bounded by an f-constant times a power with *absolute* exponent of $nA_0|\Omega|\Lambda_{\mathrm{lev}} C_S/(T\,\mathrm{gap}_{\min}\, x_0)$ (Lemma 3.14), all

quantities in hypothesis (i) of Proposition 3.21 are $\le e^{C_0 \log^2(1/T)}$ for an absolute C_0 and T small. The design quantity $\mathrm{Des}(L)$ enters only the base T of the power $T^{5\mathrm{Des}+2}$, and $\mathrm{Des}(L) \le \log_2(1/T)$ turns it into a factor of the exponent of $\log^2$. Hypothesis (ii): by Lemma 4.14(d) and Proposition 4.16, $\epsilon \le C_f \mathrm{Des}\, c_{L+1} + C_f c_{L+1}^2/T$, $c_{\mathrm{fine}} \le c_{L+1}$ and $W_{\mathrm{lev}} + \overline{W}_{\mathrm{pull}} \le C_f \mathrm{Des}^C c_{L+1}$, all $\le e^{C_0 \log^2(1/T)} c_{L+1}$; and $c_{L+1} \le e^{-1/T}$ by Lemma 4.6(d). (W-b): $K_J 2^{G+3} |\Omega| c_{\mathrm{tiny}}/A_{\min} \le e^{2C_0 \log^2(1/T)} e^{-1/T} \le \frac{1}{8}$; the matrix Y is nonzero only at bank coordinates of levers that carry a mass of the pre-row, of size $\le 4\rho s_1 \max_t |b_t^\theta(j')| \le 4 s_1 A_0/T$, and $v_l(j') \ge 1/\mathrm{Des}(L)$, so $K_0 \|Y\|_\infty \le C_f \mathrm{Des}\, s_1/T \le \frac{1}{2}$. Hypothesis (iii) is the choice of N_w.

(iii) Proposition 3.21, for $T \le T_*$, which holds for $L \ge l_f$. □

Remark 4.20 (The cost of the non-explicit margin). In Theorem 3.26 the gaps of the switching carriers are $\ge c_f T^4/(L\,\mathrm{Des})$, and $K_w \le T^{-C'}$. Here the protected carriers have gaps of the order of the Puiseux margin, so K_w is only $\le e^{C_0 \log^2(1/T)}$, and $s_{\mathrm{late}} \ge e^{-35 C_0 \log^2(1/T)}$. This is still far above $e^{-1/(2T)}$, and ($\mathrm{W_{exp}}$) makes the fine weight $c_{L+1} \le e^{-1/T}$ negligible against it. The estimate is robust: an exponent of K_w depending on $\mathrm{Des}(L)$ or on $|\Omega| \le 2L$ would give $e^{C \log^3(1/T)}$, still dominated. No change of the design is needed for the Puiseux margin.

Lemma 4.21 (No common shift is needed)**.** *The scales of $\mathcal{S}$ in Lemma* 4.14 *and in Proposition* 4.16 *need not lie in one cube of the normalization of [II, Proposition 5.54(c)]. For $t \in \mathcal{S}$ let $(\Delta'(t), \gamma(t)) \in \Gamma^{\#}(\kappa, a)$ be the projection of the decomposition data ([II, Proposition 5.54(a),(b)], [II, Lemma 5.53]), let $\Delta''(t) := \Delta'(t)\varkappa'/\varkappa^{\#}$ be its true shift, and build the pieces as in Lemma* 3.24 *with $\mathcal{V}_{\mathrm{proj}} := \mathcal{L}(\Delta''(t), \gamma(t))$ on $E_c(w) \setminus F^{\#}$, without normalization increment. Then all conclusions of Lemma* 4.14 *and Proposition* 4.16 *hold for every class set, and $\sqrt{\Gamma_w^{\#}(b^-, \omega^-)} \le \sqrt{\Gamma_w^{\#}(B_-, \Theta_-)} + K_{\mathrm{data}} t$, where $(B_\pm, \Theta_\pm)$ is the decomposition of g at scale t.*

Proof. We go through every place where the construction could use that all scales carry the same shift. (1) Step B: every ownership rule uses only the signs $\sigma_m = \mathrm{sign}\, \Delta_m''$ and ϵ_l, which are constant on the class ([II, Lemma 5.53]; the sign of Δ_m' is constant on the class cone), and the quantities Y_k, which are computed from coordinates fixed before k and do not involve the data; the dominance of [II, Lemma 5.61] is stated for every shift and every component of the class. The only construction built for one shift value is the Schauder completion [II, Lemma 5.63], which is not used. (2) The true shift is defined per scale; the estimates of [II, Lemma 5.58] are per scale. (3) The kind-[3] verification in the proof of Lemma 4.14 holds for every point of the cone, and the data are points of the cone. (4) Conditions (U1)–(U7) are conditions on each piece separately; (U5) concerns the set of negative active blocks, which is fixed by the class; Theorem 3.15 uses the common set Ω and the maxima of the masses over the pieces, and Theorem 3.17 averages per-piece bounds. (5) Lemma 4.17 and Lemma 4.19 do not involve the data. (6) On contacts of $E_c(w)$ the row (X1) at the point $(\Delta''(t), \gamma(t))$ gives $z\mathcal{V}_{\mathrm{proj}} \ge 0$, as it did at the cube corner. For the last estimate: without the increment the $-$ side differs from the decomposition only by the projection and the zeroing of the inactive components, of size $\le K_{\mathrm{data}} t$, and $\sqrt{\Gamma_w}$ is a seminorm. □

Lemma 4.22 (Window count)**.** *Assume the standing hypotheses. Let* $B := 4\mathrm{Des}(L)/u(w)$*, and let* $J := |\Omega| + N$*; then* $J \le 2L$*.*

(a) *The number of activity classes of* w *is* $D_{\mathrm{cl}}(w) \le (J+1)3^J \le B^{L+1}$*.*

(b) *The class thresholds of [II, Lemma 5.53] are* $A_i = K_*^i$ *with* $K_* \le B^{C_*}$*,* C_* *absolute, and the data of Lemma* 4.14(c) *satisfy* $p^*(g - g_t) \le K_{\mathrm{data}} t$ *with* $K_{\mathrm{data}} := CK_*^{J+2}K \le B^{C_*(2L+2)+C_K+2}$*, where* $K = C_f(\mathrm{Des}/u)^{C_K}$ *is the constant of [II, Proposition 5.54].*

(c) *Some class* a *has a class set* $\mathcal{S}$ *with* $|\mathcal{S}| \ge n(w)/D_{\mathrm{cl}}(w)$*, and there are absolute constants* C_1*,* L_0 *such that for* $L \ge \max(l_f, L_0)$*, with* $c_{\mathrm{flat}}(w) \ge c_f u(w)$*,*

$$\frac{|\mathcal{S}|\, c_{\mathrm{flat}}(w)}{K_{\mathrm{data}}} \ge L2^{L^3} \quad \textit{and} \quad K_{\mathrm{data}} T_{\mathrm{hi}}(w) \le \frac{2^{-L^3}}{L}, \qquad \log_B \frac{D_{\mathrm{cl}}(w) K_{\mathrm{data}}}{c_{\mathrm{flat}}(w)} \le C_1(L{+}1).$$

Proof. (a) A class records, for each of the J components, active or inactive and, if active, a sign, together with an index $i \le J$ of the threshold scale ([II, Lemma 5.53]); this gives $\le (J{+}1)3^J$. As $B \ge 16$, $9^L \le B^L$, and $2L{+}1 \le B$. (b) The thresholds satisfy $K_* \ge \max(2\mathrm{Des}, 4C_f \mathrm{Des}\, JH(L))$ with the Hoffman bound $H(L) \le C(\mathrm{Des}/u)^C$, so one may take $K_* \le B^{C_*}$ for $L \ge l_f$ ($C_f J \le B$). Zeroing the inactive components of the data at scale t costs $\le JA_{i(t)}Kt \le JK_*^{J+1}Kt$, and the projection onto the cone moves the data by $\le \frac{1}{2}A_{i(t)+1}Kt$; this gives K_{data}. (The exponent of K_{data} grows linearly in L; a bound of the form $C_f(\mathrm{Des}/u)^C$ with absolute C is not available.) (c) The $n(w)$ dyadic scales of w are distributed over the $D_{\mathrm{cl}}(w)$ classes; by Lemma 4.21 a class set needs no further pigeonhole. With $c_{\mathrm{flat}} \ge c_f u \ge B^{-2}$ for $L \ge l_f$, (a) and (b) give the bound on $\log_B$ with $C_1 := 2C_* + C_K + 6$. By Definition 4.3(S6), $n(w) \ge L2^{L^3}Q(w) = L2^{L^3}B^{\omega(L)+L^2+20}$, so $|\mathcal{S}|c_{\mathrm{flat}}/K_{\mathrm{data}} \ge L2^{L^3}B^{L^2+20-C_1(L+1)} \ge L2^{L^3}$ for $L \ge L_0(C_1)$, and $K_{\mathrm{data}}T_{\mathrm{hi}}(w) \le B^{C_1(L+1)}2^{-L^3}/(LQ(w)) \le 2^{-L^3}/L$. □

Remark 4.23 (Two ways to count). The count of [II, Theorem 5.68] pigeonholes over classes and over the cubes of the normalization; the cube count $(A_{\max}/\varepsilon)^p$ carries the number p of extreme rays of the cone, which depends on the row and is bounded only by $P(L) = 2^{R_\Gamma(L)}$, and the data constant K_{data} has an exponent growing with L. So the claim that the window factor is $C_{f,N}(\mathrm{Des}/u)^C$ with an absolute exponent is not correct, and T_{fin} does not contain Part II's factor $\bar{D}(w)^3$. Two corrections work; they are only sketched here (*sketch*). (A) Pigeonhole first over the support of a basic representation (at most N extreme rays, by Carathéodory; at most $(P(L)+1)^N$ choices) and then over ε-cubes in the corresponding N coordinates; with $P(L) \le \mathrm{Des}(L)^{1/2}$ and $(2L+1)9^L \le B$ (as $\mathrm{Des} \ge 2^{6L^3}$), $D_{\mathrm{cls}}(w) \le B^{1+N(2C+3)}$, the increment $\sum_J (a_j^{\max} - a_j(t))r_j$ lies in the cone with ℓ_1-norm $\le N\varepsilon$, and the exponent $\log_B(D_{\mathrm{cls}}^2 K_{\mathrm{data}}/c_{\mathrm{flat}})$ is $O(L+N)$, absorbed by the term L^2 in the exponent of $Q(w)$ (or by $\omega(L) \ge 2^L - 1$, which holds because the design enumerates the orthant patterns of all subsets of $[1, L]$). (B) Add $\bar{D}_{\mathrm{fin}}(w)^3$, with $\bar{D}_{\mathrm{fin}}(w) := (2L+1)3^{2L}(\mathrm{Des}(L)/u(w))^{C_\Gamma(P(L)+2L+2)}$, to $Q(w)$ at main stages, as in Part II. We use neither: by Lemma 4.21 the cube is not needed in route V, and Lemma 4.22 counts classes only; correction (A) shows that the result does not depend on this simplification.

Theorem 4.24 (Every activity class is recovered)**.** *Let* T_{fin} *be taken with option* (a)*, or with option* (b)*. Let* $N \ge 1$*,* $f \in S_{p_N^*}$ *with finite base support,* $g \in \mathcal{C}(f)$

and $\rho \in (0,1)$. Suppose that for infinitely many main stages L there are a clean sub-window w of L and an activity class a of w (of any kind: without active shift, one-signed, or mixed) with a class set of at least $n(w)/D_{\mathrm{cl}}(w)$ scales, whose pattern satisfies $1 \notin \mathcal{N}_T(\kappa)$ if option (b) *is used. Then $(f, \rho g) \in \mathrm{cl}\,\mathrm{NA}((c_0, p_N), \ell_2^2)$.*

Proof. Enumerate the given pairs (L_j, w_j) with $L_j \geq \max(l_f, L_0)$, $L_j \to \infty$, and let $\mathcal{S}_j$ be the class sets. Let $f_j := f^{\#}$ be the companion of Lemma 4.14 at (L_j, w_j, a_j), with the pieces of Lemma 4.21. By Proposition 4.16 they form a window family at f_j with fixed A_0, $A_2 = 22$ and $\gamma_{B,j} := \gamma(w_j)$, and $c_{\mathrm{flat}}(j) \geq c_f u(w_j)$. We apply Theorem 3.17 with $T_j := T_{\mathrm{hi}}(w_j)$, $T_j' := T_{\mathrm{lo}}(w_j)$, $n_j' := |\mathcal{S}_j|$, $K_j := K_{\mathrm{data}}(w_j)$ and $s_{1,j} := e^{-1/(2T_j')}$.

- (E-c) Lemma 4.14(c) with Lemma 4.22(b).
- (E-d) Lemma 4.22(c).
- (E-f) Lemma 4.19: the window conditions hold, $s_{1,j}$ satisfies (VT′), (FINE), (LATE), and $K_{w,j}(K_{w,j}s_{1,j} + c_{\mathrm{fine},j} + \mathrm{tail}_{W,j}) \leq e^{-1/(3T_j')} \leq K_j(T_j')^2$.
- (E-e) $\varepsilon_j = p^*(f_j - f) \leq C_f T_j'^3 \log(1/T_j')$ (Lemma 4.14(f)) and $K_{w,j}(\Pi_j \log(e/\Pi_j) + s_{1,j}) \leq e^{-1/(3T_j')}$, so (E-e) holds with $\theta_j := C_f T_j' \log(1/T_j') + e^{-1/(3T_j')} T_j'^{-2}$. As $1/u(w_j) \leq n(w_j) \leq \log_2(1/T_j')$, $\theta_j / c_{\mathrm{flat}}(j)^2 \leq C_f \theta_j \log_2^2(1/T_j') \to 0$.

Lemma 4.17 guarantees that the statuses used at f_j (strict non-peaks of $\Omega^{\#}$ with $\mathrm{gap}' \geq \mathrm{gap}^{\#}/2$, the protected carriers of $\mathcal{N}_T$ with $\varrho \leq 1 - \mu/8$, coarse peaks) persist at the approximants. Theorem 3.17 gives $(f, \rho g) \in \mathrm{cl}\,\mathrm{NA}((c_0, p_N), \ell_2^2)$. □

4.5. **Master Theorem VI and the density criterion.** For $N \geq 2$ and the operator T_{fin}, stage 1 is the carrier $(k, m) = (1, 2)$; for a first row $f \in S_{p_N^*}$ we write $\varrho_1(f)$ for the relative position at f of the coordinate $k = 1$ of block 2 ([I, Lemma 1.7]).

Theorem 4.25 (Master Theorem VI)**.** *Let p_N be the norm on c_0 built with T_{fin} and the block set $I = \{1, \dots, N\}$.*

- (a) (Option (a): $\|T_{\mathrm{fin}}\| = c_1 < 1$, a Martín-type norm.) *For every $N \geq 1$, every $f \in S_{p_N^*}$ with finite base support lies in $\mathcal{R}(p_N)$: $(f, g) \in \mathrm{cl}\,\mathrm{NA}((c_0, p_N), \ell_2^2)$ for every $g \in \mathcal{C}(f)$.*
- (b) (Option (b): T_{fin} admissible, $c_1 = 1$.)
  - (b1) *For $N = 1$, every $f \in S_{p_1^*}$ with finite base support lies in $\mathcal{R}(p_1)$.*
  - (b2) *For $N \geq 2$, every $f \in S_{p_N^*}$ with finite base support and $\varrho_1(f) \neq 1$ lies in $\mathcal{R}(p_N)$.*

Proof. Fix N, f, $g \in \mathcal{C}(f)$ and $\rho \in (0,1)$, and let l_f be as in the standing hypotheses of Subsection 4.3. Under option (b) and $N \geq 2$ with $\varrho_1(f) \neq 1$, enlarge l_f so that $b(w) < |\varrho_1(f) - 1|$ for every sub-window w of every main stage $L \geq l_f$; this is possible because $b(w) \leq T_{\mathrm{lo}}(w)^4 \leq 2^{-4L^3}$.

Step 1 (a clean sub-window at every main stage). Let L be a main stage. The rate objects of level L, evaluated at f, are $\omega(L)$ real numbers: the relative positions and margins and the determinantal and shift objects of [II, Definitions 3.6 and 3.7], and the rate objects of every pattern and class of level L of Definition 2.13, evaluated at the ratio vector of f (QC2). The $M(L) = \omega(L) + 1$ bands $(b(w), u(w))$ of the sub-windows of L are pairwise disjoint (Lemma 4.10(D2)), and each number lies in at most one of them; so some band contains none of them, and the corresponding sub-window $w(L)$ is clean: every rate object is $\leq b(w(L))$ or $\geq u(w(L))$.

Step 2 (a class set). Let $L \geq l_f$. Every dyadic scale of $w(L)$ has exactly one activity class for the data of (g, ρ) ([II, Lemma 5.53]), of one of the three kinds (without active shift, one-signed, mixed); the pigeonhole over the $D_{\mathrm{cl}}(w(L))$ classes gives a class a with a class set of at least $n(w(L))/D_{\mathrm{cl}}(w(L))$ scales (Lemma 4.22). Under option (b): if $N = 1$, block 2 is absent from p_1, so stage 1 belongs to no pattern of p_1; if $N \geq 2$ and $\varrho_1(f) \neq 1$, stage 1 is not near-threshold at f at any clean sub-window of a main stage $L \geq l_f$, so $1 \notin \mathcal{N}_T(\kappa)$ for the pattern κ of $(f, w(L), a)$.

Step 3. The hypothesis of Theorem 4.24 holds at every main stage $L \geq l_f$, so $(f, \rho g) \in \mathrm{cl}\,\mathrm{NA}((c_0, p_N), \ell_2^2)$. As $\rho < 1$ is arbitrary and $(f, g) = \lim_{\rho \to 1}(f, \rho g)$, $(f, g) \in \mathrm{cl}\,\mathrm{NA}$; as $g \in \mathcal{C}(f)$ is arbitrary, $f \in \mathcal{R}(p_N)$. The operator T_{fin} does not depend on N (Lemma 4.6(c)), and N was arbitrary. □

The order of choices in the proof is: T_{fin} (independent of N and of every row), then N, f, (g, ρ), l_f, a main stage $L \geq l_f$, the clean sub-window $w(L)$, the class a (read off the decomposition data of g at f), Step A, Step B, the data, N_w, s_1, the pre-row, the levers and the approximant. The weights are fixed before f; this is legitimate because the coincidence sets B_κ contain no datum of a row (Lemma 4.5(b)).

Theorem 4.26 (Density criterion)**.** *Let T_{fin} be taken with option* (a)*, or with option* (b) *and $N = 1$. For every $N \geq 1$ (resp. for $N = 1$) the following are equivalent:*

(i) $\mathrm{NA}((c_0, p_N), \ell_2^2)$ *is norm dense in* $\mathcal{L}((c_0, p_N), \ell_2^2)$*;*

(ii) $(\mathrm{LSC\text{-}trunc})_N$*: for every $f \in S_{p_N^*}$, $g \in \mathcal{C}(f)$, $\rho < 1$ and $\varepsilon > 0$ there is $f' \in S_{p_N^*}$ with finite base support such that $p_N^*(f' - f) < \varepsilon$ and* $\mathrm{dist}_{p_N^*}(\rho g, \mathcal{C}(f')) < \varepsilon$.

Proof. For $h_1, h_2 \in (c_0, p_N)^*$ the operator $(h_1, h_2) : x \mapsto (h_1(x), h_2(x))$ into ℓ_2^2 has norm $\leq p_N^*(h_1) + p_N^*(h_2)$. (ii)⇒(i): let $f \in S_{p_N^*}$, $g \in \mathcal{C}(f)$, $\rho < 1$, $\varepsilon > 0$, and let f' be given by (ii), with $g' \in \mathcal{C}(f')$ and $p_N^*(g' - \rho g) < \varepsilon$ ($\mathcal{C}(f')$ is closed). By Theorem 4.25, $f' \in \mathcal{R}(p_N)$, so $(f', g') \in \mathrm{cl}\,\mathrm{NA}$, and $\|(f', g') - (f, \rho g)\| < 2\varepsilon$. Hence $(f, \rho g) \in \mathrm{cl}\,\mathrm{NA}$ for every $\rho < 1$, so $(f, g) \in \mathrm{cl}\,\mathrm{NA}$ and $f \in \mathcal{R}(p_N)$; thus $\mathcal{R}(p_N) = S_{p_N^*}$, which is (i) by [I, Proposition 1.13] and [I, Definition 1.15]. (i)⇒(ii): by [I, Lemma 1.16], density gives norm-attaining first rows $f_n \to f$ with $\rho g \in \mathrm{Li}_n\,\mathcal{C}(f_n)$; a norm-attaining $f_n \in S_{p_N^*}$ has finite base support ([I, Propositions 1.10(c) and 1.12]). The results of [I] used here hold for T_{fin} by Lemma 4.9. □

Corollary 4.27 (Martín-type norm p)**.** *Let T_{fin} be taken with option* (a)*. If* $(\mathrm{LSC\text{-}trunc})_N$ *holds for infinitely many N, then* $\mathrm{NA}((c_0, p), \ell_2^2)$ *is dense in* $\mathcal{L}((c_0, p), \ell_2^2)$ *for the Martín-type norm p ($I = \mathbb{N}$) built with T_{fin}.*

Proof. By Theorem 4.26, $\mathrm{NA}((c_0, p_N), \ell_2^2)$ is dense for infinitely many N; the norms p_N and p are built with the same operator, since T_{fin} does not depend on N. By [I, Remark 1.4] (Martín tails, [I, Lemma 1.5], whose proof uses only $|R_m x|_m \leq m 2^{-m} q(x)$, valid for $\|T\| \leq 1$), $\mathrm{NA}((c_0, p), \ell_2^2)$ is dense. □

Remark 4.28 (What remains open). (a) *Infinite base support.* $(\mathrm{LSC\text{-}trunc})_N$ is not proved here, for any N; by Theorem 4.26 it is equivalent to the density of $\mathrm{NA}((c_0, p_N), \ell_2^2)$ for the norms built with T_{fin}. So the density of $\mathrm{NA}((c_0, p_N), \ell_2^2)$ and of $\mathrm{NA}((c_0, p), \ell_2^2)$ remains open. (b) *Option (b), $N \geq 2$, $\varrho_1(f) = 1$.* At such rows stage 1 may be an active near-threshold carrier of a block of kind (T) at clean sub-windows of infinitely many main stages, and its weight $\Phi_1 = 2^{-3}$ is rational,

so Lemma 2.22 does not apply to it. Whether 2^{-3} can be a critical value of the floor function ψ_1 on a stratum of Z_κ, identically in the transcendental parameters, is not known; normalizing the operator of option (a) does not help (Remark 4.8). (c) *Other operators.* Theorem 4.25 is proved for the designed operator T_{fin} only, not for an arbitrary admissible operator of Martín's Lemma B. No counterexample to density is claimed, and none of the verifications produced one.

Remark 4.29 (Verification). The results of Sections 2 and 3 on which Theorem 4.25 rests were checked by one referee each and then re-derived in a second, independent verification organized in four clusters (the route-V companion; the generic-weight chain of Section 2; the engineered approximant and the averaging of Section 3; the transfer to T_{fin} and the window bookkeeping); the combination in this section was checked by two referees and re-derived in the same verification. The corrections introduced in the second verification (the common pull window, the transfer interface of Lemma 4.10 and Proposition 4.11, the s-free coefficients, the gap criterion of Lemma 3.4(a), the bounds of Lemmas 3.10 and 3.14(g)) were each made and checked once; only the corrected window count was checked twice. Proposition 3.7, Lemma 3.23, Corollary 3.27 and Lemma 4.21 were checked by one referee. The following corrections were found and are incorporated.

(1) *Companion.* Carriers pushed inward in blocks of kind (U) are of kind $[1']$ (Lemma 4.14(c)); the companion is built in the order class, Step A, Step B; every non-active nearly neutral coarse strict non-peak of a (T)-block is held exactly (Remark 2.33); all pulls of the joint fixed point lie in one common window, without which a held carrier can require a negative bank (Lemma 2.29, Remark 2.30); the Puiseux point is chosen with $F_\Phi \le 1 - \frac{3}{4}\mu$; the cost carries Des^3; robust rate objects are only $\ge u/4$ after the rescaling of (U)-blocks; the inequality "$Mb \le \mathrm{gap}^{\#}/4$" is replaced by $\mathrm{gap}(f) \le 1.5\,\mathrm{gap}^{\#}$ (proof of Lemma 4.14).

(2) *Base row.* At the supports created by the companion the side condition (U3) holds up to antisymmetric fine residues, which are tolerated violations; Step A uses a bounded number of coordinates of each S_l, and the pull coordinates of the levers are taken among the unused ones in a window of ratio $2^{6G(L)}$ (Proposition 4.16).

(3) *Generic weights.* The Lipschitz constant in $b(w)$ and in (QC4) dominates the floor-status function and the floor statuses of all active coarse carriers; the level data use s-free aggregated coefficients (no forward reference to c_{L+1}); normalizations of the Łojasiewicz constants are fixed; the α-mass identity (Lemma 2.3) supplies the natural peak and $\sigma_{\mathrm{rob}} > 0$.

(4) *Approximant.* The neighbourhood lemma includes the margin of the natural peak and $|M'' - M|$ (Lemma 3.6); the bound on $|F(0)|$ in Lemma 3.10 contains the second-order effect of the fixed pulls and banks; the gap criterion is $|C' - C| \le C\,\mathrm{gap}/(2(1 - \mathrm{gap}))$, not $|C' - C| \le \mathrm{gap}_{\min}/4$ (Lemma 3.4(a)); the coefficients c_i are bounded through the Bregman identity (Lemma 3.14(g)); the averaging theorem allows a window-dependent γ_B (Theorem 3.17). The exponents of K_w were confirmed to be absolute.

(5) *Transfer and bookkeeping.* The patterns range over the blocks of stages $\le l$, and l_f exceeds the stage after which every block $\le N$ has a carrier; the inspection claim that every result of Part II survives for T_{fin} is replaced by Lemma 4.10 and Proposition 4.11 (with $s_{\max} \to \infty$ proved without

fresh-dominated stages, the bank efficiency proved for the base μ, and the complete list of lower-bound uses of weights, including the least absorber entry); the window count is corrected (Lemma 4.22, Remark 4.23); Part I is used under option (a) after reading the proofs, not only by locating the equality clause of (T-a) (Lemma 4.9); the route with zero-value absorbers does not transfer to T_{fin} and is not used (Remark 4.12).

The proof given here uses Lemma 4.21 (in Lemma 4.22(c) and in the proof of Theorem 4.24). Without it, Theorem 4.24 and Theorem 4.25 hold with sets of scales of one class and one cube, the data built at the cube corner, and the count of Remark 4.23(A), which is only sketched here; that version was verified twice in the sources (the count by two independent checks, the order class–Step A–cube–Step B by two independent checks), so Theorem 4.25 does not depend on Lemma 4.21. A further round of three referees read the present text; their corrections (among them the moduli of rate objects, the reduced matrix of the cone, the constant C_{**}, the a priori pull weight $\overline{W}_{\mathrm{pull}}$, the pull range $[r_{\mathrm{pull}}, r_0]$, the realized ratios in Lemma 4.14(a), and the chain in (W6) of Lemma 4.10) are incorporated and were checked once. Finite numerical models were used as sanity checks of several identities (the clamp equation, the α-mass identity, the bank equations of the tuning lemma); they are evidence only.

5. Infinite base support: truncation and transport

By [II, Proposition 6.48], density of $\mathrm{NA}((c_0, p_N), \ell_2^2)$ is equivalent to Lemma Z at the rows with finite base support together with $(\mathrm{LSC\text{-}trunc})_N$: for every $f \in S_{p_N^*}$, $g \in \mathcal{C}(f)$, $\rho \in (0,1)$ and $\varepsilon > 0$ there is a first row f' with finite base support, $p_N^*(f' - f) < \varepsilon$ and $\mathrm{dist}(\rho g, \mathcal{C}(f')) < \varepsilon$. For the design of Section 4.1 with option (a) the first half is Theorem 4.25; so for that design density for p_N is equivalent to $(\mathrm{LSC\text{-}trunc})_N$. From Subsection 5.3 on, the operator is $T_{\mathrm{fin}}^{\infty}$ (Definition 5.42); for it the first half of the equivalence is expected but not verified (Remark 5.57). This section treats (LSC-trunc). It proves four things.

(1) Truncating the base support is asymptotically free at finitely many scales fixed beforehand (its cost $2\Lambda_M + \varepsilon''_M$ tends to 0; lowering transfer, Lemma 5.2); inside the violation-tolerant engineered approximant of Section 3 the truncated support becomes a set of contacts whose two-sided use is a tolerated violation. Consequently (LSC-trunc) at (f, g, ρ) follows from window families with *side cushion domination* and exact d-consistency at base rows of arbitrary support (Theorems 5.10 and 5.12).

(2) For the design, at deep-raised rows the route-V data of Section 4.3 are cushion dominated, on the whole support of the companion, with the absolute constant $C_+ = 5$ (Proposition 5.19).

(3) After a cheap preparation of the base row, the d-consistency system is uniformly solvable at every row (Theorem 5.31), while Step A of the companion construction is solvable whenever a cone condition holds; the obstructions to the cone condition have a rigid structure (Lemma 5.37, Theorem 5.39, Corollary 5.40).

(4) The transport theorem (Theorem 5.51): $(\mathrm{LSC\text{-}trunc})_N$ holds at every (f, g, ρ) satisfying a joint push condition (H1*) and the cone condition (H2) at infinitely many main stages. Rows not covered satisfy, at almost

all main stages, one of the conditions (W*), (A*/U*) of Subsection 5.9; these are necessary conditions for not being covered, nothing more.

(LSC-trunc)$_N$ for every f, and hence density of $\mathrm{NA}((c_0,p_N),\ell_2^2)$ and of $\mathrm{NA}((c_0,p),\ell_2^2)$, is not proved here.

5.1. **Contact truncations and lowering transfer.**

Conventions. Throughout this section I is finite, $p=p_N$, and the base is diagonal, $Uk_s=\mu^\star_s e_s$ with $\mu^\star_s>0$ *non-increasing* and (k_s) orthonormal. A first row $f\in S_{p^*}$ has forced data $(\xi,q_0,a,w,F,z,\hat z,e,\nu)$ [I, Proposition 1.9]: $f=a+L^*w$, $q^*(a)=\|a\|_1+\nu=1$, $\nu=\|U^*a\|_H$, $e=U^*a/\nu$, $F=\operatorname{supp}a$ *arbitrary* (finite or infinite), $z=\operatorname{sign}a$ on F, $\|z\|_\infty\le 1$, $\hat z=z+Ue$. For $A\in\ell_1\setminus\{0\}$ and an admissible z ($z=\operatorname{sign}A$ on $\operatorname{supp}A$) we write $f[A,z]$ for the first row with base vector $A/q^*(A)$ and sign vector z, as in Section 3. Put $s_j:=\operatorname{sign}a_j$ for $j\in F$, $\alpha(M):=\sum_{j\in F,\,j>M}|a_j|$, $a_{>M}:=a1_{(M,\infty)}$, $a_{\le M}:=a1_{[1,M]}$ and $s(t):=(1+t^2)^{1/2}$.

Lemma 5.1 (Contact truncations)**.** *Let $f\in S_{p^*}$ and M with $a_{\le M}\neq 0$. Put $c_M:=q^*(a_{\le M})$, $a^M:=a_{\le M}/c_M$ and $f^M:=f[a_{\le M},z]$, with forced data $(a^M,z,e^M,\hat z^M,w^M,\dots)$. The coordinates $j\in F$, $j>M$, are contacts of f^M with $z_j=s_j$, and:*

(a) $|1-c_M|\le\alpha(M)+\|U^*a_{>M}\|$;
(b) $\|e^M-e\|\le 2\|U^*a_{>M}\|/\nu$ *and* $\hat z^M-\hat z=U(e^M-e)$;
(c) $f^M\to f$ *in norm as $M\to\infty$, and all forced data converge as in [I, Proposition 1.11];*
(d) $\|U^*a_{>M}\|\le\mu^\star_{M+1}\alpha(M)$.

Proof. z is admissible for $a_{\le M}$, and $z_j=s_j$ with $a^M_j=0$ for $j\in F$, $j>M$. (a) $1=\|a_{\le M}\|_1+\alpha(M)+\|U^*a\|$ and $c_M=\|a_{\le M}\|_1+\|U^*a_{\le M}\|$; the two Hilbert norms differ by at most $\|U^*a_{>M}\|$. (b) $\|x/\|x\|-y/\|y\|\|\le 2\|x-y\|/\|y\|$ with $x=U^*a_{\le M}$, $y=U^*a$; since z is unchanged, $\hat z^M-\hat z=U(e^M-e)$. (c) By (a), $a^M\to a$ in ℓ_1. The proof of the implication (ii)⇒(i) of [I, Proposition 1.12] uses only ℓ_1-convergence of the base vectors and coordinatewise convergence of the sign vectors, not norm attainment; with the constant sign vector z it gives $f^M\to f$, and [I, Proposition 1.11] gives the convergence of the forced data. (d) $\|U^*a_{>M}\|^2=\sum_{j>M}(\mu^\star_j)^2a_j^2\le(\mu^\star_{M+1})^2\alpha(M)^2$ because $\mu^\star$ is non-increasing. □

Lemma 5.2 (Lowering transfer)**.** *Let $f\in S_{p^*}$, M as in Lemma 5.1, $h\in X^*$, $r\neq 0$, and let $f+rh=A+L^*W$ with $q^*(A)\le\pi$ and $\|W\|_{V^*}\le\pi$. Put $B:=(A-a)/r$ and*

$$\Lambda_M(r,B):=\sum_{j\in F,\ j>M}\min\big(|a_j|,(-s_jrB_j)_+\big)\qquad\Big(\le\min(\alpha(M),|r|\,\|B1_{(M,\infty)}\|_1)\Big).$$

(a) $q^*(a^M+rB)\le q^*(A)+2\Lambda_M(r,B)+2\|U^*a_{>M}\|$, *and*

$$p^*(f^M+rh)\le\pi+2\Lambda_M(r,B)+2\|U^*a_{>M}\|+p^*\big(L^*(w^M-w)\big).$$

(b) (Partial lowering.) *Let Δa be supported in F with $s_j\Delta a_j\le 0$ and $|\Delta a_j|\le|a_j|$, let $f^{\mathrm{low}}:=f[a+\Delta a,z]$ with normer w^{low}, and put $y_j:=(-s_jrB_j)_+$, $\alpha'_j:=|a_j+\Delta a_j|$. Then*

$$p^*(f^{\mathrm{low}}+rh)\le\pi+2\sum_j\min\big(|\Delta a_j|,(y_j-\alpha'_j)_+\big)+2\|U^*\Delta a\|+p^*\big(L^*(w^{\mathrm{low}}-w)\big).$$

Proof. (a) $f^M + rh = A' + L^*W + L^*(w^M - w)$ with $A' := A + a^M - a = a^M + rB$. By [I, Lemma 1.6] and the triangle inequality it suffices to show $q^*(A') \le q^*(A) + 2\Lambda_M + 2\|U^* a_{>M}\|$. Write $A' = (a_{\le M} + rB) + (1/c_M - 1)a_{\le M}$; the second term has q^*-norm $|1 - c_M|$. *ℓ_1-part.* At $j \le M$ and at $j \notin F$ the coordinates of $a_{\le M} + rB$ and of $A = a + rB$ agree. For $j \in F$, $j > M$, put $\alpha := |a_j|$, $x := rB_j$, $y := (-s_j x)_+$; checking the cases $s_j x \ge 0$, $0 < y \le \alpha$ and $y > \alpha$ gives

$$|s_j\alpha + x| - |x| = \alpha - 2\min(\alpha, y),$$

i.e. $|x| = |A_j| - |a_j| + 2\min(|a_j|, y)$. Summing, $\|a_{\le M} + rB\|_1 = \|A\|_1 - \alpha(M) + 2\Lambda_M$. *Hilbert part.* $\|U^*(a_{\le M} + rB)\| = \|U^*(A - a_{>M})\| \le \|U^* A\| + \|U^* a_{>M}\|$. With Lemma 5.1(a), $q^*(A') \le q^*(A) - \alpha(M) + 2\Lambda_M + \|U^* a_{>M}\| + \alpha(M) + \|U^* a_{>M}\|$.

(b) Put $c := q^*(a + \Delta a)$ and $A' := (a + \Delta a)/c + rB$. Since the lowering keeps signs, $\|a + \Delta a\|_1 = \|a\|_1 - \|\Delta a\|_1$, so $|1 - c| \le \|\Delta a\|_1 + \|U^* \Delta a\|$. At $j \in \operatorname{supp} \Delta a$, with $x := rB_j$, $\alpha := |a_j|$ and $\alpha' = \alpha - |\Delta a_j|$, the identity above at α and at α' gives $|s_j\alpha' + x| - |s_j\alpha + x| = -|\Delta a_j| + 2\min(|\Delta a_j|, (y_j - \alpha'_j)_+)$. Hence $\|a + \Delta a + rB\|_1 = \|A\|_1 - \|\Delta a\|_1 + 2\sum_j \min(|\Delta a_j|, (y_j - \alpha'_j)_+)$, $\|U^*(a + \Delta a + rB)\| \le \|U^* A\| + \|U^* \Delta a\|$, and the conclusion follows as in (a). □

The ℓ_1-mass $\alpha(M)$ of the truncation cancels exactly, as the raise mass does in the raise transfer [II, Lemma 2.15]. What remains is the *cushion loss* Λ_M (anti-sign usage of the decomposition on the far support, capped at the cushion $|a_j|$), the Hilbert footprint $2\|U^* a_{>M}\| \le 2\mu^\star_{M+1}\alpha(M)$ and the normer change, which is bounded by [II, Lemma 2.12(iv)] in terms of $\hat{z}^M - \hat{z} = U(e^M - e)$. In (b) only usage beyond the *new* cushion α'_j is lost.

Remark 5.3 (The bound is sharp and scale limited). (a) The factor 2 in front of Λ_M in the base inequality of (a) cannot be lowered. Let $j \in F$, $j > M$, $0 < r \le |a_j|$, $B := -s_j e_j^*$ and $A := a + rB$. Then $\Lambda_M(r, B) = r$, $q^*(a + rB) \le \|a\|_1 - r + \nu + \mu_j^\star r = 1 - r + \mu_j^\star r$ and $q^*(a^M + rB) \ge \|a^M\|_1 + r + \|U^* a^M\| - \mu_j^\star r = 1 + r - \mu_j^\star r$, so $q^*(a^M + rB) - q^*(a + rB) \ge 2(1 - \mu_j^\star)\Lambda_M(r, B)$.

(b) The cushion loss is not uniformly small as $r \to 0$. Fix M and $r_0 > 0$ and put $h := -a_{>M}/r_0$. For $0 < r \le r_0$, $f + rh = (a + rh) + L^* w$ with $q^*(a + rh) \le \|a\|_1 - (r/r_0)\alpha(M) + \nu + (r/r_0)\|U^* a_{>M}\| \le 1$ (as $\|U^* a_{>M}\| \le \alpha(M)$), so $\pi = 1$, while $\Lambda_M(r, h) = (r/r_0)\alpha(M)$ is linear in r. At f^M the coordinates $j > M$ of F are contacts with $z_j = s_j$, and the base excess of rh there, $\sum_{j>M}(|rh_j| - s_j rh_j)$ in the sense of [I, Lemma 7.1(b)], equals $2r\alpha(M)/r_0$. Hence truncation is free only at finitely many scales fixed before M (Corollary 5.4), and in the engineered approximant below the truncation point is chosen after the smallest scale s_1 of the construction.

Corollary 5.4 (Truncation at finitely many scales)**.** *Let $f \in S_{p^*}$, let $\mathcal{S}$ be a finite set of nonzero reals, and for $r \in \mathcal{S}$ let $f + rh_r = (a + rB_r) + L^* W_r$ with $\max(q^*(a + rB_r), \|W_r\|_{V^*}) \le \pi_r$. For every $\eta > 0$ there is M_0 such that for all $M \ge M_0$ and $r \in \mathcal{S}$*

$$p^*(f^M + rh_r) \le \pi_r + \eta r^2 + \varepsilon''_M, \qquad \varepsilon''_M := 2\|U^* a_{>M}\| + p^*\big(L^*(w^M - w)\big) \xrightarrow[M\to\infty]{} 0.$$

Proof. $\Lambda_M(r, B_r) \le |r|\,\|B_r 1_{(M,\infty)}\|_1 \to 0$ for each of the finitely many r, since $B_r \in \ell_1$; apply Lemma 5.2(a). $\|U^* a_{>M}\| \to 0$, and $p^*(L^*(w^M - w)) \to 0$ by Lemma 5.1(c) and [I, Proposition 1.11]. □

Remark 5.5 (Heuristic: why exact local data are needed at the approximant). This remark is heuristic and is not used. For f' near f with $p^*(f'-f) = d > 0$, scales $|r| \gg (d/(1-\rho^2))^{1/2}$ are covered by the slack of ρg, but for $|r| \ll d^{1/2}$ the inequality $p^*(f'+rg') \le s(r)$ is a statement about the second-order structure of p^* at f' alone. The normer changes for every $f' \ne f$ (Lemma 5.1(b)), so statuses and levels of deep carriers at f' differ from those at f at arbitrarily small scales, and the mate must carry exact local data at f'. This is why the truncation is carried out inside an engineered approximant whose local data are controlled, rather than by approximating f first.

5.2. **The engineered approximant at arbitrary base support.** We now show that finite base support of the approximant costs nothing in the violation-tolerant route of Section 3. Fix $f \in S_{p^*}$ (F arbitrary), $\rho \in (0,1)$, η_0, $A_0, A_2 \ge 1$, $\gamma_B \in (0,1]$ and δ as in Section 3; f-constants are as there.

Lemma 5.6 (Uniform neighbourhoods at arbitrary support)**.** *Let $f \in S_{p^*}$ have arbitrary base support, and fix γ_m, $k_m^\natural$ and transfer data as in Lemma* 3.6*. There is $r_f(\eta_1) > 0$ such that every $f'' \in S_{p^*}$ with $p^*(f''-f) \le r_f(\eta_1)$ satisfies* (a) *of Lemma* 3.6 *without the clause "$F \subseteq \operatorname{supp} a''$ with $|a_j''| \ge a_{\min}/2$ on F", and* (b) *of Lemma* 3.6*.*

Proof. The proof of Lemma 3.6 uses [I, Proposition 1.11] (valid for every F), the continuity of the strict threshold inequality of [I, Lemma 1.7] and [I, Lemma 7.5], whose proof uses only [I, Proposition 1.11], the threshold lemma and dominated convergence in the blocks. The removed clause is the only place where F enters. □

Definition 5.7 (Window families at arbitrary support)**.** Let $C_+ \ge 1$. A *base row* is $f_0 \in S_{p^*}$ with $p^*(f_0 - f) \le r_f(\eta_1)/2$ (Lemma 5.6) and forced data $(a_0, z_0, e_0, \nu_0, \hat{z}_0, w^0, \dots)$, $F_0 := \operatorname{supp} a_0$ arbitrary, contact set K_0, $s_j := \operatorname{sign} a_0(j)$ on F_0. A *window family with cushion constant C_+* at f_0 consists of pieces $(g_i; b_i^\pm, \omega_i^\pm)$, $i = 1, \dots, n$, at scales $t_i \in [T, t_1]$ (Definition 3.1), supported in a finite set Ω of strict non-peaks of f_0, satisfying (U1), (U2), (U5), (U7) of Definition 3.12 and:

(U3^{inf}) *side cushion domination*: for every i and $j \in F_0$ there is $r_i(j) \ge 0$ with

$$(s_j b_i^+(j))_- \le \frac{C_+|a_0(j)|}{t_i} + r_i(j), \qquad (s_j b_i^-(j))_+ \le \frac{C_+|a_0(j)|}{t_i} + r_i(j);$$

(U4^{inf}) $b_i^\theta(j) = 0$ at every free coordinate $j \notin F_0 \cup K_0$ with $\operatorname{viol}_i(j) > 0$, and the total violation $\epsilon_i := \sum_{j \notin F_0} \operatorname{viol}_i(j) + 2\sum_{j \in F_0} r_i(j)$ plus the masses $|b_i^\pm(j)|$ at lever coordinates of type (PRα) below is at most ϵ;

(U6^{inf}) a *lever system* (Definition 5.9) is given.

Nothing is required of the sign-direction usage on F_0 or of b^θ on F_0. (Contact-like data on F_0, and data with $t_i|b_i^\pm(j)| \le |a_0(j)|$, satisfy (U3^{inf}) with $r_i(j) = 0$ and $C_+ = 1$; antisymmetric residues as in (U3) of Definition 3.12 are absorbed in $r_i(j)$.) Alternatively (U5) may be replaced by (U5′) of Lemma 3.23.

The pre-row and the approximant are defined first; the lever system acts on the pre-row.

Definition 5.8 (The truncated d-consistent approximant)**.** Given a window family at f_0 with lever coordinates Lev (Definition 5.9), choose in this order:

(i) $s_1 \in (0,1]$;
(ii) $N_w \geq \max \mathrm{Lev}$ such that

$$\mathrm{tail}_W := \max_i \big(\|b_i^{\pm} 1_{(N_w,\infty)}\|_1 + \|v_i 1_{(N_w,\infty)}\|_1\big) \leq \min\Big(T^3, \frac{c_{\mathrm{late}}\delta T}{A_0}, \frac{\delta s_1}{64\rho}\Big),$$

$$\|U^* a_0 1_{(N_w,\infty)}\| \leq s_1 T^4, \qquad \alpha_0(N_w) := \sum_{j\in F_0,\, j>N_w} |a_0(j)| \leq \theta_{\mathrm{tr}},$$

for a prescribed truncation tolerance θ_{tr} with

$$0 < \theta_{\mathrm{tr}} \leq \min\Big(s_1 T^4,\ \frac{r_f(\eta_1)}{8(1+\|U\|)^2}\Big);$$

(iii) the *mass set* W of contacts $j \in [1,N_w]\setminus F_0$ of f_0 with $\max_i |b_i^\theta(j)| > 0$, with masses $\mathsf{m}_j := 4\rho s_1 \max_i |b_i^\theta(j)|$ of sign $z_0(j)$, and the *raise set* $W_F := \{j \in F_0 \cap [1,N_w] : |a_0(j)| < 32\rho s_1 \max_i |b_i^\theta(j)|\}$, where $a_0(j)$ is raised, with its sign, to modulus $32\rho s_1 \max_i |b_i^\theta(j)|$;
(iv) the pre-row $f_1 := f[A_1, z_1]$, where A_1 is $a_0 1_{[1,N_w]}$ with the raises of W_F and the masses of W, and $z_1 := z_0 1_{[1,N'']}$ (N'' from (vi)); put $\Pi_{\mathrm{tr}} := 2\|U^* a_0 1_{(N_w,\infty)}\|/\nu_0$, $\Pi_X := 4\|U^*(A_1 - a_0 1_{[1,N_w]})\|/\nu_0$ (neither depends on N'');
(v) the *lever radius* $r := 4K_{\mathrm{lev}}(1 + 1/A_{\min})(\Pi_X + \Pi_{\mathrm{tr}} + 2T^4 s_1)$;
(vi) $N'' > N_w$ with the conditions (iv) of Definition 3.13;
(vii) P^* by the lever system, $f' := f[A(P^*), z(P^*)]$ with data $a', \hat{x}', c', w'_m, d'_m, \nu'$;
(viii) $g_i'' := b_i^\theta 1_{[1,N_w]} + \sum_m R^*_m(\omega^\theta_{i,m} - d'_m(\omega^\theta_{i,m})w'_m)$, $c_i := g_i''(\hat{x}')$, $g_i' := \rho(g_i'' - c_i a')$.

The perturbation size is $\Pi := \Pi_X + \Pi_{\mathrm{tr}} + r + \mathfrak{t}(N'')^{1/2}$. In the decompositions of the proof we use $\beta^\theta := b^\theta 1_{[1,N_w]}$ and $\beta^\pm := b^\pm 1_{[1,N_w]} \pm \frac{1}{2} v 1_{(N_w,\infty)}$, so that $\beta^+ - \beta^- = v$ and $\frac{1}{2}(\beta^+ + \beta^-) = \beta^\theta$.

Definition 5.9 (Lever systems)**.** Let f_0 be a base row and Ω a finite set of strict non-peaks of f_0. A *lever system* consists of a finite set Lev of coordinates, a finite index set Λ_P, numbers $r_0 \in (0,1]$ and $K_{\mathrm{lev}}, C_2, C_{\mathrm{lev}} \geq 1$, and, for every pre-row $f_1 = f[A_1, z_1]$ of Definition 5.8, a continuous family $P \mapsto (A(P), z(P))$, $P \in [-r_0, r_0]^{\Lambda_P}$, of admissible pairs with $(A(0), z(0)) = (A_1, z_1)$ and $A(P) - A_1$, $z(P) - z_1$ supported in Lev, such that:

(DC) if $r \leq r_0$ and $K_{\mathrm{lev}}|F(0)|_\infty \leq r/2$, where F is the map of Lemma 3.10 for this family, then F has a zero P^* with $|P^*|_\infty \leq K_{\mathrm{lev}} r$; at $f(P^*)$ every $l \in \Omega$ is a strict non-peak, the peaks of f_0 whose margin in value units exceeds $C_{\mathrm{lev}}(K_{\mathrm{lev}} r + \delta_1)$ plus the change of the threshold remain peaks with the same sign, the lever masses satisfy $\|X_{\mathrm{lev}}\| \leq C_2 r$, a carrier meeting no lever coordinate moves by at most $|u_k(\hat{x}_1) - u_k(\hat{z}_0)| + \|e(P^*) - e_1\|$ and any other by at most that plus $C_{\mathrm{lev}} K_{\mathrm{lev}} r$;
(PR) every $j \in \mathrm{Lev}$ is of one of two types: (PRα) $b_i^\theta(j) = 0$ for all i (the masses $|b_i^\pm(j)|$ are counted in ϵ); (PRβ) $j \in F_0$, and for $|P|_\infty \leq K_{\mathrm{lev}} r$ with $r \leq r_0$, $A(P)(j)$ has the sign s_j and $|A(P)(j)| \geq |A_1(j)|/4$.

The lever constants are $\Lambda_{\mathrm{lev}} := \max(K_{\mathrm{lev}}, C_2, C_{\mathrm{lev}}, 1/r_0, 1/A_{\min}, 1/C_{\min})$; W_{lev} is the total weight $\sum \lambda_k$ of the *fine* carriers k meeting Lev, and $W_{\mathrm{pull}} := 0$. Coarse carriers outside Ω that meet Lev (peaks whose signature sets contain lever coordinates) are not counted in W_{lev}; by (DC) they move by at most $C_{\mathrm{lev}} K_{\mathrm{lev}} r$ more than the other carriers.

The levers of Lemma 3.10 with (SEP) form a lever system when no TU lever is used; the system used below is that of Theorem 5.31.

Theorem 5.10 (Engineered bound at arbitrary base support)**.** *Let t_1, c^f_{flat}, $K_\sharp$, η_1, c_{late} be the f-constants of Theorem* 3.15*, with condition* (c1) *of its proof deleted. For $C_+ \geq 1$ put*

$$c_{\mathrm{flat}} := \min\Big(c^f_{\mathrm{flat}},\ \frac{\gamma_B}{8\rho A_2},\ \frac{1}{32\rho C_+}\Big).$$

Let f_0 be a base row of arbitrary support and let a window family with cushion constant C_+ and scales in $[T, t_1]$ be given at f_0, with window constant K_w of Lemma 3.14 *computed with the lever constants of* (U6^{inf})*. Assume* (W-a) *of Theorem* 3.15 *and let s_1 satisfy* (VT′)*,* (FINE) *and* (LATE) *of Theorem* 3.15*, and $K_w s_1 \leq r_0$. Then for every choice of N_w, N'' as in Definition* 5.8 *the approximant f' is norm attaining, has finite base support, is d-consistent with f_0 on Ω, $\Pi \leq K_w s_1$,*

$$p^*(f' - f_0) \leq K_w\big(\Pi \log(e/\Pi) + s_1\big) + 2(1 + \|U\|)^2 \alpha_0(N_w),$$

and for every piece i

$$p^*(f' + \tau g'_i) \leq 1 + \frac{\tau^2}{2}(1 - \delta) \qquad (0 < |\tau| \leq c_{\mathrm{flat}} t_i),$$
$$p^*(g'_i - \rho g_i) \leq K_w\big(K_w s_1 + c_{\mathrm{fine}} + \mathrm{tail}_W\big)/t_i.$$

Proof. The proof is that of Theorem 3.15 and Lemma 3.14 with the following modifications; everything not mentioned is verbatim.

(M0) Order of choices. $f \to f$-constants $\to$ window family $\to s_1 \to N_w \to$ pre-row $\to r \to N'' \to P^*$. s_1 may precede N_w because K_w, s_{late}, (VT′) and (FINE) do not involve N_w.

(M1) Neighbourhoods. Lemma 5.6 replaces Lemma 3.6; the removed clause was used only in condition (c1), which is replaced by $32\rho c_{\mathrm{flat}} C_+ \leq 1$.

(M2) Pre-row and levers. By Lemma 5.1(b) applied at f_0 (that is, by the inequality $\|x/\|x\| - y/\|y\|\| \leq 2\|x - y\|/\|y\|$, valid for all nonzero x, y; Lemma 3.2(ii) is stated only for finitely many mass coordinates), truncating a_0 beyond N_w moves e by at most Π_{tr} and moves no z, and the raises and masses move e by at most Π_X; so $\|e_1 - e_0\| \leq \Pi_X + \Pi_{\mathrm{tr}}$, and by Lemma 3.2(iii) every value moves by at most $\Pi_X + \Pi_{\mathrm{tr}} + t_k(N'')$. As $\|U^*(A_1 - a_0 1_{[1,N_w]})\| \leq 32\rho s_1 \mu_1^\star \sum_i \|b_i^\theta\|_1$, $\Pi_X \leq 128\rho s_1 \mu_1^\star n A_0/(T\nu_0)$. Since $F_l(0) = \epsilon_l[(u_l(\hat{x}_1) - u_l(\hat{z}_0)) - ((A_m(f_1) - A_m(f_0))/A_m(f_0))u_l(\hat{z}_0)]$ and $|A_m(f_1) - A_m(f_0)| \leq \sum_k \lambda_k |u_k(\hat{x}_1) - u_k(\hat{z}_0)|$, we get $|F(0)|_\infty \leq (1 + 1/A_{\min})(\Pi_X + \Pi_{\mathrm{tr}} + T^4 s_1)$, hence $K_{\mathrm{lev}}|F(0)|_\infty \leq r/4$; and $r \leq K_w s_1 \leq r_0$. So (DC) applies. In Lemma 3.14, Π is enlarged by Π_{tr}, K_J is replaced by K_{lev}, and $W_{\mathrm{pull}} = 0$. The truncation also changes ν: $a' = A(P^*)/q^*(A(P^*))$ with $q^*(A(P^*))$ lowered by about $\alpha_0(N_w)$, so the two-sided bound $|\nu' - \nu_0| \leq K_w\Pi$ of Lemma 3.14(c) is replaced by the lower bound $\nu' \geq \nu_0 - K_w\Pi$; this holds because the truncation lowers q^* and changes $\|U^* \cdot\|$ by at most $\|U^* a_0 1_{(N_w,\infty)}\| \leq s_1 T^4 \leq \Pi$, and $\alpha_0(N_w) \leq \theta_{\mathrm{tr}} \leq s_1 T^4 \leq \Pi$ changes q^* by at most Π. Only this lower bound is used downstream (in Lemma 3.14(f) and in the relative error of [I, Lemma 3.3] in Step 3), and the other clauses of Lemma 3.14 hold verbatim, with K_w enlarged by an f-constant. Finally f' lies in B_f: as in the step "Consequences of (LATE)" of the proof of Theorem 3.15,

$p^*(f'-f) \le r_f(\eta_1)/2 + K_w(\Pi\log(e/\Pi)+s_1) + 2(1+\|U\|)^2\alpha_0(N_w)$, where the middle term is $\le r_f(\eta_1)/4$ by (LATE) and the last is $\le r_f(\eta_1)/4$ by the choice of θ_{tr}; so Lemma 5.6 holds at f' and Step 4 applies. A coarse carrier meeting lever coordinates moves by at most $2(\Pi_X+\Pi_{\mathrm{tr}}+C_2 r)+t_k(N'')+C_{\mathrm{lev}}K_{\mathrm{lev}}r \le K_w\Pi$, so in Lemma 3.14(c)–(e) it is treated like a carrier meeting no lever coordinate (its margin or gap exceeds $2K_*\Pi$ for $K_w\Pi \le x_0$); only fine carriers meeting Lev enter through W_{lev}. In the cost bound (a) the truncation adds $q^*(a_0 1_{(N_w,\infty)}) \le (1+\|U\|)\alpha_0(N_w)$ to $q^*(A(P^*)-a_0)$; with the normalization and [II, Lemma 2.12(iv)] this gives the extra term $2(1+\|U\|)^2\alpha_0(N_w)$. The base form (f) uses only $\|\beta^\diamond - b^\diamond\|_1 \le \mathrm{tail}_W$, which holds.

(M3) Step 3, base. We verify (6) with the extra term $\rho|\tau|\mathrm{tail}_W$ on the right of the first inequality. Recall that on $\operatorname{supp} a'$ the summand is $2(-\operatorname{sign}(a'_j)\tau B_j - |a'_j|)_+$ with $a'_j + \tau B_j = (1-\tau\rho c_i)a'_j + \tau\rho\beta_j$ and $1-\tau\rho c_i \ge \frac12$, and that only the part of $\tau\rho\beta_j$ of sign opposite to a'_j can create excess. New classes of coordinates:

(1) $j \in F_0 \cap [1,N_w]$ not of type (PRα). Put $A''_j := A_1(j)$ (the raised value if $j \in W_F$). By (PRβ) for lever coordinates and trivially otherwise, $|A(P^*)(j)| \ge |A''_j|/4$ with the sign s_j; as $q^*(A(P^*)) \le 2$, $|a'_j| \ge |A''_j|/8 \ge |a_0(j)|/8$. θ-piece ($|\tau| \le s_1$): $|\tau\rho\beta^\theta_j| \le s_1\rho|b^\theta_i(j)| \le |A''_j|/32 \le |a'_j|/4$ by the definition of W_F: summand 0. $\pm$-pieces ($s_1 < |\tau| \le c_{\mathrm{flat}}t_i$, $\diamond = \operatorname{sign}\tau$): by (U3^{inf}) the anti-sign part of $\tau\rho\beta^\diamond_j$ is at most $\rho c_{\mathrm{flat}}C_+|a_0(j)| + \rho|\tau|r_i(j) \le |a'_j|/4 + \rho|\tau|r_i(j)$; the first part creates no excess and the second costs at most $2\rho|\tau|r_i(j)$, which is counted in $2\rho|\tau|\epsilon_i$.

(2) $j \in F_0 \cap (N_w, N'']$: a contact of f' with $z'_j = s_j$, $a'_j = 0$, $\beta^\theta_j = 0$, $\beta^\pm_j = \pm\frac12 v_j$. The summand is 0 for θ and at most $\rho|\tau||v_j|$ for $\pm$; in total at most $\rho|\tau|\mathrm{tail}_W \le \delta s_1|\tau|/64 \le \delta\tau^2/64$ for $|\tau| > s_1$: a tolerated violation.

(3) $j \in F_0$, $j > N''$: $z'_j = 0$; inside the cut-off term $\frac12\rho|\tau|V_{>N''}$ exactly as in [II, Lemma 2.30].

(4) Lever coordinates of type (PRα): $\beta^\theta_j = 0$, so the θ summand vanishes in every state of j at f', and the $\pm$ summand is at most $2\rho|\tau||b^\pm_i(j)|$, counted in ϵ_i.

Coordinates $j \notin F_0$ are treated exactly as in the proof of Theorem 3.15. Hence $\mathrm{Exc}'(\tau B^\pm) \le \frac12\rho|\tau|V_{>N''} + 2\rho|\tau|\epsilon_i + \rho|\tau|\mathrm{tail}_W$ and $\mathrm{Exc}'(\tau B^\theta) = 0$.

(M4) Step 5. The extra base term adds at most $\delta/32$ to the bracket of Step 5 of the proof of Theorem 3.15, and $1 - 2\delta + \frac{\delta}{8} + \frac{\delta}{8} + \frac{\delta}{8} + \frac{\delta}{16} + \frac{\delta}{4} + \frac{\delta}{32} \le 1-\delta$.

(M5) The remaining steps. Steps 1, 2 and 4, Lemma 3.14(d), (e), and the (U5′)-variant of Lemma 3.23 use the base only through $q^*(B^\diamond) \le (1+\|U\|)\rho(2A_0/t_i + |c_i|)$, the base form (f), and Lemma 3.2 at the truncated pre-row; they hold verbatim. f' is norm attaining with finite base support because $\operatorname{supp} A(P^*) \subseteq [1,N_w] \cup \mathrm{Lev}$ is finite and z' vanishes beyond N'' [I, Proposition 1.10(c)]. □

Remark 5.11. (a) The far support $F_0 \cap (N_w,\infty)$ of the base row becomes, inside the approximant, a set of contacts; its two-sided use by the side data is a violation of mass at most $\rho\,\mathrm{tail}_W \le \delta s_1/64$ per unit $|\tau|$, which the $\pm$ regime tolerates, while the θ regime does not see it ($\beta^\theta = 0$ there). The truncation is made after s_1, as Remark 5.3(b) requires. (b) On the kept support no lower bound on $|a_0(j)|$ is used: (U3^{inf}) and the radius $c_{\mathrm{flat}}t_i$ replace the finite-support condition $\rho c_{\mathrm{flat}}A_0 \le a_{\min}/32$. (c) c_{flat} depends on the window through γ_B and C_+ only.

Theorem 5.12 (Windowed recovery at arbitrary base support, with raise transfer)**.** *Let $f \in S_{p^*}$ have arbitrary base support, $g \in \mathcal{C}(f)$, $\rho \in (0,1)$, $\eta_0 \in (0,2]$ with $\rho^2(1+\eta_0) \le (1+\rho^2)/2$, and δ as above. Let f_j be base rows of arbitrary support with $p^*(f_j - f) \to 0$, and let $\lambda_j \ge 1$, $\lambda_j \to 1$, $\varepsilon'_j \ge 0$ satisfy*

(a) $p^*(f_j + r\rho g) \le 1 + \big(s(\lambda_j \rho r) - 1\big)/\lambda_j + \varepsilon'_j$ *for all* $r \in \mathbb{R}$.

Suppose that for every j there are $T_j \in (0, t_1]$, $n_j \ge 1$, a nonempty set $\mathcal{S}_j \subseteq \{T_j 2^{1-i} : 1 \le i \le n_j\}$ with $n'_j := |\mathcal{S}_j|$, $T'_j \in (0, \min \mathcal{S}_j]$, $\gamma_{B,j} \in (0,1]$, $C_{+,j} \ge 1$, $K_j \ge 1$, and a window family with cushion constant $C_{+,j}$ at f_j with scales $\mathcal{S}_j$ (window parameter $T := T'_j$), fixed A_0, A_2, and $\gamma_{B,j}$; let $c_{\mathrm{flat}}(j) := \min(c^f_{\mathrm{flat}}, \gamma_{B,j}/(8\rho A_2), 1/(32\rho C_{+,j}))$ and $K_{w,j}$ be the constants of Theorem 5.10 *for this family. Assume:*

(E-c) $p^*(g - g_{j,t}) \le K_j t$ *for* $t \in \mathcal{S}_j$;

(E-d) $K_j T_j / n'_j \to 0$ *and* $n'_j c_{\mathrm{flat}}(j)/K_j \to \infty$;

(E-f) (W-a) *holds at f_j, and there is $s_{1,j}$ satisfying* (VT′), (FINE), (LATE) *and $K_{w,j} s_{1,j} \le r_{0,j}$ at f_j such that, with $N_{w,j}$ chosen by Definition* 5.8 *and $\Pi_j \le K_{w,j} s_{1,j}$,*

$$K_{w,j}\big(K_{w,j} s_{1,j} + c_{\mathrm{fine},j} + \mathrm{tail}_{W,j}\big) \le K_j (T'_j)^2;$$

(E-e) $\varepsilon'_j + K_{w,j}\big(\Pi_j \log(e/\Pi_j) + s_{1,j}\big) + 2(1 + \|U\|)^2 \alpha_{0,j}(N_{w,j}) \le \theta_j (T'_j)^2$ *with* $\theta_j / c_{\mathrm{flat}}(j)^2 \to 0$.

Then $(f, \rho g) \in \mathrm{cl}\,\mathrm{NA}((c_0, p), \ell^2_2)$, along pairs $(f'_j, \bar g_j)$ whose first rows f'_j are norm attaining with finite base support. In particular (LSC-trunc) *holds at (f, g, ρ).*

Proof. Fix j large and write $n := n'_j$, $K := K_j$, $c := c_{\mathrm{flat}}(j)$, $\lambda := \lambda_j$, $T' := T'_j$, $\mathcal{S} := \mathcal{S}_j$. Choose $N_{w,j}$ with $\alpha_{0,j}(N_{w,j})$ so small that (E-e) holds (possible since θ_{tr} in Definition 5.8(ii) is arbitrary). Theorem 5.10 at f_j with $s_1 := s_{1,j}$ gives a norm-attaining f'_j with finite base support and $g'_{j,t}$ ($t \in \mathcal{S}$) such that

(A) $p^*(f'_j + \tau g'_{j,t}) \le 1 + \frac{\tau^2}{2}(1-\delta)$ for $0 < |\tau| \le ct$,

$p^*(f'_j - f_j) + \varepsilon'_j \le \theta_j T'^2$ by (E-e), and $p^*(g'_{j,t} - \rho g_{j,t}) \le KT'^2/t \le Kt$ by (E-f), hence $p^*(g'_{j,t} - \rho g) \le 2Kt$ by (E-c). By (a) and [II, Corollary 2.17], for every r,

(B) $p^*(f'_j + r g'_{j,t}) \le p^*(f_j + r\rho g) + p^*(f'_j - f_j) + |r|\, p^*(g'_{j,t} - \rho g) \le s(\rho r) + (\lambda - 1)\min\big(\frac{\lambda^2 r^2}{2}, 1\big) + \theta_j T'^2 + 2K|r|t$.

Let $\bar g_j := \frac{1}{n}\sum_{t \in \mathcal{S}} g'_{j,t}$; by convexity $p^*(f'_j + r\bar g_j) \le \frac{1}{n}\sum_t p^*(f'_j + r g'_{j,t})$. Fix $r_0 \in (0,1]$ with $r_0^2 \le \min(\delta, 1 - \rho^2)$. For j large,

$$\frac{(\lambda-1)\lambda^2}{2} + \frac{\theta_j}{c^2} + \frac{4K}{nc} \le \min\Big(\frac{\delta}{4}, \frac{1-\rho^2}{6}\Big),$$
$$\theta_j T'^2 + (\lambda - 1) \le \frac{(1-\rho^2) r_0^2}{6}, \qquad \frac{4KT_j}{n} \le \frac{(1-\rho^2) r_0}{6},$$

by (E-d), (E-e), $\lambda_j \to 1$ and $T' \le 1$.

Case $|r| \le r_0$. Let $\mathcal{S}_r := \{t \in \mathcal{S} : ct < |r|\}$; since the scales are distinct dyadic numbers, $\sum_{t \in \mathcal{S}_r} t < 2|r|/c$. If $\mathcal{S}_r = \emptyset$, every term obeys (A), and $1 + \frac{r^2}{2}(1-\delta) \le 1 + \frac{r^2}{2} - \frac{r^4}{8} \le s(r)$ as $r^2 \le \delta$. Otherwise $|r| > cT'$, so $\theta_j T'^2 \le \theta_j r^2/c^2$, and

$$p^*(f'_j + r\bar g_j) \le \max\Big\{1 + \frac{r^2}{2}(1-\delta),\ s(\rho r) + \frac{(\lambda-1)\lambda^2 r^2}{2}\Big\} + \frac{\theta_j}{c^2} r^2 + \frac{2K|r|}{n} \cdot \frac{2|r|}{c} \le s(r),$$

because $1+\frac{r^2}{2}(1-\delta)+\frac{\delta}{4}r^2 \le 1+\frac{r^2}{2}-\frac{r^4}{8}$ and $s(\rho r)+\frac{1-\rho^2}{6}r^2 \le s(r)$ by [I, Lemma 3.6(a)] ($|r|\le 1$).

Case $|r|\ge r_0$. By (B) for all t, $p^*(f_j'+r\bar g_j)\le s(\rho r)+(\lambda-1)+\theta_j T'^2+4K|r|T_j/n \le s(\rho r)+\frac{1-\rho^2}{3}r_0|r| \le s(r)$, by [I, Lemma 3.6(a)] and $\min(r^2,|r|)\ge r_0|r|$.

So $\bar g_j \in \mathcal{C}(f_j')$, and $(f_j',\bar g_j)$ attains its norm [I, Proposition 1.13(d)]. Finally $\|(f_j',\bar g_j)-(f,\rho g)\| \le p^*(f_j-f)+p^*(f_j'-f_j)+\frac{1}{n}\sum_t 2Kt \le p^*(f_j-f)+\theta_j T'^2+4KT_j/n \to 0$. As f_j' has finite base support and $\operatorname{dist}(\rho g,\mathcal{C}(f_j'))\le p^*(\rho g-\bar g_j)\to 0$, (LSC-trunc) holds at (f,g,ρ). □

Remark 5.13 (What the two theorems reduce (LSC-trunc) to). (a) With $\lambda_j=1$ and $\varepsilon_j'=p^*(f_j-f)$, (a) holds trivially; raised base rows ([II, Lemma 2.15]) satisfy (a) with $\varepsilon_j'=2\|U^*\Delta a_j\|_H+p^*(L^*(w^{(j)}-w))$ by [II, Corollary 2.17], and truncated rows f^M (Lemma 5.2) satisfy (a) for all r with $\lambda_j=1$ and $\varepsilon_j'=2\alpha(M)+\varepsilon_M''$, since $\Lambda_M\le\alpha(M)$ (apply Lemma 5.2(a) to an optimal decomposition of $f+r\rho g$). The better bound $2\max\Lambda_M+\varepsilon_M''$ holds only at the finitely many scales of Corollary 5.4 (Remark 5.3(b)). (b) $c_{\mathrm{flat}}(j)$ may depend on the window through $\gamma_{B,j}$ and $C_{+,j}$; the theorem needs exactly $n_j'c_{\mathrm{flat}}(j)/K_j\to\infty$ and $\theta_j/c_{\mathrm{flat}}(j)^2\to 0$. (c) Consequently (LSC-trunc) at (f,g,ρ) follows from the *existence* of window families with side cushion domination (U3^{inf}), a lever system (U6^{inf}) giving exact d-consistency, and (E-c)–(E-f), at base rows of arbitrary support converging to f in the sense of (a). Producing them is the infinite-support exactification problem treated in the rest of this section.

5.3. **The designed operator: cushion domination at deep-raised rows.** From now on the operator is the design of Section 4.1 over the doubly exponential diagonal base $\mu_s^\star:=2^{-2^{2^s}}$, enlarged as in Definition 5.42 below (all enlargements only increase $\mathrm{Des}(L)$ and the rate scheme, Lemma 5.43). We use the notation of Subsection 2.1, of [II, Subsection 5.6] and of the route-V companion of Section 4.3: $N\ge 1$, $f\in S_{p_N^*}$ with *arbitrary* base support F, $g\in\mathcal{C}(f)$, $\rho\in(0,1)$, $l_f=l_f(f,N,g,\rho)$, a main stage $L\ge l_f$, a sub-window w of L clean for f, $T:=T_{\mathrm{lo}}(w)$, $b:=b(w)$, $u:=u(w)$, $\eta':=T^4/(L\,\mathrm{Des}(L))$; $D(L)$ is the design quantity of [II, Subsection 5.6] with $\lambda_k\ge 1/D(L)$ for coarse carriers. $T(L)$ is the union of the supports of the coarse targets and $s_{\max}(L):=\max T(L)$. For a coarse carrier l, S_l is its signature set and $v_l:=u_l$ on S_l; beyond $s_{\max}(L)$ the only coarse carrier whose vector is nonzero on S_l is l, and $v_l(s)=\delta_l 2^{-s}/n_l>0$ there (proof of Lemma 3.25). The *near coordinates* N_l of l are the first six elements of $S_l\cap(s_{\max}(L),\infty)$, and

$$d(L):=\max_{l\le L}\max N_l,\qquad H(L):=\max_{l\le L}\max_{s\in N_l}\frac{1}{v_l(s)}.$$

Sums over coarse and fine carriers satisfy $\sum_{k>L}\lambda_k\le Cc_{L+1}$, and (W$_{\exp}$) gives $c_{L+1}\le e^{-1/T}$.

The *box function* of level L is $\mathrm{Bx}_L(j):=\sum_{l\le L}\lambda_l|u_l(j)|$; $\sum_j \mathrm{Bx}_L(j)\le 1$.

Route-V data. For a scale t of the window let $(B_\pm,\Theta_\pm)$ be the two-sided decomposition at scale t of the mate at the row considered, and $\Delta\omega(t)$ the route-V switching on Ω (all coarse strict non-peaks of the companion). The switching vector $V:=\sum_m R_m^*(\Delta\omega_m-\Delta_m w_m)$ (Definition 3.1; $R_m^*e_k=\lambda_k u_k$) is $V=\sum_l c_l u_l$ over all carriers, with $c_l=\lambda_l(\Delta\omega(l)-\Delta_m w_m(l))=:\gamma_l(t)$ for $l\in\Omega$ and $c_k=-\Delta_m\lambda_k w_m(k)$ for every other carrier k of block m (coarse peaks and fine carriers), since $\Delta\omega_m$

vanishes off Ω; $V_c := \sum_{l\le L} c_l u_l$ and $V_f := V - V_c$. By (U7), $|c_k| \le C_\Delta \lambda_k$ for $k \notin \Omega$, so $\|V_f\|_1 \le C_\Delta \sum_{k>L} \lambda_k \le C c_{L+1}$. Activity classes are those of [II, Lemma 5.53] read off the decomposition data, with the thresholds $A_i := K_*(w)^i$ defined below, and the projected class data $(\Delta'(t), \gamma(t))$ lie in the class cone $\Gamma^{\#}(\varkappa, a)$ [II, Definition 5.52]. The next lemma restates Lemma 4.21 in the present setting and adds the count of class sets.

Lemma 5.14 (Route V needs no shift normalization)**.** *In the route-V construction of Section* 4.3 *the cube pigeonhole and the normalization increment of [II, Proposition 5.54(c)–(e)] may be omitted: if for every scale t of a set $\mathcal{S}$ of one activity class the data are built from the projected class data $(\Delta'(t), \gamma(t)) \in \Gamma^{\#}(a)$ without increment, then all conclusions of the route-V companion lemma hold, with the Γ_w-bound improved to $\Gamma_w^{\#}(b^-, \omega^-)^{1/2} \le \Gamma_w^{\#}(B_-, \Theta_-)^{1/2} + K_{\mathrm{data}} t$, with K_{data} as in Proposition 5.19, and the pigeonhole over classes alone gives $|\mathcal{S}| \ge n(w)/D_{\mathrm{cl}}(w)$ with $D_{\mathrm{cl}}(w) := (J+1)3^J$, $J := |\Omega| + N$.*

Proof. We check every place where a common shift of all scales of $\mathcal{S}$ could be used. (1) The anti-alignment and ownership rules of the route-V companion use only the signs $\operatorname{sign} \Delta''_m$ and ϵ_l, which are fixed by the class ([II, Lemma 5.53]; the sign of Δ'_m is constant on the class cone), and quantities computed from coordinates fixed before the carrier; the dominance estimates of the completion are stated for every shift and every γ of the class. (2) The relation between true and reference shift ([II, Lemma 5.58]) and the trace-correction bound are per scale. (3) Inwardness (kind [3]) holds at every point of the cone; the increment was one more point of the cone. (4) The conditions (U1)–(U7) of a window family concern each piece separately, (U5) only through the set of negative active blocks, which is fixed by the class; the approximant uses the masses $\max_i |b_i^\theta|$ and one common Ω; the averaging uses convexity only. (5) The status lemmas at the approximant do not involve the data. (6) Every later use of $|\mathcal{S}|$ is a lower bound. Without the increment the $-$ side differs from the decomposition's $-$ side only by the projection and zeroing errors of [II, Lemma 5.53]; the zeroing error is $A_{i(t)} K t$ with $A_i = K_*(w)^i \le K_*(w)^{J+1}$, which is why the constant is K_{data} and not the projection constant K. This gives the Γ_w-bound since $\Gamma_w^{1/2}$ is a seminorm in the data. □

Remark 5.15 (Why the increment must be dropped; heuristic). This remark is heuristic and not used. The increment is a nonnegative combination of extreme rays of the class cone with coefficients up to the cube side, which is a fixed power of u/Des and not proportional to t. At an anti-sign support coordinate $j \in S_l \cap F$ below the raise depth it may add anti-sign switching of order (cube side)$\cdot v_l(j)$, which exceeds the cushion $|a_j|/t$ when $|a_j|$ is small compared with $t\, v_l(j)$; Lemma 5.18 controls only the decomposition's own switching.

Lemma 5.16 (Deep raise)**.** *Let $J > \max(d(L), s_{\mathrm{t}}(L))$, where $s_{\mathrm{t}}(L)$ is the depth of [II, Lemma 6.45], and put*

$$R(J) := \{j \in F : j \ge J,\ |a_j| < 8\mathrm{Bx}_L(j)\}, \qquad \Delta a := \sum_{j \in R(J)} s_j\big(8\mathrm{Bx}_L(j) - |a_j|\big) e_j^*,$$

$\lambda_r := q^*(a + \Delta a)$ and $f^r := f[a + \Delta a, z]$. Then:

(a) $\|\Delta a\|_1 \le \psi_L(J) := 8\sum_{j\ge J}\mathrm{Bx}_L(j) \to 0$ *as* $J\to\infty$, $\|U^*\Delta a\| \le \mu^\star_J\psi_L(J) \le 8\mu^\star_J$, $1 \le \lambda_r \le 1+2\|\Delta a\|_1$, *and* $|a^r_j| \ge 4\mathrm{Bx}_L(j)$ *for every* $j\in F$, $j\ge J$, *once* $\psi_L(J)\le\frac12$;
(b) *for every* $r\in\mathbb{R}$, $p^*(f^r+r\rho g) \le 1+(s(\lambda_r\rho r)-1)/\lambda_r+\varepsilon'_r$ *with* $\varepsilon'_r := 2\|U^*\Delta a\| + p^*(L^*(w^r-w)) \le C_f\mu^\star_J\log(e/\mu^\star_J)$;
(c) F, z, *the contacts and the rooms are unchanged; every value* $u_k(\hat z)$ *moves by at most* $2\|U^*\Delta a\|/\nu$. *Among the rate objects of level* L *([II, Lemma 6.45]): the rooms and the objects* (O1), (O2), (O6), (O11) *are unchanged; the support moduli* ((O10) *and the moduli* $|a_s|$ *of* (I3)) *are divided by* $\lambda_r\in[1,1+2\psi_L(J)]$; *the minors and cofactor polynomials of the enlarged class cones of* (I3) *depend only on design data and on the pattern, which is unchanged; and every other rate object of level* L *moves by at most* $M_f(L)\|U^*\Delta a\|$.

Let $J(w)$ *be the least* $J > \max(\bar d(L), s_{\mathrm t}(L))$ ($\bar d(L)$ *from Definition* 5.42) *with* $C_f\mu^\star_J\log(e/\mu^\star_J) \le \theta_w T^2$ *and* $M_f(L)\mu^\star_J\psi_L(J) \le b^2$, *where* $\theta_w := T$ (*so the first condition reads* $C_f\mu^\star_J\log(e/\mu^\star_J) \le T^3$). *Then at* f^r *every object of the rate scheme that is tiny at* f *is at most* $2b$ *and every robust one is at least* $u/2$, *and* $2^{J(w)} \le \mathrm{Des}(L) + 2\log_2(Cn(w))$.

Proof. (a) The raise has the sign of a, so $\|a+\Delta a\|_1 = \|a\|_1+\|\Delta a\|_1$ and $\|U^*(a+\Delta a)\| \ge \nu - \|U^*\Delta a\|$; as $\mu^\star$ is non-increasing, $\|U^*\Delta a\| \le \mu^\star_J\|\Delta a\|_1 \le \|\Delta a\|_1$, so $\lambda_r \ge 1$, and $\lambda_r \le 1+2\|\Delta a\|_1$. On $R(J)$, $|a_j+\Delta a_j| = 8\mathrm{Bx}_L(j)$, elsewhere on $F\cap[J,\infty)$, $|a_j| \ge 8\mathrm{Bx}_L(j)$; divide by $\lambda_r\le 2$. (b) is [II, Corollary 2.17] for the raise; the normer change is bounded by [II, Lemma 2.12(iv)] with $\delta = U(e^r-e)$, $\|e^r-e\| \le 2\|U^*\Delta a\|/\nu$ (by the inequality in the proof of Lemma 5.1(b); the raise set $R(J)$ may be infinite) and $c(\delta) \le C\epsilon\log(e/\epsilon)$ for $\epsilon := \|U\|\|e^r-e\|$ [II, (1)]. (c) The raise acts only on $F\cap[J,\infty)$ and keeps signs; the rest is [II, Lemma 6.45] (whose hypothesis $J > s_{\mathrm t}(L)$ holds), which distinguishes the unchanged objects, the support moduli (divided by λ_r) and the objects moved by $M_f(L)\|U^*\Delta a\|$; the moduli $|a_s|$ of (I3) are support moduli, and the class-cone minors of (I3) do not involve the row. The consequence for tiny and robust objects is the last part of [II, Lemma 6.45], since $\|\Delta a\|_1 \le \frac12$ for $L \ge l_f$: a tiny modulus stays $\le b$ and a robust one stays $\ge u/2$ because $\lambda_r\le 2$, and the moved objects change by at most $M_f(L)\mu^\star_J\psi_L(J) \le b^2$. For the bound on $J(w)$: both conditions hold as soon as $\mu^\star_J \le \epsilon_w$ with $\log_2(1/\epsilon_w) \le C\,\mathrm{Des}(L)\log_2(1/T)$, because b, θ_w and $1/M_f(L)$ are at least design powers of T (Section 4.1); since $\mu^\star_J = 2^{-2^{2^J}}$ this holds when $2^J \ge \log_2(C\,\mathrm{Des}(L)\log_2(1/T))$; with $\theta_w = T$, $\log_2(1/(\theta_w T^2)) = 3\log_2(1/T)$. With $\log_2(1/T) \le 3n(w)$ for L large, $2^{\bar d(L)+1} \le \mathrm{Des}(L)$ (Definition 5.42) and $2^{s_{\mathrm t}(L)+1} \le \mathrm{Des}(L)$ (the factor $B^\star(L) = 2^{s_{\mathrm t}(L)}/(\delta_{\min}(\mu^\star_{s_{\mathrm t}})^2)$ of $\mathrm{Des}(L)$ in Part II), the least admissible J satisfies $2^{J(w)} \le \mathrm{Des}(L) + 2\log_2(Cn(w))$. □

Classes with pinning thresholds. For a target coordinate $j\in T(L)\cap F$ the clean sub-window gives $|a_j| \le b$ (*tiny support*) or $|a_j| \ge u$ (*robust support*), because the moduli $|a_s|$, $s\le\bar d(L)$, belong to the rate scheme (Definition 5.42). The *enlarged class cone* is $\Gamma^{\#}(a)$ of [II, Definition 5.52] with, for every tiny support coordinate $j\in T(L)$, the contact row

$$(\mathrm{X1}^F)\qquad s_j\Big(\sum_{l\in\Omega}\gamma_l u_l(j) - \sum_m \Delta'_m \sum_{k\in P^{\mathrm c}_m} \mathrm{vs}_k\lambda_k u_k(j)\Big) \ge 0.$$

Its coefficients are non-weight design data, so the weight-freeness and genericity arguments of Section 2 apply verbatim to the enlarged cones, and their minors and cofactor polynomials are rate objects (Definition 5.42). Let $C_q := \frac{8}{3}(4/q_0 + 8)$ be the constant of [II, Lemma 6.37] at rows of the ball ($q_0^r \geq q_0/2$), $H(L)$ the Hoffman bound and C_Γ the row bound of [II, Lemma 5.53], $K = C_f(\mathrm{Des}/u)^C$ the projection constant, $\delta_{\min}(L) := \min_{l \leq L} \delta_l$, and

$$K_P(w) := \frac{8C_q 2^{J(w)} D(L)}{\delta_{\min}(L)},$$

$$K_*(w) := \max\Big\{K_*,\ 6\big(H(L)(C_\Gamma J + 1) + 1\big),\ \big(K_P(w)/K\big)^{1/2}\Big\},$$

with K_* of [II, Lemma 5.53]. Classes are read off the decomposition data at f^r with the thresholds $A_i := K_*(w)^i$. Write c_l for the coefficients of V_c built from the projected class data, and $c_l^{\mathrm{act}} := -\Delta\theta_l$ for the actual coefficients of the decomposition $(B_\pm, \Theta_\pm)$ at f^r ($\Delta B = -\sum_l \Delta\theta_l u_l$, [I, (15)]).

Lemma 5.17 (Active coefficients track the decomposition)**.** *For every $t \in \mathcal{S}$ and every coarse carrier l, either $c_l = 0$, or $|c_l - c_l^{\mathrm{act}}| \leq |c_l|/5$ and $|c_l| \geq K_*(w)^2 Kt/(4D(L))$.*

Proof. For $l \in \Omega$, $c_l = \gamma_l(t)$. An inactive component is set to 0. An active one has $|\gamma_{\mathrm{dec},l}| \geq A_{i(t)+1}Kt = K_*(w)A_{i(t)}Kt$; zeroing the inactive components and projecting onto the enlarged cone moves it by at most $H(L)(1 + C_\Gamma J A_{i(t)})Kt \leq H(L)(C_\Gamma J + 1)A_{i(t)}Kt \leq |\gamma_{\mathrm{dec},l}|/6$ (proof of [II, Lemma 5.53]), so $|\gamma_l| \geq \frac{5}{6}|\gamma_{\mathrm{dec},l}|$ and $|\gamma_l - \gamma_{\mathrm{dec},l}| \leq |\gamma_l|/5$; also $|\gamma_l| \geq \frac{5}{6}K_*(w)^2 Kt$. For a coarse peak k of block m, $c_k = -\Delta_m'' \lambda_k \mathrm{vs}_k$ with $\Delta_m'' := M_m \Delta_m$ ($w_m(k) = M_m \mathrm{vs}_k$ at a peak), and by [I, Lemma 8.27] at f^r, $c_k^{\mathrm{act}} = -\Delta'_{\mathrm{dec},m}\lambda_k \mathrm{vs}_k + e_k$ with $|e_k| \leq C_f D(L)t/u$ at peaks of relative margin $\geq u/2$. The shift components obey the same zeroing and projection bounds as the γ-components, and the true shift differs from the reference shift only by the trace correction of [II, Lemma 5.58], which is $\leq C_f \mathrm{Des}(L) c_{L+1} \ll Kt$; with $|\Delta'_m| \geq K_*(w)^2 Kt$ for an active shift, $\lambda_k \geq 1/D(L)$, $M_m \geq \frac{1}{2}$, and $K_*(w) \geq K_* \geq C_f D(L)/u$, the same arithmetic gives the claim. Inactive shifts give $c_k = 0$. □

Lemma 5.18 (No deep anti-sign set at the raised row)**.** *Let $t \in \mathcal{S}$ and put, for $j \in T(L)$, $\mathrm{tr}_j := |\sum_m (\Delta_m'' - \Delta_m') \sum_{k \in P_m^{\mathrm{c}}} \mathrm{vs}_k \lambda_k u_k(j)|$, and $\mathrm{tr}_j := 0$ otherwise. Then $\sum_j \mathrm{tr}_j \leq C_f \mathrm{Des}(L) c_{L+1} + C_f c_{L+1}^2/t$, and for every $j \in F$*

$$s_j V_c(j) \geq -\frac{4|a_j^r|}{t} - \mathrm{tr}_j.$$

Proof. The bound on $\sum \mathrm{tr}_j$ is the trace-correction estimate of [II, Lemma 5.58(b)], applied per scale. *Coefficient bound.* For coarse l, $|c_l| \leq C_f D(L)\lambda_l$: for $l \in \Omega$, $c_l = \lambda_l(\Delta\omega(l) - \Delta_m w_m(l))$ with $|\Delta\omega(l)| \leq \|D_m \Delta\omega_m\|_2/\Phi_m(l) \leq C_f D(L)$ by (U1) and Proposition 3.7(a), and $|\Delta_m| \leq C_\Delta$; for peaks $|c_k| \leq C_\Delta \lambda_k$. Hence $|V_c(j)| \leq C_f D(L) \mathrm{Bx}_L(j)$. Four cases.

(D) $j \geq J(w)$: $|a_j^r| \geq 4\mathrm{Bx}_L(j)$ by Lemma 5.16(a), so $4|a_j^r|/t \geq 16\mathrm{Bx}_L(j)/t \geq |V_c(j)|$ since $t \leq T_{\mathrm{hi}}(w) \leq 16/(C_f D(L))$.

(T) $j \in T(L)$. If the support is robust, $|a_j^r| \geq u/2$ and $4|a_j^r|/t \geq 2u/T_{\mathrm{hi}}(w) \geq C_f D(L) \geq |V_c(j)|$, since $T_{\mathrm{hi}}(w) \leq u/(C\,\mathrm{Des}(L))$. If it is tiny, the projected data satisfy ($\mathrm{X1}^F$) exactly, and $V_c(j)$ differs from the left side of ($\mathrm{X1}^F$) (divided by s_j) by the trace term, so $s_j V_c(j) \geq -\mathrm{tr}_j$.

(S) $j < J(w)$, $j \notin T(L)$, $j \in S_l$ for a coarse l. Among coarse carriers only l lives at j, so $V_c(j) = c_l v_l(j)$. Suppose $c_l \neq 0$ and $s_j c_l v_l(j) < -4|a_j^r|/t$. By Lemma 5.17, c_l^{act} has the sign of c_l and $|c_l^{\mathrm{act}}| \geq \frac{4}{5}|c_l|$; so with $\tau_l := \epsilon_l c_l^{\mathrm{act}}$ the coordinate j is of the type opposite to the switching direction and $|\tau_l| v_l(j) > \frac{16}{5}|a_j^r|/t$. As j is unraised and not in $T(L)$, [II, Lemma 6.37] at f^r (its term $\sum |r_s| \leq 8t^2$ requires $\sum_{l'' > L} \lambda_{l''} \leq t^3$, which holds) gives $|c_l^{\mathrm{act}}| \leq C_q t / v_l(j) \leq \frac{5}{4} C_q 2^{J(w)} t / \delta_{\min}(L)$, since $v_l(j) = \delta_l 2^{-j}/n_l$ and $n_l \leq \frac{5}{4}$. Hence $|c_l| \leq \frac{25}{16} C_q 2^{J(w)} t/\delta_{\min}(L)$, while Lemma 5.17 gives $|c_l| \geq K_*(w)^2 K t/(4D(L)) \geq K_P(w) t/(4D(L)) = 2C_q 2^{J(w)} t/\delta_{\min}(L)$: a contradiction.

(O) Otherwise no coarse carrier lives at j (a coordinate of S_l with $l > L$ is not in $T(L)$), and $V_c(j) = 0$. □

The deep assignment on the support. Let $f^{\#}$ be the companion of f^r produced by Step 0, the preparation (Definition 5.28) and Step A (Lemma 5.37) of the route-V construction, with the moves of Subsection 5.4 (no step (2), Lemma 5.20). Let $F_{\mathrm{low}} \subseteq F$ be the set of support coordinates lowered by these moves: far pull coordinates $j \in F$ with $s_j = \epsilon_l$, which are lowered to contacts and then pulled (so they carry, at $f^{\#}$, the pull mass $24\lambda_l v_l(j)$ of sign $-\epsilon_l$). Let Z be the set of coordinates of type ZD (Lemma 5.29(d)): coordinates $s \leq \bar{d}(L)$, private to a zero-data coarse carrier, which are not thick; at f they are off F or thin. We use the following properties of $f^{\#}$, established in Lemmas 5.29 and 5.37: ($\mathrm{P}^{\#}1$) $|a_j^{\#}| \geq |a_j^r|/2$ and $\operatorname{sign} a_j^{\#} = s_j$ for $j \in F \setminus (F_{\mathrm{low}} \cup Z)$; ($\mathrm{P}^{\#}2$) the support coordinates of $f^{\#}$ outside F are banks (of (P_2) and of the tuning pre-placements), at which the route-V data are those of a contact of the bank's sign, i.e. contact-like up to the fine residue and the base term, and pulled coordinates (pre-placed tuning pulls and Step-0 pulls at pullable far contacts), which carry the pull mass $24\lambda_l v_l(j)$ of sign $-\epsilon_l$; ($\mathrm{P}^{\#}3$) the coarse statuses of the route-V companion.

For $t \in \mathcal{S}$ let $(B_\pm, \Theta_\pm)$ be the decomposition of the mate g^r at f^r at scale t (as in [II, Subsection 6.7]), V the route-V switching vector and $A_j := 2|a_j^r|/t$ ($j \in F$). On $F \setminus (F_{\mathrm{low}} \cup Z)$ define b_1^+ by the deep assignment of [II, Lemma 6.28] with $X := V_c|_F$ and the trace slack: with $x_j := s_j B_+(j)$, $y_j := s_j(B_+(j) - V_c(j))$, let σ_j be the point of $[y_j - A_j - \mathrm{tr}_j, x_j + A_j]$ closest to 0 and $b_1^+(j) := B_+(j) - s_j\sigma_j$; the interval is nonempty because $x_j - y_j = s_j V_c(j) \geq -2A_j - \mathrm{tr}_j$ (Lemma 5.18). On F_{low} put $b_1^+(j) := \chi_j V_c(j)$, where χ_j is the projection of $s_j B_+(j)/(s_j V_c(j))$ onto $[0,1]$ if $s_j V_c(j) > 0$ and $\chi_j := 0$ otherwise. On $Z \cap F$ put $b_1^+(j) := \frac{1}{2} V(j)$ ($= \frac{1}{2} V_f(j)$, since only a zero-data coarse carrier lives at j). Off F the + side is that of the route-V companion (at coordinates of $Z \setminus F$ it is $\frac{1}{2} V_{\mathrm{fine}}(j)$, as in Lemma 3.25(b)). With $a_Z^{\#} := a^{\#} 1_{\mathbb{N} \setminus Z}$ put

$$b^+ := b_1^+ + b_{\mathrm{off}}^+ - \varkappa \frac{a_Z^{\#}}{a_Z^{\#}(\hat{z}^{\#})}, \quad \varkappa := (b_1^+ + b_{\mathrm{off}}^+)(\hat{z}^{\#}), \qquad b^- := b^+ - V;$$

$a_Z^{\#}(\hat{z}^{\#}) \geq \frac{1}{4}$ because the masses on Z are at most $2\bar{d}(L) b.$ and $\omega^\pm$ are those of the route-V companion.

Proposition 5.19 (Cushion domination at the deep-raised companion)**.** *For every $t \in \mathcal{S}$ the pairs $(b^\pm, \omega^\pm)$ represent one functional g_t at $f^{\#}$, and together with the other scales of $\mathcal{S}$ they satisfy* (U1), (U2), ($\mathrm{U4}^{\mathrm{inf}}$), (U7) *and*

(i) *for* $j \in F \setminus (F_{\mathrm{low}} \cup Z)$*:* $(s_j b^+(j))_- \le 5|a_j^\#|/t$ *and* $(s_j b^-(j))_+ \le 5|a_j^\#|/t + |V_f(j)| + \mathrm{tr}_j$*;*
(ii) *for* $j \in F_{\mathrm{low}}$*: at a pulled coordinate* $t|b^\pm(j)| \le |a_j^\#|$*;*
(iii) *for* $j \in Z$*:* $b^\pm(j) = \pm\frac{1}{2}V(j)$ *(with* $V(j) = V_f(j)$ *on* F*), so* $b^\theta(j) = 0$ *and* $\mathrm{viol}(j) \le |V(j)|$*;*
(iv) *at banked coordinates* $s \in \operatorname{supp} a^\# \setminus F$ *(banks of* (P_2) *and the base banks of the tuning pre-placements of Step A):* $b^\pm(s)$ *are contact-like of sign* $z_s^\#$ *up to the fine residue, plus the base term* $-\varkappa a_s^\#/a_Z^\#(\hat{z}^\#)$ *of modulus* $\le 4Kt|a_s^\#| \le |a_s^\#|/t$*;*
(v) *at pulled coordinates* $j \in \operatorname{supp} a^\# \setminus F$*:* $t|b^\pm(j)| \le |a_j^\#|$*;*

so $(\mathrm{U3}^{\mathrm{inf}})$ *holds on all of* $\operatorname{supp} a^\#$ *with* $C_+ := 5$ *and excess* $r(j) := |V_f(j)| + \mathrm{tr}_j$*, of total mass at most* $C_f \mathrm{Des}(L) c_{L+1} + C_f c_{L+1}^2/t$*, which is counted in* ϵ*. Moreover*

$$p^*(g - g_t) \le K_{\mathrm{data}} t, \qquad K_{\mathrm{data}} := (JK_*(w)^{J+1} + C_f)K,$$
$$\|B_+ 1_F - b_1^+\|_1 \le \frac{(1+\eta')t}{q_0^r} + Kt + \frac{7Cc_{L+1}}{t}.$$

Proof. (i) *Deep assignment.* By [I, Lemma 8.28] at f^r (with the budget of the window), $s_j B_+(j) \ge -|a_j^r|/t - \mathfrak{f}_j^+$ and $s_j B_-(j) \le |a_j^r|/t + \mathfrak{f}_j^-$, with $\sum_F(\mathfrak{f}_j^+ + \mathfrak{f}_j^-) \le (1+\eta')t/q_0^r$. The point of $[y - A - \mathrm{tr}, x + A]$ closest to 0 has modulus $(-x - A)_+ + (y - A - \mathrm{tr})_+ \le \mathfrak{f}_j^+ + \mathfrak{f}_j^- + |(\Delta B - V_c)_j|$, because $B_+ - V_c = B_- + (\Delta B - V_c)$. Hence $|B_+(j) - b_1^+(j)| \le \mathfrak{f}_j^+ + \mathfrak{f}_j^- + |(\Delta B - V_c)_j|$, $s_j b_1^+(j) = x_j - \sigma_j \ge -A_j$ and $s_j(b_1^+ - V_c)(j) = y_j - \sigma_j \le A_j + \mathrm{tr}_j$; also $|b_1^+(j)| \le A_j + |V_c(j)| + \mathrm{tr}_j$. On $Z \cap F$, $V_c(j) = 0$, and the flip bounds give $|B_+(j)| \le A_j/2 + \mathfrak{f}_j^+ + \mathfrak{f}_j^- + |(\Delta B - V_c)_j|$, so $|B_+(j) - b_1^+(j)| \le A_j + \mathfrak{f}_j^+ + \mathfrak{f}_j^- + |(\Delta B - V_c)_j| + |V_f(j)|$; since $|a_j^r| \le 2b$ on Z, $\sum_{Z\cap F} A_j \le 4\bar{d}(L)b/t \le Kt$. On F_{low}, $V_c(j) = c_l v_l(j)$ has the sign $s_j = \epsilon_l$ (a lowered far coordinate of a data-carrying carrier); the flip bounds give $-A_j/2 - \mathfrak{f}_j^+ \le s_j B_+(j) \le s_j V_c(j) + A_j/2 + \mathfrak{f}_j^- + |(\Delta B - V_c)_j|$, so $|B_+(j) - \chi_j V_c(j)| \le A_j + \mathfrak{f}_j^+ + \mathfrak{f}_j^- + |(\Delta B - V_c)_j|$; and $\sum_{F_{\mathrm{low}}} A_j \le Kt$ because $|F_{\mathrm{low}}| \le 2|\Lambda|$, the lowered moduli are at most $\max(2b, x'_{\max})$ and $x'_{\max} \le T^3$ (Definition 5.22).

(ii) *ℓ_1-bound and closeness.* $\|\Delta B - V_c\|_{\ell_1(F)} \le \sum_{l \le L} |c_l^{\mathrm{act}} - c_l|\,\|u_l\|_1 + \|\sum_{l>L} \Delta\theta_l u_l\|_1 \le Kt + 6\sum_{l>L}\lambda_l/t$, by Lemma 5.17 (whose errors are the zeroing and projection errors of [II, Lemma 5.53], summing to at most Kt) and [I, Lemma 8.10] for the fine carriers. This gives the ℓ_1-bound. The functional g_t differs from the one built in Section 4.3 from $B_+ 1_F$ by $b_1^+ - B_+ 1_F$ and the corresponding $\varkappa$-terms. The closeness of that functional to g is bounded by the zeroing cost $JA_{J+1}Kt \le JK_*(w)^{J+1}Kt$ of [II, Lemma 5.53] plus $C_f Kt$, and $(1 + \|U\|)\|B_+ 1_F - b_1^+\|_1 \le C_f Kt$ because $c_{L+1} \le e^{-1/T} \le t^2$; so $p^*(g - g_t) \le K_{\mathrm{data}} t$ after enlarging C_f. Both pairs represent g_t because $b^+ - b^- = V$ and $\omega^- - \omega^+ = \Delta\omega$ ([II, Lemma 2.22]), and $b^+(\xi^\#) = 0$ by the choice of $\varkappa$.

(iii) *Cushions.* $|\varkappa| \le Kt$: b_1^+ differs from $B_+ 1_F$ by $O(Kt)$ in ℓ_1, $|B_+(\hat{z}^r)| \le t/(2q_0^r)$ by [I, Lemma 8.7(a)], and $|\langle U^* B_+, e^\# - e^r\rangle| \le \|B_+\|_1 \|e^\# - e^r\| \le t$ since the companion moves e by at most $C\,\mathrm{Des}(L)T^3$. On $F \setminus (F_{\mathrm{low}} \cup Z)$, by (i) and $(\mathrm{P}^\#1)$, $(s_j b^+(j))_- \le A_j + 4|\varkappa||a_j^\#| \le 4|a_j^\#|/t + 4Kt|a_j^\#| \le 5|a_j^\#|/t$, and $(s_j b^-(j))_+ = (s_j(b_1^+ - V_c - V_f)(j) - \varkappa s_j a_j^\#/a_Z^\#(\hat{z}^\#))_+ \le A_j + \mathrm{tr}_j + |V_f(j)| + 4Kt|a_j^\#|$, as $4Kt^2 \le 1$. On Z the $\varkappa$-term vanishes, so $b^+(j) = \frac{1}{2}V(j)$ (and $b^+(j) = \frac{1}{2}V_{\mathrm{fine}}(j)$ off F),

$b^-(j) = -\frac{1}{2}V(j)$, $b^\theta(j) = 0$. On F_{low}: at a pulled coordinate $t|b^\pm(j)| \le t|V(j)| \le 24\lambda_l v_l(j) = |a_j^\#|$, since $t \le T_{\mathrm{hi}}(w)$ and $|c_l| \le C_f D(L)\lambda_l$. The bound on $\sum(|V_f(j)| + \mathrm{tr}_j)$ is Lemma 5.18 and $\|V_f\|_1 \le Cc_{L+1}$. At a banked coordinate $s \notin F$ the route-V data are those of a contact of the bank's sign up to the fine residue (as in Proposition 4.16), and the base term has modulus $|\varkappa||a_s^\#|/a_Z^\#(\hat{z}^\#) \le 4Kt|a_s^\#| \le |a_s^\#|/t$ because $4Kt^2 \le 1$; this gives (iv). At a pulled coordinate $j \notin F$ (a pre-placed tuning pull, or a Step-0 pull at a pullable far contact), the argument for F_{low} applies verbatim, since it does not use $j \in F$: $t|b^\pm(j)| \le t|V(j)| \le 24\lambda_l v_l(j) = |a_j^\#|$ for $t \le T_{\mathrm{hi}}(w)$; this gives (v).

(iv′) *The other conditions.* Only the base part on F changed; $\omega^\pm$ and the data off F are those of the route-V companion, so (U2) and (U7) hold as there. (U1): $t\|b^\pm\|_1 \le \sum_F tA_j + 2t\|V\|_1 + t\sum \mathrm{tr}_j + (\text{off-}F) \le 4\|a^r\|_1 + C_f(1 + D(L)t) \le A_0$; the Γ_w-bound changes by $O(Kt)$, because $\Gamma_w^{1/2}$ is a seminorm and $q_0 h(y) \le \|U\|^2\|y\|_1^2/\nu$. (U4^{inf}): the violations of the route-V companion, plus the excesses of (i), which sit at support coordinates. □

Lemma 5.20 (Fine carriers enter only through their weight)**.** *Lemma* 3.23 *and Corollary* 3.27 *hold for base rows of arbitrary support, with* $J_{\mathrm{fine}} := \mathbb{N} \setminus (F^\# \cup E_c(w))$*: step* (2) *of the route-V companion is omitted, the data on* J_{fine} *are* $b^+ := \chi V$, $b^- := -(1-\chi)V$ *with* $\chi = \frac{1}{2}$ *at free coordinates and* $\chi \in [0,1]$ *given by [I, Lemma 8.26] at contacts,* $\|V1_{J_{\mathrm{fine}}}\|_1 \le C_f \mathrm{Des}(L)c_{L+1}$*, the violation on* J_{fine} *is at most* $\|V1_{J_{\mathrm{fine}}}\|_1$ *with* $b^\theta = 0$ *at violated free coordinates, and* (U5′) *holds with* $x_0 := c_f \,\mathrm{gap}_{\min}\Phi_{\min}(L)$*. In Theorem* 5.10 *hypothesis* (U5) *may then be replaced by* (U5′) *provided* $4c_{\mathrm{fine}} \le c_{\mathrm{late}}\delta s_1$*, which holds under* ($\mathrm{W}_{\mathrm{exp}}$) *for* $s_1 \ge e^{-1/(2T)}$ *and* T *small.*

Proof. The proofs of Lemma 3.23 and Corollary 3.27 do not use finiteness of F: a coordinate of J_{fine} lies in no coarse target and in no near part $S_l \cap [1, s_{\mathrm{far}}(w)]$, so coarse carriers contribute to $V(j)$ only through $S_l \cap (s_{\mathrm{far}}(w), \infty)$, at most $C_f D(L)c_{L+1}^2$ in total, and fine carriers at most $C_\Delta \sum_{k>L}\lambda_k$. At a free j, $|b_j^+| + |b_j^-| = |V(j)|$ and $b_j^\theta = 0$; at a contact, $(z_j b_j^+)_- + (z_j b_j^-)_+ = (z_j V(j))_-$. (U5′) follows from the coarse statuses at $f^\#$ (Lemma 5.37(c)); without step (2) there is no later drift. The replacement of (U5) is Lemma 3.23, whose proof is local in the scrambled sets and does not involve F. Finally $4c_{\mathrm{fine}} \le 4C\sum_{k>L}\lambda_k \le Ce^{-1/T} \le c_{\mathrm{late}}\delta e^{-1/(2T)}$ for T small. □

Remark 5.21. At arbitrary base support the static-ownership recursion of the finite-support construction would require the signature sets of fine carriers to avoid F, which cannot be arranged. Lemma 5.20 removes the question: only the total weight c_{L+1} of the fine carriers is used, against $s_1 \ge e^{-1/(2T)}$, and this is where ($\mathrm{W}_{\mathrm{exp}}$) is essential.

5.4. **Levers inside the support and the preparation of the base row.** At finite base support every lever of the route-V construction acts at coordinates off the support. At arbitrary support the near and far coordinates of a carrier may lie in F, and the moves available there have first-order side effects on all values (common modes). This subsection collects the first-order calculus and the preparation of the base row.

Depth. Let Λ_L be the set of coarse carriers, $U_L := \bigcup_{k\in\Lambda_L} \operatorname{supp} u_k$, and let $\bar d(L)$ be the least $D \ge d(L)$ with

$$\max_{k\in\Lambda_L} \|u_k 1_{(D,\infty)}\|_1 \le 2^{-L^2}\big(|\Lambda_L| H(L)\big)^{-4}$$

(it exists since the u_k are finitely many vectors of ℓ_1), and $v_{\min}(L) := \min\{|u_k(s)| : k\in\Lambda_L,\ s\in \operatorname{supp} u_k \cap [1,\bar d(L)]\} > 0$. Coordinates $s \le \bar d(L)$ are said to be at *design depth*. By Definition 5.42, $\mathrm{Des}(L) \ge 2^{\bar d(L)} \bar d(L) H(L)/((\mu^\star_{\bar d(L)})^2 v_{\min}(L))$, so a mass change at $s \le \bar d(L)$ has normalized efficiency $(\mu^\star_s)^2/\nu \ge 1/\mathrm{Des}(L)$ at rows with $\nu \le 1$.

Definition 5.22 (Resources)**.** Fix a row f_1 (the input row of a linear system: of Step A, or the pre-row of Definition 5.8), a set Λ of coarse carriers to be tuned and a scale x of the system. For a coarse carrier l put $\mathrm{val}_l := \epsilon_l u_l(\hat x_1)$, with ϵ_l the sign of the switching of l if l carries data and a fixed sign otherwise, and $g_k := \epsilon_k \langle U^* u_k, e_1\rangle$. Let

$$x'_{\max} := \frac{u^2 T^3}{\mathrm{Des}(L)^2 |\Lambda|}, \qquad \mathrm{Far}_l(x) := \{j \in S_l : x \le v_l(j) \le x'_{\max}\}.$$

A far j is *pullable* for l if $j \notin F$ and $\epsilon_l z_1(j) > -\frac12$, or $j \in F$, $s_j = \epsilon_l$ and $|a_j| \le x'_{\max}$ (then j is first lowered to a contact). A coarse carrier is *zero-data* if its coefficient $c_l(t)$ vanishes for all $t \in \mathcal{S}$ (inactive Ω-carriers, held carriers, pushed carriers, coarse peaks of inactive blocks), and *data-carrying* otherwise. The *resources* of l are:

- (P1) l zero-data, and a near $s \in N_l$ off F or thin ($|a_s| \le b$): the composite move at s (Lemma 5.24(iii));
- (P2) (TU) l data-carrying, a near bank coordinate $s \in N_l$ which is a contact of sign ϵ_l or a thin support coordinate with $s_s = \epsilon_l$, and a pullable $j_p \in \mathrm{Far}_l(x)$ ([II, Lemma 4.15]);
- (P3) (Hilbert pair) two thick $s, s' \in N_l$ ($|a_s|, |a_{s'}| \ge u$) with robust spread $|1 - \mathfrak{a}_s/\mathfrak{a}_{s'}| \ge u$, $\mathfrak{a} := a/v_l$ ([II, Lemma 6.46(b)]);
- (P4) (single mass) a thick $s \in N_l$: a mass change at s;
- (P5) (anchor) a thick $p \in F \cap [1,\bar d(L)]$ with $p \notin U_L$.

l is *privately tunable* if it has (P1), (P2) or (P3).

The choice of $x'_{\max}$ (rather than a larger far depth) is what makes the costs of far moves $o(u^2T^2)$, as required by Theorem 5.12(E-e) with $c_{\mathrm{flat}} \ge c_f u$ (Lemma 5.46).

Lemma 5.23 (Exact identity and signed dichotomy)**.** *Let f_1 be a row with data $(a,z,e,\nu,\hat x)$ and $F = \operatorname{supp} a$. For $u \in \ell_1$ put $\varkappa_u := \langle U^*u, e\rangle/\nu$, $D_u(s) := u(s) - \varkappa_u a_s$ and $c_s(u) := D_u(s)\operatorname{sign} a_s$ for $s \in F$, $c_s(u) := u(s) z_s$ for $s \notin F$. Then for every $D \ge 1$, with $\alpha(D) := \sum_{s\in F,\, s>D} |a_s|$,*

$$u(\hat x) = \varkappa_u\big(1-\alpha(D)\big) + \sum_{s\le D} c_s(u) + \sum_{s>D} u(s) z_s, \qquad 1-\alpha(D) \ge \nu. \tag{7}$$

Consequently: if $\varkappa_u \ge 0$, then $\sum_{s\le D}(c_s(u))_- \ge \langle U^*u,e\rangle - u(\hat x) - \|u 1_{(D,\infty)}\|_1$; if $\varkappa_u \le 0$, then $\sum_{s\le D}(c_s(u))_+ \ge |\langle U^*u,e\rangle| + u(\hat x) - \|u 1_{(D,\infty)}\|_1$.

Proof. $u(\hat x) = u(z) + \langle U^*u, e\rangle = \sum_F u(s)\operatorname{sign} a_s + \sum_{F^c} u(s) z_s + \varkappa_u \nu$, and $\sum_F u(s)\operatorname{sign} a_s = \varkappa_u \|a\|_1 + \sum_F D_u(s)\operatorname{sign} a_s$; $\|a\|_1 + \nu = 1$. For $s \in F$, $s > D$, write $D_u(s)\operatorname{sign} a_s = u(s) z_s - \varkappa_u |a_s|$; then $\varkappa_u(\nu + \|a\|_1 - \alpha(D)) = \varkappa_u(1-\alpha(D))$ and $1 - \alpha(D) = \|a 1_{[1,D]}\|_1 + \nu$. If $\varkappa_u \ge 0$, $\varkappa_u(1-\alpha(D)) \ge \varkappa_u \nu = \langle U^*u, e\rangle$, and (7)

with $|z| \le 1$ gives $\sum_{s\le D} c_s(u) \le u(\hat{x}) - \langle U^*u, e\rangle + \|u1_{(D,\infty)}\|_1$, whence the bound on the negative parts; the case $\varkappa_u \le 0$ is symmetric. □

Lemma 5.24 (First-order actions)**.** *Let f_1 be a row as in Lemma* 5.23 *and $u \in \ell_1$. Up to remainders at most $C\|U^*u\|(\mu_s^\star|\mathrm{d}m|/\nu)^2$ for $|\mathrm{d}m| \le \nu/(2\mu_s^\star)$, and after renormalization by q^* (which changes neither e nor z):*

(i) *a change $\mathrm{d}m$ of a_s at $s \in F$ (sign kept) changes $u(\hat{x})$ by $((\mu_s^\star)^2/\nu)D_u(s)\,\mathrm{d}m$;*
(ii) *a new mass $\mathrm{d}m\, z_s$ ($\mathrm{d}m \ge 0$) at a contact $s \notin F$ changes $u(\hat{x})$ by $((\mu_s^\star)^2/\nu)u(s)z_s\,\mathrm{d}m$;*
(iii) *a change $\mathrm{d}z$ of z_s at $s \notin F$ changes $u(\hat{x})$ by exactly $u(s)\,\mathrm{d}z$. At a zero-data contact s the* composite move *(for $P > 0$ a bank of mass $(\nu/(\mu_s^\star)^2)Pz_s$, for $P \le 0$ the move $z_s \mapsto z_s(1+P)$) is $C^{1,1}$ in P with slope $u(s)z_s$ on both sides.*

In particular, for a single mass (P4) *at a thick $s \in N_l$, normalized so that a unit step changes* val_l *privately by* 1*, every coarse k satisfies* $\mathrm{dval}_k = (\delta_{kl} - g_k h_l)\,\mathrm{d}P$ *with the signed coefficient $h_l := \epsilon_l a_s/(\nu v_l(s))$, $|h_l| \le H(L)/\nu$; for an anchor* (P5) *the same holds without the private term. Every move of the size x at design depth moves e by at most $C\,\mathrm{Des}(L)x$ and no z outside its own coordinate.*

Proof. U^*a changes by $\mu_s^\star \mathrm{d}m\, k_s$, and the derivative of $y \mapsto y/\|y\|$ at U^*a is $(I - ee^T)/\nu$; with $e_s := \langle k_s, e\rangle = \mu_s^\star a_s/\nu$ we get $\mathrm{d}e = (\mu_s^\star \mathrm{d}m/\nu)(k_s - e_s e)$ and $\langle U^*u, k_s - e_s e\rangle = \mu_s^\star D_u(s)$. The second derivative of $y \mapsto y/\|y\|$ is at most $3\|\mathrm{d}y\|^2/\|y\|^2$. (ii) is (i) with $a_s = 0$; (iii): $\hat{x} = z + Ue$ and e does not depend on z. For the composite move both branches have derivative $u(s)z_s$ at $P = 0$, and the bank branch is smooth. For (P4): for $k \ne l$ coarse, $u_k(s) = 0$ (s lies beyond $s_{\max}(L)$ in S_l); by (i) a mass change $\mathrm{d}m$ changes $\epsilon_k u_k(\hat{x})$ by $\epsilon_k((\mu_s^\star)^2/\nu)(u_k(s) - \varkappa_{u_k} a_s)\mathrm{d}m$; with $\mathrm{d}P := \epsilon_l((\mu_s^\star)^2/\nu)v_l(s)\mathrm{d}m$ this is $(\delta_{kl} - g_k h_l)\mathrm{d}P$. The last claim is Lemma 3.2(ii) and the efficiency bound above. □

The other resources act as follows (proved in the cited places): (P1) has slope exactly 1 on both sides and affects no other coarse carrier (Lemma 3.10); (P2) shifts val_l down by at least $v_l(j_p)/2 \ge x/2$ through the pull, and the bank provides every upward correction continuously, with second-order cross effects [II, Lemma 4.15]; (P3) is private and two-sided with efficiency $\ge u/\mathrm{Des}(L)$ and cross effects $O(\mathrm{Des}^2 u^{-4}x^2)$ [II, Lemma 6.46(b)]. A far coordinate gives only a *one-sided* private move: a contact is at the end of its range, so z-moves there lower val_l only, and a bank there has negligible efficiency.

Lemma 5.25 (Offsets do not create two-sidedness)**.** *Let N be a square matrix with a left null vector y, c a column with $y^Tc \ne 0$, and F^0 a vector. If (P, P_0) solves the linear system $F^0 + NP + cP_0 = 0$, then $P_0 = -y^TF^0/(y^Tc)$. In particular, for a one-sided lever with column c (a bank, admissible only for $P_0 \ge 0$ measured from zero lever mass) the total position is forced and does not depend on any offset placed along the lever in the same row; if $y^TF^0/(y^Tc) > 0$, the linearized system has no admissible solution.*

Proof. Apply y^T. An offset m_0 placed in the row from which F^0 is computed is a move along c, so it changes F^0 by $c\,m_0$ and the remaining position by $-m_0$; the total is unchanged. □

So a bank placed *with an offset* at the row where the linear system is solved, as a block-norm lever along a degenerate direction, is one-sided and cannot replace a two-sided lever. The remedy below is different: the offsets are placed in the *base row*, whose values are the targets of the system, so that the forced position is the offset itself up to $O(r)$.

Lemma 5.26 (Exact drift identity)**.** *Let f° be a row with data $(a^\circ, z^\circ, F^\circ, \nu^\circ, e^\circ)$, and let $u = \sum_k c_k u_k$ be a finite combination of carrier vectors with $u|_{F^\circ} = \varkappa a^\circ|_{F^\circ}$. Let f^1 be a row with data (a^1, z^1) such that $z^1 = z^\circ$ on $\operatorname{supp} u$ and $(\operatorname{supp} a^1 \setminus F^\circ) \cap \operatorname{supp} u = \emptyset$. Then*

$$u(\hat{z}^1) - u(\hat{z}^\circ) = -\frac{\varkappa\nu^\circ}{2}\|e^1 - e^\circ\|^2.$$

Proof. $u(\hat{z}) = u(z) + \langle U^*u, e\rangle$ and $u(z^1) = u(z^\circ)$. $U^*u = \varkappa U^*(a^\circ 1_{F^\circ}) + U^*(u1_{(F^\circ)^c}) = \varkappa\nu^\circ e^\circ + \sum_{s\notin F^\circ} \mu_s^\star u(s)k_s$, while $e^1 = \sum_{s\in\operatorname{supp} a^1} \mu_s^\star a^1(s)k_s/\nu^1$; so $\langle U^*u, e^1\rangle = \varkappa\nu^\circ\langle e^\circ, e^1\rangle$ (the cross terms vanish by hypothesis) and $\langle U^*u, e^\circ\rangle = \varkappa\nu^\circ$. The difference is $\varkappa\nu^\circ(\langle e^\circ, e^1\rangle - 1) = -(\varkappa\nu^\circ/2)\|e^1 - e^\circ\|^2$. □

Remark 5.27 (Why the original lever set can be obstructed)*.* Lemma 5.26 allows arbitrary raises and lowerings on F°, truncations to contacts of the same sign, and masses and z-moves outside $\operatorname{supp} u$. So, at a row where a combination u of carriers is proportional to a° on the support, support moves change $u(\hat{z})$ only at second order and only in one direction (it decreases if $\varkappa > 0$), and the only two-sided first-order resources for u are moves on $\operatorname{supp} u \setminus F^\circ$. Applied to the left null vector of the d-consistency system built from single masses on the support and block-norm levers (which never act on $\operatorname{supp} u \setminus F^\circ$), the same computation, keeping the cross term, shows that the corresponding combination of the equations equals $-(\varkappa\nu_0/2)\|e - e_0\|^2$ plus the first-order effect of the window masses on $\operatorname{supp} u \setminus F_0$ plus the cut-off beyond N''; when the window term is of first order in s_1, exact d-consistency with those levers is impossible for small s_1. Theorem 5.31 avoids this by preparing the base row: the coordinates of $\operatorname{supp} u \setminus F_0$ at design depth then carry banks and become two-sided resources.

Definition 5.28 (Preparation)**.** Let $\theta_0 = \theta_0(w) := b/\mathrm{Des}(L)^{C_P}$ (Definition 5.42). The *prepared row* f^P is obtained from the row after the deep raise and Step 0, and before the Łojasiewicz step of Step A, by:

- (P_1) closing to a contact every free $s \le \bar{d}(L)$, $s \in U_L$, with tiny room $1 - |z_s| \le b$;
- (P_2) placing a bank of mass $\theta_0 z_s$ at every contact $s \le \bar{d}(L)$, $s \in U_L$;
- (P_3) raising, with its sign, every $s \in F \cap U_L$, $s \le \bar{d}(L)$, with $|a_s| < \theta_0$, to $|a_s| = \theta_0$.

Lemma 5.29 (Properties of the preparation)**.** *Let $L \ge l_f$ and w clean.*

- (a) $p^*(f^P - f^r) \le C_f \mathrm{Des}(L)\, b \log(1/b)$.
- (b) *Every value moves by at most $b + C\bar{d}(L)\theta_0 \le 2b$ (the closing moves of* (P_1) *by at most b, the banks and raises by at most $C\bar{d}(L)\theta_0$), every rate object by at most $2\,\mathrm{Lip}\cdot b$; the clean classification survives with the factor* 2.
- (c) *At a banked coordinate s the route-V data are those of a contact of sign z_s: $b^+ = \chi V' + \beta_s$, $b^- = -(1-\chi)V' + \beta_s$ with $z_s V' \ge 0$ up to the fine residue, and $|\beta_s| \le 4Kt|a_s^\#|$; hence* ($\mathrm{U3}^{\mathrm{inf}}$) *holds there with $C_+ = 5$. At raised thin coordinates the data of Proposition* 5.19 *satisfy* ($\mathrm{U3}^{\mathrm{inf}}$) *a fortiori.*

(d) *At the companion and at every pre-row of Definition* 5.8 *built at it, every* $s \le \bar{d}(L)$ *with* $s \in U_L$*, or* $s \in F$ *with* $|a_s| \ge u/2$*, is of one of the types:* TS *(thick support,* $|a_s| \ge u/2$*; an* anchor *if* $s \notin U_L$*);* $\mathrm{S_{th}}$ *(support with* $|a_s| \ge \theta_0/2$*: banked contacts, raised thin coordinates);* FR *(free with room* $\ge u/2$*; zero data or tolerated violations with* $b^\theta = 0$*);* ZD *(a coordinate of a zero-data carrier, in any state). Every other* $s \le \bar{d}(L)$ *is off* U_L *and either off* F *or thin,* $|a_s| \le 2b$*.*

Proof. (a) Banks off F and same-sign raises: the costs of [II, Lemma 2.15] and [II, Lemma 2.12(iv)], with value moves bounded as in (b) and $c(\delta) \le C\epsilon \log(e/\epsilon)$; there are at most $2\bar{d}(L)$ such moves of size $\le \theta_0$. The closing moves of ($\mathrm{P_1}$) are z-moves of size up to b at up to $\bar{d}(L)$ coordinates; by [II, Lemma 2.12(iv)] their normer change costs $C_f c(\delta)$ with value moves of order b, that is at most $C_f(|\Lambda_L| + \bar{d}(L))b\log(1/b)$. As $|\Lambda_L| + \bar{d}(L) \le \mathrm{Des}(L)$, (a) follows. (b) Lemma 5.24: each move changes $u_k(\hat{x})$ by at most $\theta_0\|u_k\|_1 \cdot C$ (masses: efficiency ≤ 1; closing a room: $|u_k(s)|(1 - |z_s|) \le b|u_k(s)|$, so the closing moves change each value by at most $b\|u_k\|_1 \le b$ in total). The rate objects are Lipschitz in the values, moduli and rooms; the clean dichotomy at depth $\bar{d}(L)$ (Lemma 5.44) is preserved with factor 2. (c) A banked s was a contact of f^r in the support of a coarse carrier. If the carrier carries data, the route-V data at s are the contact row of the class cone (row (X1), or ($\mathrm{X1}^F$)), so V' is z_s-signed up to the fine residue; the companion support outside $\operatorname{supp} a^r$ is treated as a set of contacts in the transplant of the route-V companion, which gives the stated form with the base term $\beta_s = -\varkappa a_s^\# / a_Z^\#(\hat{z}^\#)$, with the normalization of Proposition 5.19: $|\varkappa| \le Kt$ and $a_Z^\#(\hat{z}^\#) \ge \frac{1}{4}$. Then $(z_s b^+)_- \le |\beta_s|$ and $(z_s b^-)_+ \le |\beta_s|$, and $|\beta_s| \le 4Kt|a_s^\#| \le |a_s^\#|/t$ as $4Kt^2 \le 1$; this is Proposition 5.19(iv). Zero-data coordinates carry only fine residues. (d) By Lemma 5.44, at f^r every $s \in U_L \cap [1, \bar{d}(L)]$ is thick, thin, a contact, free with robust room, or free with tiny room; ($\mathrm{P_1}$)–($\mathrm{P_3}$) turn thin coordinates, contacts and tiny rooms into $\mathrm{S_{th}}$ (zero-data ones are also ZD). Step A moves TS and FR coordinates by at most $\mathrm{Des}^C x'_{\max} \ll u$ and uses $\mathrm{S_{th}}$ coordinates with data only through raises (Lemma 5.37); the pre-row moves design-depth coordinates by at most $K_w\Pi \ll \theta_0$ (Lemma 5.45); truncation and deep raise act beyond $\bar{d}(L)$. □

5.5. **Uniform solvability of the d-consistency system.** Let f_0 be the prepared companion after Step A, Ω the set of all coarse strict non-peaks of f_0 (no absorbers in route V), and f_1 a pre-row of Definition 5.8 at f_0. The *d-consistency system* is $F(P) = 0$ with $F_l(P) := \epsilon_l[u_l(\hat{x}(P)) - (A_m(P)/A_m^0)u_l(\hat{z}_0)]$ ($l \in \Omega_m$), $A_m^0 := A_m(f_0)$, as in Lemma 3.10. Put $p_l := \epsilon_l u_l(\hat{z}_0)/A_m^0$, $q_k := \epsilon_k \lambda_k w_m^0(k)$ ($k, l \in \Omega_m$), $g_k := \epsilon_k \langle U^* u_k, e_1\rangle$, $G_m := \langle U^* R_m^* w_m^0, e_1\rangle = \sum_k \lambda_k w_m^0(k)\langle U^* u_k, e_1\rangle$ (all carriers k of block m) and $\tilde{g} := g - \sum_m G_m p_m$; here p_m, q_m are the restrictions to Ω_m. Then $q_m^T p_m \le C_m^0 \le \frac{1}{2}$ (proof of Lemma 3.10), $\|q_m\|_2 \le 1$ and $\|p_m\|_2 \le 2|\Omega|^{1/2}/A_{\min}$.

Lemma 5.30 (Left combinations)**.** *For* $y \in \mathbb{R}^\Omega$ *put* $\beta_m := y^T p_m$ *and* $\Psi_y := \sum_{l\in\Omega} y_l \epsilon_l u_l - \sum_m \beta_m R_m^* w_m^0 \in \ell_1$*. Then:*

(a) $\Psi_y(\hat{z}_0) = 0$*;*

(b) *for every move* δ *of Lemma* 5.24 *at* f_1 *(or a composite of such moves), with* $c(\delta)$ *the first-order change of* F *and* $l_\delta(u)$ *the first-order change of* $u(\hat{x})$*,* $|y^T c(\delta) - l_\delta(\Psi_y)| \le C|\delta|\,\|y\|(K_w\Pi + c_{L+1})$*;*

(c) $\langle U^*\Psi_y, e_1\rangle = y^T\tilde{g}$*.*

Proof. (a) $\sum_{l\in\Omega_m} y_l\epsilon_l u_l(\hat z_0) = A^0_m\beta_m$, and $(R^*_m w^0_m)(\hat z_0) = \langle w^0_m, R^{**}_m\hat z_0\rangle = A^0_m$ because w^0_m norms $R^{**}_m\hat z_0$. (b) $\mathrm{d}F_l = \epsilon_l\mathrm{d}u_l - p_l\mathrm{d}A_m$ and, the block norm being differentiable at $R^{**}_m\hat x_1$ with norming functional w^1_m, $\mathrm{d}A_m = l_\delta(R^*_m w^1_m) + O(|\delta|^2)$; so $y^Tc(\delta) = l_\delta(\Psi_y)+\sum_m\beta_m l_\delta(R^*_m(w^0_m-w^1_m))$, and $\|R^*_m(w^0_m-w^1_m)\|_1 \le K_w\Pi+2c_{L+1}$ (Lemma 3.14, with the fine carriers bounded by their weight). (c) is the definition of $\tilde g$. □

Theorem 5.31 (The d-consistency system is uniformly solvable)**.** *Let $L \ge l_f$, w clean, f_0 the prepared companion after Step A and f_1 a pre-row at f_0 with $\Pi \le 1/K_w$. Let C be the column matrix of the d-consistency system at f_1 formed by:*

(i) *for every $l\in\Omega$ one two-sided lever: if l is zero-data with an off-F or thin near coordinate, the composite move at it (type ZD); if l carries data and has an off-F or thin near coordinate, a mass change at it (it is banked or raised by the preparation: type S_{th}); if all near coordinates of l are thick, (P3) if a pair of robust spread exists and (P4) otherwise;*

(ii) *one elementary column at every $s \le \bar d(L)$ of type TS (anchors included) or S_{th} (mass change), FR (z-move), ZD (composite move).*

Then

$$s_{\min}(\mathsf{C}) := \min_{\|y\|_2=1}\|\mathsf{C}^Ty\|_2 \;\ge\; \varepsilon_{\mathrm{DC}} := \frac{\nu_1(\mu^\star_{\bar d(L)})^2}{16\,\bar d(L)|\Omega|H(L)\big(1+8N|\Omega|^{1/2}/A_{\min}\big)},$$

so C has the right inverse $R := \mathsf{C}^T(\mathsf{C}\mathsf{C}^T)^{-1}$ with $\|R\|_{\infty\to\infty} \le K_{\mathrm{DC}} := |\Omega|^{1/2}/\varepsilon_{\mathrm{DC}} \le C_{f,N}\mathrm{Des}(L)^{C_0}$. Consequently the moves of (i), (ii) *form a lever system (Definition* 5.9*) with $K_{\mathrm{lev}} = K_{\mathrm{DC}}$, $W_{\mathrm{pull}} = 0$ and no hypothesis on Sherman–Morrison denominators, bordered scalars, spreads or anchors: if* (L-K2) *of Lemma* 5.45 *holds, F has a zero P^* with $|P^*|_\infty \le K_{\mathrm{DC}}r$, every carrier of Ω is a strict non-peak of $f(P^*)$ with* $\mathrm{nv} = \mathrm{nv}^0$*, the coarse peaks keep their statuses, the S_{th} masses stay $\ge \theta_0/4$, and the TS and FR coordinates stay robust.*

Proof. Let y be a unit vector with $\|\mathsf{C}^Ty\|_2 < \varepsilon$; we show $\varepsilon \ge \varepsilon_{\mathrm{DC}}$. Put $\gamma := y^T\tilde g$.

Step 1 (shape of the lever columns). By Lemma 5.24 and the linearization of Lemma 3.10, the lever of l, with its parameter scaled so that its private slope is 1, has the column $e_l - h_l\tilde g - p_{m(l)}q_l + E_l$, with h_l the signed common-mode coefficient ($h_l = 0$ for (P3) and ZD levers, $|h_l| \le H(L)/\nu_1$ always) and $\|E_l\| \le \varepsilon_E := C_f\mathrm{Des}(L)^C(K_w\Pi + c_{L+1}) \le \varepsilon_{\mathrm{DC}}/8$ by (L-K2). Hence $y_l = \beta_{m(l)}q_l + \gamma h_l + e'_l$ with $e'_l := (\mathsf{C}^Ty)_l - y^TE_l$, $|e'_l| \le |(\mathsf{C}^Ty)_l| + \varepsilon_E$.

Step 2 (γ is robust). Since $q^T_m p_{m'} = 0$ for $m \ne m'$, taking the scalar product with p_m gives $\beta_m(1 - q^T_m p_m) = \gamma h^Tp_m + e'^Tp_m$ (restrictions to Ω_m), so $|\beta_m| \le 4(|\gamma|\|h\|_2 + \|e'\|_2)\|p_m\|_2$, and

$$\begin{aligned} 1 = \|y\|_2 &\le 2\Big(\sum_m |\beta_m|\|q_m\|_2 + |\gamma|\|h\|_2 + \|e'\|_2\Big)\\ &\le 2\big(|\gamma|\|h\|_2 + \|e'\|_2\big)\big(1 + 8N|\Omega|^{1/2}/A_{\min}\big).\end{aligned}$$

If $\|e'\|_2 \ge 1/(4(1+8N|\Omega|^{1/2}/A_{\min}))$, then $\varepsilon \ge \varepsilon_{\mathrm{DC}}$ already. Otherwise $h \ne 0$ and $|\gamma| \ge \gamma_{\min} := 1/(4\|h\|_2(1+8N|\Omega|^{1/2}/A_{\min})) \ge \nu_1/(4|\Omega|H(L)(1+8N|\Omega|^{1/2}/A_{\min}))$; replacing y by $-y$ we may assume $\gamma > 0$.

Step 3 (the combination). By Lemma 5.30, $\Psi_y(\hat z_0) = 0$ and $\langle U^*\Psi_y, e_1\rangle = \gamma$. Split $\Psi_y = \Psi^c + \Psi^f$, Ψ^f the fine-carrier part of $-\sum_m\beta_m R^*_m w^0_m$, with $\|\Psi^f\|_1 \le \sum_m|\beta_m|\sum_{k>L}\lambda_k \le C(\gamma/\gamma_{\min})c_{L+1}$. The coefficients of Ψ^c are at most $C\gamma/\gamma_{\min}$,

so by the choice of $\bar{d}(L)$, $\|\Psi^c 1_{(\bar{d}(L),\infty)}\|_1 \le \gamma/16$ for $L \ge l_f$. At f_1, $|\Psi^c(\hat{x}_1)| \le |\Psi_y(\hat{x}_1) - \Psi_y(\hat{z}_0)| + \|\Psi^f\|_1 \le \|U^*\Psi_y\|\|e_1 - e_0\| + \|\Psi_y 1_{(N'',\infty)}\|_1 + \|\Psi^f\|_1 \le \gamma/16$ (the pre-row changes z only beyond N''), and $\langle U^*\Psi^c, e_1\rangle \ge \frac{15}{16}\gamma$.

Step 4 (a resource). Lemma 5.23 at f_1 with $u := \Psi^c$ and $D := \bar{d}(L)$ gives some $s \le \bar{d}(L)$ with $|c_s(\Psi^c)| \ge \gamma/(2\bar{d}(L))$. If $s \notin U_L$, then $\Psi^c(s) = 0$, so $s \in F_1$ and $|c_s| = \varkappa|a_s|$ with $\varkappa = \langle U^*\Psi^c, e_1\rangle/\nu_1 \le 2\gamma/\nu_1$; hence $|a_s| \ge \nu_1/(4\bar{d}(L)) \gg 2b$, and s is TS (an anchor). (Thin $s \notin U_L$ contribute at most $2\varkappa b$ each, in total $\ll \gamma/16$.) If $s \in U_L$, s has type TS, $\mathrm{S_{th}}$, FR or ZD by Lemma 5.29(d). In all cases C contains the elementary column at s.

Step 5 (contradiction). By Lemma 5.30(b) and Lemma 5.24, $|y^T c_s| \ge \mathrm{eff}_s|c_s(\Psi_y)| - \mathrm{err}$ with $\mathrm{eff}_s \ge (\mu^\star_{\bar{d}(L)})^2$ for masses ($\nu_1 \le 1$) and $\mathrm{eff}_s = 1$ for z-moves and composite moves, and $|c_s(\Psi_y) - c_s(\Psi^c)|$ negligible. So $\varepsilon > |y^T c_s| \ge (\mu^\star_{\bar{d}(L)})^2\gamma_{\min}/(4\bar{d}(L)) \ge \varepsilon_{\mathrm{DC}}$.

Step 6 (nonlinear solution). Put $G(y) := F(Ry) = F(0) + y + \mathrm{Rem}(Ry)$ on the cube $|y|_\infty \le r$. On the face $y_l = r$, $G_l \ge r - |F(0)|_\infty - C_2K_{\mathrm{DC}}^2 r^2 \ge 0$ when $K_{\mathrm{DC}}|F(0)|_\infty \le r/2$ and $C_2K_{\mathrm{DC}}^2 r \le \frac{1}{4}$, with C_2 a design constant (second derivatives of Lemma 5.24 at depth $\le \bar{d}(L)$ and of the levers, the composite moves being $C^{1,1}$); symmetrically on $y_l = -r$; the Poincaré–Miranda theorem gives a zero. The $\mathrm{S_{th}}$ masses move by at most $\mathrm{Des}(L)K_{\mathrm{DC}}r \le \theta_0/4$, TS and FR coordinates by $\ll u$; values move by at most $C_{\mathrm{lev}}K_{\mathrm{DC}}r$, below every gap and margin (Lemma 5.45). The protection (PR) holds: TS and $\mathrm{S_{th}}$ coordinates are of type (PRβ) since their masses stay above half their pre-row values, and FR and ZD coordinates are of type (PRα) (Proposition 5.19(iii) and Lemma 5.29(d)). □

Remark 5.32 (The degenerate configurations are bypassed). (a) Steps 2–5 are a uniform dichotomy: a near-null left vector of the system has a robust common mode γ, and then the identity of Lemma 5.23 forces a robust first-order two-sided resource at design depth. No rate object for Sherman–Morrison denominators, bordered scalars or spreads is needed. The *square* lever system (one lever per carrier) may still be degenerate; it is the enlarged column system that is uniformly onto. (b) The resource of Step 4 is never the lever coordinate of an Ω carrier, since there $|c_s(\Psi_y)| \le |y^T(\mathrm{lever})|/\mathrm{eff}_s + \mathrm{err}$ is small by Step 1. (c) No TU lever and no pull at scale r is used. (d) Without the preparation the resource of Step 4 may be a data-carrying contact, where only a bank is available, and a bank placed at the pre-row is one-sided along the degenerate direction (Lemma 5.25); after ($\mathrm{P_2}$) the offset lies in the base row, whose values are the targets of the system, and the forced position is $\theta_0 + O(K_{\mathrm{DC}}r) > 0$.

Lemma 5.33 (Tolerance variant)**.** *In Theorem* 5.10 *replace exact d-consistency on* Ω *by*

$$\text{(TOL)} \qquad |\mathrm{mis}_{i,m}| \le s_1 \ \text{ for every piece } i \text{ and block } m,$$
$$\mathrm{mis}_{i,m} := (d'_m - d_m)(\Delta\omega_{i,m}),$$

and add (LATE-tol) $C_f|\Omega|A_2^2K_w^2 s_1 \le c_{\mathrm{late}}\delta T^2$. *Then all conclusions of Theorem* 5.10 *hold, with* K_w *enlarged by an* f*-constant factor. In particular an approximate zero of the d-consistency system with* $|F|_\infty \le c_f s_1/|\Omega|^{1/2}$ *suffices.*

Proof. d-consistency is used in exactly four places. (u1) Step 2: without it, $\Omega_m^\diamond = \rho(\omega_m^\diamond - d'_m(\omega_m^\diamond)w'_m) + \kappa_m^\diamond w'_m + \rho(d_m^\diamond - d_m^\theta)(w'_m - w_m^0)$ with $\kappa_m^\pm = \mp\frac{\rho}{2}\mathrm{mis}_m$ and

$\kappa_m^\theta = 0$. (u2) Step 3: $W_0 = (1+\tau\kappa)(w' + t''(\omega^\diamond - d'(\omega^\diamond)w'))$ with $t'' = \tau\rho/(1+\tau\kappa)$ and $|\tau\kappa| \le \frac{1}{16}$, so the radius conditions hold with c_{flat} replaced by $\frac{16}{15}c_{\mathrm{flat}}$ (absorbed by shrinking c^f_{flat}), and $\mathfrak{l}'_m = \tau\kappa^\diamond_m + r_m\langle V^{\mathrm{an}}_m - w'_m, R_m x'\rangle/\sigma'_m$, so $|\mathfrak{l}'_m| \le \frac{\rho}{2}|\tau|s_1 + \frac{\delta}{64|I|}|\tau|s_1 \le \tau^2$ for $|\tau| > s_1$; for $|\tau| \le s_1$ the θ piece is used and $\mathfrak{l}'_m = 0$. Step 4 uses $\mathfrak{l}'_m$ only through $|\mathfrak{l}'_m| \le \tau^2$. (u3) In Lemma 3.14(c), $|d'_m(\omega) - d_m(\omega)| \le |\Omega|(A_2/t)K_w\Pi$ and $|H'_m(\omega) - (C_m/C'_m)H_m(\omega)| \le C_f|\Omega|A_2^2K_w\Pi/t^2$; in Step 5 this is a relative error at most $C_f|\Omega|A_2^2K_w^2 s_1/T^2 \le \delta/32$ by (LATE-tol); gaps move by at most $CK_w\Pi \le \mathrm{gap}/2$. (u4) In Lemma 3.14(g), $g_i'' - g_i$ acquires $\sum_m (d_m - d'_m)(\omega^\theta)R^*_m w'_m$, of ℓ_1-norm at most $N|\Omega|A_2K_w\Pi/t_i$. Finally $|\mathrm{mis}_{i,m}| \le (m/A'_m)\|D_m\Delta\omega_{i,m}\|_2\|F|_{\Omega_m}\|_2$ because $\mathrm{nv}'_l - \mathrm{nv}_l = \epsilon_l\lambda_l F_l/A'_m$. □

Theorem 5.31 gives exact zeros, so Lemma 5.33 is a robustness statement and is not used below.

5.6. **Step A: cone realization and the structure of its obstructions.** Step A of the route-V companion (Section 4.3) sets the ratios of the active switching carriers to the Łojasiewicz targets on the exact set $Z_\varkappa$, holds the nearly neutral inactive carriers of (T)-blocks exactly at 0, and keeps the other coarse strict non-peaks at their ratios. Downstream only *ratios* are used: the class cone in ratio variables ((X3) of [II, Definition 5.52]), the statuses (Lemma 2.4) and the true shift. So the block scalars may float, and we tune the ratios of all coarse strict non-peaks (this is necessary as soon as a common-mode resource is used, since a common mode moves every value at first order).

Definition 5.34 (The Step-A system)**.** Let f_1 be the row after the deep raise, Step 0 (in the variant described before Definition 5.49) and the preparation. Let Λ_A be the set of coarse strict non-peaks of f_1 together with the carriers pushed inward by Step A; for a block m let $\varkappa_m := \varkappa(\hat\zeta_m; \Omega_m)$ be the block scalar of Subsection 2.1. The *target ratios* rt_l are: $\epsilon_l r''_l$ for active carriers (r'' the targets on $Z_\varkappa$), 0 for the held nearly neutral carriers of (T)-blocks, the ratio at f_1 for the other held carriers, and the pushed ratio for pushed carriers. The *Step-A system* is $\tilde F(P) = 0$, $\tilde F_l := \mathrm{val}_l - \mathrm{rt}_l\varkappa_{m(l)}$ ($l \in \Lambda_A$), $\mathrm{val}_l = \epsilon_l u_l(\hat x)$; at f_1, $|\tilde F(f_1)|_\infty \le x$, where $x := C\,\mathrm{Des}(L)\eta'$ is the Step-A scale.

While the peak set of block m is fixed, $\varkappa_m$ is a C^1 function of the block vector $\zeta_m = (\lambda_k u_k(\hat x))_k$, positively homogeneous of degree 1 (the threshold equation is invariant under $\zeta \mapsto c\zeta$, $\bar\vartheta \mapsto c\bar\vartheta$). Put $K_m := \sum_k (\partial\varkappa_m/\partial\zeta_m(k))\lambda_k u_k \in \ell_1$ (derivatives at the row considered), $G^A_m := \langle U^* K_m, e_1\rangle$ and $\tilde g^A := g - (\mathrm{rt}_l G^A_{m(l)})_l$.

Lemma 5.35 (Euler combination)**.** *For $y \in \mathbb{R}^{\Lambda_A}$ put $\beta^A_m := \sum_{l\in\Lambda_A,\, m(l)=m} y_l \mathrm{rt}_l$ and $\Psi^A_y := \sum_l y_l\epsilon_l u_l - \sum_m \beta^A_m K_m$. Then* (a) $\Psi^A_y(\hat x_1) = y^T\tilde F(f_1) =: \varphi_y$, $|\varphi_y| \le x\|y\|_1$*;* (b) *for every move δ of Lemma 5.24, the first-order change of $y^T\tilde F$ is $l_\delta(\Psi^A_y)$, up to $O(|\delta|^2)$;* (c) $\langle U^*\Psi^A_y, e_1\rangle = y^T\tilde g^A$.

Proof. Euler's identity gives $\varkappa_m(f_1) = \sum_k \zeta_m(k)\partial\varkappa_m/\partial\zeta_m(k) = K_m(\hat x_1)$, whence (a); (b) is the chain rule, as $\mathrm{d}\tilde F_l = \epsilon_l \mathrm{d}u_l - \mathrm{rt}_l\, l_\delta(K_m)$; (c) follows from the definitions. □

Definition 5.36 (Generators, A-regularity, near-certificates)**.** The *generators* at f_1 are the normalized columns of the Step-A system (their action on $y^T\tilde F$ is $l_\delta(\Psi^A_y)$ by Lemma 5.35): *two-sided*: the levers (P1), (P3), (P4) of tuned carriers; TU levers

whose pull and base bank are pre-placed; one two-sided resource of a peak per block (optional); one elementary move at every design-depth coordinate of type TS (mass change), FR (z-move) and ZD (composite move); *one-sided* (allowed direction only): raises of $\mathrm{S_{th}}$ coordinates carrying data. Each two-sided generator contributes its column g and $-g$ to the set $\mathcal{G}$, each one-sided one its column. A *pre-placement* $\mathcal{P}$ is a set of discrete far moves, at most one per tuned carrier, at the smallest admissible depth in $[x, x'_{\max}]$: the pulls of TU levers with their compensating base banks. Put

$$b := -\tilde{F}(f_1 + \mathcal{P}), \qquad W := \mathrm{Des}(L)^{C_A} u^{-3} \max(x, \|b\|_2),$$
$$\rho_0 := C_2 W^2, \qquad Q := \mathrm{conv}(\mathcal{G} \cup \{0\}),$$

with C_2 the second-order constant of the generators (a design constant). The stage (L, w, a) is *A-regular* if $B(b, \rho_0) \subseteq (W/2)Q$ for some $\mathcal{P}$. A *near-certificate* of size ε is a unit y with $\max_{g\in\mathcal{G}}(-y^T g) \le \varepsilon$; $\varphi_y := y^T \tilde{F}(f_1 + \mathcal{P})$.

Lemma 5.37 (Cone realization of Step A)**.** *Let $L \ge l_f$.*

(a) *If $\varepsilon_{\mathrm{PS}} := \min_{\|y\|_2=1} \max_{g\in\mathcal{G}}(-y^T g) \ge u^3 \mathrm{Des}(L)^{-C_1}$ and $C_A \ge C_1 + 2$, the stage is A-regular with $\mathcal{P} = \emptyset$, for every right-hand side.*

(b) *If the stage is not A-regular for $\mathcal{P}$, there is a near-certificate y of size $\varepsilon < 4u^3 \mathrm{Des}(L)^{-C_A}$ with $\varphi_y > -\rho_0$.*

(c) *If the stage is A-regular, Step A is realized: there is a companion $f^{(1a)}$ with the conclusions of Step A of Section 4.3 (active ratios exactly r'' on $Z_{\varkappa}$, held values exactly 0, the threshold protection with the Puiseux margin, robust statuses kept), all moves at most W, and $p^*(f^{(1a)} - f_1) \le C_f \mathrm{Des}(L)^C W \log(1/W)$. The coarse statuses at $f^{(1a)}$ are those used in Lemma 5.20, and support signs are kept.*

Proof. (a) If some b' with $\|b'\| < \varepsilon_{\mathrm{PS}}$ were outside Q, a separating unit y would give $y^T b' < \min_g y^T g \le -\varepsilon_{\mathrm{PS}}$, which is impossible; so $B(0, \varepsilon_{\mathrm{PS}}) \subseteq Q$. Every $b' \in B(b, \rho_0)$ then lies in $((\|b\| + \rho_0)/\varepsilon_{\mathrm{PS}})Q \subseteq (W/2)Q$, since $\|b\|/\varepsilon_{\mathrm{PS}} \le \mathrm{Des}^{C_1} u^{-3}\|b\| \le W/4$ and $\rho_0/\varepsilon_{\mathrm{PS}} = C_2 W^2 \mathrm{Des}^{C_1} u^{-3} \le W/4$; the latter holds because $\max(x, \|b\|) \le \mathrm{Des}^C x'_{\max}$, so $W \le \mathrm{Des}^C T^3/u$, and $T \le (u/(4\,\mathrm{Des}(L)))^{20}$ (Section 4.1) gives $4C_2 W \mathrm{Des}^{C_1} \le u^3$ for L large.

(b) Pick $b'' \in B(b, \rho_0)$ outside the compact convex set $(W/2)Q$ and separate: a unit y with $y^T b'' < (W/2)\min(0, \min_g y^T g)$. Then $y^T b < \rho_0 - (W/2)\max(0, \max_g(-y^T g))$, so $\max_g(-y^T g) < 2(\rho_0 + \|b\|)/W \le 2C_2 W + 2u^3 \mathrm{Des}^{-C_A} \le 4u^3 \mathrm{Des}^{-C_A}$ for L large, and $\varphi_y = -y^T b > -\rho_0$.

(c) *Frozen statuses.* Compute the block scalars with the coarse peak set *frozen* at $f_1 + \mathcal{P}$: the threshold equations of [I, Lemma 1.7] with a fixed peak set form a smooth system whose Jacobian is bounded below by $\Phi^2_{P_m} \ge D(L)^{-2}$, so the frozen map is $C^{1,1}$ on the parameter set; fine carriers are computed truly (their status changes cost at most $C c_{L+1}\|t\| \ll \rho_0$). This is necessary: at a threshold crossing of an Ω carrier the derivative of the true block scalar jumps by an amount of order one, so the quadratic remainder bound fails without freezing. *Brouwer.* Write the frozen system as $\tilde{F}(f_1+\mathcal{P})+\mathcal{G}t+\mathrm{Rem}(t)$ with $t \ge 0$ indexed by $\mathcal{G}$ and $\|\mathrm{Rem}(t)\| \le C_2\|t\|^2$. For $b' \in \mathrm{cone}(\mathcal{G})$ let $S(b') := \mathrm{argmin}\{\|s\|_2 : \mathcal{G}s = b',\ s \ge 0\}$; S is continuous on $\mathrm{cone}(\mathcal{G})$ (a strictly convex parametric quadratic program whose feasible sets depend Hausdorff-continuously on b' by Hoffman's bound). On $\mathcal{K} := \{t \ge 0 : \|t\|_2 \le W\}$ the map $\mathcal{T}(t) := S(b - \mathrm{Rem}(t))$ is defined and continuous, since $\|\mathrm{Rem}(t)\| \le \rho_0$ puts $b - \mathrm{Rem}(t)$ into $B(b, \rho_0) \subseteq (W/2)Q$, which also gives $\|\mathcal{T}(t)\|_2 \le W/2$. By

Brouwer's theorem $t^* = \mathcal{T}(t^*)$ solves the frozen system with $t^* \geq 0$. *Statuses.* At t^* the ratios are the targets, so the frozen relative positions computed by Lemma 2.4 satisfy $\varrho_d \leq 1 - \mu/2$ for the near-threshold active carriers and the held values vanish; robust statuses move by at most $\mathrm{Des}^C W/u \ll u$. Hence the frozen statuses are consistent, the frozen block scalars solve the true threshold equations, and by uniqueness of the threshold solution [I, Lemma 1.7] they are the true ones. The Puiseux protection of Section 4.3 uses only the exactness of the realized ratios and held values, not the size of the moves. *Costs.* The moves have size at most $W \leq \mathrm{Des}^C x'_{\max} u^{-3}$ in value units; the cost bound is [II, Lemma 2.12(iv)] together with the costs of the pre-placements (Lemma 5.46). One-sided generators are used only in their allowed direction, so supports keep their signs. □

Lemma 5.38 (Bordered rank-one systems)**.** *Let $g, h, w \in \mathbb{R}^n$, $\beta > 0$ and $\mathsf{N} := I - gh^T$. If $|1 - h^T g| \geq \beta$ or $|h^T w| \geq \beta$, then every $b \in \mathbb{R}^n$ has a preimage under $(P, s) \mapsto \mathsf{N}P + ws$, linear in b, with $\|P\|_\infty + |s| \leq (1 + \|g\|_\infty \|h\|_1/\beta)(1 + \|w\|_\infty)(1 + \|h\|_1/\beta)\|b\|_\infty$. Every unit vector y with $\|y^T \mathsf{N}\|_2 \leq \beta' \leq \frac{1}{2}$ is within $2\beta'$ of $h/\|h\|_2$ or of $-h/\|h\|_2$.*

Proof. If $|1 - h^T g| \geq \beta$: $s := 0$ and $P := b + g\, h^T b/(1 - h^T g)$ (Sherman–Morrison). Otherwise $|h^T w| \geq \beta$: put $s := h^T b/(h^T w)$ and $P := b - ws$; then $h^T P = 0$ and $\mathsf{N}P + ws = P + ws = b$. The bounds are immediate. For the last claim write $y = th/\|h\|_2 + y_\perp$ with $y_\perp \perp h$; then $y^T \mathsf{N} = y^T - (y^T g)h^T$, whose component orthogonal to h is $y_\perp^T$, so $\|y_\perp\| \leq \beta'$, $|t| \geq (1 - \beta'^2)^{1/2} \geq 1 - \beta'^2$, and y is within $\beta' + \beta'^2 \leq 2\beta'$ of $\mathrm{sign}(t) h/\|h\|_2$. □

The structure of Step-A obstructions. Fix a unit $y \in \mathbb{R}^{\Lambda_A}$, work at $f_1 + \mathcal{P}$ (all derivatives taken there) and write $\Psi := \Psi_y^A$. Its coarse part is $\sum_{k \leq L} \mathrm{coef}_k u_k$, with $\mathrm{coef}_l = \epsilon_l y_l - \beta_m^A \lambda_l \partial \varkappa_m / \partial \zeta_m(l)$ for a tuned carrier l of block m and $\mathrm{coef}_k = -\beta_m^A \lambda_k \partial \varkappa_m / \partial \zeta_m(k)$ for an untuned coarse carrier (a peak); its fine part has ℓ_1-norm at most $C c_{L+1}$. For $s \in S_k$ beyond $s_{\max}(L)$ (private to k among coarse carriers) $\Psi(s) = \mathrm{coef}_k v_k(s)$ up to the fine part. Put $\gamma := \langle U^* \Psi, e_1 \rangle$ and $\kappa_\Psi := \gamma/\nu_1$, and let h be the vector of signed (P4) coefficients ($h_l := 0$ for carriers without (P4)). In route V, Λ_A contains every coarse strict non-peak, so the derivative of $\varkappa_m$ with respect to a strict non-peak coordinate is small ([II, Subsection 5.6]) and the non-peak Euler share $\tilde{X}_m := \sum_{l \in \Lambda_A,\, m(l) = m} \mathrm{rt}_l \epsilon_l \lambda_l \partial \varkappa_m / \partial \zeta_m(l)$ satisfies $|\tilde{X}_m| \leq \frac{1}{4}$. Since $\sum_l \mathrm{rt}_l \epsilon_l \mathrm{coef}_l = \beta_m^A (1 - \tilde{X}_m)$ (sum over the tuned carriers of block m), one gets $\|\mathrm{coef}_{\mathrm{tuned}}\|_2 \geq c_{\mathrm{A}} := 1/(2 + CN|\Lambda_A|^{1/2}/A_{\min})$. We write $\mathrm{tail}_D := C_f 2^{-L^2} (|\Lambda_L| H(L))^{-4} + C c_{L+1}$.

Theorem 5.39 (Structure of Step-A obstructions)**.** *Let $L \geq l_f$ and let y be a near-certificate of size $\varepsilon \leq \varepsilon_* := u^3 \mathrm{Des}(L)^{-C}$ at $f_1 + \mathcal{P}$, C a large absolute constant. Then:*

1. (no positive common mode) *if $\gamma > 0$, then $\gamma \leq 2(|\varphi_y| + \bar{d}(L)\varepsilon \mathrm{Des}(L) + \mathrm{tail}_D)$, where $|\varphi_y| \leq |\Lambda_A|^{1/2} \max(x, \|b\|_2)$;*
2. (action of the generators) *at every $s \leq \bar{d}(L)$, $|c_s(\Psi)| \leq \varepsilon \mathrm{Des}(L)$ if s carries a two-sided generator, and $c_s(\Psi) \geq -\varepsilon \mathrm{Des}(L)$ if it carries a one-sided one. Hence, for every coarse carrier k:*
   - (2a) *if S_k has a private FR or ZD coordinate in $[1, \bar{d}(L)]$, then $|\mathrm{coef}_k| \leq \varepsilon \mathrm{Des}(L)^2$; if k has a pre-placed TU lever whose base bank has mass m_k, then $|\mathrm{coef}_k| \leq (\varepsilon \mathrm{Des}(L) + |\kappa_\Psi| \mathsf{m}_k) \mathrm{Des}(L)$;*

(2b) *at every thick private* $p \in S_k \cap [1, \bar{d}(L)]$*:* $|\mathrm{coef}_k v_k(p) - \kappa_\Psi a_p| \le \varepsilon \mathrm{Des}(L)$*;*
(2c) *at every* S_{th} *private* $s' \in S_k \cap [1, \bar{d}(L)]$ *carrying data:* $\mathrm{coef}_k z_{s'} \ge -(\varepsilon \mathrm{Des}(L) + 2|\kappa_\Psi| b)\mathrm{Des}(L)$*;*

(3) (robustly negative common mode) $\gamma \le -c_*$*, where*

$$c_* := c_{\mathrm{A}} \min\{1/(4\|h\|_2),\ \nu_1/(8H(L))\}$$

(with $1/(4\|h\|_2) := \infty$ *if* $h = 0$*);*

(4) (proportional thick support) *at every* $s \le \bar{d}(L)$ *with* $|a_s| \ge u/2$*:* $|\Psi(s) - \kappa_\Psi a_s| \le \varepsilon \mathrm{Des}(L)$ *while* $|\kappa_\Psi a_s| \ge c_* u/(2\nu_1)$*; so* $\Psi(s) \ne 0$*,* $s \in U_L$ *(there is no anchor), and* s *lies in the support of a coarse carrier with a robust coefficient;*

(5) $|\Psi(s)| \le \varepsilon \mathrm{Des}(L)$ *at every FR or ZD coordinate* $s \le \bar{d}(L)$*;*

(6) *if* $\varphi_y > -\rho_0$*, the sum of* $c_s(\Psi)$ *over the design-depth coordinates carrying one-sided generators is at least* $c_*/2$*.*

Proof. (2) By Lemmas 5.35(b) and 5.24, a generator at s acts on $y^T \tilde{F}$ by eff_s times $D_\Psi(s)$ (mass change at a support coordinate), $\Psi(s)$ (z-move at a free coordinate) or $\Psi(s) z_s$ (composite move), times its orientation, with $\mathrm{eff}_s \ge 1/\mathrm{Des}(L)$; in each case the action is eff_s times $c_s(\Psi)$ or $\Psi(s)$ with $|c_s(\Psi)| \le |\Psi(s)|$ off F. A near-certificate has $y^T g \ge -\varepsilon$ for all $g \in \mathcal{G}$, which gives the two inequalities. (2a): at a private FR or ZD coordinate s', $|\mathrm{coef}_k| v_k(s') = |\Psi(s')| \le \varepsilon \mathrm{Des}(L)$ and $v_k(s') \ge v_{\min}(L) \ge 1/\mathrm{Des}(L)$; a pre-placed TU bank is a private two-sided mass generator at a support coordinate of mass m_k, where $D_\Psi = \mathrm{coef}_k v_k - \kappa_\Psi \mathsf{m}_k$ sign. (2b) is the TS generator at p. (2c): $c_{s'}(\Psi) = (\mathrm{coef}_k v_k(s') - \kappa_\Psi a_{s'}) z_{s'} \ge -\varepsilon \mathrm{Des}(L)$, with $|a_{s'}| \le 2b$ at S_{th} coordinates of $f_1 + \mathcal{P}$ and $v_k(s') \ge 1/\mathrm{Des}(L)$, $v_k > 0$ on S_k.

(1) Apply (7) at $f_1 + \mathcal{P}$ with $u := \Psi$ and $D := \bar{d}(L)$: $\varphi_y = \kappa_\Psi(1 - \alpha(D)) + \sum_{s \le D} c_s(\Psi) + \sum_{s > D} \Psi(s) z_s$. By (2), every design-depth coordinate carrying a generator has $c_s \ge -\varepsilon \mathrm{Des}(L)$; the remaining ones (the last kind of Lemma 5.29(d)) have $c_s = 0$ up to the fine part, or $c_s = -\kappa_\Psi |a_s| \ge -2|\kappa_\Psi| b$; the deep sum is at least $-\mathrm{tail}_D$ (the coefficients of Ψ are at most C_f). If $\gamma > 0$, then $\kappa_\Psi(1 - \alpha(D)) \ge \kappa_\Psi \nu_1 = \gamma$ and $2\bar{d}(L)\kappa_\Psi b \le \gamma/2$, whence (1). The bound on $|\varphi_y|$ is Lemma 5.35(a) at $f_1 + \mathcal{P}$.

(3) Put $\gamma_0 := c_{\mathrm{A}}/(4\|h\|_2)$ and suppose $\gamma > -\gamma_0$; by (1), $|\gamma| \le \gamma_0$. After the preparation, the off-F or thin near coordinates of zero-data carriers are ZD and those of data-carrying carriers are S_{th} of the data sign σ_l (Lemma 5.29(c), Lemma 5.18 case (S)). So every tuned l is of one of the kinds: (i) N_l contains a ZD coordinate, or l has a pre-placed TU lever: then $|\mathrm{coef}_l| \le \varepsilon \mathrm{Des}^2 + |\kappa_\Psi| \mathsf{m}_l \mathrm{Des}$ by (2a); (ii) $S_l \cap [1, \bar{d}(L)]$ contains two thick private coordinates s, s' of robust spread: by (2b) at both, $|\kappa_\Psi||\mathfrak{a}_s - \mathfrak{a}_{s'}| \le 2\varepsilon \mathrm{Des}^2$, so $|\kappa_\Psi| \le 8\varepsilon \mathrm{Des}^3/u^2$; (iii) otherwise, N_l contains a thick p: by (2b), $\mathrm{coef}_l = \kappa_\Psi a_p / v_l(p) + O(\varepsilon \mathrm{Des}^2) = \epsilon_l \gamma h_l + O(\varepsilon \mathrm{Des}^2)$; (iv) otherwise every near coordinate of l is S_{th} of the data sign, and $\mathrm{coef}_l \sigma_l \ge -(\varepsilon \mathrm{Des} + 2|\kappa_\Psi| b)\mathrm{Des}$ by (2c). If a kind-(ii) carrier exists, κ_Ψ is tiny, which contradicts the conclusion $\kappa_\Psi \le -c_U/2$ reached below; so then no near-certificate with $\gamma > -\gamma_0$ exists. Otherwise $\|\mathrm{coef}_{(\mathrm{i})}\|_2 + \|\mathrm{coef}_{(\mathrm{iii})}\|_2 \le |\gamma| \|h\|_2 + \mathrm{small} \le c_{\mathrm{A}}/4 + \mathrm{small}$, hence $\|\mathrm{coef}_{(\mathrm{iv})}\|_2 \ge c_{\mathrm{A}}/2$. At an S_{th} near coordinate s of a kind-(iv) carrier l, $c_s(\Psi) \ge \mathrm{coef}_l \sigma_l / H(L) - 2|\kappa_\Psi| b$ when $\mathrm{coef}_l \sigma_l \ge 0$, and $\sum_l \max(\mathrm{coef}_l \sigma_l, 0) \ge c_{\mathrm{A}}/2 - \mathrm{small}$. With (2) for all other design-depth coordinates,

$$\sum_{s \le \bar{d}(L)} c_s(\Psi) \ge \frac{c_{\mathrm{A}}}{2H(L)} - \bar{d}(L)\big(\varepsilon \mathrm{Des}^2 + 2|\kappa_\Psi| b\big) - |\Lambda_A| \,\mathrm{small} \ge \frac{c_{\mathrm{A}}}{4H(L)} =: c_U.$$

By (7), $\kappa_\Psi(1-\alpha(D)) \le |\varphi_y| - c_U + \mathrm{tail}_D \le -c_U/2$, so $\kappa_\Psi \le -c_U/2$ and $\gamma = \kappa_\Psi \nu_1 \le -c_{\mathrm{A}}\nu_1/(8H(L))$. In both cases $\gamma \le -c_*$.

(4) The TS generator at s is two-sided: $|c_s(\Psi)| = |\Psi(s) - \kappa_\Psi a_s| \le \varepsilon\mathrm{Des}$, and $|\kappa_\Psi| \ge c_*/\nu_1$ by (3). If $s \notin U_L$, $\Psi(s)$ is the fine part, which is tiny. (5) FR and ZD generators are two-sided with action $\Psi(s)$ times the orientation. (6) By (7), $\sum_{s\le D} c_s(\Psi) = \varphi_y - \kappa_\Psi(1-\alpha(D)) - (\mathrm{deep}) \ge -\rho_0 + |\kappa_\Psi|\nu_1 - \mathrm{tail}_D \ge c_* - \rho_0 - \mathrm{tail}_D$; two-sided coordinates contribute at most $\bar{d}(L)\varepsilon\mathrm{Des}$ in absolute value and the last kind of Lemma 5.29(d) at most $2\bar{d}(L)|\kappa_\Psi|b$; the rest is the one-sided sum. □

Corollary 5.40 (Sufficient conditions for A-regularity)**.** *Let $L \ge l_f$, w clean, a a class, after the preparation, with $\mathcal{P}$ the TU pre-placements. The generators positively span with $\varepsilon_{\mathrm{PS}} > \varepsilon_*$ – so the stage is A-regular for every right-hand side by Lemma* 5.37(a) *– if one of the following holds:*

(a) *every tuned carrier is privately tunable;*
(b) *there is an anchor: a thick $s \le \bar{d}(L)$ outside U_L;*
(c) *some coarse carrier k (tuned, or a peak) has, in $S_k \cap [1, \bar{d}(L)]$ and private to k, a thick coordinate together with an FR or ZD coordinate (or k has a pre-placed TU lever);*
(d) *some coarse carrier k has, in $S_k \cap [1, \bar{d}(L)]$ and private to k, a thick coordinate p and an S_{th} coordinate s' carrying data, with $\operatorname{sign} a_p = z_{s'}$;*
(e) *some coarse carrier k has two thick private coordinates in $S_k \cap [1, \bar{d}(L)]$ of robust spread.*

Moreover the stage is A-regular for the given right-hand side if

(f) *every tuned carrier is of kind (i) or (iii) of the proof of Theorem* 5.39*, and $|1 - h^T\tilde{g}^A| \ge u$, or $|(1-\tilde{X}_m)\lambda_c h^T \mathrm{rt}^{(m)} + h_c h^T \tilde{g}^A| \ge u$ for a two-sided peak resource c of a block m (h_c its common-mode coefficient, $\mathrm{rt}^{(m)}$ the target vector of block m), or $\varphi_{y_+} \ge 2\rho_0 + C\mathrm{Des}(L)^{-C_A}\|\tilde{F}(f_1)\|_2$ for $y_+ := h/\|h\|_2$.*

Proof. Suppose a near-certificate of size $\varepsilon \le \varepsilon_*$ exists; Theorem 5.39 applies. (a) The private two-sided columns are e_l up to small errors and span. (b) $\Psi(s)$ is tiny at an anchor, against (4). (c) $|\mathrm{coef}_k|$ is tiny by (2a), so $\Psi(p) = \mathrm{coef}_k v_k(p)$ is tiny at the thick p, against (4). (d) By (2b) and (3), $\mathrm{coef}_k v_k(p) = \kappa_\Psi a_p + O(\varepsilon\mathrm{Des})$ with $|\kappa_\Psi a_p|$ robust and $\kappa_\Psi < 0$, so $\operatorname{sign}\mathrm{coef}_k = -\operatorname{sign} a_p$ and $|\mathrm{coef}_k|$ is robust; (2c) then forces $z_{s'} = \operatorname{sign}\mathrm{coef}_k = -\operatorname{sign} a_p$. (e) This is kind (ii) in the proof of (3): no near-certificate exists. So in (a)–(e) $\varepsilon_{\mathrm{PS}} > \varepsilon_*$. (f) Without kind-(iv) carriers the proof of (3) shows that every near-certificate has $\gamma \le -\gamma_0$. The two-sided columns form the square system $I - \tilde{g}^A h^T$ (up to small errors) bordered by the peak columns $-(1-\tilde{X}_m)\lambda_c \mathrm{rt}^{(m)} - h_c\tilde{g}^A$, and a near-certificate y is a near-null left vector of all of them: $\|y^T(I - \tilde{g}^A h^T)\|_2 \le C\varepsilon_*|\Lambda_A|^{1/2}$. By Lemma 5.38, y is within $C\varepsilon_*|\Lambda_A|^{1/2}$ of $\pm y_+$. Now $y_+^T(I - \tilde{g}^A h^T) = (1 - h^T\tilde{g}^A)h^T/\|h\|_2$ and $-\|h\|_2\, y_+^T \mathrm{col}_c = (1-\tilde{X}_m)\lambda_c h^T \mathrm{rt}^{(m)} + h_c h^T\tilde{g}^A$; since $\|h\|_2 \le |\Lambda_A|^{1/2}H(L)/\nu_1 \le \mathrm{Des}(L)^C$, either scalar being $\ge u$ gives a two-sided column g with $|y^T g| \ge u\,\mathrm{Des}(L)^{-C} - C\varepsilon_*\mathrm{Des}(L)^C > \varepsilon_*$, which is impossible. So no near-certificate exists. Otherwise $+y_+$ has $\gamma = h^T\tilde{g}^A/\|h\|_2 > 0$, excluded by (3), so the only candidate is close to $-y_+$, and it is an obstruction (Lemma 5.37(b)) only if $\varphi_{-y_+} > -\rho_0$ up to its deviation from $-y_+$, i.e. only if $\varphi_{y_+} < \rho_0 + C\mathrm{Des}(L)^{-C_A}\|\tilde{F}(f_1)\|_2$. □

Remark 5.41 (Further cases and limits of the corollary). (a) *Reduction (not a sufficient condition by itself).* If every block containing kind-(iv) carriers has a private

two-sided peak resource, no kind-(iii) carrier with nonzero target, and kind-(iv) carriers with $|\mathrm{rt}_l| \geq u$ and $\mathrm{rt}_l \epsilon_l \sigma_l$ of one sign, then every near-certificate has $\gamma \leq -\gamma_0$: in such a block the peak coefficient is $O(\varepsilon \mathrm{Des}^2)$, hence $\beta_m^A = O(\varepsilon \mathrm{Des}^2)$, and the one-signed terms $\mathrm{rt}_l \epsilon_l \mathrm{coef}_l$ are each $O(\varepsilon \mathrm{Des}^2/u)$, so $\|\mathrm{coef}_{(\mathrm{iv})}\|$ is small. A-regularity then still needs the criterion of (f). (b) *Held carriers in a degenerate direction.* A carrier held at value 0 is zero-data. If its near set contains an off-F or thin coordinate, it is ZD after the preparation and has kind (i); if its near set is thick it has kind (ii) (then there is no near-certificate) or (iii). So held carriers in a degenerate direction are not a separate obstruction: they are either realized or produce a near-certificate of Theorem 5.39. (c) *Far moves against a certificate (sketch).* This item is a sketch and is not used. A far placement at $j \in S_k$, $x \leq v_k(j) \leq x'_{\max}$, of support of sign σ acts on $\Psi(\hat{x})$ by about $\mathrm{coef}_k v_k(j)(\sigma - z_j)$; pulls (placements of sign $-\epsilon_k$) break a certificate when $\epsilon_k \mathrm{coef}_k > 0$, and placements of sign $+\epsilon_k$ would break it when $\epsilon_k \mathrm{coef}_k < 0$, provided such placements are admissible (the mirror image of the pulls) and no new certificate appears after an overshooting pre-placement. Neither point is proved here.

5.7. **The design for arbitrary base support.**

Definition 5.42 (The design $T_{\mathrm{fin}}^{\infty}$)**.** $T_{\mathrm{fin}}^{\infty}$ is the design T_{fin} of Section 4.1 (option (a)) over the diagonal base $\mu_s^\star = 2^{-2^{2^s}}$, with the following insertions at every stage L.

- (I1) After the carriers of stages $\leq L$, $s_{\max}(L)$ and $\delta_{\min}(L)$ are fixed: $d(L)$, $H(L)$, Λ_L, U_L, $\bar{d}(L)$ and $v_{\min}(L)$ as in Subsections 5.3 and 5.4.
- (I2) $\mathrm{Des}(L)$ is replaced by $\mathrm{Des}(L) \cdot \left(2^{\bar{d}(L)} \bar{d}(L) H(L) |\Lambda_L| / ((\mu^\star_{\bar{d}(L)})^2 v_{\min}(L))\right)^{C_D}$ with an absolute $C_D \geq 2$.
- (I3) The rate scheme $\omega(L)$ is enlarged by: the moduli $|a_s|$ and the rooms $1 - |z_s|$ for $s \leq \bar{d}(L)$; the minors and cofactor polynomials of the enlarged class cones (with the contact rows $(\mathrm{X1}^F)$) for all patterns and classes of level L.
- (I4) For every sub-window w of L, $\theta_0(w) := b(w)/\mathrm{Des}(L)^{C_P}$ with an absolute $C_P \geq C_D + 4$.
- (I5) In the conditions (D1), (D2) of Section 4.1 that define the admissible interval $(0, b_{X1}(w)]$ of $b(w)$, the number b is replaced by $2\mathrm{Des}(L)^C b/u(w)$ in (D1) and by $2b$ in (D2).

Lemma 5.43 (The insertions are admissible)**.** (a) *Every inserted quantity of stage L is a function of quantities fixed earlier in the recursion of Section* 4.1*, so the recursion stays well founded.* (b) *All inserted quantities are finite and positive.* (c) *No rule refers to N or to a row; the number $2\bar{d}(L)$ of new rate objects is N-free.* (d) *The insertions only enlarge* $\mathrm{Des}(L)$*, $\omega(L)$, $Q(w)$ and $n(w)$, and only lower $T_{\mathrm{lo}}(w)$, $\eta'(w)$ and $b(w)$; every window condition used in Parts II and III is monotone under these changes, the $\omega(L)+1$ sub-windows of a main stage still have pairwise disjoint bands, and for every row some sub-window of every main stage $L \geq l_f$ is clean.* (e) $2^{\bar{d}(L)+1} \leq \mathrm{Des}(L)$*,* $1/v_{\min}(L) \leq \mathrm{Des}(L)$*, and a mass change at any $s \leq \bar{d}(L)$ has normalized efficiency at least* $1/\mathrm{Des}(L)$*.*

Proof. (a) $d(L)$, $H(L)$, $\bar{d}(L)$, U_L and $v_{\min}(L)$ depend only on the carriers of stages $\leq L$ and on $\delta_{\min}(L)$; $\mathrm{Des}(L)$ on these; $\omega(L)$ on $\bar{d}(L)$ and the patterns of level L; $\theta_0(w)$ on $b(w)$ and $\mathrm{Des}(L)$; and the admissible interval of $b(w)$ on the same data as before together with $u(w) = b(w^-)$, which is fixed before $b(w)$. Both sides of the

modified (D1), (D2) are monotone in b, so the admissible b still form an interval $(0, b_{X1}(w)]$. (b) $\bar{d}(L) < \infty$ because finitely many ℓ_1 vectors have tails tending to 0; $v_{\min}(L) > 0$ is a minimum of finitely many nonzero numbers; $\mu^{\star}_{\bar{d}(L)} > 0$. (c) is clear; for p_N the objects are evaluated on the blocks present. (d) [II, Lemma 3.22] and the remarks after it: every window condition is an upper bound on rate objects, weights and windows or a lower bound on Des and $Q(w)$; the disjointness of the bands ($u(w^+) = b(w) < u(w)$) and the existence of clean sub-windows ([II, Theorem 3.19]) do not depend on the size of $\omega(L)$. (e) follows from (I2), since $\nu \le 1$. □

Lemma 5.44 (Clean dichotomy at depth $\bar{d}(L)$)**.** *Let w be clean for f. For every $s \le \bar{d}(L)$, at f: $|a_s| \le b$ or $|a_s| \ge u$, and $1 - |z_s| \le b$ or $1 - |z_s| \ge u$. Along the construction $f \to f^r \to$ Step 0 $\to$ preparation $\to$ Step A $\to$ pre-row $\to$ approximant, every $s \le \bar{d}(L)$ keeps its type in the sense of Lemma* 5.29(d)*.*

Proof. The moduli and rooms at depth $\le \bar{d}(L)$ are rate objects (I3), so cleanness gives the dichotomy. The deep raise acts beyond $J(w) > \bar{d}(L)$ and divides the moduli at depth $\le \bar{d}(L)$ by $\lambda_r \le 2$ (Lemma 5.16). Step 0 (in the variant described before Definition 5.49) closes rooms and pushes weak peaks, moving values by at most $\mathrm{Des}^C b/u \ll u$ (Definition 5.49). The preparation moves values by at most $2b$ (Lemma 5.29(b)). Step A moves TS and FR coordinates by at most $\mathrm{Des}^C W \ll u$ and moves $\mathrm{S_{th}}$ coordinates with data only upward (Lemma 5.37). The pre-row moves design-depth coordinates by at most $K_w \Pi \ll \theta_0$ and the levers by at most $\mathrm{Des}(L) K_{\mathrm{DC}} r \le \theta_0/4$ (Lemma 5.45). □

Lemma 5.45 (Lateness conditions)**.** *Let $s_1 := \max(64\rho\epsilon/\delta, e^{-1/(2T)})$, as in Proposition* 3.21*. For T small (i.e. L large) the following hold, and they are the conditions* (L-K2) *used above:*

- (L1) $s_1 \le \theta_0 T/(64\rho A_0)$ *(so no S_{th} coordinate is in the raise set W_F of Definition* 5.8*);*
- (L2) $\mathrm{Des}(L) K_{\mathrm{DC}} r \le \theta_0/4$*;*
- (L3) $C_{\mathrm{lev}} K_{\mathrm{DC}} r \le \frac{1}{2} \min(\mathrm{gap}_{\min}$*, robust margins)*;
- (L4) $\varepsilon_E \le \varepsilon_{\mathrm{DC}}/8$ *and* $C_2 K_{\mathrm{DC}}^2 r \le \frac{1}{4}$*;*
- (L5) $K_{\mathrm{DC}} s_1 \le r_0$ *and* $K_w s_1 \le r_0$*.*

Proof. $b(w)$ is bounded below by a power of $\eta'(w) = T^4/(L\,\mathrm{Des}(L))$ whose exponent and constants are bounded by powers of $\mathrm{Des}(L)$ (the definition of $b_{X1}(w)$ through (D1), (D2) and the modulus of the threshold protection), and $\mathrm{Des}(L) \le n(w) \le \log_2(1/T)$; hence $\log(1/b) \le C \log^3(1/T)$ and $\theta_0 \ge \exp(-C \log^3(1/T))$. On the other side $r \le \exp(C \log^2(1/T)) s_1$, $K_{\mathrm{DC}} \le C_{f,N} \log^C(1/T)$, $C_{\mathrm{lev}}, K_w \le \exp(C \log^2(1/T))$, $\varepsilon_E \le C_f \mathrm{Des}^C (K_w \Pi + c_{L+1})$ with $\Pi \le K_w s_1$ and $c_{L+1} \le e^{-1/T}$ by ($\mathrm{W_{exp}}$), while $\varepsilon_{\mathrm{DC}} \ge \mathrm{Des}(L)^{-C}$ and $\mathrm{gap}_{\min}$, the robust margins and r_0 are at least $\exp(-C \log^2(1/T))$. Since $s_1 \le (1 + 64\rho C/\delta) e^{-1/(2T)} \le \exp(-C' \log^3(1/T))$ for T small, all five conditions hold. □

Lemma 5.46 (Costs)**.** *Let $c_{\mathrm{flat}}(w) \ge c_f u(w)$. The companion moves satisfy, with $T = T_{\mathrm{lo}}(w)$:*

- (a) *deep raise: $\varepsilon'_r \le \theta_w T^2$ (choice of $J(w)$);*
- (b) *Step 0 and the preparation: at most $C_f \mathrm{Des}(L)^C (b/u) \log(1/b) + C_f \mathrm{Des}(L) b \log(1/b)$ (Lemma 5.29(a));*
- (c) *Step A: at most $C_f \mathrm{Des}(L)^C W \log(1/W) \le C_f \mathrm{Des}(L)^C T^3 \log(1/T)/u$;*

(d) *the pre-placements and the lowering of far coordinates in F: at most $C\,\mathrm{Des}(L)|\Lambda|x'_{\max}\log(1/T)$, i.e. at most $Cu^2T^3\log(1/T)/\mathrm{Des}(L)$.*

Writing the sum as θ'_wT^2, $\theta'_w/c_{\mathrm{flat}}(w)^2\to 0$ along main stages.

Proof. (a) is Lemma 5.16. (b)–(d): the cost of a family of moves is bounded by [II, Lemma 2.12(iv)] (base change plus $C_fc(\delta)$ with $c(\delta)\le C\epsilon\log(e/\epsilon)$ for value moves ϵ), the raises by [II, Lemma 2.15]; a pull at depth v costs at most $C_f\lambda_l v\log(e/v)$ for the z-flip, $24\lambda_l v$ for its mass and $2\,\mathrm{Des}(L)\nu v$ for the compensating bank, and $\log(e/v)\le 6\log(1/T)$ since $v\ge x\ge T^5$; pulls are taken at the smallest pullable depth in $[x,x'_{\max}]$, and lowering an in-F far coordinate costs at most $2(1+\|U\|)|a_j|\le 2(1+\|U\|)x'_{\max}$. Dividing by $c_f^2u^2T^2$: (a) gives $\theta_w/(c_fu)^2\le T^{9/10}/c_f^2\to 0$ since $\theta_w=T$ and $u\ge T^{1/20}$; (b) gives at most $\mathrm{Des}^CT^2\log^3(1/T)/u^3$; (c) at most $\mathrm{Des}^CT\log(1/T)/u^3$; (d) at most $CT\log(1/T)/c_f^2$; all tend to 0 because $b\le T^4$ and $T\le(u/(4\,\mathrm{Des}(L)))^{20}$. □

Remark 5.47. A far depth larger than $x'_{\max}$ (for instance $T^2/(L\,\mathrm{Des}(L)|\Lambda|)$) is not admissible: the zeroth-order cost of pulls and lowerings would then be of order $T^2\log(1/T)/L$, which is not $o(u^2T^2)$, and $\theta'_w/c_{\mathrm{flat}}^2$ would diverge. The window $[x,x'_{\max}]$ still has ratio at least $2^{6G(L)}$, so it contains at least three elements of every S_l.

Lemma 5.48 (Window bookkeeping)**.** *Let $C_+:=5$ and $\gamma_B(w):=\min_m M_m\,u(w)/4$, so that $c_{\mathrm{flat}}(w)\ge c_f\,u(w)$. Then*

$$K_{\mathrm{data}}(w)\le C_f^L\big(\mathrm{Des}(L)/u(w)\big)^{CL}(\log_2 n(w))^{L+1},\qquad D_{\mathrm{cl}}(w)\le(2L+1)3^{2L},$$

and for a class set $\mathcal{S}$ with $|\mathcal{S}|\ge n(w)/D_{\mathrm{cl}}(w)$,

$$\frac{|\mathcal{S}|\,c_{\mathrm{flat}}(w)}{K_{\mathrm{data}}(w)}\to\infty,\qquad\frac{K_{\mathrm{data}}(w)T_{\mathrm{hi}}(w)}{|\mathcal{S}|}\to 0$$

along main stages $L\to\infty$.

Proof. $J\le 2L$ for $L\ge l_f$ [II, Lemma 5.53], and $K_*(w)\le C_f(\mathrm{Des}/u)^C(\mathrm{Des}(L)+2\log_2(Cn(w)))^{1/2}$ by Lemma 5.16; so $K_{\mathrm{data}}=(JK_*(w)^{J+1}+C_f)K$ has the stated bound. By the window count of Section 4.1, $n(w)\ge L2^{L^3}Q(w)$ with $Q(w)\ge(4\,\mathrm{Des}(L)/u(w))^{\omega(L)+L^2+20}$. For L large, $(\mathrm{Des}/u)^{CL+1}\le Q(w)^{1/4}$ (as $L^2\ge 4(CL+1)$), $C_f^LD_{\mathrm{cl}}(w)\le 2^{L^3/2}$, and $(\log_2 n)^{L+1}\le n^{1/4}$ because $\log_2 n\ge L^3$. Hence $|\mathcal{S}|c_{\mathrm{flat}}/K_{\mathrm{data}}\ge c_fu(w)n(w)^{3/4}/(2^{L^3/2}Q(w)^{1/4})\ge c_fn(w)^{1/4}\to\infty$ (as $n(w)\ge 2^{L^3}Q(w)$ and $Q(w)^{1/4}\ge 1/u(w)$) and $K_{\mathrm{data}}T_{\mathrm{hi}}(w)/|\mathcal{S}|\le K_{\mathrm{data}}D_{\mathrm{cl}}/n(w)\le n(w)^{-1/2}\to 0$. □

5.8. **The transport theorem.** *The variant of Step 0 used at arbitrary base support.* The companion of Proposition 2.32 does not push the active weak peaks (Step 0 there consists of the closing moves only, and the active weak peaks enter the pattern's Ω under the floor convention). At arbitrary base support we use the variant of Step 0 in which, after the closing moves, every active weak peak k of a (T)-block ($\varrho_k\in[1,1+b]$) is pushed to $\varrho_k\le 1-b$ by moves of total size $\le\mathrm{Des}(L)^Cb/u$ (condition (H1*) below), so that it becomes a strict non-peak. Steps 1–3 of the proof of Proposition 2.32 hold for this variant: at the pushed row one has $P\supseteq P_{\mathrm{rob}}$ and $\Omega\subseteq Q$, so Lemma 2.6 gives $F_\Phi(r_1)\le 1+C_{**}b+\mathrm{Des}^Cb/u$, which is absorbed in δ_0 after the enlargement of the radius of (D1) by (I5) of Definition 5.42. The

pushes are needed here, and not in Proposition 2.32, for two reasons of this section: Lemma 5.37(c) computes the block scalars with the coarse statuses frozen at f_1+P, which is consistent only if the near-threshold active carriers are strict non-peaks there; and Theorem 5.39 uses $|\tilde{X}_m| \le \frac{1}{4}$, which needs every tuned carrier to be a strict non-peak of f_1 (the derivative of $\varkappa_m$ in a peak coordinate is of order one).

Orientation. The push must lower $\mathrm{vs}_k u_k(\hat{x}) = \mathrm{vs}_k \epsilon_k \mathrm{val}_k$. By [I, Lemma 8.27], at an active weak peak k the route-V data $\gamma_k = \lambda_k(\Delta\omega(k) - \Delta_m w(k))$, with $w(k) = M\mathrm{vs}_k$ and $\Delta'_m = \Delta_m M$, satisfy $\mathrm{vs}_k \Delta\omega(k) = e_k := |\omega_+(k)| + |\omega_-(k)| \ge 0$; hence $\mathrm{vs}_k\gamma_k = \lambda_k(e_k - \Delta'_m)$, that is, $\epsilon_k = \mathrm{vs}_k \operatorname{sign}(e_k - \Delta'_m)$. (The formula $\epsilon_k = -\operatorname{sign}(\Delta''_m)\mathrm{vs}_k$ holds at robust peaks, where $\Delta\omega = 0$; it can fail at a weak peak when $0 < \Delta'_m < e_k$.) A far pull at a contact of sign ϵ_k lowers val_k, and a near bank raises val_k. Hence a far pull pushes inward iff $\epsilon_k = \mathrm{vs}_k$ ($e_k > \Delta'_m$), a near bank pushes inward iff $\epsilon_k = -\mathrm{vs}_k$ ($e_k < \Delta'_m$), and if $e_k = \Delta'_m$ the peak carries no data and the composite move (P1) is available.

Definition 5.49 (The push condition (H1*))**.** At a main stage L, a clean subwindow w and a class a, condition (H1*) holds if there are admissible push moves – pulls at pullable far coordinates of depth in $[2b\bar{\vartheta}_m\Phi_k/m, \mathrm{Des}(L)^C b/u(w)]$, near banks of weak peaks with $\epsilon_k = -\mathrm{vs}_k$, resources (P1), (P3), (P4), and moves of Lemma 5.24 at design depth admissible for their data – of total size at most $\mathrm{Des}(L)^C b/u(w)$, after which every active weak peak of every (T)-block has $\varrho \le 1 - b$, the block scalars being computed with the coarse statuses frozen as in Lemma 5.37(c). In particular, (H1*) holds if the Step-0 system, with the needed decreases as right-hand side, is A-regular in the sense of Definition 5.36 with x replaced by $2b$ and $W \le \mathrm{Des}(L)^C b/u(w)$ (Brouwer argument of Lemma 5.37(c), statuses frozen); the converse need not hold, since a feasible inequality system need not contain a ball around its right-hand side.

A condition for each weak peak separately does not suffice: a (P4) push of a weak peak k changes every value at first order by $-g_{k'}h_k$ per unit, with $|h_k|$ up to $H(L)/\nu$, so with several such pushes the system on the weak-peak rows is $I - g_W h_W^T$, which is singular when $1 - h_W^T g_W = 0$; already for one peak a (P4) push moves its own value by $(1 - g_k h_k)$ per unit. If the left null vector of $I - g_W h_W^T$ has components of one sign in the inward orientation, the push inequalities are infeasible although every peak has its own resource.

Lemma 5.50 (Sufficient conditions for (H1*))**.** (H1*) *holds in each of the following cases:*

(i) *every active weak peak k of a (T)-block has a* private *push resource of the correct orientation: if $\epsilon_k = \mathrm{vs}_k$ (i.e. $e_k > \Delta'_m$), a pullable far coordinate $j \in S_k$ with $v_k(j) \in [2b\bar{\vartheta}_m\Phi_k/m, \mathrm{Des}(L)^C b/u(w)]$; if $\epsilon_k = -\mathrm{vs}_k$ ($e_k < \Delta'_m$), a near bank; if $e_k = \Delta'_m$, the composite move (P1); or, in each case, (P3);*

(ii) *the active weak peaks W_4 that are pushed by (P4) at distinct private thick near coordinates satisfy $|1 - h_{W_4}^T g_{W_4}| \ge u$; the active weak peaks W_s pushed by shared design-depth moves are such that the matrix of first-order actions of the chosen moves on the rows of W_s (private and common-mode parts) has a right inverse of norm $\le \mathrm{Des}(L)^C/u(w)$; and all other active weak peaks have private push resources as in (i).*

Proof. (i) By the orientation discussion before Definition 5.49, each listed resource moves $\mathrm{vs}_k u_k(\hat{x})$ downwards. The needed decrease is at most $2b\bar{\vartheta}_m\Phi_k/m$ in value

units (from $\varrho_k \le 1+b$ to $\varrho_k \le 1-b$). A far pull at depth v moves the value by $2v$, so the pull of depth $v_k(j) \in [2b\bar\vartheta_m\Phi_k/m, \mathrm{Des}(L)^C b/u]$ realizes the push with a move of size $\le \mathrm{Des}(L)^C b/u$; near banks and (P1), (P3) have continuous size and are used with size $\le 2b\bar\vartheta_m\Phi_k/m$. The pull mass $24\lambda_k v_k(j)$ acts on the other values only through e, with efficiency $(\mu_j^\star)^2$, which is negligible for $j > s_{\max}(L)$; banks and raises act on other values only at second order (Lemma 3.2(i) and [II, Lemma 4.15]), by at most $C\,\mathrm{Des}^C(b/u)^2 \ll b$ in relative position. So pushing each peak to $\varrho \le 1-2b$ realizes Step 0 with moves of total size $\le \mathrm{Des}(L)^C b/u$, which keeps the Łojasiewicz start inside the radius of (D1) as enlarged by (I5) of Definition 5.42; an overshoot is bounded by the same size, so floor statuses and tiny minors at r_1 move by $O(\mathrm{Des}^C b/u)$, which (I5) accounts for. (ii) For the peaks of W_4 the column of a (P4) push at a private thick near coordinate is $e_k - h_k g$ (Lemma 5.24), so their rows form the system $I - g_{W_4}h_{W_4}^T$ up to second-order terms; by the Sherman–Morrison formula $\|(I - g_{W_4}h_{W_4}^T)^{-1}\| \le 1 + \|g_{W_4}\|\|h_{W_4}\|/u$. For the peaks of W_s the system is not rank one (a shared coordinate has first-order private entries for every peak whose support contains it), and the assumed right inverse is used instead. In both cases the rows are solved exactly with moves of size $\le \mathrm{Des}^C b/u$; the private pushes of the other peaks, oriented as in (i), then absorb the first-order cross effects of these moves one-sidedly (the system is block triangular). □

Theorem 5.51 (Transport)**.** *Let the design be $T_{\mathrm{fin}}^{\infty}$, $N \ge 1$, $f \in S_{p_N^*}$ with base support finite or infinite, $g \in \mathcal{C}(f)$ and $\rho \in (0,1)$. Suppose that for infinitely many main stages L there are a clean sub-window w of L and an activity class a (thresholds $K_*(w)^i$, no cube) with at least $n(w)/D_{\mathrm{cl}}(w)$ scales such that*

(H1*) *the push condition of Definition* 5.49 *holds, and*
(H2) *Step A is A-regular after Step 0 and the preparation (Definition* 5.36*).*

Then $(f, \rho g) \in \mathrm{cl}\,\mathrm{NA}((c_0, p_N), \ell_2^2)$, along pairs whose first rows are norm attaining with finite base support; in particular (LSC-trunc)$_N$ *holds at (f, g, ρ). If the hypothesis holds for every $g \in \mathcal{C}(f)$ and every $\rho \in (0,1)$, then $f \in \mathcal{R}(p_N)$.*

Proof. Fix N, f, g, ρ and l_f (all f-constants absorbed; companions and approximants lie in the ball of Lemma 5.6). The order of choices at a stage $L \ge l_f$ of the hypothesis is: $w \to J(w), f^r \to$ classes $\to$ Step 0 $\to$ preparation $\to$ Łojasiewicz targets $\to \mathcal{P} \to$ Step A $\to$ data $\to s_1 \to N_w \to N'' \to d$-consistency levers $\to$ approximant. Put $T := T_{\mathrm{lo}}(w)$.

Step 1 (clean sub-window). By Lemma 5.43(d) the sub-windows of L have disjoint bands for the enlarged rate scheme, and w is clean for f.

Step 2 (deep raise). $f \to f^r$ beyond $J(w)$ (Lemma 5.16): hypothesis (a) of Theorem 5.12 holds at f^r with λ_r and $\varepsilon_r' \le \theta_w T^2$; tiny and robust objects stay tiny and robust with factor 2; the budget lemmas hold at f^r for the mate $g^r := g - (g(\xi^r)/q_0^r)a^r$ on the window ([II, Subsection 6.7]).

Step 3 (classes). Two-sided decompositions of g^r at f^r at every dyadic scale of w, classes with the thresholds $K_*(w)^i$; by Lemma 5.14 a class a with a set $\mathcal{S}$ of at least $n(w)/D_{\mathrm{cl}}(w)$ scales suffices, and Lemmas 5.17 and 5.18 hold on $\mathcal{S}$.

Step 4 (companion). The variant of Step 0 described before Definition 5.49, under (H1*); the preparation (Lemma 5.29); Step A by Lemma 5.37(c) under (H2). The resulting companion $f^{\#} := f^{(1a)}$ has the Step-A conclusions of Section 4.3 for this variant (exact ratios on $Z_{\varkappa}$, exact held zeros, threshold protection, robust statuses),

the types of Lemma 5.29(d), and $p^*(f^\# - f^r) \le \theta'_w T^2$ with $\theta'_w/c_{\mathrm{flat}}(w)^2 \to 0$ (Lemma 5.46). No step (2) (Lemma 5.20).

Step 5 (data). By Proposition 5.19, Lemma 5.29(c) and Lemma 5.20, the route-V data with the deep assignment on F form, for $t \in \mathcal{S}$, a window family at $f_0 := f^\#$ with cushion constant $C_+ = 5$: (U1), (U2), (U3^{inf}) (support excesses counted in ϵ), (U4^{inf}), (U5′), (U7), and $p^*(g - g_t) \le K_{\mathrm{data}} t$. At companion support coordinates outside F, (U3^{inf}) holds by Proposition 5.19(iv) at banks (the data are those of contacts of the bank's sign, plus the base term) and by Proposition 5.19(v) at pulled coordinates.

Step 6 (levers). Theorem 5.31 provides a lever system (U6^{inf}) with $K_{\mathrm{lev}} = K_{\mathrm{DC}} \le C_{f,N}\mathrm{Des}(L)^{C_0}$, under the conditions of Lemma 5.45.

Step 7 (approximant). We use T as the window parameter, and we put $s_1 := \max(64\rho\epsilon/\delta,\, e^{-1/(2T)})$. Every new window constant (K_{DC}, $K_*(w) \le (\mathrm{Des}/u)^C(\log n(w))^{1/2}$, C_+) is at most a design power times a power of $\log(1/T)$, so $K_w \le \exp(C\log^2(1/T))$ and hypothesis (i) of Proposition 3.21 holds; the violation mass, c_{fine} and W_{lev} are at most $C_f\mathrm{Des}(L)^C c_{L+1} + C_f c_{L+1}^2/T$ with $c_{L+1} \le e^{-1/T}$ by ($\mathrm{W}_{\mathrm{exp}}$), which is hypothesis (ii); choose $N_w > \bar{d}(L)$ after s_1 with $\mathrm{tail}_W \le e^{-1/T}$ and $\alpha_0(N_w) \le \theta_{\mathrm{tr}}$, where θ_{tr} satisfies the bound of Definition 5.8(ii) and is as small as (E-e) requires. Proposition 3.21 gives (VT′), (FINE), (LATE), (W-a), and Lemma 5.45 the conditions (L-K2) and $K_w s_1 \le r_0$. In (W-a) of Theorem 3.15 the constant K_J is $K_J := K_{\mathrm{lev}} = K_{\mathrm{DC}}$; hypothesis (i) of Proposition 3.21 contains "(W-b) holds", a condition on Lemma 3.10, which in this section is replaced by (DC) of the lever system (Theorem 5.31); there is no pull, so $r_{\mathrm{pull}} = 0$ and (PULL) is void. Theorem 5.10 then yields a norm-attaining approximant f'_L with finite base support, d-consistent with $f^\#$ on Ω, with the per-piece bounds. The statuses of coarse carriers at f'_L are those of the status lemma at the approximant in Section 4.3, whose proof uses only $\mathrm{nv}' = \mathrm{nv}^\#$ on Ω, the coarse peaks being clamped at both rows (their margins are at least $u/2$, far above the value moves $C_{\mathrm{lev}}K_{\mathrm{DC}}r$ and the common modes $\mathrm{Des}^C r$), and the fine carriers through their weight; none of these uses privacy of the levers.

Step 8 (averaging). Apply Theorem 5.12 along the infinitely many stages L_j, with $f_j := f^\#(L_j)$, $\lambda_j := \lambda_r(L_j)$, $\varepsilon'_j := \varepsilon'_r + p^*(f^\# - f^r) \le (\theta_w + \theta'_w)T^2$, $\mathcal{S}_j := \mathcal{S}$, $K_j := K_{\mathrm{data}}(w)$, $T'_j := T$, $s_{1,j} := s_1$. (a) holds at $f^\#$ by Step 2 and the triangle inequality (the extra $p^*(f^\# - f^r)$ is in ε'_j). (E-c) is Step 5. (E-d) is Lemma 5.48. (E-f) and (E-e): by Proposition 3.21 the quantities there are at most $e^{-1/(3T)} \le K_j T^2$, and with Lemma 5.46, $\theta_j := \theta_w + \theta'_w + e^{-1/(3T)}T^{-2}$ satisfies $\theta_j/c_{\mathrm{flat}}(w)^2 \to 0$. Hence $(f, \rho g) \in \mathrm{cl}\,\mathrm{NA}$ along the norm-attaining rows f'_{L_j}, which have finite base support. The last sentence follows by letting $\rho \to 1$. □

Corollary 5.52 (Explicit sufficient conditions)**.** (LSC-trunc)$_N$ *holds at* (f, g, ρ) *if, for infinitely many main stages* L*, at a clean sub-window and a class as in Theorem* 5.51*, one of the conditions of Lemma* 5.50 *holds and one of the conditions* (a)–(f) *of Corollary* 5.40 *holds. In particular this is the case if, for infinitely many main stages* L*, the support* F *meets no signature set* S_l*,* $l \le L$*, beyond* $s_{\max}(L)$*; then* $f \in \mathcal{R}(p_N)$*.*

Proof. The first statement is Theorem 5.51 with Lemma 5.50, Corollary 5.40 and Lemma 5.37(a). For the second: if $F \cap S_l \cap (s_{\max}(L), \infty) = \emptyset$ for all $l \le L$, every near coordinate of a coarse carrier is off F, and every far coordinate of S_l in $[x, x'_{\max}]$ is

off F and, for a data-carrying carrier, a contact with $z = \epsilon_l$ after the closing moves of Step 0 (as in Lemma 3.25(a)), hence pullable. So every tuned carrier is privately tunable ((P1) or (P2)), which is Corollary 5.40(a). Every active weak peak k has a private push of the right orientation, which is Lemma 5.50(i): if $\epsilon_k = \mathrm{vs}_k$, a far pull, since its far coordinates are contacts of sign ϵ_k and, consecutive elements of S_k having v-ratio $2^{-2^{k+1}} \geq \mathrm{Des}(L)^{-1}$, the smallest one with $v_k(j) \geq 2b\bar{\vartheta}_m\Phi_k/m$ satisfies $v_k(j) \leq \mathrm{Des}(L)2b\bar{\vartheta}_m\Phi_k/m \leq \mathrm{Des}(L)^C b/u$; if $\epsilon_k = -\mathrm{vs}_k$, a near bank, since its near coordinates are off F and contacts of sign ϵ_k; and (P1) if it carries no data. This holds for every (g, ρ). □

Remark 5.53 (Relation to a modulus condition). If at a stage the two-sided Step-A columns have a right inverse of norm at most $\mathrm{Des}(L)^{C_0}/u$, then $\varepsilon_{\mathrm{PS}} \geq u\,\mathrm{Des}(L)^{-C_0}|\Lambda_A|^{-1/2} \geq u^3\mathrm{Des}(L)^{-C_1}$, and (H2) holds by Lemma 5.37(a) once $C_A \geq C_1 + 2$. So (H2) is weaker than such a modulus condition, and the d-consistency half of any such condition is not needed at all (Theorem 5.31). A condition requiring a push resource for each weak peak separately does not imply (H1*) (the discussion before Lemma 5.50); (H1*) is the correct hypothesis.

5.9. **The residual classes.**

Definition 5.54 (Residual conditions)**.** Let $L \geq l_f$ be a main stage, w a clean sub-window of L and a an activity class with at least $n(w)/D_{\mathrm{cl}}(w)$ scales. With $\gamma_0 := c_{\mathrm{A}}/(4\|h\|_2)$ and c_* as in Theorem 5.39:

- (W*) condition (H1*) fails at (L, w, a);
- (A*/U*) (H1*) holds, and for every pre-placement $\mathcal{P}$ there is a near-certificate y of size $< 4u^3\mathrm{Des}(L)^{-C_A}$ with $\varphi_y > -\rho_0$ and $\gamma \leq -c_*$; the certificate is of *type* (A*) if $\gamma \leq -\gamma_0$ and of *type* (U*) if $-\gamma_0 < \gamma \leq -c_{\mathrm{A}}\nu_1/(8H(L))$.

The type may depend on the pre-placement; this is why the two types form one condition.

Proposition 5.55 (Necessary conditions for not being covered)**.** *Let the design be $T^{\infty}_{\mathrm{fin}}$, $N \geq 1$, $f \in S_{p^*_N}$, $g \in \mathcal{C}(f)$ and $\rho \in (0, 1)$. If the hypothesis of Theorem* 5.51 *fails for (f, g, ρ), then for all but finitely many main stages L, at every clean sub-window w of L and every class a with at least $n(w)/D_{\mathrm{cl}}(w)$ scales,* (W*) *or* (A*/U*) *holds. In the case* (A*/U*) *the near-certificates have all the properties* (1)–(6) *of Theorem* 5.39*, none of the conditions* (a)–(e) *of Corollary* 5.40 *holds, and:*

- (i) *the thick support at design depth is approximately proportional to the certificate's carrier combination (error $\varepsilon\mathrm{Des}(L)/|\kappa_\Psi|$; here a_s are the moduli at $f_1 + P$, i.e. those of f at design depth divided by λ_r, up to the Step-0 and preparation moves): $|a_s - \Psi_y(s)/\kappa_\Psi| \leq \varepsilon\mathrm{Des}(L)/|\kappa_\Psi|$ for every $s \leq \bar{d}(L)$ with $|a_s| \geq u/2$, with one common $\kappa_\Psi \leq -c_*/\nu_1$, and every such s lies in U_L;*
- (ii) *for a certificate of type* (A*) *the coefficients of the carriers with only (P4) resources are $\epsilon_l\gamma h_l$ up to $O(\varepsilon\mathrm{Des}^2)$; for a certificate of type* (U*) *the coefficients of the carriers of kind* (iv) (*all near coordinates S_{th} of the data sign*) *have ℓ_2-norm at least $c_{\mathrm{A}}/2$ and oriented signs $\mathrm{coef}_l\sigma_l \geq -O(\varepsilon\mathrm{Des}^2)$;*
- (iii) *the one-sided generators carry a total action at least $c_*/2$.*

Proof. If the hypothesis of Theorem 5.51 fails, then for all but finitely many main stages, at every clean sub-window and admissible class, (H1*) or (H2) fails. If (H2)

fails, the stage is not A-regular for any $\mathcal{P}$, and Lemma 5.37(b) gives for every $\mathcal{P}$ a near-certificate of size $< 4u^3\mathrm{Des}^{-C_A} \le \varepsilon_*$ (for $C_A \ge C+2$) with $\varphi_y > -\rho_0$. Theorem 5.39 applies: (3) gives $\gamma \le -c_*$, so (A*/U*) holds, the type of each certificate being (A*) or (U*) according to whether $\gamma \le -\gamma_0$; (4) gives (i); the proof of (3) gives (ii) (in case $\gamma > -\gamma_0$ the kind-(iv) coefficients carry ℓ_2-mass $\ge c_{\mathrm{A}}/2$, and in case $\gamma \le -\gamma_0$ the kind-(iii) coefficients are $\epsilon_l \gamma h_l + O(\varepsilon \mathrm{Des}^2)$); (6) gives (iii). If one of Corollary 5.40(a)–(e) held, there would be no near-certificate of size $\le \varepsilon_*$. □

Remark 5.56 (Status of the residual). (a) Proposition 5.55 gives necessary conditions only; the converse is not claimed (Lemma 5.37(b) is one-directional, and a certificate for one pre-placement need not obstruct another). It is not shown that rows satisfying (W*) or (A*/U*) at all large main stages exist, nor that such rows have a mate g with $(f, \rho g) \notin \mathrm{cl}\,\mathrm{NA}$. No counterexample to density is known. (b) The certificates of (A*/U*) are sign (cone) obstructions, not rank obstructions: the d-consistency system has no such obstruction after the preparation (Theorem 5.31), but the preparation cannot help Step A, because the targets of Step A are ratios fixed by the Łojasiewicz step at the displacement scale $x \ge \mathrm{Des}(L)b$, not values of the row; moves that keep the tiny objects tiny have size at most b, so data-carrying contacts and thin coordinates give only one-sided resources at the scale x. (c) Item (i) is an infinite-depth form of the structure of [II, Lemma 6.39]: at every large stage the thick part of a at design depth is approximately proportional (error $\varepsilon\mathrm{Des}(L)/|\kappa_\Psi|$) to a fixed combination of carrier vectors with a robustly negative common mode. (d) *Heuristic directions (not used).* If the exact set $Z_\varkappa$ of the Łojasiewicz step has positive dimension at the selected point, the targets might be chosen so that $\varphi_y \le 0$ for the finitely many near-certificates, which would remove (A*/U*); placements of support of the sign $+\epsilon_k$ at far coordinates (the mirror image of the pulls) might break certificates as in Remark 5.41(c); and for (W*), raising another robust peak of the same block raises $\bar\vartheta_m$ and lowers ϱ_k, an additional push resource. None of this is proved. (e) Statements of earlier drafts that are *not* used and are not true as stated: a bank placed with an offset at the row where the d-consistency system is solved is not a two-sided lever along a degenerate direction (Lemma 5.25); a per-peak push condition does not imply (H1*); and descriptions of the non-covered class by carrier types are necessary conditions only, not characterizations.

Remark 5.57 (Dependence on earlier results). Theorems 5.10, 5.12, 5.31, 5.51 and the results of Subsections 5.3–5.6 are proved here modulo the cited results of Parts I and II and of Sections 2–4.3, in particular the route-V companion of Section 4.3 (Step 0, the Łojasiewicz step and its threshold protection, the status lemma at the approximant) and the window count of Section 4.1. These results of Sections 2–4 are used for $T_{\mathrm{fin}}^{\infty}$ through the base independence of the window theorems ([II, Lemma 3.22]; the base enters only through the bank factor in Des); this was not re-verified line by line for the base $\mu^\star$.

Sketch, not verified. We expect Theorem 4.25 to hold for $T_{\mathrm{fin}}^{\infty}$. The insertions are covered by Lemma 5.43(d). The change of base $\mu \to \mu^\star$ enters Section 4 only through (B3), (W2), (W6) of Lemma 4.10 and the tuning lemma behind Proposition 4.11; each has a $\mu^\star$-analogue in Part II (where, however, [II, Lemma 4.15] is formed with $B^\star$ at the third element beyond $s_{\max}$, while T_{fin}'s Des uses the first),

but this transfer has not been written out or verified. If, in addition, Theorem 4.25 holds for $T_{\mathrm{fin}}^{\infty}$, then, combined with [II, Proposition 6.48], the results of this section give: if every (f, g, ρ) with infinite base support satisfies the hypothesis of Theorem 5.51, then $\mathrm{NA}((c_0, p_N), \ell_2^2)$ is dense for the design $T_{\mathrm{fin}}^{\infty}$; this hypothesis is not verified for all rows.

6. Status and open problems

6.1. **What is proved.** We collect the main results with their hypotheses.

(1) *Finite base support* (Theorem 4.25). For the operator T_{fin} of Section 4.1, which does not depend on N, satisfies (T-b)–(T-d) and, with option (b), the conclusion of Martín's Lemma B, while with option (a) it has norm $c_1 < 1$ (Theorem 4.7): with option (a), for every $N \geq 1$ every first row $f \in S_{p_N^*}$ with finite base support lies in $\mathcal{R}(p_N)$. With option (b) the same holds for $N = 1$, and for $N \geq 2$ under $\varrho_1(f) \neq 1$.

(2) *Density criterion* (Theorem 4.26, Corollary 4.27). With option (a), for every $N \geq 1$: $\mathrm{NA}((c_0, p_N), \ell_2^2)$ is dense if and only if $(\mathrm{LSC\text{-}trunc})_N$ holds; and $(\mathrm{LSC\text{-}trunc})_N$ for infinitely many N implies density of $\mathrm{NA}((c_0, p), \ell_2^2)$ for the Martín-type norm p built with T_{fin}.

(3) *Tools valid beyond* T_{fin}. Status coherence for generic weights (Lemmas 2.22 and 2.24); exactification with threshold protection without (KN), for every design satisfying the QC requirements (Proposition 2.32); exact d-consistency by levers (Lemma 3.10); the uniform engineered bound and averaging at the norm-attaining approximant (Theorems 3.15 and 3.17); the lowering transfer (Lemma 5.2); and the engineered bound and windowed recovery at arbitrary base support (Theorems 5.10 and 5.12). Each is stated with the hypotheses under which it is proved.

(4) *Infinite base support* (Theorem 5.51, Corollary 5.52, Proposition 5.55). For $T_{\mathrm{fin}}^{\infty}$, $(\mathrm{LSC\text{-}trunc})_N$ holds at every (f, g, ρ) satisfying (H1*) and (H2) at a clean sub-window and a class with at least $n(w)/D_{\mathrm{cl}}(w)$ scales, at infinitely many main stages, and the triples not covered satisfy (W*) or (A*/U*) at almost all main stages (each result checked by one referee).

6.2. **Open problems.**

Problem 6.1 (Density for the designed operator)**.** Does $(\mathrm{LSC\text{-}trunc})_N$ hold for T_{fin} (option (a)) and some, or every, $N \geq 1$? Equivalently (Theorem 4.26): is $\mathrm{NA}((c_0, p_N), \ell_2^2)$ dense?

Problem 6.2 (The Step-A residual)**.** For $T_{\mathrm{fin}}^{\infty}$, are there triples (f, g, ρ) that satisfy (A*/U*) of Definition 5.54 at every clean sub-window of all large main stages? If so, is $(f, \rho g) \in \mathrm{cl}\,\mathrm{NA}((c_0, p_N), \ell_2^2)$ for such triples? A positive answer to the second question for all such triples, together with a positive answer to Problem 6.3, would give $(\mathrm{LSC\text{-}trunc})_N$ for $T_{\mathrm{fin}}^{\infty}$.

Problem 6.3 (The push residual)**.** Does the push condition (H1*) of Definition 5.49 hold at every row, or can the weak-peak pushes of Step 0 be dispensed with? By the discussion before Lemma 5.50 a condition for each weak peak separately does not suffice; Lemma 5.50 gives sufficient conditions.

Problem 6.4 (Option (b) at the first stage)**.** For T_{fin} with option (b) and $N \geq 2$, is every first row with finite base support and $\varrho_1(f) = 1$ recoverable? The question

is whether the rational weight $\Phi_1 = 2^{-3}$ can be a critical value of the floor function of stage 1 on a stratum of an exact zero set, identically in the transcendental parameters (Remark 4.28(b)).

Problem 6.5 (Other operators)**.** Does Master Theorem VI hold for an arbitrary admissible operator, in particular for the operators produced by Martín's Lemma B in [8]? The proofs here use genericity of the weights, the window structure and the design factor of T_{fin} in an essential way.

Problem 6.6 (Property B)**.** Does ℓ_2^2 have Lindenstrauss' property B [7]? By [I, Section 9], a negative answer to Problem 6.1 for one N would give a negative answer here.

6.3. **Remarks on the status.**

Remark 6.7 (The two designs)*.* Section 4 uses T_{fin} over the base $\mu_s = 2^{-s^2-1}$, and Section 5 uses $T_{\mathrm{fin}}^{\infty}$, which is T_{fin} with option (a) over the doubly exponential base $\mu_s^\star = 2^{-2^{2^s}}$ of [II, Definition 3.1] and with the insertions of Definition 5.42. The doubly exponential base is needed at infinite base support for the reason given in [II, Remark 3.3]; the base μ_s is used at finite base support because the tuning and bank estimates of the transfer list were re-derived for it. Whether Master Theorem VI, and hence the density criterion, hold for $T_{\mathrm{fin}}^{\infty}$ is not proved here. The monotonicity argument of Remark 5.57 covers the insertions (I1)–(I5). It does not cover the change of base from μ to $\mu^\star$, which enters the proof of Theorem 4.25 through (B3), (W2), (W6) of Lemma 4.10 and the tuning lemma behind Proposition 4.11, all verified for μ only. We expect the transfer, since each item has a $\mu^\star$-analogue with $B_{\mu^\star}$, but it has not been written out or verified.

Remark 6.8 (Later observations, not verified and not used)*.* The following observations come from an analysis made after the results of Section 5 were refereed; they have not been independently checked, and nothing in this paper depends on them. (a) *Sign of the data at weak peaks.* The sign of the data at an active weak peak, $\epsilon_k = \mathrm{vs}_k \operatorname{sign}(e_k - \Delta'_m)$ by [I, Lemma 8.27], is now used in Definition 5.49 and Lemma 5.50; see the discussion before Definition 5.49. (b) *Heuristic.* The same analysis suggests that the weak-peak pushes of Step 0 can be dropped if the block scalars are computed with all weak peaks frozen as non-peaks, which would make (W*) empty and leave the Step-A residual (A*/U*) as the only obstruction. This is not proved here.

Remark 6.9 (Evidence)*.* Nothing proved in Parts I–III points to a counterexample to density; none of the verifications, and none of the finite numerical models used as sanity checks (the clamp equation, the α-mass identity, the bank equations of the tuning lemma, the lowering transfer, the solvability of the prepared d-consistency system), produced one. Equally, nothing proved here settles Problem 6.1 for any N.

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